MCK EDUCATION CENTER

MANDALAY · MYANMAR

GRADE 12

MATHEMATICS

Old Questions with Solutions

Chapter 1 – 11 · 1 / 2 / 3 / 5 Marks

01 Complex Numbers

02 Mathematical Induction

03 Analytical Solid Geometry

04 Vector Algebra

05 Permutations and Combinations

06 Conic Sections

07 Trigonometric Functions

08 Logarithmic and Exponential Functions

09 Application of Differentiation

10 Integration

11 Application of Integration

PREPARED BY

Mr. Phyo Min Zaw (M.C.Sc)

930 QUESTIONS · ANSWERS INCLUDED

CH.01 Complex Numbers

Notes · Questions with Answers

NOTES

imaginary unit i

$i^2=-1$

Conjugate of z

$\bar{z}=x-y\,i=(x,-y)$

Division of Complex Numbers

$$z\bar{z} = (x+yi)(x-yi)$$
$$= x^2 + y^2$$

Let $z_1 = x_1 + y_1 i$ and $z_2 = x_2 + y_2 i$. We will calculate $\frac{z_1}{z_2}$, $(z_2 \neq 0)$ as follows:

$$\frac{z_1}{z_2} = \frac{z_1}{z_2} \cdot \frac{\bar{z_2}}{\bar{z_2}}$$
$$= \frac{(x_1 + y_1 i)}{x_2 + y_2 i} \cdot \frac{x_2 - y_2 i}{x_2 - y_2 i}$$
$$= \frac{x_1 x_2 + y_1 y_2 + (y_1 x_2 - x_1 y_2)i}{x_2^2 + y_2^2}$$
$$\frac{z_1}{z_2} = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + \frac{(y_1 x_2 - x_1 y_2)i}{x_2^2 + y_2^2}$$

absolute value of z , |z|=r

$$r=\sqrt{x^2+y^2},$$
$x=r\cos\theta,\quad y=r\sin\theta$

$$\cos\theta = \frac{x}{r},$$
$$\sin\theta = \frac{y}{r}$$

Trigonometric form of z

$z=r(\cos\theta+i\sin\theta)$

Polar form of z

$z=(r,\theta)$

Product Rule

$$z_1 z_2 = r_1(\cos\theta_1 + i\sin\theta_1)\, r_2(\cos\theta_2 + i\sin\theta_2)$$
$$= r_1 r_2 ((\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2))$$
$$= r_1 r_2 (\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2))$$
$$z_1 z_2 = r_1 r_2 (\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2))$$

Multiplicative Inverse

$$z^{-1} = \frac{1}{x^2+y^2}(x - yi)$$
$$= \frac{1}{r^2} r(\cos\theta - i\sin\theta)$$
$$= \frac{1}{r}(\cos\theta - i\sin\theta)$$
$$z^{-1} = \frac{1}{r}(\cos(-\theta) + i\sin(-\theta))$$

Division Rule

$$\frac{z_1}{z_2} = z_1 z_2^{-1}$$
$$= r_1(\cos\theta_1 + i\sin\theta_1)\,\frac{1}{r_2}(\cos(-\theta_2) + i\sin(-\theta_2))$$
$$\frac{z_1}{z_2} = \frac{r_1}{r_2}(\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2))$$

Powers of complex numbers

$z^n = r^n(\cos n\theta + i\sin n\theta)$, where n is an integer

the roots of z

$$w_k = \sqrt[n]{r}\left(\cos\frac{\theta+2k\pi}{n} + i\sin\frac{\theta+2k\pi}{n}\right),\ k=0,1,2,\ldots,n-1$$

1 MARK

1.1

Q1. The given complex number $i^9 + i^{19}$ in a + bi form is [25-Q]

A. 3 + 4i **B.** i **C.** 3i **D.** 0

Ans.

$i^9 = (i^4)^2 \cdot i = i$
$i^{19} = (i^4)^4 \cdot i^3 = -i$
$i^9 + i^{19} = i - i = 0$

Answer: D

1.2

Q2. If x + 3 + (y − 2)i = 5 + 2i, find the values of x and y. [S-Q, 25-Q]

A. x = 8 and y = 4 **B.** x = 2 and y = 4 **C.** x = 2 and y = 0 **D.** x = 8 and y = 0

Ans.
Equate real parts and imaginary parts:

$x + 3 = 5 \rightarrow x = 2$
$y - 2 = 2 \rightarrow y = 4$

Answer: B (x = 2, y = 4)

Q3. What is the value of x + y if (2x + y) + i(y − 5) = 0? [25-Q]

A. −5 **B.** $-\frac{5}{2}$ **C.** $\frac{5}{2}$ **D.** −5

Ans.
Real part = 0 and imaginary part = 0:

$y - 5 = 0 \rightarrow y = 5$
$2x + y = 0 \rightarrow 2x = -5 \rightarrow x = -\frac{5}{2}$

$x + y = -\frac{5}{2} + 5 = \frac{5}{2}$

Answer: C

Q4. If $z_1 = (2, -3)$ and $z_2 = (-2, 1)$, then $z_1 z_2$ is [26-Q]

A. (1, 8) **B.** (1, −8) **C.** (−1, 8) **D.** (−1, −8)

Ans.
$(x_1, y_1)(x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + y_1x_2)$

$z_1z_2 = (2)(-2) - (-3)(1),\ (2)(1) + (-3)(-2)$
$= (-4 + 3,\ 2 + 6)$
$= (-1, 8)$

Answer: C

1.3

Q5. If $z_1 = a + 3i$, $z_2 = -2 - 3i$ and $z_1z_2 = 7 - 9i$, then a is [S-Q, 24-Q]

A. 0 **B.** 1 **C.** 2 **D.** 4

Ans.

$z_1z_2 = (a + 3i)(-2 - 3i)$
$= -2a - 3ai - 6i - 9i^2$
$= (-2a + 9) + (-3a - 6)i$

Compare with 7 − 9i: −2a + 9 = 7, so a = 1. (Check: −3(1) − 6 = −9 ✓)
Answer: B

Q6. If $z_1 = a - 3i$, $z_2 = -2 + 3i$ and $z_1z_2 = 7 + 9i$, then a is [25-Q]

A. 1 **B.** −1 **C.** 3 **D.** 4

Ans.

$z_1z_2 = (a - 3i)(-2 + 3i)$
$= -2a + 3ai + 6i - 9i^2$
$= (-2a + 9) + (3a + 6)i$

−2a + 9 = 7, so a = 1. (Check: 3(1) + 6 = 9 ✓)
Answer: A

Q7. If $z_1 = -2 + ai$, $z_2 = 1 - 2i$ and $z_1z_2 = 4 + 7i$, then a is [24-Q]

A. 0 **B.** 1 **C.** 2 **D.** 3

Ans.

$z_1z_2 = (-2 + ai)(1 - 2i)$
$= -2 + 4i + ai - 2ai^2$
$= (-2 + 2a) + (4 + a)i$

−2 + 2a = 4, so a = 3. (Check: 4 + 3 = 7 ✓)
Answer: D

Q8. If $z_1 = 1 - bi$, $z_2 = 2 + i$ and $\overline{z_1} + \overline{z_2} = 3 + 2i$, then b is [25-Q]

A. 2 **B.** 3 **C.** 4 **D.** 5

Ans.

$\overline{z_1} = 1 + bi$, $\overline{z_2} = 2 - i$

$\overline{z_1} + \overline{z_2} = 3 + (b - 1)i = 3 + 2i$

$b - 1 = 2$, so $b = 3$.

Answer: B

Q9. If $\overline{z} = 2 - 3i$, then $z^2 - (\overline{z})^2$ is [25-Q]

A. −24 **B.** −24i **C.** 24 **D.** 24i

Ans.

$z = 2 + 3i$, so $z - \overline{z} = 6i$ and $z + \overline{z} = 4$.

$z^2 - (\overline{z})^2 = (z - \overline{z})(z + \overline{z})$

$= (6i)(4) = 24i$

Answer: D

Q10. If $z_1 = 1 - 3i$, $z_2 = -3 - bi$ and $z_1\overline{z_2} = 9 + 13i$, then b is [24-Q]

A. 2 **B.** 3 **C.** 4 **D.** 5

Ans.

$\overline{z_2} = -3 + bi$

$z_1\overline{z_2} = (1 - 3i)(-3 + bi)$

$= -3 + bi + 9i - 3bi^2$

$= (-3 + 3b) + (b + 9)i$

$b + 9 = 13$, so $b = 4$. (Check: $-3 + 12 = 9$ ✓)

Answer: C

Q11. If $z_1 = x + yi$ where x and y are real numbers, $z_2 = 4 + 5i$, $z_3 = 1 + 3i$ and $z_1 = \overline{z_2}z_3$, then y is [25-Q]

A. 7 **B.** 8 **C.** 9 **D.** 10

Ans.

$\overline{z_2} = 4 - 5i$

$z_1 = (4 - 5i)(1 + 3i) = 4 + 12i - 5i - 15i^2$

$= 19 + 7i$

So $x = 19$ and $y = 7$.

Answer: A

Q12. If $z_1 = 2 + i$, $z_2 = 3 - i$ and $\overline{z_1z_2}$ is [24-Q]

A. −7 + i **B.** 7 + i **C.** 7 − i **D.** −7 − i

Ans.

$z_1z_2 = (2 + i)(3 - i) = 6 - 2i + 3i - i^2$

$= 7 + i$

$\overline{z_1z_2} = 7 - i$

Answer: C

Q13. The complex number $\overline{(-2 + 3i)(5 + 2i)}$ lies in [26-Q]

A. first quadrant. **B.** second quadrant. **C.** third quadrant. **D.** fourth quadrant.

Ans.

$(-2 + 3i)(5 + 2i) = -10 - 4i + 15i + 6i^2$

$= -16 + 11i$

conjugate $= -16 - 11i$

$x < 0$ and $y < 0$, so it lies in the third quadrant.

Answer: C

Q14. If $z^{-1} = \sqrt{3} + i$ then the complex number z is equal to [25-Q]

A. $\frac{\sqrt{3}}{2} + \frac{1}{2}i$ **B.** $\frac{\sqrt{3}}{2} - \frac{1}{2}i$ **C.** $\frac{\sqrt{3}}{4} - \frac{1}{4}i$ **D.** $\frac{\sqrt{3}}{4} + \frac{1}{4}i$

Ans.

$z = \frac{1}{\sqrt{3} + i} = \frac{\sqrt{3} - i}{(\sqrt{3} + i)(\sqrt{3} - i)}$

$= \frac{\sqrt{3} - i}{3 + 1} = \frac{\sqrt{3}}{4} - \frac{1}{4}i$

Answer: C

Q15. If $z_1 = 8 - 9i$, $z_2 = -1 - bi$ and $\frac{z_1}{z_2} = 2 + 5i$, then b is [25-Q]

A. 1 **B.** 2 **C.** 3 **D.** 4

Ans.

From $\frac{z_1}{z_2} = 2 + 5i$: $z_1 = (2 + 5i)z_2$

$z_1 = (2 + 5i)(-1 - bi)$

$= -2 - 2bi - 5i - 5bi^2 = (-2 + 5b) + (-2b - 5)i$

Compare with $z_1 = 8 - 9i$: $-2 + 5b = 8$, so $b = 2$. (Check: $-2(2) - 5 = -9$ ✓)

Answer: B

Q16. If $z_1 = 6 - 17i$, $z_2 = 3 - bi$ and $\frac{z_1}{z_2} = 4 - 3i$, then b is [26-Q]

A. 1 **B.** 2 **C.** 3 **D.** 4

Ans.

From $\frac{z_1}{z_2} = 4 - 3i$: $z_1 = (4 - 3i)z_2$

$z_1 = (4 - 3i)(3 - bi)$

$= 12 - 4bi - 9i + 3bi^2 = (12 - 3b) + (-4b - 9)i$

Compare with $z_1 = 6 - 17i$: $12 - 3b = 6$, so $b = 2$. (Check: $-4(2) - 9 = -17$ ✓)

Answer: B

Q17. If $z_1 = a + bi$, $z_2 = 3 + i$ and $\frac{z_1}{z_2} = \frac{9}{10} + \frac{7}{10}i$ then $a + b$ is [25-Q]

A. 4 **B.** 5 **C.** 6 **D.** 7

Ans.

$z_1 = (\frac{9}{10} + \frac{7}{10}i)(3 + i)$

$= \frac{27}{10} + \frac{9}{10}i + \frac{21}{10}i + \frac{7}{10}i^2$

$= \frac{20}{10} + \frac{30}{10}i = 2 + 3i$

$a = 2$, $b = 3$, so $a + b = 5$.

Answer: B

Q18. If $z_1 = a + 3i$, $z_2 = -2 - 3i$ and $\overline{(\frac{z_1}{z_2})} = 1 + 3i$ then a is [25-Q]

A. 11 **B.** 10 **C.** −11 **D.** 10

Ans.

Take the conjugate of both sides: $\frac{z_1}{z_2} = 1 - 3i$.

$z_1 = (1 - 3i)(-2 - 3i)$

$= -2 - 3i + 6i + 9i^2$

$= -11 + 3i$

Compare with $a + 3i$: $a = -11$.

Answer: C

1.4

Q19. The trigonometric form of $z = -1$ with $-\pi < \theta \le \pi$ is [5-Q]

A. $\cos\pi + i\sin\pi$ **B.** $\cos\pi - i\sin\pi$ **C.** $\cos\pi + \sin\pi$ **D.** $\cos\pi - \sin\pi$

Ans.

$x = -1$, $y = 0$: $r = 1$ and the point lies on the negative real axis, so $\theta = \pi$.

$z = \cos\pi + i\sin\pi$

Answer: A

Q20. The trigonometric form of $z = -i$ with $-\pi < \theta \le \pi$ is [24-Q]

A. $\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2})$ **B.** $\cos\frac{\pi}{2} - i\sin\frac{\pi}{2}$ **C.** $\cos\pi + \sin\pi$ **D.** $\cos\pi - \sin\pi$

Ans.

$x = 0$, $y = -1$: $r = 1$ and the point lies on the negative imaginary axis, so $\theta = -\frac{\pi}{2}$.

$z = \cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2})$

Answer: A

Q21. The trigonometric form of $z = 1 + i$ is [24, 26-Q]

A. $\sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$ **B.** $\sqrt{2}(\cos\frac{\pi}{4} - i\sin\frac{\pi}{4})$ **C.** $2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$ **D.** $2(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3})$

Ans.

$r = \sqrt{1 + 1} = \sqrt{2}$

$\cos\theta = \frac{1}{\sqrt{2}}$, $\sin\theta = \frac{1}{\sqrt{2}}$ → $\theta = \frac{\pi}{4}$

$z = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$

Answer: A

Q22. If $z_1 = (r, \theta) = (2, \pi)$ and $z_2 = (r, \theta) = (4, \frac{\pi}{2})$, then the value of conjugate of $\frac{z_2}{z_1}$ is [25-Q]

A. 2i **B.** −2i **C.** i **D.** −i

Ans.

$\frac{z_2}{z_1} = \frac{4}{2}(\cos(\frac{\pi}{2} - \pi) + i\sin(\frac{\pi}{2} - \pi))$

$= 2(\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2}))$

$= 2(0 - i) = -2i$

Conjugate of −2i is 2i.

Answer: A

Q23. The value of the complex number $(1 + i)^{20}$ is [25-Q]

A. 1024 **B.** −1025 **C.** 1025 **D.** −1024

Ans.

$1 + i = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$. By De Moivre:

$(1+i)^{20} = (\sqrt{2})^{20}(\cos 5\pi + i\sin 5\pi)$

$= 2^{10}(-1+0) = -1024$

Answer: D

2 MARKS

1.2

Q1. Compute $(0, 1)(1, 0) + (1, 0)(0, 1)$. [24-Q]

Ans.

$(0,1)(1,0) = (0\cdot1 - 1\cdot0,\ 0\cdot0 + 1\cdot1) = (0, 1)$

$(1,0)(0,1) = (1\cdot0 - 0\cdot1,\ 1\cdot1 + 0\cdot0) = (0, 1)$

$(0,1) + (0,1) = (0, 2)$

Answer: (0, 2)

Q2. Compute $(0, 2)(1, -4)(1, 1)$. [25-Q]

Ans.

$(0,2)(1,-4) = (0 - (2)(-4),\ (0)(-4) + (2)(1)) = (8, 2)$

$(8,2)(1,1) = (8 - 2,\ 8 + 2) = (6, 10)$

Answer: (6, 10)

1.3

Q3. Calculate $\frac{2-3i}{3-i}$. [24-Q]

Ans.

$$\frac{(2-3i)}{(3-i)} = \frac{(2-3i)(3+i)}{(3-i)(3+i)} = \frac{9-7i}{10} = \frac{9}{10} - \frac{7}{10}i$$

Answer: $\frac{9}{10} - \frac{7}{10}i$

Q4. Calculate $\frac{2+3i}{3+i}$. [26-Q]

Ans.

$$\frac{(2+3i)}{(3+i)} = \frac{(2+3i)(3-i)}{(3+i)(3-i)} = \frac{9+7i}{10} = \frac{9}{10} + \frac{7}{10}i$$

Answer: $\frac{9}{10} + \frac{7}{10}i$

Q5. If $z_1 = 1 + \sqrt{3}i$, $z_2 = -1 + i$, find $\frac{z_1}{z_2}$. [26-Q]

Ans.

$$\frac{(1+\sqrt{3}i)}{(-1+i)} = \frac{(1+\sqrt{3}i)(-1-i)}{(-1+i)(-1-i)} = \frac{-1-i-\sqrt{3}i-\sqrt{3}i^2}{2} = \frac{\sqrt{3}-1}{2} - \frac{\sqrt{3}+1}{2}i$$

Answer: $\frac{\sqrt{3}-1}{2} - \frac{\sqrt{3}+1}{2}i$

Q6. Consider $w = \frac{z_1}{z_2}$, where $z_1 = 1 + \sqrt{3}i$ and $z_2 = 2 + 2i$. Find w in the form a + bi. [5-Q]

Ans.

$$w = \frac{(1+\sqrt{3}i)}{(2+2i)} = \frac{(1+\sqrt{3}i)(2-2i)}{(2+2i)(2-2i)} = \frac{2\sqrt{3}+2+2\sqrt{3}-2i}{8} = \frac{\sqrt{3}+1}{4} + \frac{\sqrt{3}-1}{4}i$$

Answer: $\frac{\sqrt{3}+1}{4} + \frac{\sqrt{3}-1}{4}i$

Q7. Consider $w = \frac{z_1}{z_2}$, where $z_1 = 3 + 3\sqrt{2}i$ and $z_2 = 2 - 2i$. Find w in the form a + bi. [24-Q]

Ans.

$$w = \frac{(3+3\sqrt{2}i)}{(2-2i)} = \frac{(3+3\sqrt{2}i)(2+2i)}{(2-2i)(2+2i)} = \frac{6-6\sqrt{2}+6\sqrt{2}+6i}{8} = \frac{3-3\sqrt{2}}{4} + \frac{3\sqrt{2}+3}{4}i$$

Answer: $\frac{3-3\sqrt{2}}{4} + \frac{3\sqrt{2}+3}{4}i$

Q8. Consider $w = \frac{z_1}{z_2}$, where $z_1 = 2 - 2\sqrt{2}i$ and $z_2 = 3 + 3\sqrt{2}i$. Find w in the form a + bi. [25-Q]

Ans.

$$w = \frac{(2-2\sqrt{2}i)}{(3+3\sqrt{2}i)} = \frac{(2-2\sqrt{2}i)(3-3\sqrt{2}i)}{(3+3\sqrt{2}i)(3-3\sqrt{2}i)} = \frac{-6-12\sqrt{2}i}{27} = -\frac{2}{9} - \frac{4\sqrt{2}}{9}i$$

Answer: $-\frac{2}{9} - \frac{4\sqrt{2}}{9}i$

Q9. Let $w = \frac{z_2}{z_1}$, where $z_1 = 2 - \sqrt{2}i$ and $z_2 = 3 + 3\sqrt{2}i$. Find w in the form of w = a + bi. [25-Q]

Ans.

$$w = \frac{(3+3\sqrt{2}i)}{(2-\sqrt{2}i)} = \frac{(3+3\sqrt{2}i)(2+\sqrt{2}i)}{(2-\sqrt{2}i)(2+\sqrt{2}i)} = \frac{9\sqrt{2}i}{6} = \frac{3\sqrt{2}}{2}i$$

Answer: $\frac{3\sqrt{2}}{2}i$

Q10. Given that $z_1 = 1 + \sqrt{3}i$ and $z_2 = -1 + i$. Find $\overline{(\frac{z_1}{z_2})}$. [25-Q]

Ans.

First find the quotient inside the bar:

$$\frac{z_1}{z_2} = \frac{1+\sqrt{3}i}{-1+i} = \frac{(1+\sqrt{3}i)(-1-i)}{(-1+i)(-1-i)} = \frac{-1-i-\sqrt{3}i+\sqrt{3}}{2} = \frac{(\sqrt{3}-1)-(\sqrt{3}+1)i}{2} = \frac{\sqrt{3}-1}{2} - \frac{\sqrt{3}+1}{2}i$$

The conjugate changes the sign of the imaginary part:

conjugate $= \frac{\sqrt{3}-1}{2} + \frac{\sqrt{3}+1}{2}i$

$\frac{\sqrt{3}-1}{2} + \frac{\sqrt{3}+1}{2}i$

Q11. Let $z_1 = 3 - 2i$ and $z_2 = 1 + i$ be the two complex numbers. Find $\frac{\overline{z_1}}{z_2}$. [24-Q]

Ans.

The conjugate of z_1 is

$\overline{z_1} = 3 + 2i$

$$\frac{\overline{z_1}}{z_2} = \frac{3+2i}{1+i} = \frac{(3+2i)(1-i)}{(1+i)(1-i)} = \frac{3-3i+2i-2i^2}{2} = \frac{5-i}{2} = \frac{5}{2} - \frac{1}{2}i$$

$\frac{5}{2} - \frac{1}{2}i$

Q12. Simplify $\overline{(\frac{2+3i}{-4-5i})}$. [5-Q]

Ans.

$\frac{(2+3i)}{(-4-5i)} = \frac{(2+3i)(-4+5i)}{(-4-5i)(-4+5i)}$

$= \frac{-23-2i}{41}$

$= -\frac{23}{41} - \frac{2}{41}i$

Taking the conjugate:

Answer: $-\frac{23}{41} + \frac{2}{41}i$

Q13. If the complex numbers are $z_1 = (1, 2)$, $z_2 = (-2, 3)$ and $z_3 = (1, -1)$, find the conjugate of $z = \frac{z_1}{z_2} + \frac{z_2}{z_3} + \frac{z_3}{z_1}$. [25-Q]

Ans.

$z_1 = 1 + 2i$, $z_2 = -2 + 3i$, $z_3 = 1 - i$.

$\frac{z_1}{z_2} = \frac{(1+2i)(-2-3i)}{13} = \frac{4}{13} - \frac{7}{13}i$

$\frac{z_2}{z_3} = \frac{(-2+3i)(1+i)}{2} = -\frac{5}{2} + \frac{1}{2}i$

$\frac{z_3}{z_1} = \frac{(1-i)(1-2i)}{5} = -\frac{1}{5} - \frac{3}{5}i$

$z = -\frac{311}{130} - \frac{83}{130}i$ (adding the three results)

Conjugate: $\bar{z} = -\frac{311}{130} + \frac{83}{130}i$

Q14. Find the complex number z such that $3iz + 2\bar{z} = 3 - 3i$. [25-Q]

Ans.

Let $z = x + yi$, $\bar{z} = x - yi$.

$3i(x + yi) + 2(x - yi) = 3xi - 3y + 2x - 2yi$

$= (2x - 3y) + (3x - 2y)i = 3 - 3i$

$2x - 3y = 3$ ...(1) $3x - 2y = -3$ ...(2)

(1)×3 − (2)×2: $-9y + 4y = 9 + 6$, so $y = -3$. Then $x = -3$.

z = −3 − 3i

Q15. Find the complex number z satisfying the equation $z + 4 = 3i\bar{z}$. [25-Q]

Ans.

Let $z = x + yi$:

$x + 4 + yi = 3i(x - yi) = 3xi + 3y$

Real: $x + 4 = 3y$; Imaginary: $y = 3x$.

So $x + 4 = 9x$, hence $x = \frac{1}{2}$ and $y = \frac{3}{2}$.

z = $\frac{1}{2} + \frac{3}{2}i$

1.4

Q16. Find the absolute value of complex number $\frac{3+4i}{1-2i}$. [26-Q]

Ans.

$|z_1/z_2| = |z_1| / |z_2|$

$|3 + 4i| = \sqrt{9+16} = 5$, $|1 - 2i| = \sqrt{1+4} = \sqrt{5}$

$\left|\frac{3+4i}{1-2i}\right| = \frac{5}{\sqrt{5}} = \sqrt{5}$

Answer: $\sqrt{5}$

Q17. Find the absolute value of the conjugate of $\frac{11+3i}{-1-7i}$. [25-Q]

Ans.

|conjugate of z| = |z|.

$|11 + 3i| = \sqrt{121+9} = \sqrt{130}$, $|-1 - 7i| = \sqrt{1+49} = \sqrt{50}$

$|z| = \sqrt{\frac{130}{50}} = \sqrt{\frac{13}{5}} = \frac{\sqrt{65}}{5}$

Answer: $\frac{\sqrt{65}}{5}$

Q18. Write down the trigonometric form of $z = -2 - 2\sqrt{3}i$ with $-\pi < \theta \le \pi$. [25-Q]

Ans.

$z = -2 - 2\sqrt{3}i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{4+12} = 4$

$\cos\theta = \frac{x}{r} = -\frac{1}{2}$, $\sin\theta = \frac{y}{r} = \frac{-\sqrt{3}}{2} \rightarrow \theta = -\frac{2\pi}{3}$

$z = 4(\cos(-\frac{2\pi}{3}) + i\sin(-\frac{2\pi}{3}))$

Answer: $4(\cos(-\frac{2\pi}{3}) + i\sin(-\frac{2\pi}{3}))$

Q19. Find the trigonometric form of $z = i^{2025} + i^{2026}$ with $-\pi < \theta \le \pi$. [25-Q]

Ans.

2025 = 4(506) + 1 and 2026 = 4(506) + 2, so $i^{2025} = i$ and $i^{2026} = i^2 = -1$.

$z = i + (-1) = -1 + i$

$r = \sqrt{1+1} = \sqrt{2}$

$\cos\theta = \frac{-1}{\sqrt{2}}$, $\sin\theta = \frac{1}{\sqrt{2}} \rightarrow \theta = \frac{3\pi}{4}$

$z = \sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

Answer: $\sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

Q20. Write down the trigonometric form of $(\sqrt{3} - i)^{-1}$. [5-Q]

Ans.

$z = \sqrt{3} - i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{3+1} = 2$

$\cos\theta = \frac{x}{r} = \frac{\sqrt{3}}{2}$, $\sin\theta = \frac{y}{r} = -\frac{1}{2} \rightarrow \theta = -\frac{\pi}{6}$

$z = 2(\cos(-\frac{\pi}{6}) + i\sin(-\frac{\pi}{6}))$

$z^{-1} = \frac{1}{r}(\cos(-\theta) + i\sin(-\theta))$:

$z^{-1} = \frac{1}{2}(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$

Answer: $\frac{1}{2}(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$

Q21. Write down the trigonometric form of $(-\sqrt{3} + i)^{-1}$. [25-Q]

Ans.

$z = -\sqrt{3} + i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{3+1} = 2$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{3}}{2}$, $\sin\theta = \frac{y}{r} = \frac{1}{2} \rightarrow \theta = \frac{5\pi}{6}$

$z = 2(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

$z^{-1} = \frac{1}{r}(\cos(-\theta) + i\sin(-\theta))$:

$z^{-1} = \frac{1}{2}(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6}))$

Answer: $\frac{1}{2}(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6}))$

Q22. Express the given complex number $(1 - i)^4$ in the form a + bi. [25-Q]

Ans.

$1 - i = \sqrt{2}(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4}))$.

$(1 - i)^4 = (\sqrt{2})^4(\cos(-\pi) + i\sin(-\pi))$

$= 4(-1 + 0i) = -4$

(Check: $(1 - i)^2 = -2i$, and $(-2i)^2 = -4$.)

Answer: −4

Q23. Find the value of z^5 where $z = -2 - 2i$. [25-Q]

Ans.

$z = -2 - 2i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{4+4} = 2\sqrt{2}$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{2}}{2}$, $\sin\theta = \frac{y}{r} = \frac{-\sqrt{2}}{2} \rightarrow \theta = -\frac{3\pi}{4}$

$z = 2\sqrt{2}(\cos(-\frac{3\pi}{4}) + i\sin(-\frac{3\pi}{4}))$

$z^5 = (2\sqrt{2})^5(\cos 5\theta + i\sin 5\theta)$

$= 128\sqrt{2}(\cos(-\frac{15\pi}{4}) + i\sin(-\frac{15\pi}{4}))$

$= 128\sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$ (adding/subtracting 2π multiples)

$= 128\sqrt{2}(\frac{\sqrt{2}}{2} + (\frac{\sqrt{2}}{2})i)$

$= 128 + 128i$

Answer: 128 + 128i

Q24. Given that $z = 2 - 2\sqrt{3}i$, find z^6. Give your answer in the form z = x + yi, where x and y are real numbers. [24-Q]

Ans.

$z = 2 - 2\sqrt{3}i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{4+12} = 4$

$\cos\theta = \frac{x}{r} = \frac{1}{2}$, $\sin\theta = \frac{y}{r} = \frac{-\sqrt{3}}{2} \rightarrow \theta = -\frac{\pi}{3}$

$z = 4(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3}))$

$z^6 = (4)^6(\cos 6\theta + i\sin 6\theta)$

$= 4096(\cos(-2\pi) + i\sin(-2\pi))$

$= 4096(\cos 0 + i\sin 0)$ (adding/subtracting 2π multiples)

$= 4096(1 + (0)i)$

$= 4096$

Answer: 4096

Q25. Given that $z = -2\sqrt{3} - 2i$, find z^6. Give your answer in the form $z = x + yi$, where x and y are real numbers. [25-Q]

Ans.

$z = -2\sqrt{3} - 2i$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{12 + 4} = 4$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{3}}{2}$, $\sin\theta = \frac{y}{r} = -\frac{1}{2}$ $\rightarrow$ $\theta = -\frac{5\pi}{6}$

$z = 4(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6}))$

$z^6 = (4)^6(\cos 6\theta + i\sin 6\theta)$
$= 4096(\cos(-5\pi) + i\sin(-5\pi))$
$= 4096(\cos(-\pi) + i\sin(-\pi))$ (adding/subtracting 2π multiples)
$= 4096(-1 + (0)i)$
$= -4096$

Answer: −4096

Q26. Evaluate $\left(\frac{\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}}{\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}}\right)^4$. Give your answer in x + yi form. [26-Q]

Ans.

Division: subtract the angles.

$\frac{\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}}{\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}} = \cos(\frac{4\pi}{3} - \frac{\pi}{6}) + i\sin(\frac{4\pi}{3} - \frac{\pi}{6})$

$= \cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}$

Power (De Moivre):

$(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6})^4 = \cos\frac{14\pi}{3} + i\sin\frac{14\pi}{3}$

$= \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$ ($14\pi/3 - 4\pi = 2\pi/3$)

$= -\frac{1}{2} + \frac{\sqrt{3}}{2}i$

Answer: $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$

1.5

Q27. Let w_0 and w_1 be the square roots of the complex number $2 + 2\sqrt{3}i$. Find their product w_0w_1. [24-Q]

Ans.

The two square roots of z are opposite numbers: $w_1 = -w_0$. (Using the formula, the angles differ by π.)

$w_0w_1 = w_0(-w_0) = -{w_0}^2 = -z$
$= -(2 + 2\sqrt{3}i)$
$= -2 - 2\sqrt{3}i$

(Check by polar form: $z = 4(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$, $w_0 = 2(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$, $w_1 = 2(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6})$; $w_0w_1 = 4(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}) = -2 - 2\sqrt{3}i$.)

Answer: $-2 - 2\sqrt{3}i$

3 MARKS

1.1

Q1. Simplify the expression of $i^{30} + i^{40} + i^{60}$. [5-Q]

Ans.

Use $i^2 = -1$ and $i^4 = 1$.

$i^{30} = (i^4)^7 \cdot i^2 = -1$
$i^{40} = (i^4)^{10} = 1$
$i^{60} = (i^4)^{15} = 1$
$i^{30} + i^{40} + i^{60} = -1 + 1 + 1 = 1$

Answer: 1

Q2. Evaluate $(\frac{1}{3} + 3i)^3$ in the form a + bi. [25-Q]

Ans.

Use $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$ with $x = \frac{1}{3}$, $y = 3i$.

$(\frac{1}{3} + 3i)^3 = (\frac{1}{3})^3 + 3(\frac{1}{3})^2(3i) + 3(\frac{1}{3})(3i)^2 + (3i)^3$

$= \frac{1}{27} + i + 3(\frac{1}{3})(9i^2) + 27i^3$

$= \frac{1}{27} + i - 9 - 27i$

$= \frac{1}{27} - 9 + (1 - 27)i$

$= \frac{-242}{27} - 26i$

Answer: $\frac{-242}{27} - 26i$

1.3

Q3. If $z_1 = 1 + 3i$ and $z_2 = -2 + i$, find the value of $z_1\overline{z_2} + \overline{z_1}z_2$. [25-Q]

Ans.

$\overline{z_1} = 1 - 3i$, $\overline{z_2} = -2 - i$.

$z_1\overline{z_2} = (1 + 3i)(-2 - i) = -2 - i - 6i - 3i^2 = 1 - 7i$

$\overline{z_1}z_2 = (1 - 3i)(-2 + i) = -2 + i + 6i - 3i^2 = 1 + 7i$

$z_1\overline{z_2} + \overline{z_1}z_2 = (1 - 7i) + (1 + 7i) = 2$

Answer: 2

Q4. Verify that $z = 1 + 2i$ is a solution of $z^2 - 5z^{-1}\overline{z} - 8i = 0$. [25-Q]

Ans.

$z = 1 + 2i$, $\overline{z} = 1 - 2i$.

$z^2 = (1 + 2i)^2 = 1 + 4i + 4i^2 = -3 + 4i$

$z^{-1}\overline{z} = \frac{1 - 2i}{1 + 2i} = \frac{(1 - 2i)^2}{(1 + 2i)(1 - 2i)} = \frac{1 - 4i - 4}{5} = \frac{-3 - 4i}{5}$

$5z^{-1}\overline{z} = -3 - 4i$

$z^2 - 5z^{-1}\overline{z} - 8i = (-3 + 4i) - (-3 - 4i) - 8i$
$= -3 + 4i + 3 + 4i - 8i = 0$

L.H.S. = R.H.S. = 0, so $z = 1 + 2i$ is a solution. (Verified)

Q5. Let $z_1 = (1, -2)$ and $z_2 = (3, 5)$ be two complex numbers. Express $\frac{\overline{z_1}}{z_2} + z_1$ in the form x + yi. [25-Q]

Ans.

$z_1 = 1 - 2i$, $\overline{z_1} = 1 + 2i$, $z_2 = 3 + 5i$.

$\frac{(1 + 2i)}{(3 + 5i)} = \frac{(1 + 2i)(3 - 5i)}{(3 + 5i)(3 - 5i)}$
$= \frac{13 + i}{34}$
$= \frac{13}{34} + \frac{1}{34}i$

$\frac{\overline{z_1}}{z_2} + z_1 = \frac{13}{34} + \frac{1}{34}i + (1 - 2i)$

$= \frac{47}{34} - \frac{67}{34}i$

Answer: $\frac{47}{34} - \frac{67}{34}i$

Q6. If $\frac{a + ib}{c + id} = x + iy$, prove that $x^2 + y^2 = \frac{a^2 + b^2}{c^2 + d^2}$. [26-Q]

Ans.

$\frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{(c + id)(c - id)} = \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2}$

So $x = \frac{ac + bd}{c^2 + d^2}$ and $y = \frac{bc - ad}{c^2 + d^2}$.

$x^2 + y^2 = \frac{(ac + bd)^2 + (bc - ad)^2}{(c^2 + d^2)^2}$

$= \frac{a^2c^2 + b^2d^2 + b^2c^2 + a^2d^2}{(c^2 + d^2)^2}$ (the cross terms 2abcd cancel)

$= \frac{(a^2 + b^2)(c^2 + d^2)}{(c^2 + d^2)^2}$

$= \frac{a^2 + b^2}{c^2 + d^2}$

Hence proved.

1.4

Q7. Find the absolute value of the inverse of complex number $z = -\sqrt{3} - i$. [25-Q]

Ans.

$|z| = \sqrt{3 + 1} = 2$

$|z^{-1}| = \frac{1}{|z|} = \frac{1}{2}$

Answer: $\frac{1}{2}$

Q8. Find the trigonometric form with $-\pi < \theta \le \pi$. (a) $z = 1 + \sqrt{3}i$ (b) $z = -1 + i$. [24-Q, 25-Q]

Ans.

(a)

$z = 1 + \sqrt{3}i$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{1 + 3} = 2$

$\cos\theta = \frac{x}{r} = \frac{1}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{\pi}{3}$

$z = 2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

(b)

$z = -1 + i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{1+1} = \sqrt{2}$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{2}}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{2}}{2} \rightarrow \theta = \frac{3\pi}{4}$

$z = \sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

(a) $2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$; (b) $\sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

Q9. Given that $z = -1 + i$, find trigonometric form with $-\pi < \theta \le \pi$. [25-Q]

Ans.

$z = -1 + i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{1+1} = \sqrt{2}$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{2}}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{2}}{2} \rightarrow \theta = \frac{3\pi}{4}$

$z = \sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

Answer: $\sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

Q10. Find the trigonometric form of the complex number $z = -\frac{1}{2}(1 - \sqrt{3}i)$ with $-\pi < \theta \le \pi$. [25-Q]

Ans.

$z = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{1/4 + 3/4} = 1$

$\cos\theta = \frac{x}{r} = -\frac{1}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{2\pi}{3}$

$z = (\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

Answer: $(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

Q11. Find the trigonometric form of the complex number $z = \frac{5}{2}(-\sqrt{3} + i)$ with $-\pi < \theta \le \pi$. [25-Q]

Ans.

$z = \frac{-5\sqrt{3}}{2} + \frac{5}{2}i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{75/4 + 25/4} = 5$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{3}}{2}$, $\sin\theta = \frac{y}{r} = \frac{1}{2} \rightarrow \theta = \frac{5\pi}{6}$

$z = 5(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

Answer: $5(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

Q12. Given that $z = -\sqrt{3} - i$, using trigonometric form of z, find z^{-1}. Check your answer by showing that $zz^{-1} = 1$. [25-Q]

Ans.

$z = -\sqrt{3} - i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{3+1} = 2$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{3}}{2}$, $\sin\theta = \frac{y}{r} = -\frac{1}{2} \rightarrow \theta = -\frac{5\pi}{6}$

$z = 2(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6}))$

$z^{-1} = \frac{1}{2}(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

Check: multiply the moduli and add the angles:

$zz^{-1} = 2\cdot\frac{1}{2}(\cos(-\frac{5\pi}{6} + (\frac{5\pi}{6})) + i\sin(0))$

$= 1(\cos 0 + i\sin 0) = 1$

$z^{-1} = \frac{1}{2}(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$, $zz^{-1} = 1$

Q13. Given that $z = -1 + \sqrt{3}i$, using trigonometric form of z, find z^{-1}. Check your answer by showing that $zz^{-1} = 1$. [25-Q]

Ans.

$z = -1 + \sqrt{3}i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{1+3} = 2$

$\cos\theta = \frac{x}{r} = -\frac{1}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{2\pi}{3}$

$z = 2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

$z^{-1} = \frac{1}{2}(\cos(-\frac{2\pi}{3}) + i\sin(-\frac{2\pi}{3}))$

Check: multiply the moduli and add the angles:

$zz^{-1} = 2\cdot\frac{1}{2}(\cos(\frac{2\pi}{3} + (-\frac{2\pi}{3})) + i\sin(0))$

$= 1(\cos 0 + i\sin 0) = 1$

$z^{-1} = \frac{1}{2}(\cos(-\frac{2\pi}{3}) + i\sin(-\frac{2\pi}{3}))$, $zz^{-1} = 1$

Q14. If a complex number $z = \frac{5}{2} + \frac{5\sqrt{3}}{2}i$, find $\frac{z}{\bar{z}}$ by using trigonometric forms. [25-Q]

Ans.

$r = \sqrt{\frac{25}{4} + \frac{75}{4}} = 5$

$\cos\theta = \frac{5/2}{5} = \frac{1}{2}$, $\sin\theta = \frac{5\sqrt{3}/2}{5} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{\pi}{3}$

$z = 5(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$, $\bar{z} = 5(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3}))$

$\frac{z}{\bar{z}} = \frac{5}{5}(\cos(\frac{\pi}{3} + \frac{\pi}{3}) + i\sin(\frac{\pi}{3} + \frac{\pi}{3}))$

$= \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$

Answer: $\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$ $(= -\frac{1}{2} + \frac{\sqrt{3}}{2}i)$

Q15. Given that $z = 1 + \sqrt{3}i$, find z^3. [26-Q]

Ans.

$z = 1 + \sqrt{3}i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{1+3} = 2$

$\cos\theta = \frac{x}{r} = \frac{1}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{\pi}{3}$

$z = 2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

$z^3 = (2)^3(\cos 3\theta + i\sin 3\theta)$
$= 8(\cos\pi + i\sin\pi)$
$= 8(\cos(-\pi) + i\sin(-\pi))$ (adding/subtracting 2π multiples)
$= 8(-1 + (0)i)$
$= -8$

Answer: −8

Q16. Given that $z = -2\sqrt{3} - 2i$. Find z^3. [26-Q]

Ans.

$z = -2\sqrt{3} - 2i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{12+4} = 4$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{3}}{2}$, $\sin\theta = \frac{y}{r} = -\frac{1}{2} \rightarrow \theta = -\frac{5\pi}{6}$

$z = 4(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6}))$

$z^3 = (4)^3(\cos 3\theta + i\sin 3\theta)$

$= 64(\cos(-\frac{5\pi}{2}) + i\sin(-\frac{5\pi}{2}))$

$= 64(\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2}))$ (adding/subtracting 2π multiples)

$= 64(0 + (-1)i)$
$= -64i$

Answer: −64i

Q17. Find z^4 for complex number $z = -\sqrt{2} + \sqrt{2}i$. [24-Q]

Ans.

$z = -\sqrt{2} + \sqrt{2}i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{2+2} = 2$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{2}}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{2}}{2} \rightarrow \theta = \frac{3\pi}{4}$

$z = 2(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

$z^4 = (2)^4(\cos 4\theta + i\sin 4\theta)$
$= 16(\cos 3\pi + i\sin 3\pi)$
$= 16(\cos(-\pi) + i\sin(-\pi))$ (adding/subtracting 2π multiples)
$= 16(-1 + (0)i)$
$= -16$

Answer: −16

Q18. Given that $z = -1 + \sqrt{3}i$, find z^5. [24-Q]

Ans.

$z = -1 + \sqrt{3}i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{1+3} = 2$

$\cos\theta = \frac{x}{r} = -\frac{1}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{2\pi}{3}$

$z = 2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

$z^5 = (2)^5(\cos 5\theta + i \sin 5\theta)$

$= 32(\cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3})$

$= 32(\cos(-\frac{2\pi}{3}) + i \sin(-\frac{2\pi}{3}))$ (adding/subtracting 2π multiples)

$= 32(-\frac{1}{2} + (\frac{-\sqrt{3}}{2})i)$

$= -16 - 16\sqrt{3}i$

Answer: $-16 - 16\sqrt{3}i$

Q19. Given that $z = 1 + \sqrt{3}i$, find z^{10}. [26-Q]

Ans.

$z = 1 + \sqrt{3}i$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{1+3} = 2$

$\cos\theta = \frac{x}{r} = \frac{1}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{\pi}{3}$

$z = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$

$z^{10} = (2)^{10}(\cos 10\theta + i \sin 10\theta)$

$= 1024(\cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3})$

$= 1024(\cos(-\frac{2\pi}{3}) + i \sin(-\frac{2\pi}{3}))$ (adding/subtracting 2π multiples)

$= 1024(-\frac{1}{2} + (\frac{-\sqrt{3}}{2})i)$

$= -512 - 512\sqrt{3}i$

Answer: $-512 - 512\sqrt{3}i$

1.5

Q20. Find the square roots of $z = 1 + i$. [24-Q]

Ans.

$|z| = \sqrt{2}$, so the root of the modulus is $\sqrt[4]{2}$; and $\cos \frac{\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$, $\sin \frac{\pi}{8} = \frac{\sqrt{2-\sqrt{2}}}{2}$.

Write the right-hand side in trigonometric form:

$z = 1 + i$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{1+1} = \sqrt{2}$

$\cos\theta = \frac{x}{r} = \frac{\sqrt{2}}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{2}}{2} \rightarrow \theta = \frac{\pi}{4}$

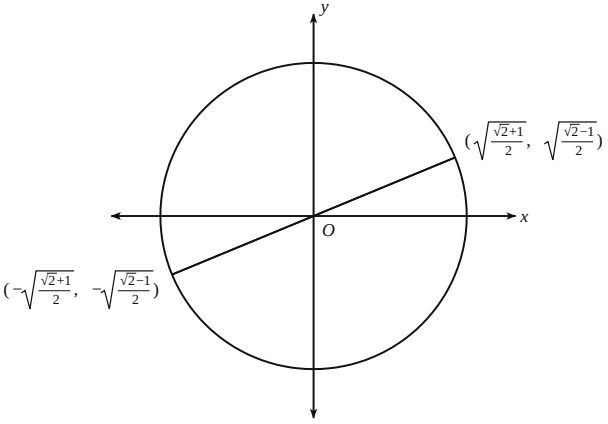
$z = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$

The 2 roots are $w_k = \sqrt[4]{2}(\cos \frac{\theta + 2k\pi}{2} + i \sin \frac{\theta + 2k\pi}{2})$, k = 0, 1, ..., 1, with $\theta = \frac{\pi}{4}$.

k = 0: $w_0 = \sqrt[4]{2}(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8}) = \sqrt{\frac{\sqrt{2}+1}{2}} + i\sqrt{\frac{\sqrt{2}-1}{2}}$

k = 1: $w_1 = \sqrt[4]{2}(\cos(-\frac{7\pi}{8}) + i \sin(-\frac{7\pi}{8})) = -\sqrt{\frac{\sqrt{2}+1}{2}} - i\sqrt{\frac{\sqrt{2}-1}{2}}$

Answer: $\pm(\sqrt{\frac{\sqrt{2}+1}{2}} + i\sqrt{\frac{\sqrt{2}-1}{2}})$



Q21. Find the square roots of $z = \sqrt{3} + i$. [24-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = \sqrt{3} + i$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{3+1} = 2$

$\cos\theta = \frac{x}{r} = \frac{\sqrt{3}}{2}$, $\sin\theta = \frac{y}{r} = \frac{1}{2} \rightarrow \theta = \frac{\pi}{6}$

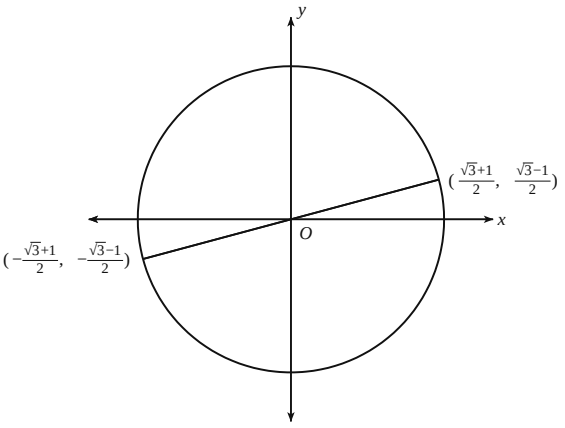
$z = 2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$

The 2 roots are $w_k = \sqrt{2}(\cos \frac{\theta + 2k\pi}{2} + i \sin \frac{\theta + 2k\pi}{2})$, k = 0, 1, ..., 1, with $\theta = \frac{\pi}{6}$.

k = 0: $w_0 = \sqrt{2}(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12}) = \frac{\sqrt{3}+1}{2} + \frac{\sqrt{3}-1}{2}i$

k = 1: $w_1 = \sqrt{2}(\cos(-\frac{11\pi}{12}) + i \sin(-\frac{11\pi}{12})) = \frac{-\sqrt{3}-1}{2} - \frac{\sqrt{3}-1}{2}i$

Answer: $\pm(\frac{\sqrt{3}+1}{2} + i\frac{\sqrt{3}-1}{2})$



Q22. Find the square roots of $z = -\sqrt{3} - i$. [24-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = -\sqrt{3} - i$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{3+1} = 2$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{3}}{2}$, $\sin\theta = \frac{y}{r} = -\frac{1}{2} \rightarrow \theta = -\frac{5\pi}{6}$

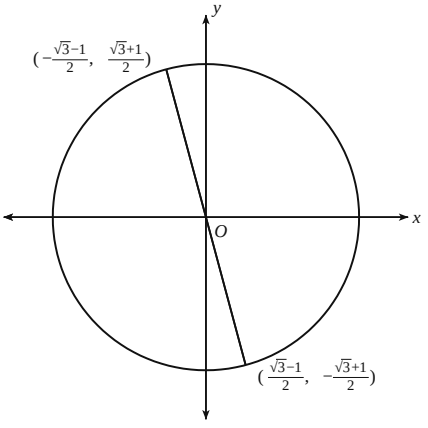
$z = 2(\cos(-\frac{5\pi}{6}) + i \sin(-\frac{5\pi}{6}))$

The 2 roots are $w_k = \sqrt{2}(\cos \frac{\theta + 2k\pi}{2} + i \sin \frac{\theta + 2k\pi}{2})$, k = 0, 1, ..., 1, with $\theta = -\frac{5\pi}{6}$.

k = 0: $w_0 = \sqrt{2}(\cos(-\frac{5\pi}{12}) + i \sin(-\frac{5\pi}{12})) = \frac{\sqrt{3}-1}{2} - \frac{\sqrt{3}+1}{2}i$

k = 1: $w_1 = \sqrt{2}(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12}) = \frac{1-\sqrt{3}}{2} + \frac{\sqrt{3}+1}{2}i$

Answer: $\pm(\frac{\sqrt{3}-1}{2} - i\frac{\sqrt{3}+1}{2})$



Q23. Find the square roots of $z = -8 - 8\sqrt{3}i$. [24-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = -8 - 8\sqrt{3}i$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{64+192} = 16$

$\cos\theta = \frac{x}{r} = -\frac{1}{2}$, $\sin\theta = \frac{y}{r} = \frac{-\sqrt{3}}{2} \rightarrow \theta = -\frac{2\pi}{3}$

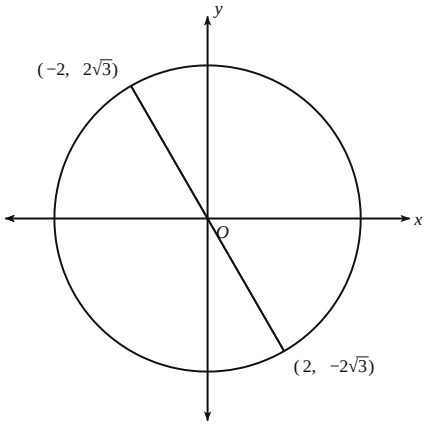
$z = 16(\cos(-\frac{2\pi}{3}) + i \sin(-\frac{2\pi}{3}))$

The 2 roots are $w_k = 4(\cos \frac{\theta + 2k\pi}{2} + i \sin \frac{\theta + 2k\pi}{2})$, k = 0, 1, ..., 1, with $\theta = -\frac{2\pi}{3}$.

$k = 0$: $w_0 = 4(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})) = 2 - 2\sqrt{3}i$

$k = 1$: $w_1 = 4(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}) = -2 + 2\sqrt{3}i$

Answer: $\pm(2 - 2\sqrt{3}i)$



Q24. Solve the equation $z^3 - 1 = 0$, where z is a complex number. [24-Q]

Ans.

$z^3 - 1 = 0 \rightarrow z^3 = 1$.

Write the right-hand side in trigonometric form:

$z = 1$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{1 + 0} = 1$

$\cos\theta = \frac{x}{r} = 1$, $\sin\theta = \frac{y}{r} = 0 \rightarrow \theta = 0$

$z = (\cos 0 + i \sin 0)$

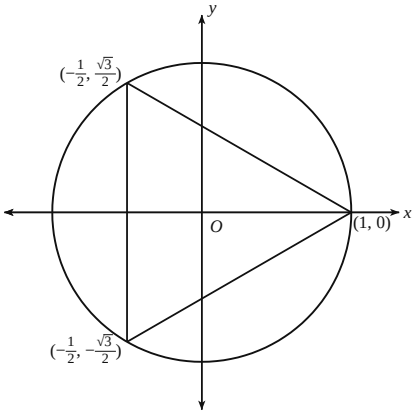
The 3 roots are $w_k = (\cos\frac{\theta + 2k\pi}{3} + i\sin\frac{\theta + 2k\pi}{3})$, $k = 0, 1, ..., 2$, with $\theta = 0$.

$k = 0$: $w_0 = (\cos 0 + i \sin 0) = 1$

$k = 1$: $w_1 = (\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$

$k = 2$: $w_2 = (\cos(-\frac{2\pi}{3}) + i\sin(-\frac{2\pi}{3})) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

Position of the roots on the xy-plane: they lie on the circle $|z| = 1$ and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: $1, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

5 MARKS

1.3

Q1. Given that $z_1 = 2 - i$ and $z_2 = -4 + 3i$. Find the inverse of z_1z_2 and $\frac{z_1}{z_2}$. [24-Q]

Ans.

(i) Inverse of z_1z_2

$z_1z_2 = (2 - i)(-4 + 3i) = -8 + 6i + 4i - 3i^2$

$= -5 + 10i$

$(z_1z_2)^{-1} = \frac{1}{-5 + 10i} = \frac{-5 - 10i}{(-5 + 10i)(-5 - 10i)} = \frac{-5 - 10i}{25 + 100}$

$= -\frac{1}{25} - \frac{2}{25}i$

(ii) Inverse of $\frac{z_1}{z_2}$ — first find z_1/z_2:

$\frac{z_1}{z_2} = \frac{(2 - i)(-4 - 3i)}{(-4 + 3i)(-4 - 3i)} = \frac{-8 - 6i + 4i + 3i^2}{25}$

$= -\frac{11}{25} - \frac{2}{25}i$

$(z_1z_2)^{-1} = -\frac{1}{25} - \frac{2}{25}i$; $\frac{z_1}{z_2} = -\frac{11}{25} - \frac{2}{25}i$

1.4

Q2. Let $z = -\sqrt{3} - i$. Using trigonometric form of z, find z^{-1}. Use your answer to show that $z^2(z^{-1})^2 = 1$. [S-Q]

Ans.

$z = -\sqrt{3} - i$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{3 + 1} = 2$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{3}}{2}$, $\sin\theta = \frac{y}{r} = -\frac{1}{2} \rightarrow \theta = -\frac{5\pi}{6}$

$z = 2(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6}))$

$z^{-1} = \frac{1}{2}(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

To show: $z^2(z^{-1})^2 = 1$

$z^2 = 4(\cos(-\frac{5\pi}{3}) + i\sin(-\frac{5\pi}{3}))$

$(z^{-1})^2 = \frac{1}{4}(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3})$

$z^2(z^{-1})^2 = 4 \cdot \frac{1}{4}(\cos(-\frac{5\pi}{3} + \frac{5\pi}{3}) + i\sin(-\frac{5\pi}{3} + \frac{5\pi}{3}))$

$= 1(\cos 0 + i \sin 0) = 1$

Hence proved.

$z^{-1} = \frac{1}{2}(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

Q3. Using trigonometric forms, evaluate the expression $\frac{(1 - i)^{40}(1 + \sqrt{3}i)^2}{(-\sqrt{3} + i)^{18}}$ and express your answer in x + yi form. [24-Q]

Ans.

$1 - i = \sqrt{2}(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4}))$

$1 + \sqrt{3}i = 2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

$-\sqrt{3} + i = 2(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6})$

$(1 - i)^{40} = 2^{20}(\cos(-10\pi) + i\sin(-10\pi)) = 2^{20}$

$(1 + \sqrt{3}i)^2 = 4(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

$(-\sqrt{3} + i)^{18} = 2^{18}(\cos 15\pi + i \sin 15\pi) = 2^{18}(-1)$

expression $= \frac{2^{20} \cdot 4}{2^{18}(-1)}(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

$= -16(-\frac{1}{2} + \frac{\sqrt{3}}{2}i)$

$= 8 - 8\sqrt{3}i$

Answer: $8 - 8\sqrt{3}i$

1.5

Q4. Find the values of a, b ∈ ℝ such that $(a - i)(6 + bi) = 7a + 6 - 15i$. Hence find the square roots of $a + bi$. [26-Q]

Ans.

Expand the left side:

$(a - i)(6 + bi) = 6a + abi - 6i - bi^2$

$= (6a + b) + (ab - 6)i$

Equate real and imaginary parts with $7a + 6 - 15i$:

$6a + b = 7a + 6 \rightarrow b = a + 6$

$ab - 6 = -15 \rightarrow ab = -9$

Substitute $b = a + 6$:

$a(a + 6) = -9$

$a^2 + 6a + 9 = 0$

$(a + 3)^2 = 0 \rightarrow a = -3$, $b = 3$

So $a + bi = -3 + 3i$. Now find its square roots.

$|z| = 3\sqrt{2}$, $\theta = \frac{3\pi}{4}$; $\cos\frac{3\pi}{8} = \frac{\sqrt{2 - \sqrt{2}}}{2}$, $\sin\frac{3\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2}$.

Write the right-hand side in trigonometric form:

$z = -3 + 3i$, $r = |z| = \sqrt{x^2 + y^2} = \sqrt{9 + 9} = 3\sqrt{2}$

$\cos\theta = \frac{x}{r} = \frac{-\sqrt{2}}{2}$, $\sin\theta = \frac{y}{r} = \frac{\sqrt{2}}{2} \rightarrow \theta = \frac{3\pi}{4}$

$z = 3\sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4})$

The 2 roots are $w_k = \sqrt{3\sqrt{2}}(\cos\frac{\theta+2k\pi}{2} + i\sin\frac{\theta+2k\pi}{2})$, k = 0, 1, ..., 1, with $\theta = \frac{3\pi}{4}$.

k = 0: $w_0 = \sqrt{3\sqrt{2}}(\cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}) = \sqrt{\frac{3\sqrt{2}-3}{2}} + i\sqrt{\frac{3\sqrt{2}+3}{2}}$

k = 1: $w_1 = \sqrt{3\sqrt{2}}(\cos(-\frac{5\pi}{8}) + i\sin(-\frac{5\pi}{8})) = -\sqrt{\frac{3\sqrt{2}-3}{2}} - i\sqrt{\frac{3\sqrt{2}+3}{2}}$

Answer: a = −3, b = 3; square roots $\pm(\sqrt{\frac{3\sqrt{2}-3}{2}} + i\sqrt{\frac{3\sqrt{2}+3}{2}})$

Q5. Find the cube roots of the complex number z = i. [25-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = i,\ r = |z| = \sqrt{x^2+y^2} = \sqrt{0+1} = 1$

$\cos\theta = \frac{x}{r} = 0,\ \sin\theta = \frac{y}{r} = 1 \rightarrow \theta = \frac{\pi}{2}$

$z = (\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$

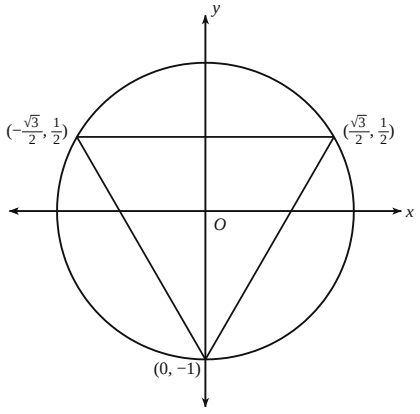
The 3 roots are $w_k = (\cos\frac{\theta+2k\pi}{3} + i\sin\frac{\theta+2k\pi}{3})$, k = 0, 1, ..., 2, with $\theta = \frac{\pi}{2}$.

k = 0: $w_0 = (\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}) = \frac{\sqrt{3}}{2} + \frac{1}{2}i$

k = 1: $w_1 = (\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}) = \frac{-\sqrt{3}}{2} + \frac{1}{2}i$

k = 2: $w_2 = (\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2})) = -i$

Position of the roots on the xy-plane: they lie on the circle |z| = 1 and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: $\frac{\sqrt{3}}{2} + \frac{1}{2}i, -\frac{\sqrt{3}}{2} + \frac{1}{2}i, -i$

Q6. Find the cube roots of z = 1 + i. [24-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = 1+i,\ r = |z| = \sqrt{x^2+y^2} = \sqrt{1+1} = \sqrt{2}$

$\cos\theta = \frac{x}{r} = \frac{\sqrt{2}}{2},\ \sin\theta = \frac{y}{r} = \frac{\sqrt{2}}{2} \rightarrow \theta = \frac{\pi}{4}$

$z = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$

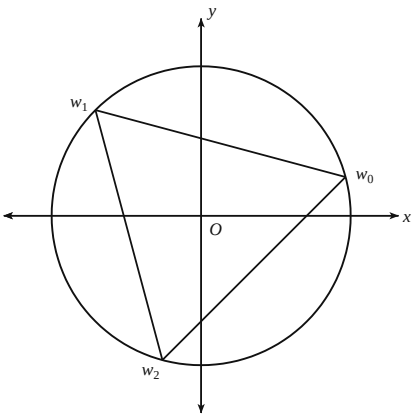
The 3 roots are $w_k = \sqrt[6]{2}(\cos\frac{\theta+2k\pi}{3} + i\sin\frac{\theta+2k\pi}{3})$, k = 0, 1, ..., 2, with $\theta = \frac{\pi}{4}$.

k = 0: $w_0 = \sqrt[6]{2}(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}) = \sqrt[6]{2}\frac{(\sqrt{6}+\sqrt{2})}{4} + i\sqrt[6]{2}\frac{(\sqrt{6}-\sqrt{2})}{4}$

k = 1: $w_1 = \sqrt[6]{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}) = -\frac{\sqrt[3]{4}}{2} + i\frac{\sqrt[3]{4}}{2}$

k = 2: $w_2 = \sqrt[6]{2}(\cos(-\frac{7\pi}{12}) + i\sin(-\frac{7\pi}{12})) = \sqrt[6]{2}\frac{(\sqrt{2}-\sqrt{6})}{4} - i\sqrt[6]{2}\frac{(\sqrt{6}+\sqrt{2})}{4}$

Position of the roots on the xy-plane: they lie on the circle $|z| = \sqrt[6]{2}$ and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: $\sqrt[6]{2}\frac{(\sqrt{6}+\sqrt{2})}{4} + i\sqrt[6]{2}\frac{(\sqrt{6}-\sqrt{2})}{4}, -\frac{\sqrt[3]{4}}{2} + i\frac{\sqrt[3]{4}}{2}, \sqrt[6]{2}\frac{(\sqrt{2}-\sqrt{6})}{4} - i\sqrt[6]{2}\frac{(\sqrt{6}+\sqrt{2})}{4}$

Q7. Find the cube roots of the complex number z = 2 + 2i. [24-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = 2+2i,\ r = |z| = \sqrt{x^2+y^2} = \sqrt{4+4} = 2\sqrt{2}$

$\cos\theta = \frac{x}{r} = \frac{\sqrt{2}}{2},\ \sin\theta = \frac{y}{r} = \frac{\sqrt{2}}{2} \rightarrow \theta = \frac{\pi}{4}$

$z = 2\sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$

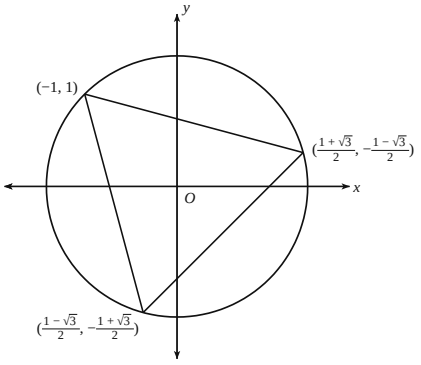
The 3 roots are $w_k = \sqrt{2}(\cos\frac{\theta+2k\pi}{3} + i\sin\frac{\theta+2k\pi}{3})$, k = 0, 1, ..., 2, with $\theta = \frac{\pi}{4}$.

k = 0: $w_0 = \sqrt{2}(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}) = \frac{\sqrt{3}+1}{2} + \frac{\sqrt{3}-1}{2}i$

k = 1: $w_1 = \sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}) = -1+i$

k = 2: $w_2 = \sqrt{2}(\cos(-\frac{7\pi}{12}) + i\sin(-\frac{7\pi}{12})) = \frac{1-\sqrt{3}}{2} - \frac{\sqrt{3}+1}{2}i$

Position of the roots on the xy-plane: they lie on the circle $|z| = \sqrt{2}$ and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: $\frac{\sqrt{3}+1}{2} + i\frac{\sqrt{3}-1}{2}, -1+i, \frac{1-\sqrt{3}}{2} - i\frac{1+\sqrt{3}}{2}$

Q8. Find the cube roots of the complex number z = 8 + 8i. [25-Q]

Ans.

$|z| = \sqrt{64+64} = 8\sqrt{2}$, so $\sqrt[3]{8\sqrt{2}} = 2^6\sqrt{2}$.

Write the right-hand side in trigonometric form:

$z = 8+8i,\ r = |z| = \sqrt{x^2+y^2} = \sqrt{64+64} = 8\sqrt{2}$

$\cos\theta = \frac{x}{r} = \frac{\sqrt{2}}{2},\ \sin\theta = \frac{y}{r} = \frac{\sqrt{2}}{2} \rightarrow \theta = \frac{\pi}{4}$

$z = 8\sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$

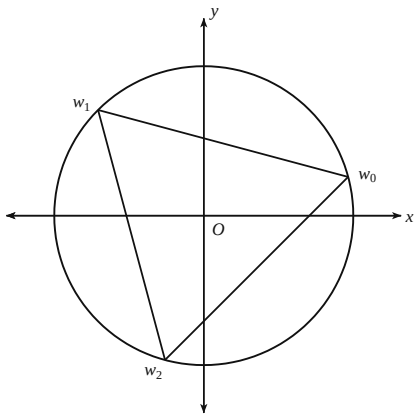
The 3 roots are $w_k = 2^6\sqrt{2}(\cos\frac{\theta+2k\pi}{3} + i\sin\frac{\theta+2k\pi}{3})$, k = 0, 1, ..., 2, with $\theta = \frac{\pi}{4}$.

k = 0: $w_0 = 2^6\sqrt{2}(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}) = \sqrt[3]{4}\frac{(\sqrt{3}+1)}{2} + i\sqrt[3]{4}\frac{(\sqrt{3}-1)}{2}$

k = 1: $w_1 = 2^6\sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}) = -\sqrt[3]{4} + i\sqrt[3]{4}$

k = 2: $w_2 = 2^6\sqrt{2}(\cos(-\frac{7\pi}{12}) + i\sin(-\frac{7\pi}{12})) = \sqrt[3]{4}\frac{(1-\sqrt{3})}{2} - i\sqrt[3]{4}\frac{(1+\sqrt{3})}{2}$

Position of the roots on the xy-plane: they lie on the circle $|z| = 2^6\sqrt{2}$ and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: $\sqrt[3]{4}\,\frac{(\sqrt{3}+1)}{2} + i\sqrt[3]{4}\,\frac{(\sqrt{3}-1)}{2}$, $-\sqrt[3]{4} + i\sqrt[3]{4}$, $\sqrt[3]{4}\,\frac{(1-\sqrt{3})}{2} - i\sqrt[3]{4}\,\frac{(1+\sqrt{3})}{2}$

Q9. Solve $z^3 = 1$. [25, 26-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = 1,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{1+0} = 1$

$\cos\theta = \frac{x}{r} = 1,\ \sin\theta = \frac{y}{r} = 0 \rightarrow \theta = 0$

$z = (\cos 0 + i \sin 0)$

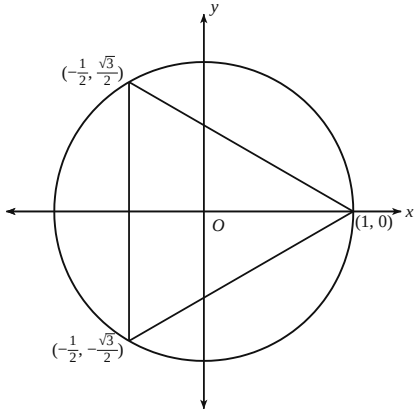
The 3 roots are $w_k = (\cos\frac{\theta + 2k\pi}{3} + i\sin\frac{\theta + 2k\pi}{3})$, $k = 0, 1, ..., 2$, with $\theta = 0$.

$k = 0$: $w_0 = (\cos 0 + i \sin 0) = 1$

$k = 1$: $w_1 = (\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$

$k = 2$: $w_2 = (\cos(-\frac{2\pi}{3}) + i\sin(-\frac{2\pi}{3})) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

Position of the roots on the xy-plane: they lie on the circle $|z| = 1$ and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: $1, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} - \frac{\sqrt{3}}{2}i$

Q10. Solve $z^3 = -1$. [25-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = -1,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{1+0} = 1$

$\cos\theta = \frac{x}{r} = -1,\ \sin\theta = \frac{y}{r} = 0 \rightarrow \theta = \pi$

$z = (\cos\pi + i \sin\pi)$

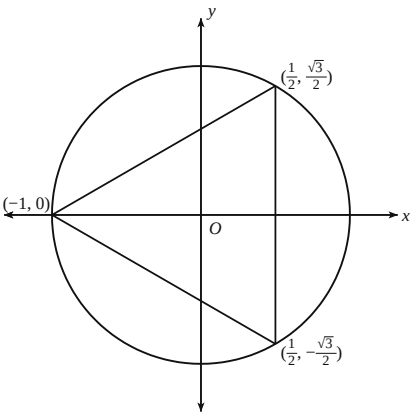
The 3 roots are $w_k = (\cos\frac{\theta + 2k\pi}{3} + i\sin\frac{\theta + 2k\pi}{3})$, $k = 0, 1, ..., 2$, with $\theta = \pi$.

$k = 0$: $w_0 = (\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$

$k = 1$: $w_1 = (\cos\pi + i\sin\pi) = -1$

$k = 2$: $w_2 = (\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})) = \frac{1}{2} - \frac{\sqrt{3}}{2}i$

Position of the roots on the xy-plane: they lie on the circle $|z| = 1$ and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: $-1, \frac{1}{2} + \frac{\sqrt{3}}{2}i, \frac{1}{2} - \frac{\sqrt{3}}{2}i$

Q11. Solve $z^4 = 1$. [24-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = 1,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{1+0} = 1$

$\cos\theta = \frac{x}{r} = 1,\ \sin\theta = \frac{y}{r} = 0 \rightarrow \theta = 0$

$z = (\cos 0 + i \sin 0)$

The 4 roots are $w_k = (\cos\frac{\theta + 2k\pi}{4} + i\sin\frac{\theta + 2k\pi}{4})$, $k = 0, 1, ..., 3$, with $\theta = 0$.

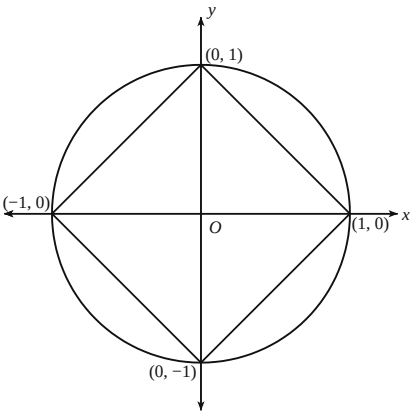
$k = 0$: $w_0 = (\cos 0 + i \sin 0) = 1$

$k = 1$: $w_1 = (\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}) = i$

$k = 2$: $w_2 = (\cos\pi + i\sin\pi) = -1$

$k = 3$: $w_3 = (\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2})) = -i$

Position of the roots on the xy-plane: they lie on the circle $|z| = 1$ and are equally spaced, $\frac{\pi}{2}$ apart.



Answer: $\pm 1, \pm i$

Q12. Solve $z^4 = -1$. [26-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = -1,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{1+0} = 1$

$\cos\theta = \frac{x}{r} = -1,\ \sin\theta = \frac{y}{r} = 0 \rightarrow \theta = \pi$

$z = (\cos\pi + i \sin\pi)$

The 4 roots are $w_k = (\cos\frac{\theta + 2k\pi}{4} + i\sin\frac{\theta + 2k\pi}{4})$, $k = 0, 1, ..., 3$, with $\theta = \pi$.

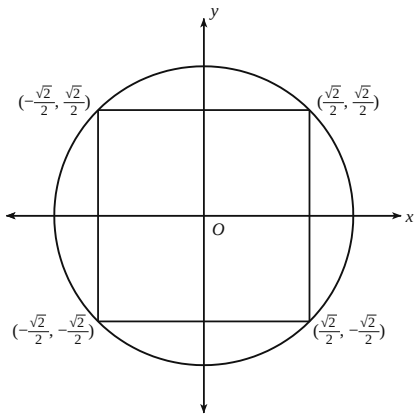
$k = 0$: $w_0 = (\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$

$k = 1$: $w_1 = (\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}) = \frac{-\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$

$k = 2$: $w_2 = (\cos(-\frac{3\pi}{4}) + i\sin(-\frac{3\pi}{4})) = \frac{-\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$

$k = 3$: $w_3 = (\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$

Position of the roots on the xy-plane: they lie on the circle $|z| = 1$ and are equally spaced, $\frac{\pi}{2}$ apart.



Answer: $\pm \frac{\sqrt{2}}{2} \pm \frac{\sqrt{2}}{2}\,i$

Q13. Solve $z^4 = -4$, where z is a complex number. [25-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = -4,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{16 + 0} = 4$

$\cos\theta = \frac{x}{r} = -1,\ \sin\theta = \frac{y}{r} = 0 \rightarrow \theta = \pi$

$z = 4(\cos\pi + i\sin\pi)$

The 4 roots are $w_k = \sqrt{2}(\cos\frac{\theta + 2k\pi}{4} + i\sin\frac{\theta + 2k\pi}{4})$, k = 0, 1, ..., 3, with $\theta = \pi$.

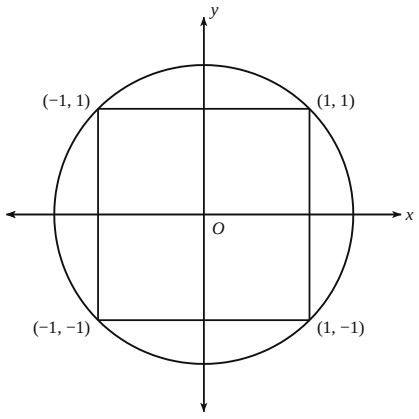
k = 0: $w_0 = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}) = 1 + i$

k = 1: $w_1 = \sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}) = -1 + i$

k = 2: $w_2 = \sqrt{2}(\cos(-\frac{3\pi}{4}) + i\sin(-\frac{3\pi}{4})) = -1 - i$

k = 3: $w_3 = \sqrt{2}(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})) = 1 - i$

Position of the roots on the xy-plane: they lie on the circle $|z| = \sqrt{2}$ and are equally spaced, $\frac{\pi}{2}$ apart.



Answer: $\pm 1 \pm i$

Q14. Solve the equation $z^4 = 4 + 4\sqrt{3}i$. [25-Q]

Ans.

$|z| = \sqrt{16 + 48} = 8$, so the fourth root of the modulus is $\sqrt[4]{8} = 2^{3/4}$; and $2^{3/4}\cos\frac{\pi}{12} = \sqrt[4]{2}\,\frac{\sqrt{3}+1}{2}$.

Write the right-hand side in trigonometric form:

$z = 4 + 4\sqrt{3}i,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{16 + 48} = 8$

$\cos\theta = \frac{x}{r} = \frac{1}{2},\ \sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{\pi}{3}$

$z = 8(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

The 4 roots are $w_k = \sqrt[4]{8}(\cos\frac{\theta + 2k\pi}{4} + i\sin\frac{\theta + 2k\pi}{4})$, k = 0, 1, ..., 3, with $\theta = \frac{\pi}{3}$.

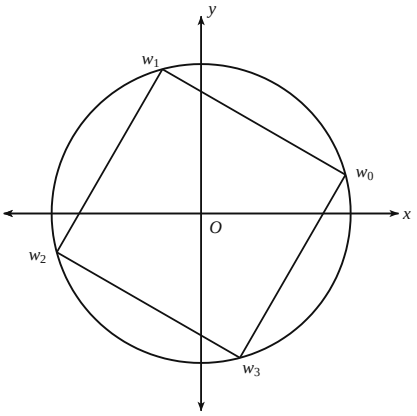
k = 0: $w_0 = \sqrt[4]{8}(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}) = \sqrt[4]{2}\,\frac{(\sqrt{3}+1)}{2} + i\sqrt[4]{2}\,\frac{(\sqrt{3}-1)}{2}$

k = 1: $w_1 = \sqrt[4]{8}(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}) = -\sqrt[4]{2}\,\frac{(\sqrt{3}-1)}{2} + i\sqrt[4]{2}\,\frac{(\sqrt{3}+1)}{2}$

k = 2: $w_2 = \sqrt[4]{8}(\cos(-\frac{11\pi}{12}) + i\sin(-\frac{11\pi}{12})) = -\sqrt[4]{2}\,\frac{(\sqrt{3}+1)}{2} - i\sqrt[4]{2}\,\frac{(\sqrt{3}-1)}{2}$

k = 3: $w_3 = \sqrt[4]{8}(\cos(-\frac{5\pi}{12}) + i\sin(-\frac{5\pi}{12})) = \sqrt[4]{2}\,\frac{(\sqrt{3}-1)}{2} - i\sqrt[4]{2}\,\frac{(\sqrt{3}+1)}{2}$

Position of the roots on the xy-plane: they lie on the circle $|z| = \sqrt[4]{8}$ and are equally spaced, $\frac{\pi}{2}$ apart.



Answer: $\pm(\sqrt[4]{2}\,\frac{(\sqrt{3}+1)}{2} + i\sqrt[4]{2}\,\frac{(\sqrt{3}-1)}{2}),\ \pm(-\sqrt[4]{2}\,\frac{(\sqrt{3}-1)}{2} + i\sqrt[4]{2}\,\frac{(\sqrt{3}+1)}{2})$

Q15. Solve $z^4 = -8 + 8\sqrt{3}i$, where z is a complex number. [25-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = -8 + 8\sqrt{3}i,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{64 + 192} = 16$

$\cos\theta = \frac{x}{r} = -\frac{1}{2},\ \sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{2\pi}{3}$

$z = 16(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

The 4 roots are $w_k = 2(\cos\frac{\theta + 2k\pi}{4} + i\sin\frac{\theta + 2k\pi}{4})$, k = 0, 1, ..., 3, with $\theta = \frac{2\pi}{3}$.

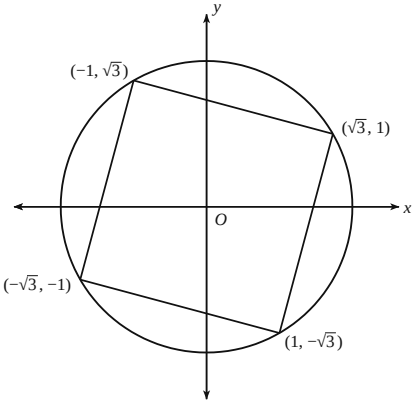
k = 0: $w_0 = 2(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}) = \sqrt{3} + i$

k = 1: $w_1 = 2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}) = -1 + \sqrt{3}i$

k = 2: $w_2 = 2(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6})) = -\sqrt{3} - i$

k = 3: $w_3 = 2(\cos(-\frac{\pi}{3}) + i\sin(-\frac{\pi}{3})) = 1 - \sqrt{3}i$

Position of the roots on the xy-plane: they lie on the circle $|z| = 2$ and are equally spaced, $\frac{\pi}{2}$ apart.



Answer: $\pm(\sqrt{3} + i),\ \pm(-1 + \sqrt{3}i)$

Q16. Solve the equation $z^4 = 32(1 + \sqrt{3}i)$, where z is a complex number. [25-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = 32 + 32\sqrt{3}i,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{1024 + 3072} = 64$

$\cos\theta = \frac{x}{r} = \frac{1}{2},\ \sin\theta = \frac{y}{r} = \frac{\sqrt{3}}{2} \rightarrow \theta = \frac{\pi}{3}$

$z = 64(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$

The 4 roots are $w_k = 2\sqrt{2}(\cos\frac{\theta + 2k\pi}{4} + i\sin\frac{\theta + 2k\pi}{4})$, k = 0, 1, ..., 3, with $\theta = \frac{\pi}{3}$.

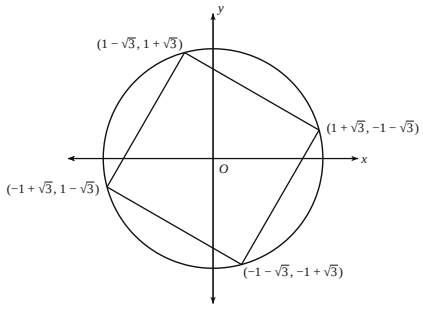
k = 0: $w_0 = 2\sqrt{2}(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}) = (\sqrt{3}+1) + (\sqrt{3}-1)i$

k = 1: $w_1 = 2\sqrt{2}(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}) = (1-\sqrt{3}) + (\sqrt{3}+1)i$

k = 2: $w_2 = 2\sqrt{2}(\cos(-\frac{11\pi}{12}) + i\sin(-\frac{11\pi}{12})) = (-\sqrt{3}-1) + (1-\sqrt{3})i$

k = 3: $w_3 = 2\sqrt{2}(\cos(-\frac{5\pi}{12}) + i\sin(-\frac{5\pi}{12})) = (\sqrt{3}-1) - (\sqrt{3}+1)i$

Position of the roots on the xy-plane: they lie on the circle $|z| = 2\sqrt{2}$ and are equally spaced, $\frac{\pi}{2}$ apart.



Answer: $\pm(\sqrt{3}+1+i(\sqrt{3}-1))$, $\pm(1-\sqrt{3}+i(\sqrt{3}+1))$

Q17. Solve the equation $z^3 + 8i = 0$. [25-Q]

Ans.

$z^3 + 8i = 0 \rightarrow z^3 = -8i$.

Write the right-hand side in trigonometric form:

$z = -8i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{0+64} = 8$

$\cos\theta = \frac{x}{r} = 0$, $\sin\theta = \frac{y}{r} = -1 \rightarrow \theta = -\frac{\pi}{2}$

$z = 8(\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2}))$

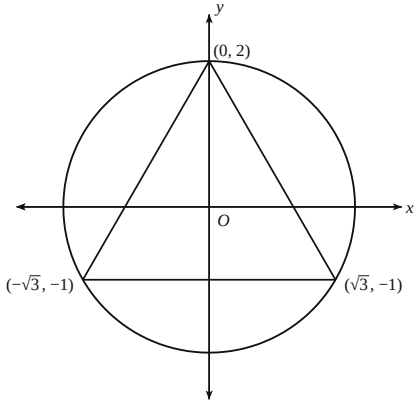
The 3 roots are $w_k = 2(\cos\frac{\theta+2k\pi}{3} + i\sin\frac{\theta+2k\pi}{3})$, k = 0, 1, ..., 2, with $\theta = -\frac{\pi}{2}$.

k = 0: $w_0 = 2(\cos(-\frac{\pi}{6}) + i\sin(-\frac{\pi}{6})) = \sqrt{3} - i$

k = 1: $w_1 = 2(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}) = 2i$

k = 2: $w_2 = 2(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6})) = -\sqrt{3} - i$

Position of the roots on the xy-plane: they lie on the circle |z| = 2 and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: $2i$, $\sqrt{3} - i$, $-\sqrt{3} - i$

Q18. Find all the solutions of $z^3 = 8$ and indicate their position on the xy-coordinate plane. [25-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = 8$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{64+0} = 8$

$\cos\theta = \frac{x}{r} = 1$, $\sin\theta = \frac{y}{r} = 0 \rightarrow \theta = 0$

$z = 8(\cos 0 + i\sin 0)$

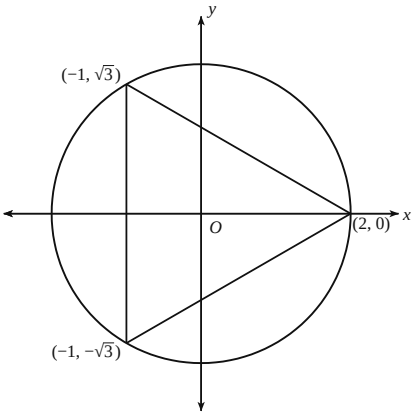
The 3 roots are $w_k = 2(\cos\frac{\theta+2k\pi}{3} + i\sin\frac{\theta+2k\pi}{3})$, k = 0, 1, ..., 2, with $\theta = 0$.

k = 0: $w_0 = 2(\cos 0 + i\sin 0) = 2$

k = 1: $w_1 = 2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}) = -1 + \sqrt{3}i$

k = 2: $w_2 = 2(\cos(-\frac{2\pi}{3}) + i\sin(-\frac{2\pi}{3})) = -1 - \sqrt{3}i$

Position of the roots on the xy-plane: they lie on the circle |z| = 2 and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: 2, $-1 + \sqrt{3}i$, $-1 - \sqrt{3}i$

Q19. Find all the solutions of $z^3 + 27i = 0$ and indicate their position on the xy-coordinate plane. [24-Q]

Ans.

$z^3 + 27i = 0 \rightarrow z^3 = -27i$.

Write the right-hand side in trigonometric form:

$z = -27i$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{0+729} = 27$

$\cos\theta = \frac{x}{r} = 0$, $\sin\theta = \frac{y}{r} = -1 \rightarrow \theta = -\frac{\pi}{2}$

$z = 27(\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2}))$

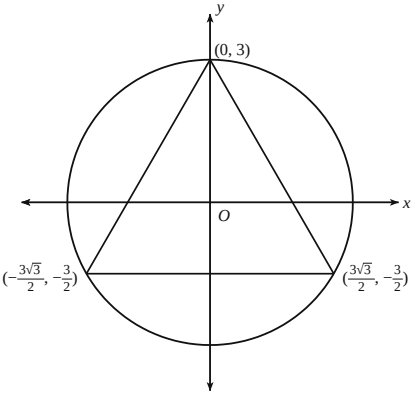
The 3 roots are $w_k = 3(\cos\frac{\theta+2k\pi}{3} + i\sin\frac{\theta+2k\pi}{3})$, k = 0, 1, ..., 2, with $\theta = -\frac{\pi}{2}$.

k = 0: $w_0 = 3(\cos(-\frac{\pi}{6}) + i\sin(-\frac{\pi}{6})) = \frac{3\sqrt{3}}{2} - \frac{3}{2}i$

k = 1: $w_1 = 3(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}) = 3i$

k = 2: $w_2 = 3(\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6})) = \frac{-3\sqrt{3}}{2} - \frac{3}{2}i$

Position of the roots on the xy-plane: they lie on the circle |z| = 3 and are equally spaced, $\frac{2\pi}{3}$ apart.



Answer: $3i$, $\frac{3\sqrt{3}}{2} - \frac{3}{2}i$, $-\frac{3\sqrt{3}}{2} - \frac{3}{2}i$

Q20. Find all the solutions of $z^4 + 16 = 0$ and indicate their position on the xy-coordinate plane. [25-Q]

Ans.

$z^4 + 16 = 0 \rightarrow z^4 = -16$.

Write the right-hand side in trigonometric form:

$z = -16$, $r = |z| = \sqrt{x^2+y^2} = \sqrt{256+0} = 16$

$\cos\theta = \frac{x}{r} = -1$, $\sin\theta = \frac{y}{r} = 0 \rightarrow \theta = \pi$

$z = 16(\cos\pi + i\sin\pi)$

The 4 roots are $w_k = 2(\cos\frac{\theta+2k\pi}{4} + i\sin\frac{\theta+2k\pi}{4})$, k = 0, 1, ..., 3, with $\theta = \pi$.

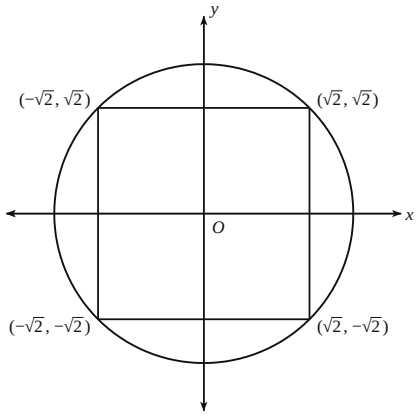
k = 0: $w_0 = 2(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}) = \sqrt{2} + \sqrt{2}i$

k = 1: $w_1 = 2(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}) = -\sqrt{2} + \sqrt{2}i$

k = 2: $w_2 = 2(\cos(-\frac{3\pi}{4}) + i\sin(-\frac{3\pi}{4})) = -\sqrt{2} - \sqrt{2}i$

k = 3: $w_3 = 2(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})) = \sqrt{2} - \sqrt{2}i$

Position of the roots on the xy-plane: they lie on the circle |z| = 2 and are equally spaced, $\frac{\pi}{2}$ apart.



Answer: $\pm\sqrt{2} \pm \sqrt{2}i$

Q21. Find the four fourth roots of −16 and indicate their positions on the xy-coordinate plane. [26-Q]

Ans.

Write the right-hand side in trigonometric form:

$z = -16,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{256 + 0} = 16$

$\cos\theta = \frac{x}{r} = -1,\ \sin\theta = \frac{y}{r} = 0 \rightarrow \theta = \pi$

$z = 16(\cos\pi + i\sin\pi)$

The 4 roots are $w_k = 2(\cos\frac{\theta + 2k\pi}{4} + i\sin\frac{\theta + 2k\pi}{4})$, k = 0, 1, ..., 3, with θ = π.

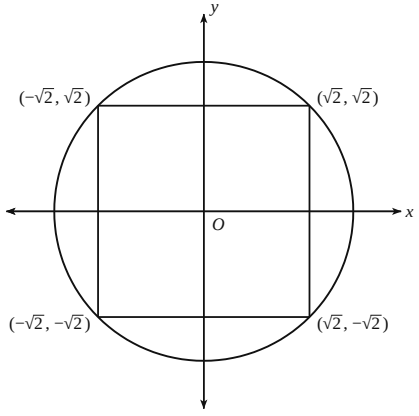
k = 0: $w_0 = 2(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}) = \sqrt{2} + \sqrt{2}i$

k = 1: $w_1 = 2(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}) = -\sqrt{2} + \sqrt{2}i$

k = 2: $w_2 = 2(\cos(-\frac{3\pi}{4}) + i\sin(-\frac{3\pi}{4})) = -\sqrt{2} - \sqrt{2}i$

k = 3: $w_3 = 2(\cos(-\frac{\pi}{4}) + i\sin(-\frac{\pi}{4})) = \sqrt{2} - \sqrt{2}i$

Position of the roots on the xy-plane: they lie on the circle |z| = 2 and are equally spaced, $\frac{\pi}{2}$ apart.



Answer: $\pm\sqrt{2} \pm \sqrt{2}i$

Q22. Find all the solutions of $z^4 - 16i = 0$ and indicate their position on the xy-coordinate plane. [25-Q]

Ans.

Using $\cos\frac{\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$, $\sin\frac{\pi}{8} = \frac{\sqrt{2-\sqrt{2}}}{2}$ (half-angle formulas).

$z^4 - 16i = 0 \rightarrow z^4 = 16i$.

Write the right-hand side in trigonometric form:

$z = 16i,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{0 + 256} = 16$

$\cos\theta = \frac{x}{r} = 0,\ \sin\theta = \frac{y}{r} = 1 \rightarrow \theta = \frac{\pi}{2}$

$z = 16(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$

The 4 roots are $w_k = 2(\cos\frac{\theta + 2k\pi}{4} + i\sin\frac{\theta + 2k\pi}{4})$, k = 0, 1, ..., 3, with $\theta = \frac{\pi}{2}$.

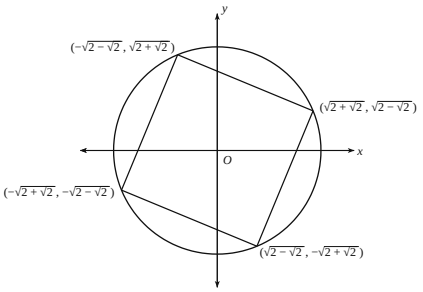
k = 0: $w_0 = 2(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}) = 2\cos\frac{\pi}{8} + 2i\sin\frac{\pi}{8} = \sqrt{2+\sqrt{2}} + i\sqrt{2-\sqrt{2}}$

k = 1: $w_1 = 2(\cos\frac{5\pi}{8} + i\sin\frac{5\pi}{8}) = -\sqrt{2-\sqrt{2}} + i\sqrt{2+\sqrt{2}}$

k = 2: $w_2 = 2(\cos(-\frac{7\pi}{8}) + i\sin(-\frac{7\pi}{8})) = -\sqrt{2+\sqrt{2}} - i\sqrt{2-\sqrt{2}}$

k = 3: $w_3 = 2(\cos(-\frac{3\pi}{8}) + i\sin(-\frac{3\pi}{8})) = \sqrt{2-\sqrt{2}} - i\sqrt{2+\sqrt{2}}$

Position of the roots on the xy-plane: they lie on the circle |z| = 2 and are equally spaced, $\frac{\pi}{2}$ apart.



Answer: $\pm(\sqrt{2+\sqrt{2}} + i\sqrt{2-\sqrt{2}}),\ \pm(-\sqrt{2-\sqrt{2}} + i\sqrt{2+\sqrt{2}})$

Q23. Find all the solutions of $z^6 + 1 = 0$ and indicate their positions on the xy-coordinate plane. [25-Q]

Ans.

$z^6 + 1 = 0 \rightarrow z^6 = -1$.

Write the right-hand side in trigonometric form:

$z = -1,\ r = |z| = \sqrt{x^2 + y^2} = \sqrt{1 + 0} = 1$

$\cos\theta = \frac{x}{r} = -1,\ \sin\theta = \frac{y}{r} = 0 \rightarrow \theta = \pi$

$z = (\cos\pi + i\sin\pi)$

The 6 roots are $w_k = (\cos\frac{\theta + 2k\pi}{6} + i\sin\frac{\theta + 2k\pi}{6})$, k = 0, 1, ..., 5, with θ = π.

k = 0: $w_0 = (\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}) = \frac{\sqrt{3}}{2} + \frac{1}{2}i$

k = 1: $w_1 = (\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}) = i$

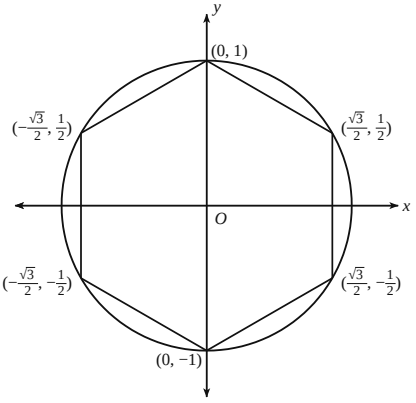
k = 2: $w_2 = (\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}) = \frac{-\sqrt{3}}{2} + \frac{1}{2}i$

k = 3: $w_3 = (\cos(-\frac{5\pi}{6}) + i\sin(-\frac{5\pi}{6})) = \frac{-\sqrt{3}}{2} - \frac{1}{2}i$

k = 4: $w_4 = (\cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2})) = -i$

k = 5: $w_5 = (\cos(-\frac{\pi}{6}) + i\sin(-\frac{\pi}{6})) = \frac{\sqrt{3}}{2} - \frac{1}{2}i$

Position of the roots on the xy-plane: they lie on the circle |z| = 1 and are equally spaced, $\frac{\pi}{3}$ apart.



Answer: $\pm i, \pm\frac{\sqrt{3}}{2} \pm \frac{1}{2}i$

CH.02 Mathematical Induction

Notes · Questions with Answers

NOTES

Principle of Mathematical Induction

For each natural number n, let P(n) be a statement depending on n. Suppose that:

1. The statement is true for n=1.
2. For any natural number k, if the statement is true for n=k, then the statement is true for n=k+1.

Then the statement P(n) is true for all natural numbers n.

Two Steps of the Method

Initial step: Prove that the statement is true for n=1.
Inductive step: Assume that the statement is true for n=k, and use this assumption to prove that the statement is true for n=k+1.

3 MARKS

2.1

Q1. Use the mathematical induction principle to prove that $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ for all natural numbers n. [25-Q]

Ans.

Let P(n) denote the statement: $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

(1) For n = 1:

L.H.S = 1

R.H.S = $\frac{1(1+1)}{2} = 1$

L.H.S = R.H.S

The statement is true for n = 1.
(2) Assume that the statement is true for n = k, that is

$1 + 2 + \dots + k = \frac{k(k+1)}{2}$

We will show that the statement is true for n = k + 1, that is we have to show that

$1 + 2 + \dots + k + (k+1) = \frac{(k+1)(k+2)}{2}$

L.H.S $= (1 + 2 + \dots + k) + (k+1)$

$= \frac{k(k+1)}{2} + (k+1)$

$= \frac{k(k+1) + 2(k+1)}{2}$

$= \frac{(k+1)(k+2)}{2}$

= R.H.S

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q2. Prove that 2 + 4 + 6 + ... + 2n = n(n+1) for all natural numbers n by using the principle of mathematical induction. [24-Q]

Ans.

Let P(n) denote the statement: 2 + 4 + 6 + ... + 2n = n(n+1)
(1) For n = 1:

L.H.S = 2
R.H.S = 1(1+1) = 2
L.H.S = R.H.S

The statement is true for n = 1.
(2) Assume that the statement is true for n = k, that is

$2 + 4 + \dots + 2k = k(k+1)$

We will show that the statement is true for n = k + 1, that is we have to show that

$2 + 4 + \dots + 2k + 2(k+1) = (k+1)(k+2)$

L.H.S $= (2 + 4 + \dots + 2k) + 2(k+1)$

$= k(k+1) + 2(k+1)$

$= (k+1)(k+2)$

= R.H.S

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q3. Prove that $\frac{2}{5} + \frac{4}{5} + \frac{6}{5} + \dots + \frac{2n}{5} = \frac{1}{5}n(n+1)$ for all natural numbers n by the use of mathematical induction principle. [24-Q]

Ans.

Let P(n) denote the statement: $\frac{2}{5} + \frac{4}{5} + \dots + \frac{2n}{5} = \frac{1}{5}n(n+1)$

(1) For n = 1:

L.H.S = $\frac{2}{5}$

R.H.S = $\frac{1}{5}(1)(1+1) = \frac{2}{5}$

L.H.S = R.H.S

The statement is true for n = 1.
(2) Assume that the statement is true for n = k, that is

$\frac{2}{5} + \frac{4}{5} + \dots + \frac{2k}{5} = \frac{1}{5}k(k+1)$

We will show that the statement is true for n = k + 1, that is we have to show that

$\frac{2}{5} + \frac{4}{5} + \dots + \frac{2k}{5} + \frac{2(k+1)}{5} = \frac{1}{5}(k+1)(k+2)$

L.H.S $= (\frac{2}{5} + \frac{4}{5} + \dots + \frac{2k}{5}) + \frac{2(k+1)}{5}$

$= \frac{1}{5}k(k+1) + \frac{2(k+1)}{5}$

$= \frac{1}{5}(k+1)(k+2)$

= R.H.S

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q4. Use the mathematical induction principle to prove that $2{\cdot}1 + 2{\cdot}2 + 2{\cdot}3 + \dots + 2{\cdot}n = n^2 + n$, for all natural numbers n. [24-Q]

Ans.

Let P(n) denote the statement: $2{\cdot}1 + 2{\cdot}2 + \dots + 2{\cdot}n = n^2 + n$
(1) For n = 1:

L.H.S = $2{\cdot}1 = 2$
R.H.S = $1^2 + 1 = 2$
L.H.S = R.H.S

The statement is true for n = 1.
(2) Assume that the statement is true for n = k, that is

$2{\cdot}1 + 2{\cdot}2 + \dots + 2{\cdot}k = k^2 + k$

We will show that the statement is true for n = k + 1, that is we have to show that

$2{\cdot}1 + 2{\cdot}2 + \dots + 2{\cdot}k + 2(k+1) = (k+1)^2 + (k+1)$

L.H.S $= (k^2 + k) + 2(k+1)$

$= k^2 + 3k + 2$

$= (k^2 + 2k + 1) + (k + 1)$

$= (k+1)^2 + (k+1)$

= R.H.S

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q5. Prove that $1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$, for all natural numbers n by the use of mathematical induction principle. [25-Q]

Ans.

Let P(n) denote the statement: $1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$

(1) For n = 1:

L.H.S = 1

R.H.S = $\frac{1(3-1)}{2} = 1$

L.H.S = R.H.S

The statement is true for n = 1.
(2) Assume that the statement is true for n = k, that is

$1 + 4 + 7 + \dots + (3k-2) = \frac{k(3k-1)}{2}$

We will show that the statement is true for n = k + 1, that is we have to show that

$1 + 4 + 7 + \dots + (3k-2) + (3(k+1)-2) = \frac{(k+1)(3(k+1)-1)}{2}$

L.H.S $= (1 + 4 + \dots + (3k-2)) + (3k+1)$

$= \frac{k(3k-1)}{2} + (3k+1)$

$= \frac{3k^2 - k + 6k + 2}{2}$

$= \frac{3k^2 + 5k + 2}{2}$

$= \frac{(k+1)(3k+2)}{2}$

$= \frac{(k+1)(3(k+1)-1)}{2}$

= R.H.S

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q6. Use the mathematical induction principle to prove that $1 + 5 + 9 + \dots + (4n-3) = 2n^2 - n$, for all natural numbers n. [25-Q]

Ans.

Let P(n) denote the statement: $1 + 5 + 9 + \dots + (4n-3) = 2n^2 - n$

(1) For n = 1:

L.H.S = 1

R.H.S = $2(1)^2 - 1 = 1$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$1 + 5 + 9 + \dots + (4k-3) = 2k^2 - k$

We will show that the statement is true for n = k + 1, that is we have to show that

$1 + 5 + 9 + \dots + (4k-3) + (4(k+1)-3) = 2(k+1)^2 - (k+1)$

$$\begin{aligned}\text{L.H.S} &= (2k^2 - k) + (4k+1)\\ &= 2k^2 + 3k + 1\\ &= (2k^2 + 4k + 2) - (k + 1)\\ &= 2(k+1)^2 - (k+1)\\ &= \text{R.H.S}\end{aligned}$$

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q7. Prove that $\frac{1}{1\cdot5} + \frac{1}{5\cdot9} + \frac{1}{9\cdot13} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$ for $n \geq 1$ by using the principle of mathematical induction. [26-Q]

Ans.

Let P(n) denote the statement: $\frac{1}{1\cdot5} + \frac{1}{5\cdot9} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$

(1) For n = 1:

L.H.S = $\frac{1}{1\cdot5} = \frac{1}{5}$

R.H.S = $\frac{1}{4(1)+1} = \frac{1}{5}$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$\frac{1}{1\cdot5} + \dots + \frac{1}{(4k-3)(4k+1)} = \frac{k}{4k+1}$

We will show that the statement is true for n = k + 1, that is we have to show that

$\frac{1}{1\cdot5} + \dots + \frac{1}{(4k-3)(4k+1)} + \frac{1}{(4k+1)(4k+5)} = \frac{k+1}{4k+5}$

$$\begin{aligned}\text{L.H.S} &= \frac{k}{4k+1} + \frac{1}{(4k+1)(4k+5)}\\ &= \frac{k(4k+5)+1}{(4k+1)(4k+5)}\\ &= \frac{4k^2+5k+1}{(4k+1)(4k+5)}\\ &= \frac{(4k+1)(k+1)}{(4k+1)(4k+5)}\\ &= \frac{k+1}{4k+5}\\ &= \frac{k+1}{4(k+1)+1}\\ &= \text{R.H.S}\end{aligned}$$

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q8. Use the mathematical induction principle to prove that $1 + 3 + 3^2 + \dots + 3^{n-1} = \frac{3^n - 1}{2}$ for all natural numbers n. [26-Q]

Ans.

Let P(n) denote the statement: $1 + 3 + 3^2 + \dots + 3^{n-1} = \frac{3^n - 1}{2}$

(1) For n = 1:

L.H.S = 1

R.H.S = $\frac{3^1 - 1}{2} = 1$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$1 + 3 + \dots + 3^{k-1} = \frac{3^k - 1}{2}$

We will show that the statement is true for n = k + 1, that is we have to show that

$1 + 3 + \dots + 3^{k-1} + 3^k = \frac{3^{k+1} - 1}{2}$

$$\begin{aligned}\text{L.H.S} &= \frac{3^k - 1}{2} + 3^k\\ &= \frac{3^k - 1 + 2\cdot3^k}{2}\\ &= \frac{3\cdot3^k - 1}{2}\\ &= \frac{3^{k+1} - 1}{2}\\ &= \text{R.H.S}\end{aligned}$$

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q9. Prove that $\frac{1}{2} + (\frac{1}{2})^2 + \dots + (\frac{1}{2})^n = 1 - (\frac{1}{2})^n$ for all natural numbers n by using the principle of mathematical induction. [24-Q]

Ans.

Let P(n) denote the statement: $\frac{1}{2} + (\frac{1}{2})^2 + \dots + (\frac{1}{2})^n = 1 - (\frac{1}{2})^n$

(1) For n = 1:

L.H.S = $\frac{1}{2}$

R.H.S = $1 - \frac{1}{2} = \frac{1}{2}$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$\frac{1}{2} + (\frac{1}{2})^2 + \dots + (\frac{1}{2})^k = 1 - (\frac{1}{2})^k$

We will show that the statement is true for n = k + 1, that is we have to show that

$\frac{1}{2} + \dots + (\frac{1}{2})^k + (\frac{1}{2})^{k+1} = 1 - (\frac{1}{2})^{k+1}$

Note that $(\frac{1}{2})^k = 2(\frac{1}{2})^{k+1}$.

$$\begin{aligned}\text{L.H.S} &= 1 - (\tfrac{1}{2})^k + (\tfrac{1}{2})^{k+1}\\ &= 1 - 2(\tfrac{1}{2})^{k+1} + (\tfrac{1}{2})^{k+1}\\ &= 1 - (\tfrac{1}{2})^{k+1}\\ &= \text{R.H.S}\end{aligned}$$

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q10. Prove that $\frac{2}{3^1} + \frac{2}{3^2} + \frac{2}{3^3} + \dots + \frac{2}{3^n} = 1 - \frac{1}{3^n}$ for all natural numbers n by using the principle of mathematical induction. [25-Q]

Ans.

Let P(n) denote the statement: $\frac{2}{3} + \frac{2}{3^2} + \dots + \frac{2}{3^n} = 1 - \frac{1}{3^n}$

(1) For n = 1:

L.H.S = $\frac{2}{3}$

R.H.S = $1 - \frac{1}{3} = \frac{2}{3}$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$\frac{2}{3} + \frac{2}{3^2} + \dots + \frac{2}{3^k} = 1 - \frac{1}{3^k}$

We will show that the statement is true for n = k + 1, that is we have to show that

$\frac{2}{3} + \dots + \frac{2}{3^k} + \frac{2}{3^{k+1}} = 1 - \frac{1}{3^{k+1}}$

$$\begin{aligned}\text{L.H.S} &= 1 - \frac{1}{3^k} + \frac{2}{3^{k+1}}\\ &= 1 - \frac{3}{3^{k+1}} + \frac{2}{3^{k+1}}\\ &= 1 - \frac{1}{3^{k+1}}\\ &= \text{R.H.S}\end{aligned}$$

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

2.2

Q11. Prove that $5^n - 1$ is divisible by 4 for all natural numbers n by using mathematical induction. [26-Q]

Ans.

Let P(n) denote the statement: $5^n - 1$ is divisible by 4

(1) For n = 1:

$5^1 - 1 = 4 = 4 \cdot 1$

which is divisible by 4.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$5^k - 1$ is divisible by 4, that is $5^k - 1 = 4m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$5^{k+1} - 1$ is divisible by 4.

$$\begin{aligned} 5^{k+1} - 1 &= 5 \cdot 5^k - 1 \\ &= 5(5^k - 1) + 4 \\ &= 5(4m) + 4 \\ &= 4(5m + 1) \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q12. Use the mathematical induction principle to prove that $6^n - 1$ is divisible by 5 for all natural numbers n. [25-Q]

Ans.

Let P(n) denote the statement: $6^n - 1$ is divisible by 5

(1) For n = 1:

$6^1 - 1 = 5$

which is divisible by 5.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$6^k - 1$ is divisible by 5, that is $6^k - 1 = 5m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$6^{k+1} - 1$ is divisible by 5.

$$\begin{aligned} 6^{k+1} - 1 &= 6 \cdot 6^k - 1 \\ &= 6(6^k - 1) + 5 \\ &= 6(5m) + 5 \\ &= 5(6m + 1) \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q13. Use the mathematical induction principle to prove that $7^n - 1$ is a multiple of 3 for all natural numbers n. [24-Q]

Ans.

Let P(n) denote the statement: $7^n - 1$ is a multiple of 3

(1) For n = 1:

$7^1 - 1 = 6 = 3 \cdot 2$

which is a multiple of 3.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$7^k - 1$ is a multiple of 3, that is $7^k - 1 = 3m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$7^{k+1} - 1$ is a multiple of 3.

$$\begin{aligned} 7^{k+1} - 1 &= 7 \cdot 7^k - 1 \\ &= 7(7^k - 1) + 6 \\ &= 7(3m) + 6 \\ &= 3(7m + 2) \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q14. Use the mathematical induction principle to prove that $5^n + 3$ is divisible by 4 for all natural numbers n. [25-Q]

Ans.

Let P(n) denote the statement: $5^n + 3$ is divisible by 4

(1) For n = 1:

$5^1 + 3 = 8 = 4 \cdot 2$

which is divisible by 4.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$5^k + 3$ is divisible by 4, that is $5^k + 3 = 4m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$5^{k+1} + 3$ is divisible by 4.

$$\begin{aligned} 5^{k+1} + 3 &= 5 \cdot 5^k + 3 \\ &= 5(5^k + 3) - 15 + 3 \\ &= 5(4m) - 12 \\ &= 4(5m - 3) \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q15. Use the mathematical induction principle to prove that $3^{2n} - 1$ is divisible by 8 for all natural numbers n. [25-Q, 26-Q]

Ans.

Let P(n) denote the statement: $3^{2n} - 1$ is divisible by 8

(1) For n = 1:

$3^2 - 1 = 8$

which is divisible by 8.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$3^{2k} - 1$ is divisible by 8, that is $3^{2k} - 1 = 8m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$3^{2(k+1)} - 1$ is divisible by 8.

$$\begin{aligned} 3^{2(k+1)} - 1 &= 3^2 \cdot 3^{2k} - 1 \\ &= 9(3^{2k} - 1) + 9 - 1 \\ &= 9(8m) + 8 \\ &= 8(9m + 1) \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q16. Prove that $n^3 + 2n$ is divisible by 3 for all natural numbers n by using the mathematical induction. [24-Q]

Ans.

Let P(n) denote the statement: $n^3 + 2n$ is divisible by 3

(1) For n = 1:

$1^3 + 2(1) = 3$

which is divisible by 3.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$k^3 + 2k$ is divisible by 3, that is $k^3 + 2k = 3m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$(k+1)^3 + 2(k+1)$ is divisible by 3.

$$\begin{aligned} (k+1)^3 + 2(k+1) &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \\ &= (k^3 + 2k) + 3k^2 + 3k + 3 \\ &= 3m + 3(k^2 + k + 1) \\ &= 3(m + k^2 + k + 1) \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

2.3

Q17. Consider the sequence of real numbers defined by the relations $x_1 = 1$ and $x_{n+1} = \sqrt{1 + 2x_n}$ for all $n \geq 1$. Use the mathematical induction principle to prove that $x_n < 4$ for all $n \geq 1$. [5-Q]

Ans.

Let P(n) denote the statement: $x_n < 4$.

(1) For n = 1: $x_1 = 1 < 4$, so the statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is $x_k < 4$. We will show that $x_{k+1} < 4$.

$$\begin{aligned} x_{k+1} &= \sqrt{1 + 2x_k} \\ &< \sqrt{1 + 2(4)} \\ &= \sqrt{9} = 3 < 4 \end{aligned}$$

(Since $x_k < 4$ gives $2x_k < 8$ and $1 + 2x_k < 9$; the square root is increasing.) Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, $x_n < 4$ for all $n \geq 1$.

Q18. Given $n^4 < 10^n$ for a fixed positive integer $n \geq 2$, prove that $(n+1)^4 < 10^{n+1}$ using principle of mathematical induction. [5-Q]

Ans.

Let P(n) denote the statement: $n^4 < 10^n$ for $n \geq 2$

(1) For n = 2:

L.H.S $= 2^4 = 16$

R.H.S $= 10^2 = 100$

$16 < 100$, so the statement is true.

The statement is true for n = 2.

(2) Assume that the statement is true for n = k, that is

$k^4 < 10^k$ $(k \geq 2)$

We will show that the statement is true for n = k + 1, that is we have to show that

$(k+1)^4 < 10^{k+1}$

$$\begin{aligned} (k+1)^4 &= k^4\left(1 + \frac{1}{k}\right)^4 \\ &\leq k^4\left(1 + \frac{1}{2}\right)^4 \text{ (since } k \geq 2) \end{aligned}$$

$= \frac{81}{16} k^4$

$< 10k^4$

$< 10 \cdot 10^k = 10^{k+1}$

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all integers n ≥ 2.

Q19. Use the mathematical induction principle to prove that $2n < 2^n - 1$ for all natural numbers n ≥ 3. [25-Q]

Ans.
Let P(n) denote the statement: $2n < 2^n - 1$
(1) For n = 3:

L.H.S = 2(3) = 6
R.H.S = $2^3 - 1 = 7$
6 < 7, so the statement is true.

The statement is true for n = 3.
(2) Assume that the statement is true for n = k, that is

$2k < 2^k - 1 \ (k \geq 3)$

We will show that the statement is true for n = k + 1, that is we have to show that
$2(k+1) < 2^{k+1} - 1$

$2(k+1) = 2k + 2$
$< (2^k - 1) + 2 = 2^k + 1$
$\leq 2^k + 2^k - 1$ (since $2 \leq 2^k$)
$= 2^{k+1} - 1$

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n ≥ 3.

Q20. Use the principle of mathematical induction to prove that $2n + 1 < n^2$ for all natural numbers n ≥ 3. [25-Q]

Ans.
Let P(n) denote the statement: $2n + 1 < n^2$
(1) For n = 3:

L.H.S = 2(3) + 1 = 7
R.H.S = $3^2 = 9$
7 < 9, so the statement is true.

The statement is true for n = 3.
(2) Assume that the statement is true for n = k, that is

$2k + 1 < k^2 \ (k \geq 3)$

We will show that the statement is true for n = k + 1, that is we have to show that
$2(k+1) + 1 < (k+1)^2$

$2(k+1) + 1 = (2k + 1) + 2$
$< k^2 + 2$
$< k^2 + 2k + 1$ (since $2 < 2k + 1$)
$= (k+1)^2$

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n ≥ 3.

Q21. Use the principle of mathematical induction to prove that $4n^2 > n + 11$ for all natural numbers n ≥ 2. [25-Q]

Ans.
Let P(n) denote the statement: $4n^2 > n + 11$
(1) For n = 2:

L.H.S = $4(2)^2 = 16$
R.H.S = 2 + 11 = 13
16 > 13, so the statement is true.

The statement is true for n = 2.
(2) Assume that the statement is true for n = k, that is

$4k^2 > k + 11 \ (k \geq 2)$

We will show that the statement is true for n = k + 1, that is we have to show that
$4(k+1)^2 > (k+1) + 11$

$4(k+1)^2 = 4k^2 + 8k + 4$
$> (k + 11) + 8k + 4$
$= 9k + 15$
$> k + 12 = (k+1) + 11$

(The last step holds because $9k + 15 > k + 12$ for k ≥ 2.)
Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n ≥ 2.

Q22. Prove that $2n < 2^n$ for all natural numbers n ≥ 3 by using the mathematical induction. [25*-Q]

Ans.
Let P(n) denote the statement: $2n < 2^n$
(1) For n = 3:

L.H.S = 6
R.H.S = $2^3 = 8$
6 < 8, so the statement is true.

The statement is true for n = 3.
(2) Assume that the statement is true for n = k, that is

$2k < 2^k \ (k \geq 3)$

We will show that the statement is true for n = k + 1, that is we have to show that
$2(k+1) < 2^{k+1}$

$2(k+1) = 2k + 2$
$< 2^k + 2$
$< 2^k + 2^k$ (since $2 < 2^k$)
$= 2^{k+1}$

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n ≥ 3.

Q23. Use the mathematical induction principle to prove that $2^n > n$ for all natural numbers n. [25*-Q]

Ans.
Let P(n) denote the statement: $2^n > n$
(1) For n = 1:

L.H.S = $2^1 = 2$
R.H.S = 1
2 > 1, so the statement is true.

The statement is true for n = 1.
(2) Assume that the statement is true for n = k, that is
$2^k > k$
We will show that the statement is true for n = k + 1, that is we have to show that
$2^{k+1} > k + 1$

$2^{k+1} = 2 \cdot 2^k$
$> 2k$
$= k + k$
$\geq k + 1$ (since $k \geq 1$)

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

5 MARKS

2.1

Q1. Use the mathematical induction principle to prove that $1 + 4 + 7 + \ldots + (3n-2) = \frac{n(3n-1)}{2}$ for all natural numbers n. [25-Q]

Ans.
Let P(n) denote the statement: $1 + 4 + 7 + \ldots + (3n-2) = \frac{n(3n-1)}{2}$
(1) For n = 1:

L.H.S = 1
R.H.S = $\frac{1(3-1)}{2} = 1$
L.H.S = R.H.S

The statement is true for n = 1.
(2) Assume that the statement is true for n = k, that is

$1 + 4 + 7 + \ldots + (3k-2) = \frac{k(3k-1)}{2}$

We will show that the statement is true for n = k + 1, that is we have to show that

$1 + 4 + 7 + \ldots + (3k-2) + (3(k+1)-2) = \frac{(k+1)(3(k+1)-1)}{2}$

L.H.S $= (1 + 4 + \ldots + (3k-2)) + (3k+1)$
$= \frac{k(3k-1)}{2} + (3k+1)$
$= \frac{3k^2 - k + 6k + 2}{2}$
$= \frac{3k^2 + 5k + 2}{2}$
$= \frac{(k+1)(3k+2)}{2}$
$= \frac{(k+1)(3(k+1)-1)}{2}$
= R.H.S

Therefore, the statement is true for n = k + 1.
(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q2. Prove that $1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for all natural numbers n by using the

principle of mathematical induction. [24-Q]

Ans.

Let P(n) denote the statement: $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

(1) For n = 1:

L.H.S $= 1^2 = 1$

R.H.S $= \frac{1 \cdot 2 \cdot 3}{6} = 1$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$

We will show that the statement is true for n = k + 1, that is we have to show that

$1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$

L.H.S $= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$

$= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6}$

$= \frac{(k+1)(2k^2 + 7k + 6)}{6}$

$= \frac{(k+1)(k+2)(2k+3)}{6}$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q3. Prove that $2^2 + 4^2 + 6^2 + \dots + (2n)^2 = \frac{2}{3} n(n+1)(2n+1)$ for all natural numbers n by using the principle of mathematical induction. [25*-Q]

Ans.

Let P(n) denote the statement: $2^2 + 4^2 + \dots + (2n)^2 = \frac{2}{3} n(n+1)(2n+1)$

(1) For n = 1:

L.H.S $= 2^2 = 4$

R.H.S $= \frac{2}{3}(1)(2)(3) = 4$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$2^2 + 4^2 + \dots + (2k)^2 = \frac{2}{3} k(k+1)(2k+1)$

We will show that the statement is true for n = k + 1, that is we have to show that

$2^2 + \dots + (2k)^2 + (2k+2)^2 = \frac{2}{3}(k+1)(k+2)(2k+3)$

L.H.S $= \frac{2}{3} k(k+1)(2k+1) + 4(k+1)^2$

$= \frac{2}{3}(k+1)[k(2k+1) + 6(k+1)]$

$= \frac{2}{3}(k+1)(2k^2 + 7k + 6)$

$= \frac{2}{3}(k+1)(k+2)(2k+3)$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q4. Prove that $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}$ by using the principle of mathematical induction for all natural numbers n. [24-Q]

Ans.

Let P(n) denote the statement: $1^2 + 3^2 + \dots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}$

(1) For n = 1:

L.H.S $= 1^2 = 1$

R.H.S $= \frac{1 \cdot 1 \cdot 3}{3} = 1$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$1^2 + 3^2 + \dots + (2k-1)^2 = \frac{k(2k-1)(2k+1)}{3}$

We will show that the statement is true for n = k + 1, that is we have to show that

$1^2 + \dots + (2k-1)^2 + (2k+1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}$

L.H.S $= \frac{k(2k-1)(2k+1)}{3} + (2k+1)^2$

$= \frac{(2k+1)[k(2k-1) + 3(2k+1)]}{3}$

$= \frac{(2k+1)(2k^2 + 5k + 3)}{3}$

$= \frac{(2k+1)(k+1)(2k+3)}{3}$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q5. Prove that $1^3 + 3^3 + 5^3 + \dots + (2n-1)^3 = n^2(2n^2 - 1)$ for all natural numbers n by the use of the mathematical induction principle. [25-Q]

Ans.

Let P(n) denote the statement: $1^3 + 3^3 + \dots + (2n-1)^3 = n^2(2n^2 - 1)$

(1) For n = 1:

L.H.S $= 1^3 = 1$

R.H.S $= 1^2(2 - 1) = 1$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$1^3 + 3^3 + \dots + (2k-1)^3 = k^2(2k^2 - 1)$

We will show that the statement is true for n = k + 1, that is we have to show that

$1^3 + \dots + (2k-1)^3 + (2k+1)^3 = (k+1)^2(2(k+1)^2 - 1)$

Check: $(k+1)^2(2(k+1)^2 - 1) = (k^2+2k+1)(2k^2+4k+1) = 2k^4 + 8k^3 + 11k^2 + 6k + 1$.

L.H.S $= k^2(2k^2 - 1) + (2k+1)^3$

$= 2k^4 - k^2 + 8k^3 + 12k^2 + 6k + 1$

$= 2k^4 + 8k^3 + 11k^2 + 6k + 1$

$= (k^2 + 2k + 1)(2k^2 + 4k + 1)$

$= (k+1)^2(2(k+1)^2 - 1)$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q6. Prove that $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$, for all natural numbers n by the use of mathematical induction principle. [25-Q, 25*-Q]

Ans.

Let P(n) denote the statement: $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$

(1) For n = 1:

L.H.S $= \frac{1}{1 \cdot 3} = \frac{1}{3}$

R.H.S $= \frac{1}{2(1)+1} = \frac{1}{3}$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$\frac{1}{1 \cdot 3} + \dots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1}$

We will show that the statement is true for n = k + 1, that is we have to show that

$\frac{1}{1 \cdot 3} + \dots + \frac{1}{(2k-1)(2k+1)} + \frac{1}{(2k+1)(2k+3)} = \frac{k+1}{2k+3}$

L.H.S $= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)}$

$= \frac{k(2k+3) + 1}{(2k+1)(2k+3)}$

$= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)}$

$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$

$= \frac{k+1}{2k+3}$

$= \frac{k+1}{2(k+1)+1}$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q7. Prove that $\frac{1}{10} + \frac{1}{40} + \frac{1}{88} + \dots + \frac{1}{(3n-1)(3n+2)} = \frac{n}{6n+4}$ for all $n \in Z^+$. [5-Q]

Ans.

Let P(n) denote the statement: $\frac{1}{10} + \frac{1}{40} + \frac{1}{88} + \dots + \frac{1}{(3n-1)(3n+2)} = \frac{n}{6n+4}$

(1) For n = 1:

L.H.S $= \frac{1}{(3-1)(3+2)} = \frac{1}{2\cdot5} = \frac{1}{10}$

R.H.S $= \frac{1}{6+4} = \frac{1}{10}$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$$\frac{1}{10} + \ldots + \frac{1}{(3k-1)(3k+2)} = \frac{k}{6k+4}$$

We will show that the statement is true for n = k + 1, that is we have to show that

$$\frac{1}{10} + \ldots + \frac{1}{(3k-1)(3k+2)} + \frac{1}{(3k+2)(3k+5)} = \frac{k+1}{6(k+1)+4}$$

L.H.S $= \frac{k}{6k+4} + \frac{1}{(3k+2)(3k+5)}$

$= \frac{k}{2(3k+2)} + \frac{1}{(3k+2)(3k+5)}$

$= \frac{k(3k+5)+2}{2(3k+2)(3k+5)}$

$= \frac{3k^2+5k+2}{2(3k+2)(3k+5)}$

$= \frac{(3k+2)(k+1)}{2(3k+2)(3k+5)}$

$= \frac{k+1}{2(3k+5)}$

$= \frac{k+1}{6k+10}$

$= \frac{k+1}{6(k+1)+4}$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q8. Prove that $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \ldots + \frac{1}{1+2+3+\ldots+n} = \frac{2n}{n+1}$ by the use of the mathematical induction principle. [25-Q]

Ans.

Let P(n) denote the statement: $1 + \frac{1}{1+2} + \frac{1}{1+2+3} + \ldots + \frac{1}{1+2+3+\ldots+n} = \frac{2n}{n+1}$

(1) For n = 1:

L.H.S = 1

R.H.S $= \frac{2(1)}{1+1} = 1$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$$1 + \frac{1}{1+2} + \ldots + \frac{1}{1+2+\ldots+k} = \frac{2k}{k+1}$$

We will show that the statement is true for n = k + 1, that is we have to show that

$$1 + \frac{1}{1+2} + \ldots + \frac{1}{1+2+\ldots+k} + \frac{1}{1+2+\ldots+(k+1)} = \frac{2(k+1)}{k+2}$$

Use $1 + 2 + \ldots + (k+1) = \frac{(k+1)(k+2)}{2}$, so $\frac{1}{1+2+\ldots+(k+1)} = \frac{2}{(k+1)(k+2)}$.

L.H.S $= \frac{2k}{k+1} + \frac{1}{1+2+\ldots+(k+1)}$

$= \frac{2k}{k+1} + \frac{2}{(k+1)(k+2)}$

$= \frac{2k(k+2)+2}{(k+1)(k+2)}$

$= \frac{2(k^2+2k+1)}{(k+1)(k+2)}$

$= \frac{2(k+1)^2}{(k+1)(k+2)}$

$= \frac{2(k+1)}{k+2}$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q9. Prove that $1 + 2 + 2^2 + \ldots + 2^{n-1} = 2^n - 1$ by using the mathematical induction for all natural numbers n. [24-Q]

Ans.

Let P(n) denote the statement: $1 + 2 + 2^2 + \ldots + 2^{n-1} = 2^n - 1$

(1) For n = 1:

L.H.S = 1

R.H.S $= 2^1 - 1 = 1$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$$1 + 2 + \ldots + 2^{k-1} = 2^k - 1$$

We will show that the statement is true for n = k + 1, that is we have to show that

$$1 + 2 + \ldots + 2^{k-1} + 2^k = 2^{k+1} - 1$$

L.H.S $= (2^k - 1) + 2^k$

$= 2\cdot2^k - 1$

$= 2^{k+1} - 1$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q10. Prove that $1\cdot2 + 2\cdot2^2 + 3\cdot2^3 + \ldots + n\cdot2^n = (n-1)2^{n+1} + 2$ by using the mathematical induction for all natural numbers n. [24-Q]

Ans.

Let P(n) denote the statement: $1\cdot2 + 2\cdot2^2 + 3\cdot2^3 + \ldots + n\cdot2^n = (n-1)2^{n+1} + 2$

(1) For n = 1:

L.H.S $= 1\cdot2 = 2$

R.H.S $= (1-1)2^2 + 2 = 2$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$$1\cdot2 + 2\cdot2^2 + \ldots + k\cdot2^k = (k-1)2^{k+1} + 2$$

We will show that the statement is true for n = k + 1, that is we have to show that

$$1\cdot2 + \ldots + k\cdot2^k + (k+1)2^{k+1} = k\cdot2^{k+2} + 2$$

L.H.S $= (k-1)2^{k+1} + 2 + (k+1)2^{k+1}$

$= 2^{k+1}[(k-1) + (k+1)] + 2$

$= 2k\cdot2^{k+1} + 2$

$= k\cdot2^{k+2} + 2$

$= ((k+1)-1)2^{(k+1)+1} + 2$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q11. Prove that $\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \ldots + \frac{1}{3^n} = \frac{3^n - 1}{2(3^n)}$ for all natural numbers n by using the principle of mathematical induction. [25-Q]

Ans.

Let P(n) denote the statement: $\frac{1}{3} + \frac{1}{9} + \ldots + \frac{1}{3^n} = \frac{3^n - 1}{2(3^n)}$

(1) For n = 1:

L.H.S $= \frac{1}{3}$

R.H.S $= \frac{3-1}{2(3)} = \frac{1}{3}$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$$\frac{1}{3} + \frac{1}{9} + \ldots + \frac{1}{3^k} = \frac{3^k - 1}{2(3^k)}$$

We will show that the statement is true for n = k + 1, that is we have to show that

$$\frac{1}{3} + \ldots + \frac{1}{3^k} + \frac{1}{3^{k+1}} = \frac{3^{k+1} - 1}{2(3^{k+1})}$$

L.H.S $= \frac{3^k - 1}{2(3^k)} + \frac{1}{3^{k+1}}$

$= \frac{3(3^k - 1)}{2(3^{k+1})} + \frac{2}{2(3^{k+1})}$

$= \frac{3^{k+1} - 3 + 2}{2(3^{k+1})}$

$= \frac{3^{k+1} - 1}{2(3^{k+1})}$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q12. Prove that $1^2 - 2^2 + 3^2 - 4^2 + \ldots + (-1)^{n-1}n^2 = (-1)^{n-1}\frac{n(n+1)}{2}$ by using the mathematical induction principle for all natural numbers n. (Hint: $(-1)^{k-1} = -(-1)^k$.) [26-Q]

Ans.

Let P(n) denote the statement: $1^2 - 2^2 + 3^2 - \ldots + (-1)^{n-1}n^2 = (-1)^{n-1}\frac{n(n+1)}{2}$

(1) For n = 1:

L.H.S $= 1^2 = 1$

R.H.S $= (-1)^0 \frac{1\cdot 2}{2} = 1$

L.H.S = R.H.S

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$1^2 - 2^2 + \ldots + (-1)^{k-1}k^2 = (-1)^{k-1}\frac{k(k+1)}{2}$

We will show that the statement is true for n = k + 1, that is we have to show that

$1^2 - 2^2 + \ldots + (-1)^{k-1}k^2 + (-1)^k(k+1)^2 = (-1)^k\frac{(k+1)(k+2)}{2}$

L.H.S $= (-1)^{k-1}\frac{k(k+1)}{2} + (-1)^k(k+1)^2$

$= -(-1)^k\frac{k(k+1)}{2} + (-1)^k(k+1)^2$

$= (-1)^k(k+1)[-\frac{k}{2} + (k+1)]$

$= (-1)^k(k+1)\frac{-k+2k+2}{2}$

$= (-1)^k\frac{(k+1)(k+2)}{2}$

$= (-1)^{(k+1)-1}\frac{(k+1)((k+1)+1)}{2}$

= R.H.S

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

2.2

Q13. Use the mathematical induction principle to prove that $3^n - 1$ is divisible by 2 for all natural numbers n. [25-Q]

Ans.

Let P(n) denote the statement: $3^n - 1$ is divisible by 2

(1) For n = 1:

$3^1 - 1 = 2$

which is divisible by 2.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$3^k - 1$ is divisible by 2, that is $3^k - 1 = 2m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$3^{k+1} - 1$ is divisible by 2.

$3^{k+1} - 1 = 3\cdot 3^k - 1$

$= 3(3^k - 1) + 2$

$= 3(2m) + 2$

$= 2(3m + 1)$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q14. Prove that $4^{2n} - 1$ is divisible by 15 for all natural numbers n by using the mathematical induction. [24-Q]

Ans.

Let P(n) denote the statement: $4^{2n} - 1$ is divisible by 15

(1) For n = 1:

$4^2 - 1 = 15$

which is divisible by 15.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$4^{2k} - 1$ is divisible by 15, that is $4^{2k} - 1 = 15m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$4^{2(k+1)} - 1$ is divisible by 15.

$4^{2(k+1)} - 1 = 16\cdot 4^{2k} - 1$

$= 16(4^{2k} - 1) + 15$

$= 16(15m) + 15$

$= 15(16m + 1)$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q15. Prove that $n^3 + 2n$ is divisible by 3 for all natural numbers n by using the mathematical induction. [24-Q]

Ans.

Let P(n) denote the statement: $n^3 + 2n$ is divisible by 3

(1) For n = 1:

$1^3 + 2(1) = 3$

which is divisible by 3.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$k^3 + 2k$ is divisible by 3, that is $k^3 + 2k = 3m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$(k+1)^3 + 2(k+1)$ is divisible by 3.

$(k+1)^3 + 2(k+1) = k^3 + 3k^2 + 3k + 1 + 2k + 2$

$= (k^3 + 2k) + 3k^2 + 3k + 3$

$= 3m + 3(k^2 + k + 1)$

$= 3(m + k^2 + k + 1)$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q16. Prove that $10\cdot 3^n + 6^{2n}$ is divisible by 11 for all integers $n \geq 0$ by using the mathematical induction. [24-Q]

Ans.

Let P(n) denote the statement: $10\cdot 3^n + 6^{2n}$ is divisible by 11

(1) For n = 0:

$10\cdot 3^0 + 6^0 = 10 + 1 = 11$

which is divisible by 11.

The statement is true for n = 0.

(2) Assume that the statement is true for n = k, that is

$10\cdot 3^k + 6^{2k}$ is divisible by 11, that is $10\cdot 3^k + 6^{2k} = 11m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$10\cdot 3^{k+1} + 6^{2(k+1)}$ is divisible by 11.

$10\cdot 3^{k+1} + 6^{2(k+1)} = 3(10\cdot 3^k) + 36\cdot 6^{2k}$

$= 3(10\cdot 3^k + 6^{2k}) + 33\cdot 6^{2k}$

$= 3(11m) + 33\cdot 6^{2k}$

$= 11(3m + 3\cdot 6^{2k})$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all integers $n \geq 0$.

Q17. Show by the principle of mathematical induction that $2^{2n-1} + 5^{2n-1}$ is divisible by 7 for all natural numbers n. [25-Q]

Ans.

Let P(n) denote the statement: $2^{2n-1} + 5^{2n-1}$ is divisible by 7

(1) For n = 1:

$2^1 + 5^1 = 7$

which is divisible by 7.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$2^{2k-1} + 5^{2k-1}$ is divisible by 7, that is $2^{2k-1} + 5^{2k-1} = 7m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$2^{2k+1} + 5^{2k+1}$ is divisible by 7.

$2^{2k+1} + 5^{2k+1} = 4\cdot 2^{2k-1} + 25\cdot 5^{2k-1}$

$= 4(2^{2k-1} + 5^{2k-1}) + 21\cdot 5^{2k-1}$

$= 4(7m) + 21\cdot 5^{2k-1}$

$= 7(4m + 3\cdot 5^{2k-1})$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q18. Use the mathematical induction principle to prove that $x^{2n} - y^{2n}$ is divisible by x + y for all natural numbers n. [26-Q]

Ans.

Let P(n) denote the statement: $x^{2n} - y^{2n}$ is divisible by (x + y)

(1) For n = 1:

$x^2 - y^2 = (x - y)(x + y)$

which is divisible by (x + y).

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$x^{2k} - y^{2k}$ is divisible by (x + y), that is $x^{2k} - y^{2k} = (x + y)m$ for some integer m.

We will show that the statement is true for n = k + 1, that is we have to show that

$x^{2k+2} - y^{2k+2}$ is divisible by (x + y).

$x^{2k+2} - y^{2k+2} = x^2\cdot x^{2k} - y^2\cdot y^{2k}$

$= x^2(x^{2k} - y^{2k}) + x^2\cdot y^{2k} - y^2\cdot y^{2k}$

$= x^2(x+y)m + y^{2k}(x^2 - y^2)$

$= x^2(x+y)m + y^{2k}(x-y)(x+y)$

$= (x+y)[x^2m + y^{2k}(x-y)]$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n.

Q19. Use the mathematical induction principle to prove that x + y is a factor of $x^{2n+1} + y^{2n+1}$ for $n \geq 1$. [25-Q]

Ans.

Let P(n) denote the statement: x + y is a factor of $x^{2n+1} + y^{2n+1}$

(1) For n = 1:

$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$

so x + y is a factor of $x^3 + y^3$.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$x^{2k+1} + y^{2k+1}$ is divisible by (x + y), that is $x^{2k+1} + y^{2k+1} = (x + y)m$ for some polynomial m.

We will show that the statement is true for n = k + 1, that is we have to show that

$x^{2k+3} + y^{2k+3}$ is divisible by (x + y).

$$\begin{aligned} x^{2k+3} + y^{2k+3} &= x^2 \cdot x^{2k+1} + y^2 \cdot y^{2k+1} \\ &= x^2(x^{2k+1} + y^{2k+1}) - x^2 \cdot y^{2k+1} + y^2 \cdot y^{2k+1} \\ &= x^2(x+y)m - y^{2k+1}(x^2 - y^2) \\ &= x^2(x+y)m - y^{2k+1}(x-y)(x+y) \\ &= (x+y)[x^2 m - y^{2k+1}(x-y)] \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for n ≥ 1.

2.3

Q20. Consider the sequence of real numbers defined by the relations $x_1 = 1$ and $x_{n+1} = \sqrt{1 + 2x_n}$ for all n ≥ 1. Use the mathematical induction principle to prove that $x_n < 4$ for all n ≥ 1. [25*-Q]

Ans.

Let P(n) denote the statement: $x_n < 4$.

(1) For n = 1: $x_1 = 1 < 4$, so the statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is $x_k < 4$. We will show that $x_{k+1} < 4$.

$$\begin{aligned} x_{k+1} &= \sqrt{1 + 2x_k} \\ &< \sqrt{1 + 2(4)} \\ &= \sqrt{9} = 3 < 4 \end{aligned}$$

(Since $x_k < 4$ gives $2x_k < 8$ and $1 + 2x_k < 9$; the square root is increasing.) Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, $x_n < 4$ for all n ≥ 1.

Q21. Use the mathematical induction principle to prove that $2^n \geq 1 + n$ for all natural numbers n ≥ 1. [24-Q]

Ans.

Let P(n) denote the statement: $2^n \geq 1 + n$

(1) For n = 1:

L.H.S $= 2^1 = 2$

R.H.S $= 1 + 1 = 2$

$2 \geq 2$, so the statement is true.

The statement is true for n = 1.

(2) Assume that the statement is true for n = k, that is

$2^k \geq 1 + k$

We will show that the statement is true for n = k + 1, that is we have to show that

$2^{k+1} \geq 1 + (k+1)$

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k \\ \geq 2(1 + k) &= 2 + 2k \\ &\geq 2 + k \text{ (since } k \geq 0) \\ &= 1 + (k+1) \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n ≥ 1.

Q22. Use the mathematical induction principle to prove that $n! > 3^n$ for all natural numbers n ≥ 7. [25-Q, 5-Q]

Ans.

Let P(n) denote the statement: $n! > 3^n$

(1) For n = 7:

L.H.S $= 7! = 5040$

R.H.S $= 3^7 = 2187$

$5040 > 2187$, so the statement is true.

The statement is true for n = 7.

(2) Assume that the statement is true for n = k, that is

$k! > 3^k$ (k ≥ 7)

We will show that the statement is true for n = k + 1, that is we have to show that $(k+1)! > 3^{k+1}$

$$\begin{aligned} (k+1)! &= (k+1) \cdot k! \\ &> (k+1) \cdot 3^k \\ &\geq 3 \cdot 3^k \text{ (since } k + 1 \geq 8 > 3) \\ &= 3^{k+1} \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n ≥ 7.

Q23. Use the mathematical induction principle to prove that $4n < 2^n$ for all natural numbers n ≥ 5. [5-Q, 26-Q]

Ans.

Let P(n) denote the statement: $4n < 2^n$

(1) For n = 5:

L.H.S $= 4(5) = 20$

R.H.S $= 2^5 = 32$

$20 < 32$, so the statement is true.

The statement is true for n = 5.

(2) Assume that the statement is true for n = k, that is

$4k < 2^k$ (k ≥ 5)

We will show that the statement is true for n = k + 1, that is we have to show that $4(k+1) < 2^{k+1}$

$$\begin{aligned} 4(k+1) &= 4k + 4 \\ &< 2^k + 4 \\ &< 2^k + 2^k \text{ (since } 4 < 2^k \text{ for } k \geq 5) \\ &= 2^{k+1} \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n ≥ 5.

Q24. Prove that $4n < n^2 - 5$ for all natural numbers n ≥ 6 by using the mathematical induction. [26-Q]

Ans.

Let P(n) denote the statement: $4n < n^2 - 5$

(1) For n = 6:

L.H.S $= 4(6) = 24$

R.H.S $= 6^2 - 5 = 31$

$24 < 31$, so the statement is true.

The statement is true for n = 6.

(2) Assume that the statement is true for n = k, that is

$4k < k^2 - 5$ (k ≥ 6)

We will show that the statement is true for n = k + 1, that is we have to show that $4(k+1) < (k+1)^2 - 5$

$$\begin{aligned} 4(k+1) &= 4k + 4 \\ < (k^2 - 5) + 4 &= k^2 - 1 \\ &< k^2 + 2k - 4 \text{ (since } 2k - 4 > -1) \\ &= (k+1)^2 - 5 \end{aligned}$$

Therefore, the statement is true for n = k + 1.

(3) Hence, by the principle of mathematical induction, the statement P(n) is true for all natural numbers n ≥ 6.

CH.03 Analytical Solid Geometry

Notes · Questions with Answers

NOTES

Coordinates of a point in space
A point P in space is determined by an ordered triple (x, y, z) of real numbers, measured along three mutually perpendicular axes (the x-axis, y-axis and z-axis) through the origin O.

Coordinate planes
The xy-plane (equation z=0), yz-plane (equation x=0) and zx-plane (equation y=0) are the three planes formed by pairs of axes. A plane parallel to the xy-plane has equation z=c; similarly x=a is parallel to the yz-plane and y=b is parallel to the zx-plane.

Directed values of a line segment
For a line segment PQ with $P(x_1,y_1,z_1)$ and $Q(x_2,y_2,z_2)$, the directed values $\langle l, m, n\rangle$ of PQ are $\langle x_2-x_1, y_2-y_1, z_2-z_1\rangle$. k is called a parameter, and (kl, km, kn) are also directed values of the same line for any real number k.

Skew lines
In space, a pair of lines that are neither parallel nor intersecting are called skew lines.

Plane through three points
A plane is determined by three points that are not on the same line. If A, B, C are such points with $\langle AB\rangle=\langle l_1,m_1,n_1\rangle$ and $\langle AC\rangle=\langle l_2,m_2,n_2\rangle$, the plane has Cartesian equation ax+by+cz=d.

Sphere
A sphere is the set of all points (x, y, z) whose distance from a fixed center (x_1, y_1, z_1) equals a fixed radius r.

Distance between two points
$$PQ=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

Directed values of PQ
$\langle PQ\rangle=\langle l, m, n\rangle=\langle x_2-x_1, y_2-y_1, z_2-z_1\rangle$

Length of segment PQ
$$PQ=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}=\sqrt{l^2+m^2+n^2}$$

Coordinates of point R on line PQ
$(x, y, z)=(x_1+kl, y_1+km, z_1+kn)$, for a real parameter k

Equation of a line (symmetric form)
$$\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$$

Parallel lines
Two lines are parallel if and only if their directed values are multiples of each other by some real number.

cos∠PAQ
$$\cos\angle PAQ=\frac{l_1l_2+m_1m_2+n_1n_2}{\sqrt{l_1^2+m_1^2+n_1^2}\sqrt{l_2^2+m_2^2+n_2^2}}$$

Perpendicular lines
Two lines are perpendicular if and only if they intersect and $l_1l_2+m_1m_2+n_1n_2=0$ for any directed values $\langle l_1, m_1, n_1\rangle$ and $\langle l_2, m_2, n_2\rangle$ of the lines.

Cartesian form of a plane
ax+by+cz=d

where $a=m_1n_2-m_2n_1$, $b=n_1l_2-n_2l_1$, $c=l_1m_2-l_2m_1$, for two non-parallel directed values $\langle l_1,m_1,n_1\rangle$, $\langle l_2,m_2,n_2\rangle$ in the plane, and $d=ax_1+by_1+cz_1$ for a point (x_1,y_1,z_1) on the plane.

Line perpendicular to a plane
Any line with directed values $\langle ka, kb, kc\rangle$, for some parameter k, is perpendicular to the plane ax+by+cz=d.

Radius of a sphere
$$\sqrt{(x-x_1)^2+(y-y_1)^2+(z-z_1)^2}=r$$

Equation of a sphere
$$(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=r^2$$

with center (x_1, y_1, z_1) and radius r.

1 MARK

3.1

Q1. The equation of the plane containing the point (3, 4, 5) and parallel to the xy−plane is [24-Q]

A. x = 3 **B.** y = 4 **C.** z = 5 **D.** y = −3

Ans.

A plane parallel to the xy−plane has the form z = c. It passes through (3, 4, 5), so c = 5.
Answer: C (z = 5)

Q2. The equation of the line passing through the points (3, 2, −7) and perpendicular to the yz plane is [25-Q]

A. x = 3 **B.** y = 2, z = 7 **C.** x = 3, y = 2 **D.** y = 2, z = −7

Ans.

A line perpendicular to the yz−plane is parallel to the x−axis, so y and z stay constant: y = 2, z = −7.
Answer: D

Q3. The distance between two points (3, 2, −2) and (7, 4, 2) is [24-Q]

A. 5 **B.** 6 **C.** 7 **D.** 8

Ans.

$$d=\sqrt{(7-3)^2+(4-2)^2+(2-(-2))^2}$$
$$=\sqrt{16+4+16}$$
$$=\sqrt{36}=6$$

Answer: B

Q4. The distance from the point (4, 3, 0) to the z−axis is [25-Q]

A. 2 **B.** 3 **C.** 4 **D.** 5

Ans.

Distance from (a, b, c) to the z−axis is $\sqrt{a^2+b^2}$.

$$d=\sqrt{4^2+3^2}$$
$$=\sqrt{25}=5$$

Answer: D

Q5. The distance from the point (3, 7, −5) to the z−axis is [25-Q]

A. $\sqrt{74}$ **B.** $\sqrt{58}$ **C.** $\sqrt{34}$ **D.** $\sqrt{43}$

Ans.

$$d=\sqrt{3^2+7^2}$$
$$=\sqrt{9+49}=\sqrt{58}$$

Answer: B

3.4

Q6. If the points (−1, 3, 2) and (4, 5, 0) lie on the plane ax + by − 3z = 2, then the value of b is [25-Q]

A. 1 **B.** 2 **C.** 3 **D.** 4

Ans.

Substitute both points:

$$-a+3b-6=2 \rightarrow -a+3b=8$$
$$4a+5b=2$$

From the first, a = 3b − 8. Then 4(3b − 8) + 5b = 2, so 17b = 34 and b = 2.
Answer: B

Q7. The plane passing through the point (a, b, c) and parallel to the plane x + y + z = 0 is [24-Q]

A. x + y + z = a + b + c **B.** x + y + z + (a + b + c) = 0 **C.** x + y + z + abc = 0
D. ax + by + cz = 0

Ans.

Parallel planes share the normal (1, 1, 1): x + y + z = d. Putting in (a, b, c): d = a + b + c.
Answer: A

Q8. The plane passing through the point (1, 2, 3) and parallel to the plane x + y + z = 0 is [24-Q]

A. x + y + z = 6 **B.** x + y + z + 6 = 0 **C.** x + y + z = 5 **D.** x + 2y + 3z = 0

Ans.

x + y + z = d with d = 1 + 2 + 3 = 6.
Answer: A

Q9. The plane passing through the point (1, 2, −1) and parallel to the plane x − y + z = 2 is [24-Q]

A. x + y + z = 2 **B.** x + y − z − 4 = 0 **C.** $x-y+z+\sqrt{6}=0$ **D.** x − y + z = −2

Ans.

x − y + z = d with d = 1 − 2 + (−1) = −2.
Answer: D

Q10. The equation of the plane through the origin and parallel to the plane 3x − 4y + 5z = 6 is [25*-Q]

A. 3x − 4y + 5z = 6 **B.** 3x + 4y − 5z = 6 **C.** 3x − 4y − 5z = 6 **D.** 3x − 4y + 5z = 0

Ans.

Same normal (3, −4, 5); through the origin d = 0: 3x − 4y + 5z = 0.
Answer: D

Q11. The equation of the plane that is parallel to the plane 4x − 3y + 2z = 5 and passing through the point (5, 1, −7) is [26-Q]

A. 4x − 3y + 2z = 3 **B.** 4x + 3y + 2z = 3 **C.** 4x − 3y + 2z = −3 **D.** 3x − 4y + 2z = −3

Ans.

4x − 3y + 2z = d with d = 4(5) − 3(1) + 2(−7) = 20 − 3 − 14 = 3.
Answer: A

3.5

Q12. The equation of the sphere is $x^2 + y^2 + z^2 + 6x - 2y + 2z - 5 = 0$. Then the radius of the sphere is [26-Q]

A. 3 **B.** 4 **C.** 5 **D.** 6

Ans.

Complete the square:

$(x+3)^2 + (y-1)^2 + (z+1)^2 = 5 + 9 + 1 + 1 = 16$

Center $(-3, 1, -1)$, $r = 4$.

Answer: B

2 MARKS

3.1

Q1. Given P = (1, 2, −3) and Q = (5, −2, 5), find the distance between the midpoint of PQ and the origin (0, 0, 0). [25-Q]

Ans.

$$\text{Midpoint M} = \left(\frac{1+5}{2}, \frac{2+(-2)}{2}, \frac{-3+5}{2}\right) = (3, 0, 1)$$

$$\text{OM} = \sqrt{3^2 + 0^2 + 1^2} = \sqrt{10}$$

Answer: $\sqrt{10}$

Q2. Given A = (−1, 2, 5) and B = (3, 2, 1), find the distance between the midpoint of AB and the origin (0, 0, 0). [25*-Q]

Ans.

$$\text{Midpoint M} = \left(\frac{-1+3}{2}, \frac{2+2}{2}, \frac{5+1}{2}\right) = (1, 2, 3)$$

$$\text{OM} = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$$

Answer: $\sqrt{14}$

Q3. The coordinates of the points A and B are (−1, 3, 14) and (2k, 3k, 13) respectively. Given that the distance from A to B is $\sqrt{163}$ units, and that k is an integer, find the value of k. [5-Q]

Ans.

$$AB^2 = (2k+1)^2 + (3k-3)^2 + (13-14)^2 = 163$$

$$4k^2 + 4k + 1 + 9k^2 - 18k + 9 + 1 = 163$$

$$13k^2 - 14k - 152 = 0$$

$$(k - 4)(13k + 38) = 0$$

So $k = 4$ or $k = -\frac{38}{13}$. Since k is an integer, $k = 4$.

Answer: k = 4

3.2

Q4. Given that P(1, 2, 3) and Q(1, 1, 3), find the coordinates of point R(x, y, z) on the line PQ with respect to the point P and the parameter k = 2. [26-Q]

Ans.

Coordinate form: $(x, y, z) = (x_1 + kl, y_1 + km, z_1 + kn)$ where $\langle l, m, n\rangle = \langle PQ\rangle = \langle 1-1, 1-2, 3-3\rangle = \langle 0, -1, 0\rangle$.

$$(x, y, z) = (1 + 2(0), 2 + 2(-1), 3 + 2(0)) = (1, 0, 3)$$

Answer: R(1, 0, 3)

Q5. Find the coordinates of the point R on the line segment joining the points P(1, 2, 3) and Q(7, 5, −3) such that $PR = \frac{2}{3}PQ$. [25-Q]

Ans.

Directed values: $\langle PQ\rangle = (6, 3, -6)$ and $k = \frac{2}{3}$.

$$R = \left(1 + \frac{2}{3}(6), 2 + \frac{2}{3}(3), 3 + \frac{2}{3}(-6)\right)$$

$$= (1 + 4, 2 + 2, 3 - 4)$$

$$= (5, 4, -1)$$

Answer: R(5, 4, −1)

3.3

Q6. Suppose that the directed values of lines AB and AC are $\langle 2, 3, -1\rangle$ and $\langle -9, 5, -3\rangle$ respectively. Find ∠BAC. [25*-Q]

Ans.

$$\cos \angle BAC = \frac{l_1l_2 + m_1m_2 + n_1n_2}{\sqrt{l_1^2+m_1^2+n_1^2}\sqrt{l_2^2+m_2^2+n_2^2}}$$

$$= \frac{2(-9) + 3(5) + (-1)(-3)}{\sqrt{14}\sqrt{115}}$$

$$= \frac{-18 + 15 + 3}{\sqrt{14}\sqrt{115}} = 0$$

∠BAC = 90° (or $\frac{\pi}{2}$). The lines are perpendicular.

3.4

Q7. Find the coordinate form of the equation of the line passing through the point (1, 5, 2) and perpendicular to the plane $2x - 3y + z = 5$. [24-Q, 26-Q]

Ans.

Directed values of the line perpendicular to the plane $2x - 3y + z = 5$ are $\langle 2, -3, 1\rangle$. Then equation of the line is

$$\frac{x-1}{2} = \frac{y-5}{-3} = \frac{z-2}{1}$$

So coordinates of the points on the line are of the form (coordinate form)

$$x = 1 + 2s$$
$$y = 5 - 3s$$
$$z = 2 + s$$

3.5

Q8. The equation of the sphere is $x^2 + y^2 + z^2 - 4x + 2y + 6z - 11 = 0$. The center $C(x_1, y_1, z_1)$ and radius of the sphere are r respectively, then find $r(x_1 + y_1 + z_1)$. [24-Q]

Ans.

$(x-2)^2 + (y+1)^2 + (z+3)^2 = 11 + 4 + 1 + 9 = 25$

Center $(2, -1, -3)$, $r = 5$.

$r(x_1 + y_1 + z_1) = 5(2 - 1 - 3) = -10$

Answer: −10

Q9. The equation of the sphere is $x^2 + y^2 + z^2 - 4y - 14z - 91 = 0$. If the center and radius of the sphere are (a, b, c) and r, respectively, then find the value of r(a − b + c). [25-Q]

Ans.

$x^2 + (y-2)^2 + (z-7)^2 = 91 + 4 + 49 = 144$

Center (0, 2, 7), r = 12.

$r(a - b + c) = 12(0 - 2 + 7) = 60$

Answer: 60

Q10. Find the diameter of the sphere $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$. [25-Q]

Ans.

$(x+2)^2 + (y-3)^2 + (z+1)^2 = -6 + 4 + 9 + 1 = 8$

$r = \sqrt{8} = 2\sqrt{2}$, so the diameter = 2r.

Answer: $4\sqrt{2}$

Q11. Find the equation of the sphere on the join of the two points (1, −2, 3) and (3, 4, 1) as diameter. [25-Q]

Ans.

Center = midpoint = (2, 1, 2).

$r^2 = (3-2)^2 + (4-1)^2 + (1-2)^2 = 1 + 9 + 1 = 11$

Answer: $(x-2)^2 + (y-1)^2 + (z-2)^2 = 11$

Q12. Find the equation of the sphere that passes through the point (−1, 7, 0) and is centered at the point (3, 4, 3). [25-Q]

Ans.

$r^2 = (-1-3)^2 + (7-4)^2 + (0-3)^2 = 16 + 9 + 9 = 34$

Answer: $(x-3)^2 + (y-4)^2 + (z-3)^2 = 34$

Q13. Find the equation of the sphere with center (6, −7, −3) and touching the xy−plane. [25-Q]

Ans.

The line passing through the center C(6, −7, −3) and perpendicular to the xy−plane (z = 0) has directed values $\langle 0, 0, 1\rangle$. So coordinates of the points on the line are of the form

$$x = 6$$
$$y = -7$$
$$z = -3 + s$$

If one of these points P is on the xy−plane z = 0, then

$$-3 + s = 0$$
$$s = 3$$

Therefore the point of intersection is P(6, −7, 0). Thus we have the radius

$$CP = \sqrt{(6-6)^2 + (-7+7)^2 + (0+3)^2} = \sqrt{9} = 3$$

Therefore the equation of the sphere is

$(x-6)^2 + (y+7)^2 + (z+3)^2 = 9$

3 MARKS

3.1

Q1. Show that the points (−2, 4, 10), (2, 2, 12) and (0, 10, 12) form a right triangle. [26-Q]

Ans.

Let A(−2, 4, 10), B(2, 2, 12), C(0, 10, 12).

$AB^2 = 4^2 + (-2)^2 + 2^2 = 24$

$AC^2 = 2^2 + 6^2 + 2^2 = 44$

$BC^2 = (-2)^2 + 8^2 + 0^2 = 68$

Since $AB^2 + AC^2 = 24 + 44 = 68 = BC^2$, the triangle is right-angled at A (Pythagoras).

Q2. Prove that A(4, 2, 2), C(16, 4, 4) and B(10, 3, 3) are collinear. [25-Q]

Ans.

$AB = \sqrt{6^2 + 1^2 + 1^2} = \sqrt{38}$

$BC = \sqrt{6^2 + 1^2 + 1^2} = \sqrt{38}$

$AC = \sqrt{12^2 + 2^2 + 2^2} = \sqrt{152} = 2\sqrt{38}$

Since $AB + BC = \sqrt{38} + \sqrt{38} = 2\sqrt{38} = AC$, the points A, B, C are collinear (B is the midpoint of AC).

Q3. Show that the points (1, 2, 2), (0, 3, 1) and (3, 0, 4) are collinear. [26-Q]

Ans.

Let P(1, 2, 2), Q(0, 3, 1), R(3, 0, 4).

$PQ = \sqrt{1 + 1 + 1} = \sqrt{3}$

$PR = \sqrt{4 + 4 + 4} = 2\sqrt{3}$

$QR = \sqrt{9 + 9 + 9} = 3\sqrt{3}$

Since $PQ + PR = \sqrt{3} + 2\sqrt{3} = 3\sqrt{3} = QR$, the three points are collinear (P lies between Q and R).

3.2

Q4. Determine the following forms of the equation of the lines passing through the points (2, −1, 5) and (−1, 3, 2). (a) a coordinate form, (b) a Cartesian form, and (c) a vector form. [25-Q]

Ans.

Given P(2, −1, 5) and Q(−1, 3, 2), we have $\langle PQ\rangle = \langle l, m, n\rangle = \langle -3, 4, -3\rangle$.

(a) Coordinate form of the line PQ (with respect to the point P and parameter k):

$(x, y, z) = (2 - 3k, -1 + 4k, 5 - 3k)$

(b) Cartesian form (equation of the line PQ):

$$\frac{x-2}{-3} = \frac{y+1}{4} = \frac{z-5}{-3}$$

(c) Vector form:

$$r = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} + k \begin{pmatrix} -3 \\ 4 \\ -3 \end{pmatrix}$$

Q5. Find the coordinates of the point R on the line segment joining the points P(1, 3, 2) and Q(7, −2, 5) such that $PR = \frac{2}{3} PQ$. [26-Q]

Ans.

$\langle PQ\rangle = (6, -5, 3)$, $k = \frac{2}{3}$.

$R = (1 + \frac{2}{3}(6), 3 + \frac{2}{3}(-5), 2 + \frac{2}{3}(3))$

$= (1 + 4, 3 - \frac{10}{3}, 2 + 2)$

$= (5, -\frac{1}{3}, 4)$

Answer: R(5, $-\frac{1}{3}$, 4)

Q6. Find the points of intersection of the line joining the two points (1, 3, 5) and (3, 5, −4) with the xy-plane. [24-Q]

Ans.

Given P(1, 3, 5) and Q(3, 5, −4), we have $\langle PQ\rangle = \langle 2, 2, -9\rangle$. So coordinates of the points on the line PQ are

$(x, y, z) = (1 + 2k, 3 + 2k, 5 - 9k)$

If the point is on the xy-plane z = 0, then

$5 - 9k = 0$

$k = \frac{5}{9}$

$x = 1 + 2(\frac{5}{9}) = \frac{19}{9}$

$y = 3 + 2(\frac{5}{9}) = \frac{37}{9}$

Therefore the point of intersection is

$(\frac{19}{9}, \frac{37}{9}, 0)$

Q7. Find the parametric equations of the lines passing through the points A(2, 4, −3) and B(3, −1, 1). At what point does the line intersect the xy-plane? [25-Q]

Ans.

Given A(2, 4, −3) and B(3, −1, 1), we have $\langle AB\rangle = \langle 1, -5, 4\rangle$. So the parametric equations (coordinate form) of the line AB are

$x = 2 + k$

$y = 4 - 5k$

$z = -3 + 4k$

If the point is on the xy-plane z = 0, then

$-3 + 4k = 0$

$k = \frac{3}{4}$

$x = 2 + \frac{3}{4} = \frac{11}{4}$

$y = 4 - 5(\frac{3}{4}) = \frac{1}{4}$

Therefore the line meets the xy-plane at

$(\frac{11}{4}, \frac{1}{4}, 0)$

3.4

Q8. Find the Cartesian equation of the plane containing the points A(0, 1, 2), B(2, 3, 1) and C(−1, 2, 3). [24-Q]

Ans.

Directed values of two lines in the plane: $\langle AB\rangle = \langle l_1, m_1, n_1\rangle = \langle 2, 2, -1\rangle$ and $\langle AC\rangle = \langle l_2, m_2, n_2\rangle = \langle -1, 1, 1\rangle$.

Plane ax + by + cz = d, where $a = m_1n_2 - m_2n_1$, $b = n_1l_2 - n_2l_1$, $c = l_1m_2 - l_2m_1$:

$a = (2)(1) - (1)(-1) = 3$

$b = (-1)(-1) - (1)(2) = -1$

$c = (2)(1) - (-1)(2) = 4$

$d = ax_1 + by_1 + cz_1$ using A(0,1,2):

$d = (3)(0) + (-1)(1) + (4)(2) = 7$

Answer: 3x − y + 4z = 7

Q9. Find the equation of the plane containing A(2, 1, 6), B(5, −2, 0), and C(4, −5, −2). [24, 25-Q]

Ans.

Directed values of two lines in the plane: $\langle AB\rangle = \langle l_1, m_1, n_1\rangle = \langle 3, -3, -6\rangle$ and $\langle AC\rangle = \langle l_2, m_2, n_2\rangle = \langle 2, -6, -8\rangle$.

Plane ax + by + cz = d, where $a = m_1n_2 - m_2n_1$, $b = n_1l_2 - n_2l_1$, $c = l_1m_2 - l_2m_1$:

$a = (-3)(-8) - (-6)(-6) = -12$

$b = (-6)(2) - (-8)(3) = 12$

$c = (3)(-6) - (2)(-3) = -12$

Divide by 12 (common factor): a = 1, b = −1, c = 1.

$d = ax_1 + by_1 + cz_1$ using A(2,1,6):

$d = (1)(2) + (-1)(1) + (1)(6) = 7$

Answer: x − y + z = 7

Q10. Find the equation of the plane containing A(2, 5, −3), B(−2, −3, 5), and C(5, 3, −3). [25*-Q]

Ans.

Directed values of two lines in the plane: $\langle AB\rangle = \langle l_1, m_1, n_1\rangle = \langle -4, -8, 8\rangle$ and $\langle AC\rangle = \langle l_2, m_2, n_2\rangle = \langle 3, -2, 0\rangle$.

Plane ax + by + cz = d, where $a = m_1n_2 - m_2n_1$, $b = n_1l_2 - n_2l_1$, $c = l_1m_2 - l_2m_1$:

$a = (-8)(0) - (-2)(8) = 16$

$b = (8)(3) - (0)(-4) = 24$

$c = (-4)(-2) - (3)(-8) = 32$

Divide by 8 (common factor): a = 2, b = 3, c = 4.

$d = ax_1 + by_1 + cz_1$ using A(2,5,−3):

$d = (2)(2) + (3)(5) + (4)(-3) = 7$

Answer: 2x + 3y + 4z = 7

Q11. Find the Cartesian form of the plane equation passing through the point (8, 3, 3) and perpendicular to the line $\frac{x-1}{6} = \frac{y+3}{1} = \frac{z+8}{-8}$. [25-Q]

Ans.

Directed values of the given line are $\langle 6, 1, -8\rangle$. This line is perpendicular to the required plane.

So equation of the required plane is

$6x + y - 8z = d$

Since the plane passes through (8, 3, 3), we get

$6(8) + 3 - 8(3) = d$

$d = 27$

Therefore the Cartesian form of the plane equation is

6x + y − 8z = 27

3.5

Q12. Find the equation of the sphere on the join of (2, 7, −4) and (4, 5, −1) as diameter. [25*-Q]

Ans.

Center = midpoint = $(\frac{2+4}{2}, \frac{7+5}{2}, \frac{-4-1}{2}) = (3, 6, -\frac{5}{2})$.

$r^2 = (4-3)^2 + (5-6)^2 + (-1+\frac{5}{2})^2 = 1 + 1 + \frac{9}{4} = \frac{17}{4}$

Answer: $(x-3)^2 + (y-6)^2 + (z+\frac{5}{2})^2 = \frac{17}{4}$

Q13. Find the equation of the plane tangent to the sphere $(x-2)^2 + (y+1)^2 + (z-1)^2 = 9$ at the point (4, 1, 2). [25-Q]

Ans.

Directed values of the line joining the center C(2, −1, 1) of the sphere and the given point P(4, 1, 2) are ⟨CP⟩ = ⟨2, 2, 1⟩. Line CP is perpendicular to the tangent plane.
Thus equation of the plane is

$2x + 2y + z = d$

Since P(4, 1, 2) is on this plane, so we get

$2(4) + 2(1) + 1(2) = d$
$d = 12$

The equation of the plane is
2x + 2y + z = 12

Q14. Find the equation of the plane tangent to the sphere $(x-2)^2 + (y-1)^2 + (z+1)^2 = 15$ at the point (3, 4, 2). [26-Q]

Ans.
Directed values of the line joining the center C(2, 1, −1) of the sphere and the given point P(3, 4, 2) are ⟨CP⟩ = ⟨1, 3, 3⟩. Line CP is perpendicular to the tangent plane.
Thus equation of the plane is

$x + 3y + 3z = d$

Since P(3, 4, 2) is on this plane, so we get

$1(3) + 3(4) + 3(2) = d$
$d = 21$

The equation of the plane is
x + 3y + 3z = 21
Note: the point (3, 4, 2) gives 1 + 9 + 9 = 19 on the left side of the sphere equation, not 15, so it is not exactly on the printed sphere; the book key (x + 3y + 3z = 21) is obtained by the method above.

5 MARKS

3.3

Q1. Point A has coordinates (7, −1, 20) and the line l is defined by the equation: x = 3 + λ, y = −1 + 2λ, z = λ. Point B lies on the line l such that AB is perpendicular to the line l. Find the coordinates of the point B. [5-Q]

Ans.
Directed values of the line l are ⟨1, 2, 1⟩. Since B is on the line l, B(3 + λ, −1 + 2λ, λ), where A(7, −1, 20).
Directed values of the line AB are

⟨AB⟩ = ⟨−4 + λ, 2λ, −20 + λ⟩

Since AB is perpendicular to the line l, $l_1l_2 + m_1m_2 + n_1n_2 = 0$, that is

$(-4 + \lambda) + 2(2\lambda) + (-20 + \lambda) = 0$
$-24 + 6\lambda = 0$
$\lambda = 4$

Hence B = (7, 7, 4).
B(7, 7, 4)

Q2. Find the equation of the line passing through the point (7, −1, 8) and perpendicular to the line (x, y, z) = (3 + k, −1 + 2k, k). Find also the point of intersection of the two lines. [26-Q]

Ans.
Directed values of the given line are ⟨1, 2, 1⟩. Let B be the point of intersection of the two lines, so B(3 + k, −1 + 2k, k).
Directed values of the line AB are

⟨AB⟩ = ⟨−4 + k, 2k, −8 + k⟩

Since AB is perpendicular to the line l, $l_1l_2 + m_1m_2 + n_1n_2 = 0$, that is

$(-4 + k) + 2(2k) + (-8 + k) = 0$
$-12 + 6k = 0$
$k = 2$

So B = (5, 3, 2) and directed values of the required line are
Directed values of the required line are ⟨AB⟩ = ⟨−2, 4, −6⟩ and it passes through A(7, −1, 8).
Equation of the line is

(x, y, z) = (7 − 2t, −1 + 4t, 8 − 6t)

The point of intersection is B(5, 3, 2).

Q3. Find the equation of the line passing through the point (−1, 5, 4) and perpendicular to the line (x, y, z) = (1 + 2k, −2 + k, 1 − k). Find also the point of intersection of the two lines. [25*-Q]

Ans.
Directed values of the given line are ⟨2, 1, −1⟩. Let B be the point of intersection of the two lines, so B(1 + 2k, −2 + k, 1 − k).
Directed values of the line AB are

⟨AB⟩ = ⟨2 + 2k, −7 + k, −3 − k⟩

Since AB is perpendicular to the line l, $l_1l_2 + m_1m_2 + n_1n_2 = 0$, that is

$2(2 + 2k) + (-7 + k) - 1(-3 - k) = 0$
$6k = 0$
$k = 0$

So B = (1, −2, 1) and directed values of the required line are
Directed values of the required line are ⟨AB⟩ = ⟨2, −7, −3⟩ and it passes through A(−1, 5, 4).
Equation of the line is

(x, y, z) = (−1 + 2t, 5 − 7t, 4 − 3t)

The point of intersection is B(1, −2, 1).

Q4. Point A has coordinates (5, 2, 2) and the line l passes through the points (2, 5, −1) and (1, 9, −3). Point B lies on the line l such that AB is perpendicular to the line l. Find the distance between points A and B. [24-Q]

Ans.
Given P(2, 5, −1) and Q(1, 9, −3) on the line l, we have ⟨PQ⟩ = ⟨−1, 4, −2⟩. So coordinates of the points on l are

(x, y, z) = (2 − k, 5 + 4k, −1 − 2k)

Since B is on the line l, B(2 − k, 5 + 4k, −1 − 2k) and A(5, 2, 2).
Directed values of the line AB are

⟨AB⟩ = ⟨−3 − k, 3 + 4k, −3 − 2k⟩

Since AB is perpendicular to the line l, $l_1l_2 + m_1m_2 + n_1n_2 = 0$, that is

$-1(-3 - k) + 4(3 + 4k) - 2(-3 - 2k) = 0$
$21 + 21k = 0$
$k = -1$

Hence B = (3, 1, 1) and ⟨AB⟩ = ⟨−2, −1, −1⟩. The distance

$AB = \sqrt{(-2)^2 + (-1)^2 + (-1)^2}$
$= \sqrt{6}$

AB = $\sqrt{6}$ units

Q5. Point A has coordinates (7, −1, 8) and the line l is defined by the Cartesian equation: $\frac{x-3}{1} = \frac{y+1}{2} = z$. Point B lies on the line l such that AB is perpendicular to the line l. Find the distance between points A and B. [25-Q]

Ans.
Let $\frac{x-3}{1} = \frac{y+1}{2} = \frac{z}{1} = k$. Then the coordinates of the points on the line l are (x, y, z) = (3 + k, −1 + 2k, k), and directed values of l are ⟨1, 2, 1⟩. Since B is on the line l, B(3 + k, −1 + 2k, k).
Directed values of the line AB are

⟨AB⟩ = ⟨−4 + k, 2k, −8 + k⟩

Since AB is perpendicular to the line l, $l_1l_2 + m_1m_2 + n_1n_2 = 0$, that is

$(-4 + k) + 2(2k) + (-8 + k) = 0$
$-12 + 6k = 0$
$k = 2$

Hence B = (5, 3, 2) and ⟨AB⟩ = ⟨−2, 4, −6⟩. The distance

$AB = \sqrt{(-2)^2 + (4)^2 + (-6)^2}$
$= \sqrt{56}$
$= 2\sqrt{14}$

AB = $2\sqrt{14}$ units

3.4

Q6. Find the equation of the line that passes through the point (−1, 3, 2) and perpendicular to the plane 3x − 2y − z = 3. Find the point of intersection of the line and the given plane. [26-Q]

Ans.
Directed values of the line perpendicular to the plane 3x − 2y − z = 3 are ⟨3, −2, −1⟩. Then equation of the line is

$\frac{x+1}{3} = \frac{y-3}{-2} = \frac{z-2}{-1}$

So coordinates of the points on the line are of the form

$x = -1 + 3s$
$y = 3 - 2s$
$z = 2 - s$

If one of these points P is on the plane, then

$3(-1 + 3s) - 2(3 - 2s) - (2 - s) = 3$
$-3 + 9s - 6 + 4s - 2 + s = 3$
$14s = 14$
$s = 1$

$x = -1 + 3 = 2$
$y = 3 - 2 = 1$
$z = 2 - 1 = 1$

Therefore the point of intersection is (2, 1, 1).

3.5

Q7. Find the equation of the sphere with center (5, −6, −2) and touching the plane 3x − y − 2z = 17. [24-Q]

Ans.
The equation of the line that passes through the center C(5, −6, −2) and perpendicular to the plane is

$\frac{x-5}{3} = \frac{y+6}{-1} = \frac{z+2}{-2}$

So coordinates of the points on the line are of the form

$x = 5 + 3s$
$y = -6 - s$
$z = -2 - 2s$

If one of these points P is on the plane $3x - y - 2z = 17$, then

$$\begin{aligned} 3(5+3s) - (-6-s) - 2(-2-2s) &= 17 \\ 25 + 14s &= 17 \\ 14s &= -8 \\ s &= -\frac{4}{7} \end{aligned}$$

Therefore the point of intersection is $P(\frac{23}{7}, -\frac{38}{7}, -\frac{6}{7})$. Thus we have the radius

$$\begin{aligned} CP &= \sqrt{(\frac{23}{7} - 5)^2 + (-\frac{38}{7} + 6)^2 + (-\frac{6}{7} + 2)^2} \\ &= \sqrt{\frac{144}{49} + \frac{16}{49} + \frac{64}{49}} \\ &= \sqrt{\frac{32}{7}} \end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{(x-5)^2 + (y+6)^2 + (z+2)^2 = \frac{32}{7}}$$

Q8. Find the equation of the sphere with center (1, 2, −1) and touching the plane $2x + y + z = 9$. [24-Q]

Ans.

The equation of the line that passes through the center C(1, 2, −1) and perpendicular to the plane is

$$\frac{x-1}{2} = \frac{y-2}{1} = \frac{z+1}{1}$$

So coordinates of the points on the line are of the form

$$\begin{aligned} x &= 1 + 2s \\ y &= 2 + s \\ z &= -1 + s \end{aligned}$$

If one of these points P is on the plane $2x + y + z = 9$, then

$$\begin{aligned} 2(1+2s) + (2+s) + (-1+s) &= 9 \\ 3 + 6s &= 9 \\ 6s &= 6 \\ s &= 1 \end{aligned}$$

Therefore the point of intersection is P(3, 3, 0). Thus we have the radius

$$\begin{aligned} CP &= \sqrt{(3-1)^2 + (3-2)^2 + (0+1)^2} \\ &= \sqrt{4+1+1} \\ &= \sqrt{6} \end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{(x-1)^2 + (y-2)^2 + (z+1)^2 = 6}$$

Q9. Find the equation of the sphere with center (3, 6, −4) and touching the plane $2x - 2y - z = 10$. [24-Q]

Ans.

The equation of the line that passes through the center C(3, 6, −4) and perpendicular to the plane is

$$\frac{x-3}{2} = \frac{y-6}{-2} = \frac{z+4}{-1}$$

So coordinates of the points on the line are of the form

$$\begin{aligned} x &= 3 + 2s \\ y &= 6 - 2s \\ z &= -4 - s \end{aligned}$$

If one of these points P is on the plane $2x - 2y - z = 10$, then

$$\begin{aligned} 2(3+2s) - 2(6-2s) - (-4-s) &= 10 \\ -2 + 9s &= 10 \\ 9s &= 12 \\ s &= \frac{4}{3} \end{aligned}$$

Therefore the point of intersection is $P(\frac{17}{3}, \frac{10}{3}, -\frac{16}{3})$. Thus we have the radius

$$\begin{aligned} CP &= \sqrt{(\frac{17}{3} - 3)^2 + (\frac{10}{3} - 6)^2 + (-\frac{16}{3} + 4)^2} \\ &= \sqrt{\frac{64}{9} + \frac{64}{9} + \frac{16}{9}} \\ &= \sqrt{16} = 4 \end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{(x-3)^2 + (y-6)^2 + (z+4)^2 = 16}$$

Q10. Find the equation of the sphere with center (0, 2, 0) and touching the plane $2x - 4y + 4z = -10$. [24-Q]

Ans.

The equation of the line that passes through the center C(0, 2, 0) and perpendicular to the plane is

$$\frac{x}{2} = \frac{y-2}{-4} = \frac{z}{4}$$

So coordinates of the points on the line are of the form

$$\begin{aligned} x &= 2s \\ y &= 2 - 4s \\ z &= 4s \end{aligned}$$

If one of these points P is on the plane $2x - 4y + 4z = -10$, then

$$\begin{aligned} 2(2s) - 4(2-4s) + 4(4s) &= -10 \\ -8 + 36s &= -10 \\ 36s &= -2 \\ s &= -\frac{1}{18} \end{aligned}$$

Therefore the point of intersection is $P(-\frac{1}{9}, \frac{20}{9}, -\frac{2}{9})$. Thus we have the radius

$$\begin{aligned} CP &= \sqrt{(-\frac{1}{9} - 0)^2 + (\frac{20}{9} - 2)^2 + (-\frac{2}{9} - 0)^2} \\ &= \sqrt{\frac{1}{81} + \frac{4}{81} + \frac{4}{81}} \\ &= \sqrt{\frac{1}{9}} = \frac{1}{3} \end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{x^2 + (y-2)^2 + z^2 = \frac{1}{9}}$$

Q11. Find the equation of the sphere with center (1, 2, −1) and touching the plane $2x + y + z = 5$. [24-Q]

Ans.

The equation of the line that passes through the center C(1, 2, −1) and perpendicular to the plane is

$$\frac{x-1}{2} = \frac{y-2}{1} = \frac{z+1}{1}$$

So coordinates of the points on the line are of the form

$$\begin{aligned} x &= 1 + 2s \\ y &= 2 + s \\ z &= -1 + s \end{aligned}$$

If one of these points P is on the plane $2x + y + z = 5$, then

$$\begin{aligned} 2(1+2s) + (2+s) + (-1+s) &= 5 \\ 3 + 6s &= 5 \\ 6s &= 2 \\ s &= \frac{1}{3} \end{aligned}$$

Therefore the point of intersection is $P(\frac{5}{3}, \frac{7}{3}, -\frac{2}{3})$. Thus we have the radius

$$\begin{aligned} CP &= \sqrt{(\frac{5}{3} - 1)^2 + (\frac{7}{3} - 2)^2 + (-\frac{2}{3} + 1)^2} \\ &= \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{1}{9}} \\ &= \sqrt{\frac{2}{3}} \end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{(x-1)^2 + (y-2)^2 + (z+1)^2 = \frac{2}{3}}$$

Q12. Find the equation of the sphere with center (0, 1, 0) and touching the plane $x + 2y - 2z = 11$. [25-Q]

Ans.

The equation of the line that passes through the center C(0, 1, 0) and perpendicular to the plane is

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z}{-2}$$

So coordinates of the points on the line are of the form

$$\begin{aligned} x &= s \\ y &= 1 + 2s \\ z &= -2s \end{aligned}$$

If one of these points P is on the plane $x + 2y - 2z = 11$, then

$$\begin{aligned} s + 2(1+2s) - 2(-2s) &= 11 \\ 2 + 9s &= 11 \\ 9s &= 9 \\ s &= 1 \end{aligned}$$

Therefore the point of intersection is P(1, 3, −2). Thus we have the radius

$$\begin{aligned} CP &= \sqrt{(1-0)^2 + (3-1)^2 + (-2-0)^2} \\ &= \sqrt{1+4+4} \\ &= \sqrt{9} = 3 \end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{x^2 + (y-1)^2 + z^2 = 9}$$

Q13. Find the equation of the sphere with center (2, 2, 2) and touching the plane $x - 2y + 2z = -5$. [25-Q]

Ans.
The equation of the line that passes through the center C(2, 2, 2) and perpendicular to the plane is

$$\frac{x-2}{1} = \frac{y-2}{-2} = \frac{z-2}{2}$$

So coordinates of the points on the line are of the form

$x = 2 + s$
$y = 2 - 2s$
$z = 2 + 2s$

If one of these points P is on the plane $x - 2y + 2z = -5$, then

$$\begin{aligned}(2+s) - 2(2-2s) + 2(2+2s) &= -5\\ 2 + 9s &= -5\\ 9s &= -7\\ s &= -\frac{7}{9}\end{aligned}$$

Therefore the point of intersection is $P(\frac{11}{9}, \frac{32}{9}, \frac{4}{9})$. Thus we have the radius

$$\begin{aligned}CP &= \sqrt{(\frac{11}{9}-2)^2 + (\frac{32}{9}-2)^2 + (\frac{4}{9}-2)^2}\\ &= \sqrt{\frac{49}{81} + \frac{196}{81} + \frac{196}{81}}\\ &= \sqrt{\frac{49}{9}} = \frac{7}{3}\end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{(x-2)^2 + (y-2)^2 + (z-2)^2 = \frac{49}{9}}$$

Q14. Find the equation of the sphere with center (1, 2, −4) and touching the plane $4x - 2y + 4z = -20$. [25-Q]

Ans.
The equation of the line that passes through the center C(1, 2, −4) and perpendicular to the plane is

$$\frac{x-1}{4} = \frac{y-2}{-2} = \frac{z+4}{4}$$

So coordinates of the points on the line are of the form

$x = 1 + 4s$
$y = 2 - 2s$
$z = -4 + 4s$

If one of these points P is on the plane $4x - 2y + 4z = -20$, then

$$\begin{aligned}4(1+4s) - 2(2-2s) + 4(-4+4s) &= -20\\ -16 + 36s &= -20\\ 36s &= -4\\ s &= -\frac{1}{9}\end{aligned}$$

Therefore the point of intersection is $P(\frac{5}{9}, \frac{20}{9}, -\frac{40}{9})$. Thus we have the radius

$$\begin{aligned}CP &= \sqrt{(\frac{5}{9}-1)^2 + (\frac{20}{9}-2)^2 + (-\frac{40}{9}+4)^2}\\ &= \sqrt{\frac{16}{81} + \frac{4}{81} + \frac{16}{81}}\\ &= \sqrt{\frac{4}{9}} = \frac{2}{3}\end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{(x-1)^2 + (y-2)^2 + (z+4)^2 = \frac{4}{9}}$$

Q15. Find the equation of the sphere with center (2, 1, −1) and touching the plane $x + 3y + 2z = 17$. [25-Q]

Ans.
The equation of the line that passes through the center C(2, 1, −1) and perpendicular to the plane is

$$\frac{x-2}{1} = \frac{y-1}{3} = \frac{z+1}{2}$$

So coordinates of the points on the line are of the form

$x = 2 + s$
$y = 1 + 3s$
$z = -1 + 2s$

If one of these points P is on the plane $x + 3y + 2z = 17$, then

$$\begin{aligned}(2+s) + 3(1+3s) + 2(-1+2s) &= 17\\ 3 + 14s &= 17\\ 14s &= 14\\ s &= 1\end{aligned}$$

Therefore the point of intersection is P(3, 4, 1). Thus we have the radius

$$\begin{aligned}CP &= \sqrt{(3-2)^2 + (4-1)^2 + (1+1)^2}\\ &= \sqrt{1+9+4}\\ &= \sqrt{14}\end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{(x-2)^2 + (y-1)^2 + (z+1)^2 = 14}$$

Q16. Find the equation of the sphere with center (1, 1, 1) and touching the plane $x - 2y + 2z = -5$. [25*-Q]

Ans.
The equation of the line that passes through the center C(1, 1, 1) and perpendicular to the plane is

$$\frac{x-1}{1} = \frac{y-1}{-2} = \frac{z-1}{2}$$

So coordinates of the points on the line are of the form

$x = 1 + s$
$y = 1 - 2s$
$z = 1 + 2s$

If one of these points P is on the plane $x - 2y + 2z = -5$, then

$$\begin{aligned}(1+s) - 2(1-2s) + 2(1+2s) &= -5\\ 1 + 9s &= -5\\ 9s &= -6\\ s &= -\frac{2}{3}\end{aligned}$$

Therefore the point of intersection is $P(\frac{1}{3}, \frac{7}{3}, -\frac{1}{3})$. Thus we have the radius

$$\begin{aligned}CP &= \sqrt{(\frac{1}{3}-1)^2 + (\frac{7}{3}-1)^2 + (-\frac{1}{3}-1)^2}\\ &= \sqrt{\frac{4}{9} + \frac{16}{9} + \frac{16}{9}}\\ &= \sqrt{4} = 2\end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{(x-1)^2 + (y-1)^2 + (z-1)^2 = 4}$$

Q17. Find the equation of the sphere with center (1, 1, 2) and touching the plane $2x - 2y + z = 5$. [25*-Q]

Ans.
The equation of the line that passes through the center C(1, 1, 2) and perpendicular to the plane is

$$\frac{x-1}{2} = \frac{y-1}{-2} = \frac{z-2}{1}$$

So coordinates of the points on the line are of the form

$x = 1 + 2s$
$y = 1 - 2s$
$z = 2 + s$

If one of these points P is on the plane $2x - 2y + z = 5$, then

$$\begin{aligned}2(1+2s) - 2(1-2s) + (2+s) &= 5\\ 2 + 9s &= 5\\ 9s &= 3\\ s &= \frac{1}{3}\end{aligned}$$

Therefore the point of intersection is $P(\frac{5}{3}, \frac{1}{3}, \frac{7}{3})$. Thus we have the radius

$$\begin{aligned}CP &= \sqrt{(\frac{5}{3}-1)^2 + (\frac{1}{3}-1)^2 + (\frac{7}{3}-2)^2}\\ &= \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}}\\ &= \sqrt{1} = 1\end{aligned}$$

Therefore the equation of the sphere is

$$\mathbf{(x-1)^2 + (y-1)^2 + (z-2)^2 = 1}$$

CH.04 Vector Algebra

Notes · Questions with Answers

NOTES

Coordinates of a point in space

A point P in space is determined by an ordered triple (x, y, z) of real numbers, measured along three mutually perpendicular axes (the x-axis, y-axis and z-axis) through a fixed origin O.

Standard unit vectors

The standard unit vectors $\hat{i}, \hat{j}, \hat{k}$ in three dimensions are $\hat{i}=\begin{pmatrix}1\\0\\0\end{pmatrix}$, $\hat{j}=\begin{pmatrix}0\\1\\0\end{pmatrix}$, $\hat{k}=\begin{pmatrix}0\\0\\1\end{pmatrix}$, representing the unit vectors in the positive x, y and z directions respectively.

Position vector

A vector whose initial point is the origin O is called a position vector. The position vector of a point P(a, b, c) is $\vec{OP}=\begin{pmatrix}a\\b\\c\end{pmatrix}=a\hat{i}+b\hat{j}+c\hat{k}$, with magnitude $|\vec{OP}|=\sqrt{a^2+b^2+c^2}$.

Vector AB from two points

For points $A(x_1,y_1,z_1)$ and $B(x_2,y_2,z_2)$, the vector $\vec{AB}=\vec{OB}-\vec{OA}=\begin{pmatrix}x_2-x_1\\y_2-y_1\\z_2-z_1\end{pmatrix}$ is called the position vector of B relative to A, and $|\vec{AB}|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$.

Zero vector

The zero vector in three dimensions is $0=\begin{pmatrix}0\\0\\0\end{pmatrix}$.

Vector algebra (addition, subtraction, scalar multiplication)

For $\vec{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$, $\vec{a}+\vec{b}=\begin{pmatrix}x_1+x_2\\y_1+y_2\\z_1+z_2\end{pmatrix}$, $\vec{a}-\vec{b}=\begin{pmatrix}x_1-x_2\\y_1-y_2\\z_1-z_2\end{pmatrix}$ and $k\vec{a}=\begin{pmatrix}kx_1\\ky_1\\kz_1\end{pmatrix}$ for any scalar k.

Equal vectors

Two vectors are equal if they have the same magnitude and direction. If $\vec{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$, then $\vec{a}=\vec{b}$ if and only if $x_1=x_2$, $y_1=y_2$ and $z_1=z_2$.

Parallel vectors

Two vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if $\vec{a}=k\vec{b}$ for some scalar k.

Unit vector

A vector with magnitude 1 is called a unit vector. For a nonzero vector $\vec{a}$, the unit vector in the same direction of $\vec{a}$, denoted $\hat{a}$ ($\hat{a}$), is given by $\hat{a}=\frac{\vec{a}}{|\vec{a}|}$.

Collinear points

Points A, B, C are collinear if the vectors $\vec{AB}$ and $\vec{BC}$ (or $\vec{AC}$) are parallel and share a common point.

Scalar (dot) product

For $\vec{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$, the scalar product is $\vec{a}\cdot\vec{b}=x_1x_2+y_1y_2+z_1z_2$, and also $\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$.

Perpendicular vectors

Two nonzero vectors $\vec{a}$ and $\vec{b}$ are perpendicular if and only if $\vec{a}\cdot\vec{b}=0$.

Vector (cross) product

The vector product is defined only for three-dimensional vectors. For $\vec{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$, $\vec{a}\times\vec{b}=\begin{pmatrix}y_1z_2-z_1y_2\\z_1x_2-x_1z_2\\x_1y_2-y_1x_2\end{pmatrix}$. The vector $\vec{a}\times\vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$, with magnitude $|\vec{a}\times\vec{b}|=|\vec{a}||\vec{b}|\sin\theta$.

Direction vector of a line

A direction vector of a straight line is a vector parallel to the line.

Normal vector of a plane

A normal vector of a plane is a vector perpendicular to every line lying in the plane.

Magnitude of a position vector

$|\vec{OP}|=\sqrt{a^2+b^2+c^2}$ for $\vec{OP}=\begin{pmatrix}a\\b\\c\end{pmatrix}$

Unit vector

$\hat{a}=\frac{\vec{a}}{|\vec{a}|}$

Scalar (Dot) Product

$\vec{a}\cdot\vec{b}=x_1x_2+y_1y_2+z_1z_2=|\vec{a}||\vec{b}|\cos\theta$

Angle between two vectors

$\cos\theta=\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

Vector (Cross) Product

For $\vec{a}=\begin{pmatrix}x_1\\y_1\\z_1\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}x_2\\y_2\\z_2\end{pmatrix}$,

$\vec{a}\times\vec{b}=\begin{pmatrix}y_1z_2-z_1y_2\\z_1x_2-x_1z_2\\x_1y_2-y_1x_2\end{pmatrix}$

Area of a parallelogram

$\alpha(OACB)=|\vec{a}\times\vec{b}|$

for a parallelogram OACB with adjacent sides $\vec{OA}=\vec{a}$ and $\vec{OB}=\vec{b}$.

Area of a triangle

$\alpha(\Delta OAB)=\frac{1}{2}|\vec{a}\times\vec{b}|$

Vector equation of a line

$\vec{r}=\vec{a}+t\vec{b}$, $t\in\mathbb{R}$

where R is any point on the line with $\vec{OR}=\vec{r}$, A is a known fixed point on the line with $\vec{OA}=\vec{a}$, and $\vec{b}$ is the direction vector of the line.

Vector equation of a line (column form)

$$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}+t\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}$$

Parametric equations of a line

$x = a_1+tb_1$

$y = a_2+tb_2$

$z = a_3+tb_3$

where $t\in\mathbb{R}$ is the parameter.

Cartesian equation of a line

$$\frac{x-a_1}{b_1}=\frac{y-a_2}{b_2}=\frac{z-a_3}{b_3}$$

Vector equation of a plane (parametric form)

$\vec{r}=\vec{a}+t_1\vec{d}_1+t_2\vec{d}_2$

where A is a known point on the plane with $\vec{OA}=\vec{a}$, and $\vec{d}_1$, $\vec{d}_2$ are two non-parallel direction vectors lying in the plane.

Vector equation of a plane (normal form)

$\vec{r}\cdot\vec{n}=\vec{a}\cdot\vec{n}=d$

where $\vec{n}$ is a normal vector to the plane and A is a known point on the plane with $\vec{OA}=\vec{a}$.

Cartesian equation of a plane

$ax+by+cz=d$

where $\vec{n}=\begin{pmatrix}a\\b\\c\end{pmatrix}$ is the normal vector of the plane, and $d=\vec{a}\cdot\vec{n}$ for a point A on the plane with $\vec{OA}=\vec{a}$.

1 MARK

4.1

Q1. If $\vec{a}=\begin{pmatrix}-1\\1\\4\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}2\\0\\-2\end{pmatrix}$, then $\vec{a}-\frac{3}{2}\vec{b}$ is [24-Q]

A. $\begin{pmatrix}-4\\1\\7\end{pmatrix}$ **B.** $\begin{pmatrix}2\\-1\\1\end{pmatrix}$ **C.** $\begin{pmatrix}2\\0\\1\end{pmatrix}$ **D.** $\begin{pmatrix}-1\\1\\2\end{pmatrix}$

Ans.

$$\vec{a}-\frac{3}{2}\vec{b}=\begin{pmatrix}-1\\1\\4\end{pmatrix}-\frac{3}{2}\begin{pmatrix}2\\0\\-2\end{pmatrix}$$
$$=\begin{pmatrix}-1\\1\\4\end{pmatrix}-\begin{pmatrix}3\\0\\-3\end{pmatrix}$$
$$=\begin{pmatrix}-4\\1\\7\end{pmatrix}$$

Answer: A

Q2. If $\vec{a}=\begin{pmatrix}2\\2\\1\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}3\\0\\-9\end{pmatrix}$, then $\vec{a}-\frac{2}{3}\vec{b}$ is [25-Q]

A. $\begin{pmatrix}0\\2\\7\end{pmatrix}$ **B.** $\begin{pmatrix}0\\2\\-5\end{pmatrix}$ **C.** $\begin{pmatrix}3\\0\\-5\end{pmatrix}$ **D.** $\begin{pmatrix}0\\3\\-6\end{pmatrix}$

Ans.

$$\vec{a}-\frac{2}{3}\vec{b}=\begin{pmatrix}2\\2\\1\end{pmatrix}-\begin{pmatrix}2\\0\\-6\end{pmatrix}$$
$$=\begin{pmatrix}0\\2\\7\end{pmatrix}$$

Answer: A

Q3. If $\vec{a}=\begin{pmatrix}1\\2\\7\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}-6\\4\\2\end{pmatrix}$, then $\vec{a}-\frac{1}{2}\vec{b}$ is [26-Q]

A. $\begin{pmatrix}-2\\0\\4\end{pmatrix}$ **B.** $\begin{pmatrix}-4\\0\\6\end{pmatrix}$ **C.** $\begin{pmatrix}4\\0\\6\end{pmatrix}$ **D.** $\begin{pmatrix}-2\\0\\6\end{pmatrix}$

Ans.

$$\vec{a}-\frac{1}{2}\vec{b}=\begin{pmatrix}1\\2\\7\end{pmatrix}-\begin{pmatrix}-3\\2\\1\end{pmatrix}$$
$$=\begin{pmatrix}4\\0\\6\end{pmatrix}$$

Answer: C

Q4. Given that $\vec{p}=\begin{pmatrix}1\\0\\-1\end{pmatrix}$ and $\vec{q}=\begin{pmatrix}-2\\1\\0\end{pmatrix}$, then $2\vec{p}+\vec{q}$ is [25-Q]

A. $\begin{pmatrix}0\\-1\\-2\end{pmatrix}$ **B.** $\begin{pmatrix}-1\\0\\1\end{pmatrix}$ **C.** $\begin{pmatrix}0\\1\\-2\end{pmatrix}$ **D.** $\begin{pmatrix}-1\\1\\-1\end{pmatrix}$

Ans.

$$2\vec{p}+\vec{q}=\begin{pmatrix}2\\0\\-2\end{pmatrix}+\begin{pmatrix}-2\\1\\0\end{pmatrix}$$
$$=\begin{pmatrix}0\\1\\-2\end{pmatrix}$$

Answer: C

Q5. If $\vec{a}=\begin{pmatrix}1\\1\\x\end{pmatrix}$ is parallel to $\vec{b}=\begin{pmatrix}y\\2\\-2\end{pmatrix}$, the sum of x and y is [25-Q]

A. 1 **B.** 2 **C.** 3 **D.** 4

Ans.

$\vec{a}$ is parallel to $\vec{b}$, so $\vec{a}=k\vec{b}$ for some scalar k:

$$\begin{pmatrix}1\\1\\x\end{pmatrix}=k\begin{pmatrix}y\\2\\-2\end{pmatrix}$$
$$1=ky,\quad 1=2k,\quad x=-2k$$

From $1=2k$, $k=\frac{1}{2}$. Then $y=\frac{1}{k}=2$ and $x=-2k=-1$.

$x+y=-1+2=1$

Answer: A

Q6. Given that $\vec{a}=\begin{pmatrix}-1\\2\\u\end{pmatrix}$ is parallel to $\vec{b}=\begin{pmatrix}v\\-2\\1\end{pmatrix}$, then $u-v$ is [25-Q]

A. −2 **B.** −1 **C.** 0 **D.** 1

Ans.

$\vec{a}=k\vec{b}$ for some scalar k:

$$\begin{pmatrix}-1\\2\\u\end{pmatrix}=k\begin{pmatrix}v\\-2\\1\end{pmatrix}$$
$$-1=kv,\quad 2=-2k,\quad u=k$$

From $2=-2k$, $k=-1$. Then $u=-1$ and $-1=(-1)v$ gives $v=1$.

$u-v=-1-1=-2$

Answer: A

4.2

Q7. Given that $\vec{a}=\begin{pmatrix}2\\2\\3\end{pmatrix}$, $\vec{b}=\begin{pmatrix}-1\\2\\1\end{pmatrix}$ and $\vec{c}=\begin{pmatrix}3\\1\\0\end{pmatrix}$. If $\vec{a}+\lambda\vec{b}$ is perpendicular to $\vec{c}$, then the value of λ is [24-Q]

A. −1 **B.** $-\frac{1}{2}$ **C.** 8 **D.** −8

Ans.

Perpendicular vectors have zero dot product.

$$\vec{a}+\lambda\vec{b}=\begin{pmatrix}2-\lambda\\2+2\lambda\\3+\lambda\end{pmatrix}$$
$$(\vec{a}+\lambda\vec{b})\cdot\vec{c}=3(2-\lambda)+1(2+2\lambda)+0=0$$
$$8-\lambda=0$$
$$\lambda=8$$

Answer: C

4.3

Q8. If $\vec{a}=3\hat{i}+2\hat{j}+5\hat{k}$ and $\vec{b}=\hat{i}-4\hat{j}+2\hat{k}$, then $\vec{a}\times\vec{b}$ is [24-Q]

A. $\begin{pmatrix}24\\1\\14\end{pmatrix}$ **B.** $\begin{pmatrix}24\\-1\\14\end{pmatrix}$ **C.** $\begin{pmatrix}24\\1\\-14\end{pmatrix}$ **D.** $\begin{pmatrix}24\\-1\\-14\end{pmatrix}$

Ans.

$$\vec{a}=\begin{pmatrix}3\\2\\5\end{pmatrix},\ \vec{b}=\begin{pmatrix}1\\-4\\2\end{pmatrix}$$
$$\vec{a}\times\vec{b}=\begin{pmatrix}3\\2\\5\end{pmatrix}\times\begin{pmatrix}1\\-4\\2\end{pmatrix}$$
$$=\begin{pmatrix}(2)(2)-(5)(-4)\\(5)(1)-(3)(2)\\(3)(-4)-(2)(1)\end{pmatrix}$$
$$=\begin{pmatrix}24\\-1\\-14\end{pmatrix}$$

Answer: D

Q9. If $\vec{a} = -3\hat{i} - 2\hat{j} + 5\hat{k}$ and $\vec{b} = -\hat{i} + 4\hat{j} + 2\hat{k}$, then $-\vec{a} \times \vec{b}$ is [25, 25*-Q]

A. $\begin{pmatrix} 24 \\ 1 \\ 14 \end{pmatrix}$ **B.** $\begin{pmatrix} 24 \\ -1 \\ 14 \end{pmatrix}$ **C.** $\begin{pmatrix} 24 \\ 1 \\ -14 \end{pmatrix}$ **D.** $\begin{pmatrix} 24 \\ -1 \\ -14 \end{pmatrix}$

Ans.

$\vec{a} = \begin{pmatrix} -3 \\ -2 \\ 5 \end{pmatrix}, \vec{b} = \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}$

$\vec{a} \times \vec{b} = \begin{pmatrix} -3 \\ -2 \\ 5 \end{pmatrix} \times \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} (-2)(2) - (5)(4) \\ (5)(-1) - (-3)(2) \\ (-3)(4) - (-2)(-1) \end{pmatrix}$

$= \begin{pmatrix} -24 \\ 1 \\ -14 \end{pmatrix}$

$-\vec{a} \times \vec{b} = -\begin{pmatrix} -24 \\ 1 \\ -14 \end{pmatrix} = \begin{pmatrix} 24 \\ -1 \\ 14 \end{pmatrix}$

Answer: B

Q10. If $\vec{a} = 3\hat{i} + \hat{k}$ and $\vec{b} = -\hat{i} + 5\hat{j} + \frac{19}{3}\hat{k}$, then $\vec{a} \times \vec{b}$ is [25*-Q]

A. $\begin{pmatrix} 5 \\ -20 \\ 15 \end{pmatrix}$ **B.** $\begin{pmatrix} -5 \\ 20 \\ 15 \end{pmatrix}$ **C.** $\begin{pmatrix} -5 \\ -20 \\ 15 \end{pmatrix}$ **D.** $\begin{pmatrix} 5 \\ 20 \\ -15 \end{pmatrix}$

Ans.

$\vec{a} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}, \vec{b} = \begin{pmatrix} -1 \\ 5 \\ \frac{19}{3} \end{pmatrix}$

$\vec{a} \times \vec{b} = \begin{pmatrix} 0 \cdot \frac{19}{3} - 1 \cdot 5 \\ 1 \cdot (-1) - 3 \cdot \frac{19}{3} \\ 3 \cdot 5 - 0 \cdot (-1) \end{pmatrix}$

$= \begin{pmatrix} -5 \\ -1 - 19 \\ 15 \end{pmatrix}$

$= \begin{pmatrix} -5 \\ -20 \\ 15 \end{pmatrix}$

Answer: C

Q11. If $\vec{a} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, then $\vec{a} \times \vec{b}$ is [26-Q]

A. $\begin{pmatrix} -5 \\ 3 \\ 2 \end{pmatrix}$ **B.** $\begin{pmatrix} 5 \\ -3 \\ 2 \end{pmatrix}$ **C.** $\begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}$ **D.** $\begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix}$

Ans.

$\vec{a} \times \vec{b} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

$= \begin{pmatrix} (0)(-1) - (2)(2) \\ (2)(1) - (1)(-1) \\ (1)(2) - (0)(1) \end{pmatrix}$

$= \begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix}$

Answer: D

Q12. If $\vec{a} = -3\hat{i} - 2\hat{j} + 5\hat{k}$ and $\vec{b} = -\hat{i} + 4\hat{j} + 2\hat{k}$, then $|-\vec{a} \times \vec{b}|$ is [25-Q]

A. $\sqrt{714}$ **B.** $\sqrt{274}$ **C.** $\sqrt{724}$ **D.** $\sqrt{773}$

Ans.

From above, $\vec{a} \times \vec{b} = \begin{pmatrix} -24 \\ 1 \\ -14 \end{pmatrix}$, so $-\vec{a} \times \vec{b} = \begin{pmatrix} 24 \\ -1 \\ 14 \end{pmatrix}$.

$|-\vec{a} \times \vec{b}| = \sqrt{24^2 + (-1)^2 + 14^2}$

$= \sqrt{576 + 1 + 196} = \sqrt{773}$

Answer: D

Q13. If θ is the angle between any two vectors $\vec{a}$ and $\vec{b}$, then $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$ when θ is equal to [25-Q]

A. 0 **B.** $\frac{\pi}{4}$ **C.** $\frac{\pi}{2}$ **D.** π

Ans.

$|\vec{a}||\vec{b}||\cos\theta| = |\vec{a}||\vec{b}||\sin\theta|$

$|\cos\theta| = |\sin\theta| \rightarrow \tan\theta = 1$

For $0 \le \theta \le \pi$ this gives $\theta = \frac{\pi}{4}$ (or $\frac{3\pi}{4}$ with $|\tan\theta| = 1$); the option available is $\frac{\pi}{4}$.

Answer: B

4.4

Q14. The Cartesian form of the equation of the plane $\vec{r} \cdot (2\hat{i} + 3\hat{j} - \hat{k}) = 10$ is [S-Q]

A. $2x + 3y - z = 10$ **B.** $2x + 3y - z = \sqrt{14}$ **C.** $2x + 3y - z = -\sqrt{14}$ **D.** $2x + 3y + z + 10 = 0$

Ans.

Let $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. Then $\vec{r} \cdot \vec{n} = d$ gives:

$2x + 3y - z = 10$

Answer: A

2 MARKS

4.1

Q1. Let $\vec{a} = \begin{pmatrix} 5 \\ -2 \\ 4 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 3 \\ -6 \\ 1 \end{pmatrix}$. Find the vector $2\vec{a} - \vec{b}$. [24-Q]

Ans.

$2\vec{a} - \vec{b} = \begin{pmatrix} 10 \\ -4 \\ 8 \end{pmatrix} - \begin{pmatrix} 3 \\ -6 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} 7 \\ 2 \\ 7 \end{pmatrix}$

Answer: $\begin{pmatrix} 7 \\ 2 \\ 7 \end{pmatrix}$

Q2. If P is (3, 1, 2) and Q is (1, 1, 3), find $|\vec{PQ}|$. [24-Q]

Ans.

$\vec{PQ} = \vec{OQ} - \vec{OP}$

$= \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} 1-3 \\ 1-1 \\ 3-2 \end{pmatrix}$

$= \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$

$|\vec{PQ}| = \sqrt{(-2)^2 + 0^2 + 1^2}$

$= \sqrt{5}$

Answer: $\sqrt{5}$

Q3. Point A and point B has position vectors $\vec{a} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix}$. Find $|\vec{AB}|$. [25-Q]

Ans.

$\vec{AB} = \vec{OB} - \vec{OA}$

$= \vec{b} - \vec{a}$

$= \begin{pmatrix} 2 \\ 3 \\ -3 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} 2-1 \\ 3+1 \\ -3-0 \end{pmatrix}$

$= \begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix}$

$|\vec{AB}| = \sqrt{1^2 + 4^2 + (-3)^2}$

$= \sqrt{26}$

Answer: $\sqrt{26}$

Q4. The two vectors $\hat{i} + \hat{j}$ and $3\hat{i} - \hat{j} + 4\hat{k}$ represent the two sides $\vec{AB}$ and $\vec{AC}$ of ΔABC respectively. Find the length of median through A. [25*-Q]

Ans.

Let M be the midpoint of BC. Then $\vec{BM} = \frac{1}{2}\vec{BC}$ and

$\vec{AM} = \vec{AB}+\vec{BM}$

$= \vec{AB}+\frac{1}{2}\vec{BC}$

$= \vec{AB}+\frac{1}{2}(\vec{AC}-\vec{AB})$

$= \frac{1}{2}(\vec{AB}+\vec{AC})$

$\vec{AM} = \frac{1}{2}\left(\begin{pmatrix}1\\1\\0\end{pmatrix}+\begin{pmatrix}3\\-1\\4\end{pmatrix}\right)$

$= \frac{1}{2}\begin{pmatrix}1+3\\1-1\\0+4\end{pmatrix}$

$= \frac{1}{2}\begin{pmatrix}4\\0\\4\end{pmatrix}$

$= \begin{pmatrix}2\\0\\2\end{pmatrix}$

$|\vec{AM}| = \sqrt{2^2+0^2+2^2}$

$= \sqrt{8}$

$= 2\sqrt{2}$

Answer: $2\sqrt{2}$

4.2

Q5. Determine the angle between two vectors $\begin{pmatrix}1\\1\end{pmatrix}$ and $\begin{pmatrix}1\\-1\end{pmatrix}$. [24-Q]

Ans.

Let $\vec{a}=\begin{pmatrix}1\\1\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}1\\-1\end{pmatrix}$.

$|\vec{a}|=\sqrt{1^2+1^2}=\sqrt{2}$ and $|\vec{b}|=\sqrt{1^2+(-1)^2}=\sqrt{2}$.

$\vec{a}\cdot\vec{b}=\begin{pmatrix}1\\1\end{pmatrix}\cdot\begin{pmatrix}1\\-1\end{pmatrix}=1\times1+1\times(-1)=0$.

$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

$= \frac{0}{\sqrt{2}\times\sqrt{2}}$

$\theta = 90°$ (or $\frac{\pi}{2}$)

Q6. Find the angle between two vectors $\begin{pmatrix}3\\4\end{pmatrix}$ and $\begin{pmatrix}4\\3\end{pmatrix}$. [25-Q]

Ans.

Let $\vec{a}=\begin{pmatrix}3\\4\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}4\\3\end{pmatrix}$.

$|\vec{a}|=\sqrt{3^2+4^2}=5$ and $|\vec{b}|=\sqrt{4^2+3^2}=5$.

$\vec{a}\cdot\vec{b}=\begin{pmatrix}3\\4\end{pmatrix}\cdot\begin{pmatrix}4\\3\end{pmatrix}=3\times4+4\times3=24$.

$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

$= \frac{24}{5\times5} = \frac{24}{25}$

$\theta = \cos^{-1}\left(\frac{24}{25}\right) \approx 16.26°$

Answer: $\cos^{-1}\left(\frac{24}{25}\right)$ **or 16.26°**

Q7. Determine the value of q so that $\vec{a} = \begin{pmatrix}2\\q\\1\end{pmatrix}$ and $\vec{b} = \begin{pmatrix}4\\-2\\-2\end{pmatrix}$ are perpendicular. [26-Q]

Ans.

Let $\vec{a}=\begin{pmatrix}2\\q\\1\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}4\\-2\\-2\end{pmatrix}$. Since $\vec{a}$ and $\vec{b}$ are perpendicular, $\vec{a}\cdot\vec{b}=0$.

$\vec{a}\cdot\vec{b} = \begin{pmatrix}2\\q\\1\end{pmatrix}\cdot\begin{pmatrix}4\\-2\\-2\end{pmatrix}$

$= 2(4)+q(-2)+1(-2)$

$= 8-2q-2$

$= 6-2q$

$6-2q=0$

$q=3$

Answer: q = 3

4.3

Q8. Given that $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a}$ and $\vec{b}$ are perpendicular, evaluate $|(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})|$. [24-Q]

Ans.

$(\vec{a}-\vec{b})\times(\vec{a}+\vec{b}) = \vec{a}\times\vec{a}+\vec{a}\times\vec{b}-\vec{b}\times\vec{a}-\vec{b}\times\vec{b}$

$= 0+\vec{a}\times\vec{b}+\vec{a}\times\vec{b}-0$

$= 2(\vec{a}\times\vec{b})$

$|(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})| = 2|\vec{a}||\vec{b}|\sin 90° = 2(2)(3)(1) = 12$

Answer: 12

Q9. Given that $\vec{a}$ and $\vec{b}$ are two vectors such that $|-\vec{a}\times\vec{b}| = \vec{a}\cdot\vec{b}$. Find the angle between $\vec{a}$ and $\vec{b}$. [25-Q]

Ans.

$|\vec{a}||\vec{b}|\sin\theta = |\vec{a}||\vec{b}|\cos\theta$

$\tan\theta = 1$

$\theta = \frac{\pi}{4}$

4.4

Q10. What is the vector equation of the plane containing points P(a, b, c), Q(x, y, z) and R(u, v, w)? [24-Q]

Ans.

Take P as the known point and $\vec{PQ}$, $\vec{PR}$ as two direction vectors lying in the plane:

$\vec{PQ} = \begin{pmatrix}x-a\\y-b\\z-c\end{pmatrix}$, $\vec{PR} = \begin{pmatrix}u-a\\v-b\\w-c\end{pmatrix}$

$\vec{r} = \begin{pmatrix}a\\b\\c\end{pmatrix}+t_1\begin{pmatrix}x-a\\y-b\\z-c\end{pmatrix}+t_2\begin{pmatrix}u-a\\v-b\\w-c\end{pmatrix}$

Answer: $\vec{r} = \begin{pmatrix}\mathbf{a}\\\mathbf{b}\\\mathbf{c}\end{pmatrix}+\mathbf{t_1}\begin{pmatrix}\mathbf{x-a}\\\mathbf{y-b}\\\mathbf{z-c}\end{pmatrix}+\mathbf{t_2}\begin{pmatrix}\mathbf{u-a}\\\mathbf{v-b}\\\mathbf{w-c}\end{pmatrix}$

Q11. Find the vector equation of the line that passes through the point (−2, 3, 1) and perpendicular to the plane 2x − 3y + 5z = 7. [26-Q]

Ans.

The normal vector of the plane, $\vec{n} = \begin{pmatrix}2\\-3\\5\end{pmatrix}$, is the direction vector of the line.

$\vec{r} = \begin{pmatrix}-2\\3\\1\end{pmatrix}+t\begin{pmatrix}2\\-3\\5\end{pmatrix}$

Answer: $\vec{r} = \begin{pmatrix}\mathbf{-2}\\\mathbf{3}\\\mathbf{1}\end{pmatrix}+\mathbf{t}\begin{pmatrix}\mathbf{2}\\\mathbf{-3}\\\mathbf{5}\end{pmatrix}$

3 MARKS

4.1

Q1. Let $\vec{a} = \begin{pmatrix}1\\2\\5\end{pmatrix}$ and $\vec{c} = \begin{pmatrix}-1\\p\\q\end{pmatrix}$. Find the values of p and q such that $\vec{c}$ is parallel to $\vec{a}$. [24-Q]

Ans.

Since $\vec{c}$ is parallel to $\vec{a}$, $\vec{c}=k\vec{a}$ for some scalar k.

$\begin{pmatrix}-1\\p\\q\end{pmatrix} = k\begin{pmatrix}1\\2\\5\end{pmatrix}$

$-1=k$, $p=2k$, $q=5k$

From $-1=k$, we get $k=-1$. Thus $p=2k=-2$ and $q=5k=-5$.

Answer: p = −2, q = −5

Q2. Given that $\vec{a} = \begin{pmatrix}2\\0\\2\end{pmatrix}$ and $\vec{b} = \begin{pmatrix}3\\1\\3\end{pmatrix}$, find the value of scalar q such that $\vec{a}+q\vec{b}$ is parallel to the vector $\begin{pmatrix}3\\2\\3\end{pmatrix}$. [25*-Q]

Ans.

$\vec{a}+q\vec{b} = \begin{pmatrix}2\\0\\2\end{pmatrix}+q\begin{pmatrix}3\\1\\3\end{pmatrix}$

$\vec{a}+q\vec{b} = \begin{pmatrix}2\\0\\2\end{pmatrix}+q\begin{pmatrix}3\\1\\3\end{pmatrix}$

$= \begin{pmatrix} 3q+2 \\ q \\ 3q+2 \end{pmatrix}$

Since $\vec{a}+q\vec{b}$ is parallel to $\begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix}$, $\vec{a}+q\vec{b}=k\begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix}$.

$\begin{pmatrix} 3q+2 \\ q \\ 3q+2 \end{pmatrix} = k\begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix}$

$3q+2=3k$, $q=2k$, $3q+2=3k$
From the second equation, $q=2k$. Substituting in the first: $2+3(2k)=3k$
$k=-\frac{2}{3}$ and $q=-\frac{4}{3}$

Answer: q = $-\frac{4}{3}$

Q3. In a parallelogram PQRS, P is (−2, 2, 2), Q is (4, 0, −4), and S is (6, 2, 8). Find the coordinates of R. [24-Q]

Ans.

Let R be (x, y, z). Since PQRS is a parallelogram, $\vec{PQ}$ is parallel to $\vec{SR}$ and they have the same length and direction, so:

$$\vec{SR} = \vec{PQ}$$
$$\vec{OR}-\vec{OS} = \vec{OQ}-\vec{OP}$$
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}-\begin{pmatrix} 6 \\ 2 \\ 8 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ -4 \end{pmatrix}-\begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$$
$$= \begin{pmatrix} 4+2 \\ 0-2 \\ -4-2 \end{pmatrix}$$
$$= \begin{pmatrix} 6 \\ -2 \\ -6 \end{pmatrix}$$

So $x-6=6 \rightarrow x=12$, $y-2=-2 \rightarrow y=0$, $z-8=-6 \rightarrow z=2$.
Therefore R is (12, 0, 2).
Answer: R(12, 0, 2)

Q4. The points A(3, 1, 2), B(−1, 1, 5) and C(7, 2, 3) are vertices of a parallelogram ABCD. Find the coordinates of D. [24-Q]

Ans.

Let D be (x, y, z). Since ABCD is a parallelogram, $\vec{AB}$ is parallel to $\vec{DC}$ and they have the same length and direction, so:

$$\vec{AB} = \vec{DC}$$
$$\vec{OB}-\vec{OA} = \vec{OC}-\vec{OD}$$
$$\begin{pmatrix} -1 \\ 1 \\ 5 \end{pmatrix}-\begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \\ 3 \end{pmatrix}-\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$
$$= \begin{pmatrix} -4 \\ 0 \\ 3 \end{pmatrix}$$
$$= \begin{pmatrix} 7-x \\ 2-y \\ 3-z \end{pmatrix}$$

So $-4=7-x \rightarrow x=11$, $0=2-y \rightarrow y=2$, $3=3-z \rightarrow z=0$.
Therefore D is (11, 2, 0).
Answer: D(11, 2, 0)

Q5. In parallelogram ABCD, A is (1, 0, 0), B is (0, −1, 2) and C is (3, 2, 1). Find the coordinates of D. [25(2)-Q]

Ans.

Let D be (x, y, z). Since ABCD is a parallelogram, $\vec{AB}$ is parallel to $\vec{DC}$ and they have the same length and direction, so:

$$\vec{AB} = \vec{DC}$$
$$\vec{OB}-\vec{OA} = \vec{OC}-\vec{OD}$$
$$\begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}-\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}-\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$
$$= \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$$
$$= \begin{pmatrix} 3-x \\ 2-y \\ 1-z \end{pmatrix}$$

So $-1=3-x \rightarrow x=4$, $-1=2-y \rightarrow y=3$, $2=1-z \rightarrow z=-1$.
Therefore D is (4, 3, −1).
Answer: D(4, 3, −1)

Q6. In parallelogram ABCD, A is (2, −1, 4), B is (5, 1, 2) and C is (1, 5, 5). Find the coordinates of D. [25-Q]

Ans.

Let D be (x, y, z). Since ABCD is a parallelogram, $\vec{AB}$ is parallel to $\vec{DC}$ and they have the same length and direction, so:

$$\vec{AB} = \vec{DC}$$
$$\vec{OB}-\vec{OA} = \vec{OC}-\vec{OD}$$
$$\begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix}-\begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 5 \end{pmatrix}-\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$
$$= \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$$
$$= \begin{pmatrix} 1-x \\ 5-y \\ 5-z \end{pmatrix}$$

So $3=1-x \rightarrow x=-2$, $2=5-y \rightarrow y=3$, $-2=5-z \rightarrow z=7$.
Therefore D is (−2, 3, 7).
Answer: D(−2, 3, 7)

Q7. In a parallelogram PQRS, P is (−1, 1, 1), Q is (2, 0, −2), and S is (3, 1, 4). Find the coordinates of R. [25-Q]

Ans.

Let R be (x, y, z). Since PQRS is a parallelogram, $\vec{PQ}$ is parallel to $\vec{SR}$ and they have the same length and direction, so:

$$\vec{SR} = \vec{PQ}$$
$$\vec{OR}-\vec{OS} = \vec{OQ}-\vec{OP}$$
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}-\begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ -2 \end{pmatrix}-\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$
$$= \begin{pmatrix} 2+1 \\ 0-1 \\ -2-1 \end{pmatrix}$$
$$= \begin{pmatrix} 3 \\ -1 \\ -3 \end{pmatrix}$$

So $x-3=3 \rightarrow x=6$, $y-1=-1 \rightarrow y=0$, $z-4=-3 \rightarrow z=1$.
Therefore R is (6, 0, 1).
Answer: R(6, 0, 1)

Q8. Points A, B and C have position vectors $\vec{a} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$ and $\vec{c} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ respectively. Find the position vector of the point D such that ABCD is a parallelogram. [25-Q]

Ans.

For parallelogram ABCD, $\vec{AB}$ is parallel to $\vec{DC}$ with the same length and direction, so:

$$\vec{AB} = \vec{DC}$$
$$\vec{OB}-\vec{OA} = \vec{OC}-\vec{OD}$$
$$\vec{b}-\vec{a} = \vec{c}-\vec{d}$$
$$\vec{d} = \vec{a}+\vec{c}-\vec{b}$$
$$= \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}+\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}-\begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$$
$$= \begin{pmatrix} -2+2-3 \\ 1+1-1 \\ 3+1-2 \end{pmatrix}$$
$$= \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}$$

Answer: d = $\begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}$

Q9. Points A, B, C and D have position vectors $\vec{a} = \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix}$, $\vec{c} = \begin{pmatrix} 7 \\ 8 \\ -3 \end{pmatrix}$ and $\vec{d} = \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}$ respectively. Point E is the midpoint of BC. Show that ABED is a parallelogram. [26-Q]

Ans.

Since E is the midpoint of BC,

$\vec{e} = \frac{1}{2}(\vec{b}+\vec{c})$

$= \frac{1}{2}\left(\begin{pmatrix}5\\0\\3\end{pmatrix}+\begin{pmatrix}7\\8\\-3\end{pmatrix}\right)$

$= \frac{1}{2}\begin{pmatrix}12\\8\\0\end{pmatrix}$

$= \begin{pmatrix}6\\4\\0\end{pmatrix}$

$\vec{AB} = \vec{OB}-\vec{OA}$

$= \vec{b}-\vec{a} = \begin{pmatrix}5\\0\\3\end{pmatrix}-\begin{pmatrix}3\\-1\\1\end{pmatrix}$

$= \begin{pmatrix}2\\1\\2\end{pmatrix}$

$\vec{DE} = \vec{OE}-\vec{OD}$

$= \vec{e}-\vec{d} = \begin{pmatrix}6\\4\\0\end{pmatrix}-\begin{pmatrix}4\\3\\-2\end{pmatrix}$

$= \begin{pmatrix}2\\1\\2\end{pmatrix}$

$\vec{AD} = \vec{OD}-\vec{OA}$

$= \vec{d}-\vec{a} = \begin{pmatrix}4\\3\\-2\end{pmatrix}-\begin{pmatrix}3\\-1\\1\end{pmatrix}$

$= \begin{pmatrix}1\\4\\-3\end{pmatrix}$

$\vec{BE} = \vec{OE}-\vec{OB}$

$= \vec{e}-\vec{b} = \begin{pmatrix}6\\4\\0\end{pmatrix}-\begin{pmatrix}5\\0\\3\end{pmatrix}$

$= \begin{pmatrix}1\\4\\-3\end{pmatrix}$

Since $\vec{AB}=\vec{DE}$ and $\vec{AD}=\vec{BE}$, the opposite sides are equal and parallel.
Therefore ABED is a parallelogram. (Proved)

Q10. Find the unit vector in the same direction as $\vec{a} = \begin{pmatrix}2\\-2\\1\end{pmatrix}$. [24-Q]

Ans.

$|\vec{a}| = \sqrt{2^2+(-2)^2+1^2}$

$= \sqrt{9}$

$= 3$

The unit vector of $\vec{a}$ is $\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{3}\begin{pmatrix}2\\-2\\1\end{pmatrix} = \begin{pmatrix}\frac{2}{3}\\-\frac{2}{3}\\\frac{1}{3}\end{pmatrix}$

Answer: $\begin{pmatrix}\frac{2}{3}\\-\frac{2}{3}\\\frac{1}{3}\end{pmatrix}$

4.2

Q11. Find the angle between two vectors $\begin{pmatrix}-5\\12\end{pmatrix}$ and $\begin{pmatrix}-3\\4\end{pmatrix}$. [25-Q]

Ans.

Let $\vec{a} = \begin{pmatrix}-5\\12\end{pmatrix}$ and $\vec{b} = \begin{pmatrix}-3\\4\end{pmatrix}$.

$|\vec{a}| = \sqrt{(-5)^2+12^2} = 13$ and $|\vec{b}| = \sqrt{(-3)^2+4^2} = 5$.

$\vec{a}\cdot\vec{b} = \begin{pmatrix}-5\\12\end{pmatrix}\cdot\begin{pmatrix}-3\\4\end{pmatrix} = (-5)\times(-3)+12\times4 = 63$.

$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = \frac{63}{13\times5} = \frac{63}{65}$

$\theta = \cos^{-1}\left(\frac{63}{65}\right) \approx 14.25°$

Answer: $\cos^{-1}\left(\frac{63}{65}\right)$ or 14.25°

Q12. The position vectors of the points A and B are given by $\vec{OA} = \begin{pmatrix}5\\-2\\4\end{pmatrix}$ and $\vec{OB} = \begin{pmatrix}-1\\3\\-1\end{pmatrix}$. Find the angle OBA. [25-Q]

Ans.

∠OBA is the angle between $\vec{BO}$ and $\vec{BA}$ at B.

$\vec{BO} = -\vec{OB}$

$= -\begin{pmatrix}-1\\3\\-1\end{pmatrix}$

$= \begin{pmatrix}1\\-3\\1\end{pmatrix}$

$\vec{BA} = \vec{OA}-\vec{OB}$

$= \begin{pmatrix}5\\-2\\4\end{pmatrix}-\begin{pmatrix}-1\\3\\-1\end{pmatrix}$

$= \begin{pmatrix}5+1\\-2-3\\4+1\end{pmatrix}$

$= \begin{pmatrix}6\\-5\\5\end{pmatrix}$

$|\vec{BO}| = \sqrt{1^2+(-3)^2+1^2} = \sqrt{11}$ and $|\vec{BA}| = \sqrt{6^2+(-5)^2+5^2} = \sqrt{86}$.

$\vec{BO}\cdot\vec{BA} = \begin{pmatrix}1\\-3\\1\end{pmatrix}\cdot\begin{pmatrix}6\\-5\\5\end{pmatrix} = 1\times6+(-3)\times(-5)+1\times5 = 26$.

$\cos\theta = \frac{\vec{BO}\cdot\vec{BA}}{|\vec{BO}||\vec{BA}|} = \frac{26}{\sqrt{11}\times\sqrt{86}}$

$\theta = \cos^{-1}\left(\frac{26}{\sqrt{946}}\right) \approx 32.29°$

Answer: $\cos^{-1}\left(\frac{26}{\sqrt{946}}\right)$ or 32.29°

Q13. Given that $\vec{a} = \begin{pmatrix}2\\-1\\1\end{pmatrix}$, $\vec{b} = \begin{pmatrix}1\\-1\\-2\end{pmatrix}$ and $\vec{c} = \begin{pmatrix}-3\\3\\-1\end{pmatrix}$, find the value of scalar k such that $\vec{a}$ is perpendicular to $k\vec{b} + \vec{c}$. [25*-Q]

Ans.

Let $\vec{a} = \begin{pmatrix}2\\-1\\1\end{pmatrix}$, $\vec{b} = \begin{pmatrix}1\\-1\\-2\end{pmatrix}$ and $\vec{c} = \begin{pmatrix}-3\\3\\-1\end{pmatrix}$. Since $\vec{a}$ is perpendicular to $k\vec{b}+\vec{c}$, $\vec{a}\cdot(k\vec{b}+\vec{c})=0$.

$k\vec{b}+\vec{c} = k\begin{pmatrix}1\\-1\\-2\end{pmatrix}+\begin{pmatrix}-3\\3\\-1\end{pmatrix}$

$= \begin{pmatrix}k\\-k\\-2k\end{pmatrix}+\begin{pmatrix}-3\\3\\-1\end{pmatrix}$

$= \begin{pmatrix}k-3\\-k+3\\-2k-1\end{pmatrix}$

$\vec{a}\cdot(k\vec{b}+\vec{c}) = 2(k-3)+(-1)(-k+3)+1(-2k-1)$

$= k-10$

$= 0$

$k = 10$

Answer: k = 10

Q14. Given $|\vec{a}| = 4$, $|\vec{b}| = 1$ and $(\vec{a}-\vec{b})\cdot(\vec{a}+5\vec{b}) = 11$. Show that $\vec{a}$ and $\vec{b}$ are perpendicular vectors. [24-Q]

Ans.

$(\vec{a}-\vec{b})\cdot(\vec{a}+5\vec{b}) = \vec{a}\cdot\vec{a}+5\vec{a}\cdot\vec{b}-\vec{b}\cdot\vec{a}-5\vec{b}\cdot\vec{b}$

$= |\vec{a}|^2 + 4\vec{a}\cdot\vec{b} - 5|\vec{b}|^2$

$= 16 + 4\vec{a}\cdot\vec{b} - 5 = 11$

So $4\vec{a}\cdot\vec{b} = 0$, i.e. $\vec{a}\cdot\vec{b} = 0$. Since $\vec{a}$ and $\vec{b}$ are non-zero, $\vec{a}$ and $\vec{b}$ are perpendicular. (Proved)

Q15. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be the three vectors such that $|\vec{a}| = 3$, $|\vec{b}| = 4$, $|\vec{c}| = 5$ and each one of them being perpendicular to the sum of the other two, find $|\vec{a} + \vec{b} + \vec{c}|$. [25-Q]

Ans.

Each vector is perpendicular to the sum of the other two:

$\vec{a}\cdot(\vec{b} + \vec{c}) = 0 \rightarrow \vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c} = 0$

$\vec{b}\cdot(\vec{a} + \vec{c}) = 0 \rightarrow \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} = 0$

$\vec{c}\cdot(\vec{a} + \vec{b}) = 0 \rightarrow \vec{a}\cdot\vec{c} + \vec{b}\cdot\vec{c} = 0$

Adding gives $2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) = 0$, and then each of $\vec{a}\cdot\vec{b}$, $\vec{b}\cdot\vec{c}$, $\vec{c}\cdot\vec{a}$ is 0.

$$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$$
$$= 9 + 16 + 25 + 0 = 50$$
$$|\vec{a} + \vec{b} + \vec{c}| = 5\sqrt{2}$$

Answer: $5\sqrt{2}$

4.3

Q16. Let $\vec{a} = \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 6 \\ 2 \\ -3 \end{pmatrix}$, $\vec{c} = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$. Which vector has greater magnitude $\vec{a} \times \vec{b}$ or $\vec{b} \times \vec{c}$? [24-Q]

Ans.

$$\vec{a}\times\vec{b} = \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix} \times \begin{pmatrix} 6 \\ 2 \\ -3 \end{pmatrix} = \begin{pmatrix} (-2)(-3)-(5)(2) \\ (5)(6)-(2)(-3) \\ (2)(2)-(-2)(6) \end{pmatrix} = \begin{pmatrix} -4 \\ 36 \\ 16 \end{pmatrix}$$

$$\vec{b}\times\vec{c} = \begin{pmatrix} 6 \\ 2 \\ -3 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} (2)(-2)-(-3)(1) \\ (-3)(0)-(6)(-2) \\ (6)(1)-(2)(0) \end{pmatrix} = \begin{pmatrix} -1 \\ 12 \\ 6 \end{pmatrix}$$

$$|\vec{a}\times\vec{b}| = \sqrt{(-4)^2+36^2+16^2} = \sqrt{1568} = 28\sqrt{2}$$

$$|\vec{b}\times\vec{c}| = \sqrt{(-1)^2+12^2+6^2} = \sqrt{181}$$

$|\vec{a}\times\vec{b}| \approx 39.6$ and $|\vec{b}\times\vec{c}| \approx 13.5$

Answer: $\vec{a}\times\vec{b}$ has greater magnitude.

Q17. Find a vector of magnitude 10 units that is perpendicular to both $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$, when $\vec{a} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 1 \\ -3 \\ 4 \end{pmatrix}$. [25-Q]

Ans.

$$\vec{a}+\vec{b} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -3 \\ 4 \end{pmatrix} = \begin{pmatrix} 2+1 \\ -2-3 \\ 1+4 \end{pmatrix} = \begin{pmatrix} 3 \\ -5 \\ 5 \end{pmatrix}$$

$$\vec{a}-\vec{b} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ -3 \\ 4 \end{pmatrix} = \begin{pmatrix} 2-1 \\ -2+3 \\ 1-4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}$$

A vector perpendicular to both $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ is $(\vec{a}+\vec{b})\times(\vec{a}-\vec{b})$.

$$(\vec{a}+\vec{b})\times(\vec{a}-\vec{b}) = \begin{pmatrix} 3 \\ -5 \\ 5 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} = \begin{pmatrix} (-5)(-3)-(5)(1) \\ (5)(1)-(3)(-3) \\ (3)(1)-(-5)(1) \end{pmatrix} = \begin{pmatrix} 10 \\ 14 \\ 8 \end{pmatrix}$$

$$|(\vec{a}+\vec{b})\times(\vec{a}-\vec{b})| = \sqrt{10^2+14^2+8^2} = \sqrt{360} = 6\sqrt{10}$$

A vector of magnitude 10 is $10\times \frac{(\vec{a}+\vec{b})\times(\vec{a}-\vec{b})}{|(\vec{a}+\vec{b})\times(\vec{a}-\vec{b})|} = 10\times \frac{1}{6\sqrt{10}} \begin{pmatrix} 10 \\ 14 \\ 8 \end{pmatrix}$

$$= \frac{20}{6\sqrt{10}} \begin{pmatrix} 5 \\ 7 \\ 4 \end{pmatrix} = \frac{10}{3\sqrt{10}} \begin{pmatrix} 5 \\ 7 \\ 4 \end{pmatrix}$$

$$= \frac{1}{3}\sqrt{10} \begin{pmatrix} 5 \\ 7 \\ 4 \end{pmatrix}$$

Answer: $\pm \frac{1}{3}\sqrt{10} \begin{pmatrix} 5 \\ 7 \\ 4 \end{pmatrix}$

Q18. The points A(1, 2, 3), B(2, 5, −1) and C(3, 0, 1) are vertices of a parallelogram ABCD. Find the area of the parallelogram. [24-Q, 25-Q]

Ans.

$$\vec{AB} = \vec{OB}-\vec{OA} = \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2-1 \\ 5-2 \\ -1-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}$$

$$\vec{BC} = \vec{OC}-\vec{OB} = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} = \begin{pmatrix} 3-2 \\ 0-5 \\ 1+1 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ 2 \end{pmatrix}$$

$\vec{AB}$ and $\vec{BC}$ are adjacent sides, so the area of the parallelogram is $|\vec{AB}\times\vec{BC}|$.

$$\vec{AB}\times\vec{BC} = \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} \times \begin{pmatrix} 1 \\ -5 \\ 2 \end{pmatrix} = \begin{pmatrix} (3)(2)-(-4)(-5) \\ (-4)(1)-(1)(2) \\ (1)(-5)-(3)(1) \end{pmatrix} = \begin{pmatrix} -14 \\ -6 \\ -8 \end{pmatrix}$$

$$|\vec{AB}\times\vec{BC}| = \sqrt{(-14)^2+(-6)^2+(-8)^2} = \sqrt{296} = 2\sqrt{74}$$

Answer: $2\sqrt{74}$ square units

Q19. Find the area of the parallelogram determined by the vectors $\vec{a} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 2 \\ -7 \\ 1 \end{pmatrix}$. [25-Q]

Ans.

The area of the parallelogram determined by $\vec{a}$ and $\vec{b}$ is $|\vec{a}\times\vec{b}|$.

$$\vec{a}\times\vec{b} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \times \begin{pmatrix} 2 \\ -7 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} (-1)(1)-(3)(-7) \\ (3)(2)-(1)(1) \\ (1)(-7)-(-1)(2) \end{pmatrix}$$

$$= \begin{pmatrix} 20 \\ 5 \\ -5 \end{pmatrix}$$

$|\vec{a}\times\vec{b}| = \sqrt{20^2+5^2+(-5)^2}$

$= \sqrt{450}$

$= 15\sqrt{2}$

Answer: $15\sqrt{2}$ square units

Q20. Find the area of triangle ABC with vertices A(1, 1, 1), B(1, 2, 3) and C(2, 3, 1). [24-Q]

Ans.

$\vec{AB} = \vec{OB}-\vec{OA}$

$$= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1-1 \\ 2-1 \\ 3-1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

$\vec{AC} = \vec{OC}-\vec{OA}$

$$= \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2-1 \\ 3-1 \\ 1-1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$\vec{AB}\times\vec{AC} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} (1)(0)-(2)(2) \\ (2)(1)-(0)(0) \\ (0)(2)-(1)(1) \end{pmatrix}$$

$$= \begin{pmatrix} -4 \\ 2 \\ -1 \end{pmatrix}$$

$|\vec{AB}\times\vec{AC}| = \sqrt{(-4)^2+2^2+(-1)^2}$

$= \sqrt{21}$

Area of triangle ABC $= \frac{1}{2}|\vec{AB}\times\vec{AC}| = \frac{1}{2}(\sqrt{21}) = \frac{\sqrt{21}}{2}$ square units.

Answer: $\frac{\sqrt{21}}{2}$ square units

Q21. Find the area of triangle ABC with vertices A(2, −1, 4), B(5, 1, 2) and C(3, 1, 4). [25-Q]

Ans.

$\vec{AB} = \vec{OB}-\vec{OA}$

$$= \begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} 5-2 \\ 1+1 \\ 2-4 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$$

$\vec{AC} = \vec{OC}-\vec{OA}$

$$= \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$$

$$= \begin{pmatrix} 3-2 \\ 1+1 \\ 4-4 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$\vec{AB}\times\vec{AC} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} (2)(0)-(-2)(2) \\ (-2)(1)-(3)(0) \\ (3)(2)-(2)(1) \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ -2 \\ 4 \end{pmatrix}$$

$|\vec{AB}\times\vec{AC}| = \sqrt{4^2+(-2)^2+4^2}$

$= \sqrt{36}$

$= 6$

Area of triangle ABC $= \frac{1}{2}|\vec{AB}\times\vec{AC}| = \frac{1}{2}(6) = 3$ square units.

Answer: 3 square units

Q22. Find the area of triangle ABC with vertices A(1, 1, 2), B(2, 3, 5) and C(1, 5, 5). [25-Q]

Ans.

$\vec{AB} = \vec{OB}-\vec{OA}$

$$= \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 2-1 \\ 3-1 \\ 5-2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$\vec{AC} = \vec{OC}-\vec{OA}$

$$= \begin{pmatrix} 1 \\ 5 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1-1 \\ 5-1 \\ 5-2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 4 \\ 3 \end{pmatrix}$$

$$\vec{AB}\times\vec{AC} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 0 \\ 4 \\ 3 \end{pmatrix}$$

$$= \begin{pmatrix} (2)(3)-(3)(4) \\ (3)(0)-(1)(3) \\ (1)(4)-(2)(0) \end{pmatrix}$$

$$= \begin{pmatrix} -6 \\ -3 \\ 4 \end{pmatrix}$$

$|\vec{AB}\times\vec{AC}| = \sqrt{(-6)^2+(-3)^2+4^2}$

$= \sqrt{61}$

Area of triangle ABC $= \frac{1}{2}|\vec{AB}\times\vec{AC}| = \frac{1}{2}(\sqrt{61}) = \frac{\sqrt{61}}{2}$ square units.

Answer: $\frac{\sqrt{61}}{2}$ square units

Q23. Find the area of the triangle ABC with vertices A(1, −1, 0), B(2, 1, −1) and C(−1, 1, 2). [26-Q]

Ans.

$\vec{AB} = \vec{OB}-\vec{OA}$

$$= \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 2-1 \\ 1+1 \\ -1-0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$\vec{AC} = \vec{OC}-\vec{OA}$

$$= \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} -1-1 \\ 1+1 \\ 2-0 \end{pmatrix}$$

$= \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$

$\vec{AB} \times \vec{AC} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} (2)(2)-(-1)(2) \\ (-1)(-2)-(1)(2) \\ (1)(2)-(2)(-2) \end{pmatrix}$

$= \begin{pmatrix} 6 \\ 0 \\ 6 \end{pmatrix}$

$|\vec{AB} \times \vec{AC}| = \sqrt{6^2+0^2+6^2}$

$= \sqrt{72}$

$= 6\sqrt{2}$

Area of triangle ABC $= \frac{1}{2} |\vec{AB} \times \vec{AC}| = \frac{1}{2}(6\sqrt{2}) = 3\sqrt{2}$ square units.

Answer: $3\sqrt{2}$ square units

4.4

Q24. Find the vector equation and the Cartesian equation of the line that passes through the point (3, −1, 2) and perpendicular to the plane $4x - 2y + 3z = 10$. [24-Q]

Ans.

The line is perpendicular to the plane, so its direction vector is the normal vector of the plane.

$\vec{d} = \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}$

Since $\vec{r} = \vec{a} + t\vec{d}$, the vector equation of the line is $\vec{r} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}$.

The parametric equations are:

$x = 3+4t \rightarrow t = \frac{x-3}{4}$

$y = -1-2t \rightarrow t = \frac{y+1}{-2}$

$z = 2+3t \rightarrow t = \frac{z-2}{3}$

The Cartesian equation of the line is $\frac{x-3}{4} = \frac{y+1}{-2} = \frac{z-2}{3}$.

Answer: $\vec{r} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}$; $\frac{x-3}{4} = \frac{y+1}{-2} = \frac{z-2}{3}$

Q25. Find the vector equation of the line passing through the point (2, −4, 3) and perpendicular to the plane $x + 4y - 2z = 5$. [25-Q]

Ans.

The line is perpendicular to the plane, so its direction vector is the normal vector of the plane.

$\vec{d} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$

Since $\vec{r} = \vec{a} + t\vec{d}$, the vector equation of the line is $\vec{r} = \begin{pmatrix} 2 \\ -4 \\ 3 \end{pmatrix} + t \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$.

Answer: $\vec{r} = \begin{pmatrix} 2 \\ -4 \\ 3 \end{pmatrix} + t \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$

Q26. The vector equation of the line L_1 is $\vec{r} = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix} + t \begin{pmatrix} 4 \\ 2 \\ 3 \end{pmatrix}$, where $t \in \mathbb{R}$. The points A(3, p, 5) and B(q, 0, 2), where p and q are constants, lie on the line L_1. Find the value of p and the value of q. [25-Q]

Ans.

From the vector equation, the parametric equations are:

$x = 3+4t$, $y = 2+2t$, $z = 5+3t$

For A(3, p, 5): $3 = 3+4t$ gives $t = 0$. Then $p = 2+2(0) = 2$.

For B(q, 0, 2): $2 = 5+3t$ gives $t = -1$. Check y: $2+2(-1) = 0$ ✓. Then $q = 3+4(-1) = -1$.

Answer: p = 2, q = −1

Q27. Find the vector equation of the plane containing points P(2, −5, 4), Q(−5, 2, 4) and R(−2, 3, −1). [24-Q]

Ans.

Let $\vec{a} = \vec{OP} = \begin{pmatrix} 2 \\ -5 \\ 4 \end{pmatrix}$.

$\vec{d}_1 = \vec{PQ} = \vec{OQ} - \vec{OP}$

$= \begin{pmatrix} -5 \\ 2 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -5 \\ 4 \end{pmatrix}$

$= \begin{pmatrix} -5-2 \\ 2+5 \\ 4-4 \end{pmatrix}$

$= \begin{pmatrix} -7 \\ 7 \\ 0 \end{pmatrix}$

$\vec{d}_2 = \vec{PR} = \vec{OR} - \vec{OP}$

$= \begin{pmatrix} -2 \\ 3 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ -5 \\ 4 \end{pmatrix}$

$= \begin{pmatrix} -2-2 \\ 3+5 \\ -1-4 \end{pmatrix}$

$= \begin{pmatrix} -4 \\ 8 \\ -5 \end{pmatrix}$

Therefore the vector equation of the plane is

$\vec{r} = \begin{pmatrix} 2 \\ -5 \\ 4 \end{pmatrix} + t_1 \begin{pmatrix} -7 \\ 7 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} -4 \\ 8 \\ -5 \end{pmatrix}$

Answer: $\vec{r} = \begin{pmatrix} 2 \\ -5 \\ 4 \end{pmatrix} + t_1 \begin{pmatrix} -7 \\ 7 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} -4 \\ 8 \\ -5 \end{pmatrix}$

Q28. Find the vector equation of the plane containing the points M(2, 3, −2), N(1, −2, 3) and P(4, 0, 1). [26-Q]

Ans.

Let $\vec{a} = \vec{OM} = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}$.

$\vec{d}_1 = \vec{MN} = \vec{ON} - \vec{OM}$

$= \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}$

$= \begin{pmatrix} 1-2 \\ -2-3 \\ 3+2 \end{pmatrix}$

$= \begin{pmatrix} -1 \\ -5 \\ 5 \end{pmatrix}$

$\vec{d}_2 = \vec{MP} = \vec{OP} - \vec{OM}$

$= \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}$

$= \begin{pmatrix} 4-2 \\ 0-3 \\ 1+2 \end{pmatrix}$

$= \begin{pmatrix} 2 \\ -3 \\ 3 \end{pmatrix}$

Therefore the vector equation of the plane is

$\vec{r} = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix} + t_1 \begin{pmatrix} -1 \\ -5 \\ 5 \end{pmatrix} + t_2 \begin{pmatrix} 2 \\ -3 \\ 3 \end{pmatrix}$

Answer: $\vec{r} = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix} + t_1 \begin{pmatrix} -1 \\ -5 \\ 5 \end{pmatrix} + t_2 \begin{pmatrix} 2 \\ -3 \\ 3 \end{pmatrix}$

Q29. Find the vector equation of the plane passing through the point (6, 0, −2) and contains the line $x = 4 - 2t$, $y = 3 + 5t$, $z = 7 + 4t$. [25-Q]

Ans.

Let $\vec{a} = \vec{OA} = \begin{pmatrix} 6 \\ 0 \\ -2 \end{pmatrix}$. The direction vector of the line, $\vec{d}_1 = \begin{pmatrix} -2 \\ 5 \\ 4 \end{pmatrix}$, lies in the plane.

B(4, 3, 7) is a point on the line (t=0), hence on the plane. So a second direction vector in the plane is

$\vec{d}_2 = \vec{AB} = \vec{OB} - \vec{OA}$

$= \begin{pmatrix} 4 \\ 3 \\ 7 \end{pmatrix} - \begin{pmatrix} 6 \\ 0 \\ -2 \end{pmatrix}$

$= \begin{pmatrix} 4-6 \\ 3-0 \\ 7+2 \end{pmatrix}$

$= \begin{pmatrix} -2 \\ 3 \\ 9 \end{pmatrix}$

Therefore the vector equation of the plane is

$\vec{r}=\begin{pmatrix}6\\0\\-2\end{pmatrix}+t_1\begin{pmatrix}-2\\5\\4\end{pmatrix}+t_2\begin{pmatrix}-2\\3\\9\end{pmatrix}$

Answer: $\vec{r}=\begin{pmatrix}\mathbf{6}\\\mathbf{0}\\\mathbf{-2}\end{pmatrix}+\mathbf{t_1}\begin{pmatrix}\mathbf{-2}\\\mathbf{5}\\\mathbf{4}\end{pmatrix}+\mathbf{t_2}\begin{pmatrix}\mathbf{-2}\\\mathbf{3}\\\mathbf{9}\end{pmatrix}$

Q30. A plane containing the point A(3, −9, 1) and has a normal vector $\begin{pmatrix}4\\2\\-2\end{pmatrix}$. Find the equation of the plane in its Cartesian form. [24-Q]

Ans.

The normal vector is $\vec{n}=\begin{pmatrix}4\\2\\-2\end{pmatrix}$ and $\vec{a}=\vec{OA}=\begin{pmatrix}3\\-9\\1\end{pmatrix}$.

$d=\vec{a}\cdot\vec{n}=\begin{pmatrix}3\\-9\\1\end{pmatrix}\cdot\begin{pmatrix}4\\2\\-2\end{pmatrix}=3(4)+(-9)(2)+1(-2)=-8.$

The Cartesian equation of the plane is 4x+2y−2z=−8.
Dividing by 2:
Answer: 2x+y−z+4=0

Q31. Find the Cartesian equation of the plane a normal vector $\begin{pmatrix}2\\3\\1\end{pmatrix}$ and which passes through the point (1, 1, 1). [25*-Q]

Ans.

The normal vector is $\vec{n}=\begin{pmatrix}2\\3\\1\end{pmatrix}$ and $\vec{a}=\vec{OA}=\begin{pmatrix}1\\1\\1\end{pmatrix}$.

$d=\vec{a}\cdot\vec{n}=\begin{pmatrix}1\\1\\1\end{pmatrix}\cdot\begin{pmatrix}2\\3\\1\end{pmatrix}=1(2)+1(3)+1(1)=6.$

The Cartesian equation of the plane is 2x+3y+z=6.
Answer: 2x+3y+z=6

Q32. Find the Cartesian equation of the plane passing through the point (1, 2, −1) and parallel to the plane $\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1\\1\\1\end{pmatrix}+t_1\begin{pmatrix}0\\1\\-1\end{pmatrix}+t_2\begin{pmatrix}-1\\-1\\2\end{pmatrix}$. [25-Q]

Ans.

A parallel plane has the same normal vector.

$\vec{n}=\vec{d_1}\times\vec{d_2}$

$=\begin{pmatrix}0\\1\\-1\end{pmatrix}\times\begin{pmatrix}-1\\-1\\2\end{pmatrix}$

$=\begin{pmatrix}(1)(2)-(-1)(-1)\\(-1)(-1)-(0)(2)\\(0)(-1)-(1)(-1)\end{pmatrix}$

$=\begin{pmatrix}1\\1\\1\end{pmatrix}$

The required plane passes through $\vec{a}=\vec{OA}=\begin{pmatrix}1\\2\\-1\end{pmatrix}$.

$d=\vec{a}\cdot\vec{n}=\begin{pmatrix}1\\2\\-1\end{pmatrix}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}=1(1)+2(1)+(-1)(1)=2.$

The Cartesian equation of the plane is x+y+z=2.
Answer: x+y+z=2

Q33. Find the Cartesian equation of the plane given by the vector equation $\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1\\2\\5\end{pmatrix}+t_1\begin{pmatrix}-2\\3\\1\end{pmatrix}+t_2\begin{pmatrix}2\\2\\5\end{pmatrix}$. [26-Q]

Ans.

$\vec{n}=\vec{d_1}\times\vec{d_2}$

$=\begin{pmatrix}-2\\3\\1\end{pmatrix}\times\begin{pmatrix}2\\2\\5\end{pmatrix}$

$=\begin{pmatrix}(3)(5)-(1)(2)\\(1)(2)-(-2)(5)\\(-2)(2)-(3)(2)\end{pmatrix}$

$=\begin{pmatrix}13\\12\\-10\end{pmatrix}$

$d=\vec{a}\cdot\vec{n}=\begin{pmatrix}1\\2\\5\end{pmatrix}\cdot\begin{pmatrix}13\\12\\-10\end{pmatrix}=1(13)+2(12)+5(-10)=-13.$

The Cartesian equation of the plane is 13x+12y−10z=−13.
Answer: 13x+12y−10z+13=0

5 MARKS

4.3

Q1. (i) Calculate $\vec{a}\times\vec{b}$ when $\vec{a}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}1\\0\\-1\end{pmatrix}$. (ii) Find a unit vector n̂ that is perpendicular to both $\vec{a}$ and $\vec{b}$. [24-Q]

Ans.

(i)

$\vec{a}\times\vec{b}=\begin{pmatrix}2\\-1\\3\end{pmatrix}\times\begin{pmatrix}1\\0\\-1\end{pmatrix}$

$=\begin{pmatrix}(-1)(-1)-(3)(0)\\(3)(1)-(2)(-1)\\(2)(0)-(-1)(1)\end{pmatrix}$

$=\begin{pmatrix}1\\5\\1\end{pmatrix}$

(ii) $\vec{a}\times\vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$. Divide it by its magnitude:

$|\vec{a}\times\vec{b}|=\sqrt{1^2+5^2+1^2}$

$=\sqrt{27}$

$=3\sqrt{3}$

$\hat{n}=\frac{1}{3\sqrt{3}}\begin{pmatrix}1\\5\\1\end{pmatrix}$

Answer: (i) $\begin{pmatrix}\mathbf{1}\\\mathbf{5}\\\mathbf{1}\end{pmatrix}$**; (ii)** $\frac{\mathbf{1}}{\mathbf{3\sqrt{3}}}\begin{pmatrix}\mathbf{1}\\\mathbf{5}\\\mathbf{1}\end{pmatrix}$ **(or its negative)**

Q2. The points A(4, 7, 8), B(2, 3, 4) and C(−1, −2, 1) are vertices of a parallelogram ABCD. Find the coordinates of D and the area of the parallelogram ABCD. [25-Q]

Ans.

(i)

Let D be (x, y, z). Since ABCD is a parallelogram, $\vec{AB}$ is parallel to $\vec{DC}$ and they have the same length and direction, so:

$\vec{AB}=\vec{DC}$

$\vec{OB}-\vec{OA}=\vec{OC}-\vec{OD}$

$\begin{pmatrix}2\\3\\4\end{pmatrix}-\begin{pmatrix}4\\7\\8\end{pmatrix}=\begin{pmatrix}-1\\-2\\1\end{pmatrix}-\begin{pmatrix}x\\y\\z\end{pmatrix}$

$=\begin{pmatrix}-2\\-4\\-4\end{pmatrix}$

$=\begin{pmatrix}-1-x\\-2-y\\1-z\end{pmatrix}$

So $-2=-1-x\rightarrow x=1$, $-4=-2-y\rightarrow y=2$, $-4=1-z\rightarrow z=5$.
Therefore D is (1, 2, 5).

(ii) $\vec{AB}$ and $\vec{BC}$ are adjacent sides of the parallelogram.

$\vec{AB}=\vec{OB}-\vec{OA}$

$=\begin{pmatrix}2\\3\\4\end{pmatrix}-\begin{pmatrix}4\\7\\8\end{pmatrix}$

$=\begin{pmatrix}2-4\\3-7\\4-8\end{pmatrix}$

$=\begin{pmatrix}-2\\-4\\-4\end{pmatrix}$

$\vec{BC}=\vec{OC}-\vec{OB}$

$=\begin{pmatrix}-1\\-2\\1\end{pmatrix}-\begin{pmatrix}2\\3\\4\end{pmatrix}$

$=\begin{pmatrix}-1-2\\-2-3\\1-4\end{pmatrix}$

$=\begin{pmatrix}-3\\-5\\-3\end{pmatrix}$

$$\vec{AB}\times\vec{BC} = \begin{pmatrix}-2\\-4\\-4\end{pmatrix}\times\begin{pmatrix}-3\\-5\\-3\end{pmatrix}$$
$$= \begin{pmatrix}(-4)(-3)-(-4)(-5)\\(-4)(-3)-(-2)(-3)\\(-2)(-5)-(-4)(-3)\end{pmatrix}$$
$$= \begin{pmatrix}-8\\6\\-2\end{pmatrix}$$
$$|\vec{AB}\times\vec{BC}| = \sqrt{(-8)^2+6^2+(-2)^2}$$
$$= \sqrt{104}$$
$$= 2\sqrt{26}$$

Area $=|\vec{AB}\times\vec{BC}|=2\sqrt{26}$ square units

Answer: D(1, 2, 5); area = $2\sqrt{26}$ square units

Q3. The points A(2, −1, 4), B(5, 1, 2) and C(3, 1, 4) are vertices of a parallelogram ABCD. (i) Find the coordinates of D. (ii) Calculate the area of the parallelogram. [25-Q]

Ans.

(i)

Let D be (x, y, z). Since ABCD is a parallelogram, $\vec{AB}$ is parallel to $\vec{DC}$ and they have the same length and direction, so:

$$\vec{AB} = \vec{DC}$$
$$\vec{OB}-\vec{OA} = \vec{OC}-\vec{OD}$$
$$\begin{pmatrix}5\\1\\2\end{pmatrix}-\begin{pmatrix}2\\-1\\4\end{pmatrix} = \begin{pmatrix}3\\1\\4\end{pmatrix}-\begin{pmatrix}x\\y\\z\end{pmatrix}$$
$$= \begin{pmatrix}3\\2\\-2\end{pmatrix}$$
$$= \begin{pmatrix}3-x\\1-y\\4-z\end{pmatrix}$$

So $3=3-x\rightarrow x=0$, $2=1-y\rightarrow y=-1$, $-2=4-z\rightarrow z=6$.

Therefore D is (0, −1, 6).

(ii) $\vec{AB}$ and $\vec{BC}$ are adjacent sides of the parallelogram.

$$\vec{AB} = \vec{OB}-\vec{OA}$$
$$= \begin{pmatrix}5\\1\\2\end{pmatrix}-\begin{pmatrix}2\\-1\\4\end{pmatrix}$$
$$= \begin{pmatrix}5-2\\1+1\\2-4\end{pmatrix}$$
$$= \begin{pmatrix}3\\2\\-2\end{pmatrix}$$

$$\vec{BC} = \vec{OC}-\vec{OB}$$
$$= \begin{pmatrix}3\\1\\4\end{pmatrix}-\begin{pmatrix}5\\1\\2\end{pmatrix}$$
$$= \begin{pmatrix}3-5\\1-1\\4-2\end{pmatrix}$$
$$= \begin{pmatrix}-2\\0\\2\end{pmatrix}$$

$$\vec{AB}\times\vec{BC} = \begin{pmatrix}3\\2\\-2\end{pmatrix}\times\begin{pmatrix}-2\\0\\2\end{pmatrix}$$
$$= \begin{pmatrix}(2)(2)-(-2)(0)\\(-2)(-2)-(3)(2)\\(3)(0)-(2)(-2)\end{pmatrix}$$
$$= \begin{pmatrix}4\\-2\\4\end{pmatrix}$$
$$|\vec{AB}\times\vec{BC}| = \sqrt{4^2+(-2)^2+4^2}$$
$$= \sqrt{36}$$
$$= 6$$

Area $=|\vec{AB}\times\vec{BC}|=6$ square units

Answer: D(0, −1, 6); area = 6 square units

4.4

Q4. Find the vector equation of the line passing through the point (1, 2, 3) and perpendicular to the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{7-z}{2}$. Find also the distance between the given point (1, 2, 3) and the point of intersection of two lines. [25-Q]

Ans.

The given line: $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2} = t$, so its points are (6 + 3t, 7 + 2t, 7 − 2t) and its direction is $\vec{d} = \begin{pmatrix}3\\2\\-2\end{pmatrix}$.

Let A(1, 2, 3) and let B be the point of intersection.

$$\vec{AB} = \begin{pmatrix}5+3t\\5+2t\\4-2t\end{pmatrix}$$
$$\vec{AB}\cdot\vec{d} = 3(5+3t) + 2(5+2t) - 2(4-2t) = 0$$
$$15 + 9t + 10 + 4t - 8 + 4t = 0$$
$$17 + 17t = 0$$
$$t = -1$$

B = (6 − 3, 7 − 2, 7 + 2) = (3, 5, 9), and $\vec{AB} = \begin{pmatrix}2\\3\\6\end{pmatrix}$.

$$\vec{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + s\begin{pmatrix}2\\3\\6\end{pmatrix}$$

$|\vec{AB}| = \sqrt{4+9+36} = 7$

Answer: $\vec{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + s\begin{pmatrix}2\\3\\6\end{pmatrix}$; distance = 7 units

Q5. Find the equation of the plane through A(1, 1, −1), B(6, 4, −5) and C(−4, −2, 3): (i) in vector form (ii) in Cartesian form. [25-Q]

Ans.

$$\vec{AB} = \vec{OB}-\vec{OA}$$
$$= \begin{pmatrix}6\\4\\-5\end{pmatrix}-\begin{pmatrix}1\\1\\-1\end{pmatrix}$$
$$= \begin{pmatrix}6-1\\4-1\\-5+1\end{pmatrix}$$
$$= \begin{pmatrix}5\\3\\-4\end{pmatrix}$$

$$\vec{AC} = \vec{OC}-\vec{OA}$$
$$= \begin{pmatrix}-4\\-2\\3\end{pmatrix}-\begin{pmatrix}1\\1\\-1\end{pmatrix}$$
$$= \begin{pmatrix}-4-1\\-2-1\\3+1\end{pmatrix}$$
$$= \begin{pmatrix}-5\\-3\\4\end{pmatrix}$$

Here $\vec{AC}=-\vec{AB}$, so $\vec{AB}$ and $\vec{AC}$ are parallel. Since A is common, A, B and C are collinear.

Check: $\vec{AB}\times\vec{AC} = \begin{pmatrix}5\\3\\-4\end{pmatrix}\times\begin{pmatrix}-5\\-3\\4\end{pmatrix} = \begin{pmatrix}0\\0\\0\end{pmatrix}$.

Three collinear points do not determine a unique plane, so there is no single vector (or Cartesian) equation; infinitely many planes contain the line through A, B and C.

Answer: Collinear points; infinite planes exist.

Q6. Find the Cartesian equation of the plane with vector equation $\vec{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + t_1\begin{pmatrix}1\\1\\2\end{pmatrix} + t_2\begin{pmatrix}2\\-1\\5\end{pmatrix}$. [26-Q]

Ans.

$$\vec{n} = \vec{d_1}\times\vec{d_2}$$
$$= \begin{pmatrix}1\\1\\2\end{pmatrix}\times\begin{pmatrix}2\\-1\\5\end{pmatrix}$$
$$= \begin{pmatrix}(1)(5)-(2)(-1)\\(2)(2)-(1)(5)\\(1)(-1)-(1)(2)\end{pmatrix}$$

$= \begin{pmatrix} 7 \\ -1 \\ -3 \end{pmatrix}$

$d = \vec{a} \cdot \vec{n} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ -1 \\ -3 \end{pmatrix} = 1(7)+2(-1)+3(-3) = -4.$

The Cartesian equation of the plane is $7x-y-3z=-4$.

Answer: 7x−y−3z+4=0

Q7. Find the vector equation of the plane containing points A(2, −1, 0), B(−4, 2, 2) and C(−9, −1, 2). Then find also the Cartesian equation of that plane. [26-Q]

Ans.

Let $\vec{a} = \vec{OA} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$.

$\vec{d}_1 = \vec{AB} = \vec{OB} - \vec{OA}$

$= \begin{pmatrix} -4 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} -4-2 \\ 2+1 \\ 2-0 \end{pmatrix}$

$= \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix}$

$\vec{d}_2 = \vec{AC} = \vec{OC} - \vec{OA}$

$= \begin{pmatrix} -9 \\ -1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} -9-2 \\ -1+1 \\ 2-0 \end{pmatrix}$

$= \begin{pmatrix} -11 \\ 0 \\ 2 \end{pmatrix}$

The vector equation of the plane is

$\vec{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + t_1 \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix} + t_2 \begin{pmatrix} -11 \\ 0 \\ 2 \end{pmatrix}$

The Cartesian equation: $\vec{n} = \vec{d}_1 \times \vec{d}_2$

$\vec{n} = \vec{d}_1 \times \vec{d}_2$

$= \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix} \times \begin{pmatrix} -11 \\ 0 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} (3)(2)-(2)(0) \\ (2)(-11)-(-6)(2) \\ (-6)(0)-(3)(-11) \end{pmatrix}$

$= \begin{pmatrix} 6 \\ -10 \\ 33 \end{pmatrix}$

$d = \vec{a} \cdot \vec{n} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -10 \\ 33 \end{pmatrix} = 2(6)+(-1)(-10)+0(33) = 22.$

The Cartesian equation of the plane is $6x-10y+33z=22$.

Answer: $\vec{r} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} + t_1 \begin{pmatrix} -6 \\ 3 \\ 2 \end{pmatrix} + t_2 \begin{pmatrix} -11 \\ 0 \\ 2 \end{pmatrix}$ **; 6x−10y+33z=22**

Q8. Find the magnitude of unit vector perpendicular to the plane containing the points P(1, −1, 0), Q(2, 1, −1) and R(−1, 1, 2). [25-Q]

Ans.

$\vec{PQ} = \vec{OQ} - \vec{OP}$

$= \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} 2-1 \\ 1+1 \\ -1-0 \end{pmatrix}$

$= \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

$\vec{PR} = \vec{OR} - \vec{OP}$

$= \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} -1-1 \\ 1+1 \\ 2-0 \end{pmatrix}$

$= \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$

A normal vector of the plane is $\vec{n} = \vec{PQ} \times \vec{PR}$.

$\vec{n} = \vec{PQ} \times \vec{PR} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$

$= \begin{pmatrix} (2)(2)-(-1)(2) \\ (-1)(-2)-(1)(2) \\ (1)(2)-(2)(-2) \end{pmatrix}$

$= \begin{pmatrix} 6 \\ 0 \\ 6 \end{pmatrix}$

$|\vec{n}| = \sqrt{6^2+0^2+6^2}$

$= \sqrt{72}$

$= 6\sqrt{2}$

The unit vector $\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{1}{6\sqrt{2}} \begin{pmatrix} 6 \\ 0 \\ 6 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

$|\hat{n}| = \frac{1}{\sqrt{2}} \sqrt{1^2+0^2+1^2} = \frac{1}{\sqrt{2}} \times \sqrt{2} = 1$

Any unit vector has magnitude 1.

Answer: 1

CH.05 Permutations and Combinations

Notes · Questions with Answers

NOTES

Permutation

A **permutation** is an arrangement of objects in a specific order. If there are n objects, then a permutation of any such r objects, with $r \le n$, is said to be a permutation of n objects taken r at a time. The number of permutations of n distinct objects taken r at a time is usually denoted by ${}^nP_r$.

Factorial

If n is a positive integer, n! (n factorial) stands for the product of the first n positive integers n, (n − 1), (n − 2), ..., 1. We define 0! to be 1.

Combination

A **combination** is a selection, in which the order does not matter, of objects from a collection. The number of combinations of n distinct objects taken r at a time is denoted by ${}^nC_r$.

Disjoint cases

Two cases of a counting problem are **disjoint** if they have no arrangement in common. If every possible arrangement falls into some of the cases, the Addition Principle can be applied.

The Multiplication Principle

Suppose a task can be performed in m ways, and no matter how the task has been performed another task can then be performed in n ways. Then the number of ways to perform these two tasks in succession is mn.
The principle can be extended to any finite number of successive tasks.

The Addition Principle

Suppose there are k pairwise disjoint cases, into which the elements in a counting problem fall. If there are n_1 elements in the first case, n_2 elements in the second case, n_3 elements in the third case, ..., and n_k elements in the kth case, then the number of elements in the problem is
$n_1 + n_2 + n_3 + \dots + n_k$

Factorial notation

If n is a positive integer,

$n! = n(n-1)(n-2)\dots 1$

$0! = 1$

Notice that

$n! = n \cdot (n-1)!$

$n! = n \cdot (n-1) \cdot (n-2)!$

Permutations nPr

If n and r are positive integers with $r \le n$,
${}^nP_r = n(n-1)(n-2)\dots(n-(r-1))$ (r factors)

We define ${}^nP_0 = 1$ for a nonnegative integer n. Generally, for integers n and r with $0 \le r \le n$,

$${}^nP_r = \frac{n!}{(n-r)!}$$

Notice that ${}^nP_n = n!$.

Combinations nCr

Each combination of r objects gives rise to r! permutations, so ${}^nC_r \cdot r! = {}^nP_r$. If n and r are integers with $0 \le r \le n$, then

$${}^nC_r = \frac{{}^nP_r}{r!} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 1}$$

Properties of nCr

For integers n and r with $0 \le r \le n$,
${}^nC_1 = n$, ${}^nC_n = 1$ and ${}^nC_r = {}^nC_{n-r}$

Permutations with repetitions

Suppose we are given n objects, in which n_1 objects are of the first kind, n_2 objects are of the second kind, n_3 objects are of the third kind, ..., and n_r objects are of the rth kind, such that $n_1 + n_2 + \dots + n_r = n$. Then the number of permutations of all these n objects is

$$\frac{n!}{n_1!\, n_2! \dots n_r!}$$

The Exclusion Principle

Count what you are not interested in and subtract it from the total.

Number of subsets of a finite set

1. If a finite set contains n elements, then it has 2^n subsets.
2. If n is an integer with $n \ge 0$, then ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$.

1 MARK

5.1

Q1. A girl has 5 tops, 6 skirts and 4 caps from which to choose an outfit. In how many ways can she select one top, one skirt and one cap? [25-Q]

A. 40 **B.** 30 **C.** 100 **D.** 120

Ans.
Multiplication principle: the choices are made one after another.

$5 \cdot 6 \cdot 4 = 120$

Answer: D

Q2. What is the number of arrangements if a die is rolled 4 times? [25*-Q]

A. 37 **B.** 36 **C.** 216 **D.** 1296

Ans.
Each roll has 6 possible outcomes, and the 4 rolls are independent.

$6 \cdot 6 \cdot 6 \cdot 6 = 6^4 = 1296$

Answer: D

Q3. A boy has 3 pockets. The number of ways that he can put 6 marbles in his pocket is [25-Q]

A. 729 **B.** 216 **C.** 18 **D.** 9

Ans.
Each of the 6 marbles can go into any of the 3 pockets.

$3^6 = 729$

Answer: A

Q4. How many 2−digit positive integers are there such that the sum of their two digits is divisible by 4? [24-Q]

A. 21 **B.** 22 **C.** 23 **D.** 24

Ans.
The digit sum of a 2−digit number is between 1 and 18, so it must be 4, 8, 12 or 16. Use the addition principle:

Sum 4: 13, 22, 31, 40 → 4
Sum 8: 17, 26, 35, 44, 53, 62, 71, 80 → 8
Sum 12: 39, 48, 57, 66, 75, 84, 93 → 7
Sum 16: 79, 88, 97 → 3
Total = 4 + 8 + 7 + 3 = 22

Answer: B

Q5. If (n + 4)! = 42(n + 2)!, the value of n is [26-Q]

A. 2 **B.** 1 **C.** 5 **D.** 3

Ans.
$(n+4)! = (n+4)(n+3)(n+2)!$

$(n+4)(n+3)(n+2)! = 42(n+2)!$

$(n+4)(n+3) = 42 = 7 \cdot 6$

$n + 3 = 6 \rightarrow n = 3$

Answer: D

5.2

Q6. Evaluate ${}^{10}P_1 + {}^{10}P_0$. [25-Q]

A. 12 **B.** 10 **C.** 11 **D.** 20

Ans.

${}^{10}P_1 = 10$, ${}^{10}P_0 = 1$

${}^{10}P_1 + {}^{10}P_0 = 10 + 1 = 11$

Answer: C

Q7. If ${}^5P_2 + {}^7P_0 = 2c - 1$, then the value of c is [25-Q]

A. 11 **B.** 12 **C.** 22 **D.** 32

Ans.

${}^5P_2 = 5 \cdot 4 = 20$, ${}^7P_0 = 1$

$20 + 1 = 2c - 1$

$2c = 22$

$c = 11$

Answer: A

Q8. How many ways can eight runners in an Olympic race finish in first, second and third places? [26-Q]

A. 8 **B.** 56 **C.** 24 **D.** 336

Ans.

Order matters: ${}^nP_r = \frac{n!}{(n-r)!}$.

${}^8P_3 = 8 \cdot 7 \cdot 6 = 336$

Answer: D

Q9. In how many ways can all the letters of the word ORANGE be arranged, without repeating any letters? [24-Q]

A. 730 **B.** 270 **C.** 720 **D.** 780

Ans.
ORANGE has 6 different letters.

${}^6P_6 = 6! = 720$

Answer: C

Q10. In how many ways can all the letters of the word QUESTION be arranged, without repeating any letters? [25-Q]

A. 40230 **B.** 40500 **C.** 43200 **D.** 40320

Ans.

QUESTION has 8 different letters.

$8! = 40320$

Answer: D

Q11. Five people go to the theater. They sit in a row with eight seats. In how many ways can this be done if they can sit anywhere? [25-Q]

A. 120 **B.** 40230 **C.** 6720 **D.** 7620

Ans.

${}^8P_5 = 8{\cdot}6{\cdot}5{\cdot}4 = 6720$

Answer: C

5.3

Q12. ${}^{10}P_2 + {}^{10}P_0 + {}^{10}C_{10}$ is [26-Q]

A. 91 **B.** 92 **C.** 101 **D.** 110

Ans.

${}^{10}P_2 = 90,\ {}^{10}P_0 = 1,\ {}^{10}C_{10} = 1$

$90 + 1 + 1 = 92$

Answer: B

Q13. ${}^{12}C_5{\cdot}{}^7C_4 =$ [25-Q]

A. ${}^{12}C_6{\cdot}{}^6C_1$ **B.** ${}^{12}C_4{\cdot}{}^7C_3$ **C.** ${}^{12}C_4{\cdot}{}^8C_5$ **D.** 12220

Ans.

${}^{12}C_5{\cdot}{}^7C_4 = \frac{12!}{5!7!}\cdot\frac{7!}{4!3!} = \frac{12!}{5!4!3!}$

${}^{12}C_4{\cdot}{}^8C_5 = \frac{12!}{4!8!}\cdot\frac{8!}{5!3!} = \frac{12!}{4!5!3!}$

Both are equal.

Answer: C

Q14. The number of ways of selection of 4 people from a group of 10 people is [24-Q]

A. 202 **B.** 270 **C.** 210 **D.** 120

Ans.

${}^{10}C_4 = \frac{10{\cdot}9{\cdot}8{\cdot}7}{4{\cdot}3{\cdot}2{\cdot}1} = 210$

Answer: C

Q15. After a sport tournament, each player shakes hands with every other player once. If there are 36 handshakes in total, the number of players at the tournament is [5-Q]

A. 18 **B.** 8 **C.** 10 **D.** 9

Ans.

${}^nC_2 = 36$

$\frac{n(n-1)}{2} = 36$

$n^2 - n - 72 = 0$

$(n - 9)(n + 8) = 0$

$n = 9\ (n > 0)$.

Answer: D

Q16. The number of diagonals that can be drawn in a regular octagon is [24-Q]

A. 20 **B.** 28 **C.** 56 **D.** 48

Ans.

Join any 2 of the 8 vertices: ${}^8C_2$ lines; subtract the 8 sides.

${}^8C_2 - 8 = 28 - 8 = 20$

Answer: A

Q17. A coin is tossed 6 times. The number of outcomes consisting of 3 heads and 3 tails (in any order) is [25*-Q]

A. 720 **B.** 36 **C.** 6 **D.** 20

Ans.

Choose which 3 of the 6 tosses are heads:

${}^6C_3 = 20$

Answer: D

Q18. In how many ways can 12 people be divided into two groups, one with 7 people and the other with 5 people? [26-Q]

A. 420 **B.** 720 **C.** 792 **D.** 1080

Ans.

${}^{12}C_7 = {}^{12}C_5 = 792$

Answer: C

Q19. The number of ways a committee consisting of 3 men and 2 women, can be chosen from 7 men and 5 women is [25*-Q]

A. 45 **B.** 350 **C.** 4200 **D.** 230

Ans.

${}^7C_3{\cdot}{}^5C_2 = 35{\cdot}10 = 350$

Answer: B

Q20. The number of ways of selection of two numbers from the set {1, 2, ..., 12} whose sum is divisible by 3 is [24-Q]

A. 66 **B.** 16 **C.** 6 **D.** 22

Ans.

Group the numbers by remainder on division by 3: {3, 6, 9, 12} (remainder 0), {1, 4, 7, 10} (remainder 1), {2, 5, 8, 11} (remainder 2). A sum is divisible by 3 when both are from remainder 0, or one from remainder 1 and one from remainder 2.

${}^4C_2 + {}^4C_1{\cdot}{}^4C_1 = 6 + 16 = 22$

Answer: D

Q21. How many two–element subsets {x, y} of the set {0, 1, 2, 3, ..., 9} are there that x × y is divisible by 7? [25-Q]

A. 8 **B.** 9 **C.** 17 **D.** 18

Ans.

x × y is divisible by 7 if one of them is 0 or 7.

Subsets containing 0: {0, k}, k = 1, ..., 9 → 9

Subsets containing 7 (without 0): {7, k}, k = 1, 2, 3, 4, 5, 6, 8, 9 → 8

Total = 9 + 8 = 17

Answer: C

Q22. Suppose there are 4 black cars and 7 white cars. If all the cars are distinguishable, in how many ways can 3 cars of the same colour be chosen? [24-Q]

A. 37 **B.** 38 **C.** 39 **D.** 40

Ans.

${}^4C_3 + {}^7C_3 = 4 + 35 = 39$

Answer: C

Q23. Suppose there are 6 black cars and 5 white cars. If all the cars are distinguishable, in how many ways can 3 cars of the same colour be chosen? [24-Q]

A. 37 **B.** 30 **C.** 39 **D.** 40

Ans.

${}^6C_3 + {}^5C_3 = 20 + 10 = 30$

Answer: B

5.4

Q24. The number of numbers containing four digits are [24-Q]

A. 5040 **B.** 9000 **C.** 53024 **D.** 6561

Ans.

The first digit cannot be 0 (9 choices); each of the other 3 digits has 10 choices.

$9{\cdot}10{\cdot}10{\cdot}10 = 9000$

Answer: B

Q25. The number of permutations of the word 'SQUARE' have the letters Q and U next to each other is [25-Q]

A. 720 **B.** 240 **C.** 110 **D.** 620

Ans.

Treat QU as one block: 5 objects (QU, S, A, R, E) can be arranged in 5! ways, and Q, U can be ordered in 2! ways inside the block.

$5!{\cdot}2! = 120{\cdot}2 = 240$

Answer: B

Q26. How many numbers greater than 200, 000 by arranging all digits of number 242302? [25-Q]

A. 120 **B.** 100 **C.** 130 **D.** 20

Ans.

The digits are 2, 2, 2, 4, 3, 0 (three 2's). Total arrangements:

$\frac{6!}{3!} = 120$

A number starting with 0 is below 200,000 (not a 6–digit number). Arrangements with 0 first:

$\frac{5!}{3!} = 20$

$120 - 20 = 100$

(All other first digits 2, 3, 4 give a number greater than 200,000.)

Answer: B

2 MARKS

5.1

Q1. How many 3–digit numbers can be formed by using the digits 1 to 9 if no digits are repeated? [25*-Q]

Ans.

First digit: 9 choices; second: 8 choices; third: 7 choices.

$9{\cdot}8{\cdot}7 = 504$

Answer: 504

Q2. How many 6–digit numbers can be formed from the digits 0, 1, 3, 5, 7 and 9 which are divisible by 10 and no digits is repeated? [25*-Q]

Ans.
Divisible by 10 means the last digit is 0. The other 5 digits (1, 3, 5, 7, 9) fill the first 5 places in any order:

$5! = 120$

Answer: 120

Q3. How many integers from 1 through 100 contain the digit 6 exactly once? [26-Q]

Ans.
Count by cases:

1–digit: 6 → 1
2–digit, 6 in the tens place: 60, 61, ..., 69 except 66 → 9
2–digit, 6 in the units place: tens digit is 1, 2, ..., 9 except 6 → 8

Total $= 1 + 9 + 8 = 18$

(100 does not contain 6.)
Answer: 18

5.2

Q4. If ${}^{m+n}P_2 = 56$, ${}^{m-n}P_2 = 12$, find m + 2n. [25-Q]

Ans.
${}^{x}P_2 = x(x-1)$.

$(m+n)(m+n-1) = 56 = 8\cdot7 \rightarrow m + n = 8$
$(m-n)(m-n-1) = 12 = 4\cdot3 \rightarrow m - n = 4$

Adding: $2m = 12$, $m = 6$; then $n = 2$.

$m + 2n = 6 + 4 = 10$

Answer: 10

Q5. In how many ways can a president, a treasurer and a secretary for a committee be selected from a group of 9 people? [24-Q]

Ans.
${}^{9}P_3 = 9\cdot8\cdot7 = 504$

Answer: 504

Q6. In how many ways can a president, a treasurer and a secretary for a committee be selected from a group of 10 people? [5-Q, 24-Q, 26-Q]

Ans.
${}^{10}P_3 = 10\cdot9\cdot8 = 720$

Answer: 720

Q7. A mathematics debating team consists of 4 speakers. How many ways can the captain and vice–captain be chosen? [25-Q]

Ans.
${}^{4}P_2 = 4\cdot3 = 12$

Answer: 12

Q8. How many ways are there to rearrange the eight letters in the word MAJORITY? [25-Q]

Ans.
The 8 letters M, A, J, O, R, I, T, Y are all different.

$8! = 40320$

Answer: 40320

5.3

Q9. Solve the equation ${}^{n}C_{n-1} = n^2 - 110$ for n. [24-Q]

Ans.
${}^{n}C_{n-1} = {}^{n}C_1 = n$.

$n = n^2 - 110$
$n^2 - n - 110 = 0$
$(n - 11)(n + 10) = 0$

Answer: n = 11 (n = −10 is rejected)

Q10. If ${}^{6}C_3 = n\cdot{}^{4}C_1$, find ${}^{n}C_2$ where n is a positive integer. [25-Q]

Ans.
${}^{6}C_3 = 20$, ${}^{4}C_1 = 4$
$20 = 4n \rightarrow n = 5$
${}^{5}C_2 = \frac{5\cdot4}{2} = 10$

Answer: 10

Q11. In how many ways can a committee of 3 people be selected from a group of 10 people. [25-Q]

Ans.
${}^{10}C_3 = \frac{10\cdot9\cdot8}{3\cdot2\cdot1} = 120$

Answer: 120

Q12. In how many ways can a committee of 4 people be selected from a group of 10 people. [26-Q]

Ans.
${}^{10}C_4 = \frac{10\cdot9\cdot8\cdot7}{4\cdot3\cdot2\cdot1} = 210$

Answer: 210

Q13. How many diagonals have a 12 sided polygon? [26-Q]

Ans.
Lines joining any 2 of the 12 vertices, minus the 12 sides:

${}^{12}C_2 - 12 = 66 - 12 = 54$

Answer: 54

Q14. A boat's crew consists of 8 men, of whom 2 can only row on one side and 2 only on the other. In how many ways can a selection be made so that 4 men may row on each side? [24-Q]

Ans.
2 men are fixed on each side, so each side needs 2 more from the 4 remaining men. Choose 2 of the 4 for the first side; the other 2 go to the other side.

${}^{4}C_2 = 6$

Answer: 6

Q15. In a meeting of three representatives from each of 6 companies, each representative shook hands with every person other than those from his or her own company. How many handshakes took place in the meeting? [25-Q]

Ans.
There are $6 \times 3 = 18$ people. All possible handshakes between any 2 people: ${}^{18}C_2 = 153$.

Subtract the handshakes inside the same company: $6 \times {}^{3}C_2 = 18$.

$153 - 18 = 135$

Answer: 135

5.4

Q16. In how many ways can a permutation of all the letter of the word POSITIVITY be formed? [24-Q]

Ans.
POSITIVITY has 10 letters: I appears 3 times, T appears 2 times, the others (P, O, S, V, Y) once each.

$\frac{10!}{3!2!} = \frac{3628800}{12} = 302400$

Answer: 302,400

Q17. The English alphabet has 5 vowels and 21 consonants. How many words with 2 different vowels and 2 different consonants can be formed from the alphabet? [24-Q]

Ans.
Choose the letters, then arrange the 4 chosen letters in 4! ways:

${}^{5}C_2 \cdot {}^{21}C_2 \cdot 4! = 10\cdot210\cdot24 = 50400$

Answer: 50,400

Q18. Find the number of all 5–digit positive integers written with 0, 1, 2, 3, 4 such that none of the digit is repeated. [25-Q]

Ans.
The first digit cannot be 0, so it has 4 choices; the remaining 4 digits are then arranged from the other 4 digits in 4! ways.

$4\cdot4! = 4\cdot24 = 96$

(or $5! - 4! = 120 - 24 = 96$)

Answer: 96

Q19. How many four–digit positive integers can be formed from 0, 1, 3, 4, 5, 6, 8 and 9 such that the digits are distinct and are in increasing order? [24-Q]

Ans.
Digits in increasing order means each choice of 4 different digits gives exactly one number (arranged in increasing order). A four–digit number cannot begin with 0, and in increasing order 0 would come first, so 0 cannot be used. Choose 4 from the 7 digits 1, 3, 4, 5, 6, 8, 9:

${}^{7}C_4 = 35$

Answer: 35

Q20. In how many ways can 5 girls and 3 boys be seated in a row so that no two boys are together? [25-Q]

Ans.
Seat the 5 girls first: 5! ways. This makes 6 gaps (including both ends); choose 3 of them for the boys and arrange the boys: ${}^{6}P_3$.

$5! \cdot {}^{6}P_3 = 120\cdot120 = 14400$

Answer: 14,400

Q21. A bag contains 12 different chocolates, 4 different mints and 6 different toffees. Three sweets are chosen. How many ways are there of choosing not all chocolates? [25-Q]

Ans.
Total sweets = 12 + 4 + 6 = 22. Subtract the selections of 3 chocolates from all selections:

${}^{22}C_3 - {}^{12}C_3 = 1540 - 220 = 1320$

Answer: 1320

Q22. How many permutations of all the letters of the word EXCELLENCE do not have N and X in the first two letters? [25-Q]

Ans.

EXCELLENCE has 10 letters: E × 4, C × 2, L × 2, X, N. (Here we count the arrangements in which neither N nor X is in the first two places.)

Put N and X in two of the last 8 places: ${}^{8}P_{2} = 56$ ways. The remaining 8 places are filled by E, E, E, E, C, C, L, L:

$$\frac{8!}{4!2!2!} = 420$$

$56 \cdot 420 = 23520$

Check: total = $\frac{10!}{4!2!2!} = 37800$; arrangements with N, X in the first two places = 2·420 = 840. (If the question means "N and X not both in the first two places" the answer is 37800 − 840 = 36960.)

Answer: 23,520

3 MARKS

5.2

Q1. How many 5−digit telephone numbers can be constructed using the digits 0 to 9, if each number starts with 67 and no digit appears more than once? [5-Q]

Ans.

The first two digits are fixed as 6 and 7. The remaining 3 places are filled from the other 8 digits (no repetition):

${}^{8}P_{3} = 8 \cdot 7 \cdot 6 = 336$

Answer: 336

Q2. A railway carriage can seat 8 people. In how many ways can 6 people be seated in the carriage: (i) if any seat can be occupied? (ii) if two particular seats have to be occupied? [25-Q]

Ans.

(i) Choose and arrange 6 people in 8 seats:

${}^{8}P_{6} = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 20160$

(ii) The two particular seats are filled first by 2 of the 6 people: ${}^{6}P_{2}$ ways. The other 4 people then take any 4 of the remaining 6 seats: ${}^{6}P_{4}$ ways.

${}^{6}P_{2} \cdot {}^{6}P_{4} = 30 \cdot 360 = 10800$

Answer: (i) 20160; (ii) 10800

Q3. There are 6 boys who enter a boat with 8 seats, 4 on each side. How many ways can 2 boys A and B sit on the port side and another boy W sit on the starboard side? [25-Q, 25*-Q]

Ans.

A and B take 2 of the 4 port seats: ${}^{4}P_{2} = 12$. W takes 1 of the 4 starboard seats: 4 ways. The other 3 boys take any 3 of the remaining 5 seats: ${}^{5}P_{3} = 60$.

$12 \cdot 4 \cdot 60 = 2880$

Answer: 2880

5.3

Q4. Show that ${}^{12}C_{5} \cdot {}^{7}C_{4} = {}^{12}C_{4} \cdot {}^{8}C_{5}$. [24-Q]

Ans.

$$\text{L.H.S} = {}^{12}C_{5} \cdot {}^{7}C_{4} = \frac{12!}{5!7!} \cdot \frac{7!}{4!3!} = \frac{12!}{5!4!3!}$$

$$\text{R.H.S} = {}^{12}C_{4} \cdot {}^{8}C_{5} = \frac{12!}{4!8!} \cdot \frac{8!}{5!3!} = \frac{12!}{4!5!3!}$$

L.H.S = R.H.S. (Proved)

Q5. If ${}^{n}P_{3} = 6 \times {}^{n}C_{4}$, find n. [24-Q]

Ans.

$$n(n-1)(n-2) = 6 \cdot \frac{n(n-1)(n-2)(n-3)}{4!}$$

$$1 = \frac{6(n-3)}{24} = \frac{n-3}{4}$$

$$n - 3 = 4$$

(dividing by $n(n-1)(n-2) \neq 0$)

Answer: n = 7

Q6. If ${}^{n}C_{r} : {}^{n}C_{r+1} : {}^{n}C_{r+2} = 1 : 2 : 3$, then find n and r. [25-Q]

Ans.

Use ${}^{n}C_{r+1} = {}^{n}C_{r} \cdot \frac{n-r}{r+1}$.

$$\frac{{}^{n}C_{r+1}}{{}^{n}C_{r}} = \frac{n-r}{r+1} = 2 \rightarrow n - r = 2r + 2 \rightarrow n = 3r + 2$$

$$\frac{{}^{n}C_{r+2}}{{}^{n}C_{r+1}} = \frac{n-r-1}{r+2} = \frac{3}{2} \rightarrow 2n - 2r - 2 = 3r + 6 \rightarrow 2n = 5r + 8$$

Substitute n = 3r + 2: 6r + 4 = 5r + 8, so r = 4 and n = 14. Check: ${}^{14}C_{4} : {}^{14}C_{5} : {}^{14}C_{6} = 1001 : 2002 : 3003 = 1 : 2 : 3$.

Answer: n = 14, r = 4

Q7. A music class consists of 3 piano players, 6 guitarists and 5 violinists. A band of 1 piano player, 4 guitarists and 3 violinists must be chosen to play at a school concert. In how many ways can the band be chosen? [24-Q]

Ans.

${}^{3}C_{1} \cdot {}^{6}C_{4} \cdot {}^{5}C_{3} = 3 \cdot 15 \cdot 10 = 450$

Answer: 450

Q8. A box contains 4 black balls and 7 red balls. If all the balls are distinguishable, in how many ways can 3 balls of the same color be chosen? [24-Q]

Ans.

${}^{4}C_{3} + {}^{7}C_{3} = 4 + 35 = 39$

Answer: 39

Q9. Find the number of ways of selecting 9 balls from 6 red balls, 5 white balls and 5 blue balls if each selection consists of 3 balls of each colour. [25*-Q]

Ans.

${}^{6}C_{3} \cdot {}^{5}C_{3} \cdot {}^{5}C_{3} = 20 \cdot 10 \cdot 10 = 2000$

Answer: 2000

Q10. A football team consists of a goalkeeper, two defense players, four middle field players and four forwards. Three players are chosen to collect a medal at the closing ceremony of a competition. How many selections are possible if one midfield player, one defense player and one forward must be chosen? [25-Q]

Ans.

${}^{4}C_{1} \cdot {}^{2}C_{1} \cdot {}^{4}C_{1} = 4 \cdot 2 \cdot 4 = 32$

Answer: 32

Q11. A group of 15 friends go on a trip in 3 cars, which respectively can contain 6, 5 and 4 people (each a driver). The owners of cars are members of the group and want to drive their own car. In how many ways can the remaining 12 members be divided between 3 cars? [5-Q]

Ans.

The 3 owners fill 1 seat each, so the cars still have 5, 4 and 3 seats. Divide the 12 members into groups of 5, 4, 3:

${}^{12}C_{5} \cdot {}^{7}C_{4} \cdot {}^{3}C_{3} = 792 \cdot 35 \cdot 1 = 27720$

(or $\frac{12!}{5!4!3!}$).

Answer: 27720

Q12. There are 9 different books. In how many ways can the books be given to 3 children, if the youngest wants to receive 4 books, the elder 3 and the eldest 2 respectively? [26-Q]

Ans.

${}^{9}C_{4} \cdot {}^{5}C_{3} \cdot {}^{2}C_{2} = 126 \cdot 10 \cdot 1 = 1260$

Answer: 1260

Q13. In how many ways can a 7−member team be chosen, from among 9 boys and 4 girls, in such a way that team has at least 3 girls? [25*-Q]

Ans.

At least 3 girls means exactly 3 or exactly 4 girls.

3 girls, 4 boys: ${}^{4}C_{3} \cdot {}^{9}C_{4} = 4 \cdot 126 = 504$

4 girls, 3 boys: ${}^{4}C_{4} \cdot {}^{9}C_{3} = 1 \cdot 84 = 84$

Total = 504 + 84 = 588

Answer: 588

5.4

Q14. Out of 30 persons, 5 are teachers. In how many ways can 5 persons be selected so as to include at least one teacher? [25-Q]

Ans.

At least one teacher = (all selections) − (selections with no teacher). There are 25 non−teachers.

${}^{30}C_{5} - {}^{25}C_{5} = 142506 - 53130 = 89376$

Answer: 89,376

Q15. Six papers are set in an examination of which two are mathematical. In how many different orders can the papers be arranged, so that the two mathematical papers are not consecutive. [24-Q]

Ans.

All orders: 6! = 720. Orders with the two mathematical papers together (as one block): 5!·2! = 240.

$720 - 240 = 480$

Answer: 480

Q16. How many five−digit codes can be formed by using each of the digits 1, 2, 3, 4, 5 exactly once do not end in 25? [25-Q]

Ans.

All codes: 5! = 120. Codes ending in 25: the other 3 digits fill the first 3 places in 3! = 6 ways.

$120 - 6 = 114$

Answer: 114

Q17. In how many 3–digit even numbers greater than 300 can be formed using the digits 1, 2, 3 and 4 without repeating any digit? [26-Q]

Ans.

Greater than 300, so the hundreds digit is 3 or 4.

Hundreds digit 3: units digit 2 or 4 (2 ways), tens digit from the remaining 2 digits (2 ways) → 4

Hundreds digit 4: units digit 2 (1 way), tens digit from the remaining 2 digits (2 ways) → 2

Total $= 4 + 2 = 6$

Answer: 6

Q18. How many permutations of the letters H, E, X, A, G, O, N are there if (i) the 3 vowels are placed together? (ii) the 3 vowels are not placed together? [26-Q]

Ans.

The 7 letters are all different; the vowels are E, A, O.

(i) Treat the 3 vowels as one block: 5 objects in 5! ways, vowels inside in 3! ways.

$5! \cdot 3! = 120 \cdot 6 = 720$

(ii) Total − together:

$7! - 720 = 5040 - 720 = 4320$

Answer: (i) 720; (ii) 4320

Q19. In how many different ways can the letters of the word THOUGHTS be arranged so that the vowels always come together? [26-Q]

Ans.

THOUGHTS: T × 2, H × 2, O, U, G, S (8 letters). Vowels O, U form one block (2! orders inside). The block with T, T, H, H, G, S gives 7 objects:

$$\frac{7!}{2! \cdot 2!} \cdot 2! = 1260 \cdot 2 = 2520$$

Answer: 2520

Q20. How many permutations are there of the letters of the word PROGRAM, if they do not end in 2R's? [25-Q]

Ans.

PROGRAM has 7 letters with R twice.

All: $\frac{7!}{2!} = 2520$

Ending in RR: the other 5 different letters fill the first 5 places: 5! = 120.

$2520 - 120 = 2400$

Answer: 2400

Q21. How many words can be made with the letters of the word CONFIDENT in which both N not come together? [25-Q]

Ans.

CONFIDENT has 9 letters with N twice.

All: $\frac{9!}{2!} = 181440$

N, N together (one block): $8! = 40320$

$181440 - 40320 = 141120$

Answer: 141,120

Q22. In how many ways can all the letters of the word SUCCESSOR be arranged in a line such that the two C's are not together in the arrangements? [25-Q]

Ans.

SUCCESSOR has 9 letters: S × 3, C × 2, U, E, O, R.

All: $\frac{9!}{3! \cdot 2!} = 30240$

C, C together (one block): $\frac{8!}{3!} = 6720$

$30240 - 6720 = 23520$

Answer: 23,520

Q23. Find the number of permutations of the letters of the word EXCELLENCE in such a way that there are exactly 6 letters between two L's. [24-Q]

Ans.

With exactly 6 letters between the two L's, the L's occupy places (1, 8), (2, 9) or (3, 10): 3 ways (the L's are identical). The other 8 letters E, E, E, E, X, C, C, N fill the remaining places:

$3 \cdot \frac{8!}{4! \cdot 2!} = 3 \cdot 840 = 2520$

Answer: 2520

5.4 (outside syllabus)

Q24. In how many ways can 8 differently coloured beads be threaded on a string? [25-Q]

Ans.

Arranging 8 objects in a circle: (8 − 1)! ways. A string can be turned over, so each arrangement and its mirror image are the same:

$\frac{(8-1)!}{2} = \frac{5040}{2} = 2520$

Answer: 2520

Q25. Determine the number of positive integers that are factors of the number $2^5 \times 3^2 \times 5^4$. [25-Q]

Ans.

A factor has the form $2^a 3^b 5^c$ with $0 \le a \le 5$, $0 \le b \le 2$, $0 \le c \le 4$.

$(5+1)(2+1)(4+1) = 6 \cdot 3 \cdot 5 = 90$

Answer: 90

5 MARKS

5.3

Q1. On a certain examination, the student must answer 8 out of the 12 question, including exactly 5 of the first 6. In how many ways may he write the examination. [24-Q]

Ans.

Exactly 5 of the first 6, so 3 of the last 6:

${}^6C_5 \cdot {}^6C_3 = 6 \cdot 20 = 120$

Answer: 120

Q2. A badminton club consists of 10 boys and 7 girls. How many mixed double teams (one boy and one girl) are possible? In how many ways can a mixed double match be arranged? [25-Q]

Ans.

Teams:

${}^{10}C_1 \cdot {}^7C_1 = 70$

Match: choose 2 boys and 2 girls, then pair them in 2 ways (each boy with one girl):

${}^{10}C_2 \cdot {}^7C_2 \cdot 2 = 45 \cdot 21 \cdot 2 = 1890$

Answer: 70 teams; 1890 matches

Q3. A committee of three boys and three girls is to be selected from a class of 14 boys and 17 girls. In how many ways can the committee be selected if (i) Maung Myint Myat has to be in the committee? (ii) The girl must include either Ma Aye or Ma Win, but not both? [25-Q]

Ans.

(i) Maung Myint Myat is already chosen; choose 2 more boys from 13 and 3 girls from 17:

${}^{13}C_2 \cdot {}^{17}C_3 = 78 \cdot 680 = 53040$

(ii) Girls: one of Ma Aye / Ma Win (2 ways) and 2 other girls from the remaining 15; boys: 3 from 14:

${}^{14}C_3 \cdot (2 \cdot {}^{15}C_2) = 364 \cdot 210 = 76440$

Answer: (i) 53040; (ii) 76440

Q4. In a small village, there are 87 families, of which 52 families have at most 2 children. In a rural development programme, 20 families are to be chosen for assistance, of which at least 18 families must have at most 2 children. In how many ways can the choice be made? [5-Q]

Ans.

52 families have at most 2 children; the other 35 have more. At least 18 of the 20 must have at most 2 children: 18, 19 or 20 of them.

18 families: ${}^{52}C_{18} \cdot {}^{35}C_2$

19 families: ${}^{52}C_{19} \cdot {}^{35}C_1$

20 families: ${}^{52}C_{20}$

Answer: $({}^{52}C_{18} \cdot {}^{35}C_2) + ({}^{52}C_{19} \cdot {}^{35}C_1) + {}^{52}C_{20}$

5.4

Q5. A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done when the committee consists of: (i) exactly 3 girls? (ii) at least 3 girls? (iii) at most 3 girls? [25-Q]

Ans.

(i) 3 girls and 4 boys:

${}^4C_3 \cdot {}^9C_4 = 4 \cdot 126 = 504$

(ii) 3 girls or 4 girls:

$504 + {}^4C_4 \cdot {}^9C_3 = 504 + 84 = 588$

(iii) At most 3 girls = all − (4 girls):

${}^{13}C_7 - {}^4C_4 \cdot {}^9C_3 = 1716 - 84 = 1632$

Answer: (i) 504; (ii) 588; (iii) 1632

Q6. A group consists of 4 boys and 7 girls. In how many ways can a team of 5 members be selected if the team has (i) no boys, (ii) at least one boy and two girls, (iii) at least 3 boys? [25-Q]

Ans.

(i) All 5 are girls:

${}^7C_5 = 21$

(ii) At least one boy and at least two girls (5 members): (1 boy, 4 girls), (2 boys, 3 girls), (3 boys, 2 girls).

${}^4C_1 \cdot {}^7C_4 + {}^4C_2 \cdot {}^7C_3 + {}^4C_3 \cdot {}^7C_2$

$= 4 \cdot 35 + 6 \cdot 35 + 4 \cdot 21$

$= 140 + 210 + 84 = 434$

(iii) 3 boys or 4 boys:

$^4C_3 \cdot {}^7C_2 + {}^4C_4 \cdot {}^7C_1 = 84 + 7 = 91$

Answer: (i) 21; (ii) 434; (iii) 91

Q7. An art gallery has a total of 11 paintings by a particular artist; 5 are oil paintings and 6 are water colour paintings. The art gallery is asked to provide a special exhibition of the artists' work but is restricted to showing only 5 of the paintings. Calculate the number of ways in which the 5 paintings can be selected for the exhibition if (i) there are no restrictions, (ii) there must be exactly 3 water colour paintings, (iii) there must be more water colour paintings than oil paintings. [26-Q]

Ans.

(i)

$^{11}C_5 = 462$

(ii) 3 water colour and 2 oil:

$^6C_3 \cdot {}^5C_2 = 20 \cdot 10 = 200$

(iii) More water colour than oil in 5 paintings: 3, 4 or 5 water colour.

$^6C_3 \cdot {}^5C_2 + {}^6C_4 \cdot {}^5C_1 + {}^6C_5 \cdot {}^5C_0$

$= 200 + 75 + 6 = 281$

Answer: (i) 462; (ii) 200; (iii) 281

Q8. A bag contains 3 red, 4 green and 5 white cards, which are considered distinguishable. The order does not matter. How many ways are there to fill a box with 3 cards, so that at least two colours are used? [25-Q]

Ans.

At least two colours = all selections − single-colour selections.

All: $^{12}C_3 = 220$

Single colour: $^3C_3 + {}^4C_3 + {}^5C_3 = 1 + 4 + 10 = 15$

$220 - 15 = 205$

Answer: 205

Q9. How many different 4-digit codes can be formed using all the digits 1, 2, 3, 4 if (i) there is no restriction? (ii) repetition is not allowed? (iii) repetition is not allowed, and 1 is either the first or the last digit? [24-Q]

Ans.

(i) Each of the 4 places has 4 choices:

$4^4 = 256$

(ii)

$4! = 24$

(iii) 1 is first (3! ways) or last (3! ways); it cannot be both:

$3! + 3! = 12$

Answer: (i) 256; (ii) 24; (iii) 12

Q10. How many different 4-digit codes can be formed using all the digits 5, 6, 7, 8 if (i) there is no restriction? (ii) repetition is not allowed? (iii) repetition is not allowed, and 5 is either the first or last digit? [25-Q]

Ans.

(i)

$4^4 = 256$

(ii)

$4! = 24$

(iii)

$3! + 3! = 12$

Answer: (i) 256; (ii) 24; (iii) 12

Q11. From the digits 2, 3, 4, 5, 6, (i) how many numbers greater than 4000 can be formed? (ii) how many 4 digits numbers would be even? (digits are not repeated) [25-Q]

Ans.

Four-digit numbers, no digit repeated.

(i) Greater than 4000: the first digit is 4, 5 or 6 (3 ways); the other 3 places are filled from the remaining 4 digits: $^4P_3 = 24$.

$3 \cdot 24 = 72$

(ii) Even: the last digit is 2, 4 or 6 (3 ways); the other 3 places: $^4P_3 = 24$.

$3 \cdot 24 = 72$

Answer: (i) 72; (ii) 72

Q12. How many different 4-digit codes can be formed using all the digits 0, 1, 2, 3 if (i) there is no restriction? (ii) repetition is not allowed? (iii) repetition is not allowed, and 0 is either the first or the last digit? [26-Q]

Ans.

(i)

$4^4 = 256$

(ii)

$4! = 24$

(iii) 0 first: 3! = 6; 0 last: 3! = 6.

$6 + 6 = 12$

Answer: (i) 256; (ii) 24; (iii) 12

Q13. How many different 5-digit codes can be formed using the digits 0, 1, 2, 3, 4 if (i) there is no restriction? (ii) repetition is not allowed? (iii) repetition is not allowed and 1 is either the first or the last digit? [25-Q]

Ans.

(i)

$5^5 = 3125$

(ii)

$5! = 120$

(iii) 1 first: 4! = 24; 1 last: 4! = 24.

$24 + 24 = 48$

Answer: (i) 3125; (ii) 120; (iii) 48

Q14. How many 4-digit positive even integers with distinct digits can be constructed with 0, 1, 2, 3, 4, 5? [25*-Q]

Ans.

Split by the last digit.

Last digit 0: the other 3 places from 1, 2, 3, 4, 5: $^5P_3$

Last digit 2 or 4 (2 ways): first digit not 0 and not the last digit → 4 ways; middle two places from the remaining 4 digits: $^4P_2$

Total

Answer: 156

Q15. How many integers are there between 100 and 1000 such that at least one of the digits is 4? [25-Q]

Ans.

These are the 3-digit numbers 101 to 999 (100 and 1000 excluded; neither contains 4, so we may count 100 to 999: 900 numbers).

Without any 4: $8 \cdot 9 \cdot 9 = 648$ (first digit: not 0, not 4; others: not 4)

$900 - 648 = 252$

Answer: 252

Q16. How many words, with or without meaning can be made from the letters of the word MONDAY, assuming that no letter is repeated, if (a) 4 letters are used at a time, (b) all letters are used at a time, and (c) all letters are used but first letter is a vowel? [26-Q]

Ans.

MONDAY has 6 different letters; the vowels are O, A.

(a) $^6P_4 = 6 \cdot 5 \cdot 4 \cdot 3 = 360$

(b) $6! = 720$

(c) first letter O or A (2 ways), the other 5 letters in 5! ways: $2 \cdot 120 = 240$

Answer: (a) 360; (b) 720; (c) 240

Q17. Calculate the total number of different permutations of all the letters A, B, C, D, E, F when (i) there are no restriction (ii) the letters A and B are to be adjacent to one another (iii) the first letter is A, B or C and the last letter is D, E or F. [25-Q]

Ans.

(i)

$6! = 720$

(ii) AB as one block (2! orders) with 4 other letters:

$2! \cdot 5! = 240$

(iii) First letter: 3 ways; last letter: 3 ways; the middle 4 places: 4! ways.

$3 \cdot 3 \cdot 24 = 216$

Answer: (i) 720; (ii) 240; (iii) 216

Q18. How many arrangements of the word REMAND are possible if (i) they have RE together in order? (ii) they have REM together in any order? (iii) R, E and M are not to be together? [25*-Q]

Ans.

REMAND has 6 different letters.

(i) "RE" as one block (fixed order): 5 objects.

$5! = 120$

(ii) R, E, M as one block (3! orders inside) with A, N, D:

$3!\cdot 4! = 144$

(iii) Total − (R, E, M together):

$720 - 144 = 576$

Answer: (i) 120; (ii) 144; (iii) 576

Q19. How many permutations of the letters S, U, N, D, A, Y are there if (i) the 2 vowels are placed together? (ii) the 2 vowels are not placed together? (iii) consonants and vowels do not appear alternately? [24-Q]

Ans.

The 6 letters are different; vowels U, A and consonants S, N, D, Y.

(i) UA block (2! inside), 5 objects:

$2!\cdot 5! = 240$

(ii)

$720 - 240 = 480$

(iii) There are 4 consonants but only 2 vowels, so the letters can never alternate (C V C V C V needs 3 of each). Hence every permutation qualifies:

$6! = 720$

Answer: (i) 240; (ii) 480; (iii) 720

Q20. How many permutations of the letters S, I, N, G, A, P, O, R, E are there if (i) the 4 vowels are placed together? (ii) the 4 vowels are not placed together? (iii) consonants and vowels do not appear alternately? [25-Q]

Ans.

9 different letters: vowels I, A, O, E (4) and consonants S, N, G, P, R (5).

(i) Vowel block (4! inside) with 5 consonants → 6 objects:

$4!\cdot 6! = 24\cdot 720 = 17280$

(ii)

$9! - 17280 = 362880 - 17280 = 345600$

(iii) Alternating is possible only as C V C V C V C V C: consonants in 5! ways, vowels in 4! ways:

$5!\cdot 4! = 2880$

$9! - 2880 = 360000$

Answer: (i) 17,280; (ii) 345,600; (iii) 360,000

Q21. How many words can be made with the letters of the word EDUCATION if (i) C, A and T always appear together? (ii) all vowels are in alphabetical order but may or may not appear together? [24-Q]

Ans.

EDUCATION has 9 different letters; the vowels are E, U, A, I, O (5 of them).

(i) CAT as one block (3! orders inside) with the other 6 letters:

$3!\cdot 7! = 6\cdot 5040 = 30240$

(ii) All 9! arrangements; the 5 vowels can be in 5! orders but only 1 is alphabetical (A, E, I, O, U):

$\frac{9!}{5!} = 3024$

Answer: (i) 30,240; (ii) 3024

Q22. How many permutation can be made of all letters of the word ENGINEERING? In how many of these will the three N's stand together and in how many will they stand first? [25-Q]

Ans.

ENGINEERING has 11 letters: E × 3, N × 3, G × 2, I × 2, R.

$$\text{All: } \frac{11!}{3!\cdot 3!\cdot 2!\cdot 2!} = \frac{39916800}{144} = 277200$$

$$\text{N N N together (one block): } \frac{9!}{3!\cdot 2!\cdot 2!} = \frac{362880}{24} = 15120$$

$$\text{N N N first: } \frac{8!}{3!\cdot 2!\cdot 2!} = \frac{40320}{24} = 1680$$

Answer: 277,200; 15,120; 1680

Q23. In how many ways can the letters of the word PINEAPPLE be arranged if (i) the words start with I and end with N, (ii) the vowels are all together, (iii) there are always 5 letters between 2 E's? [24-Q]

Ans.

PINEAPPLE has 9 letters: P × 3, E × 2, I, N, A, L. Vowels: I, E, A, E.

(i) I first, N last; the other 7 letters P, P, P, E, E, A, L fill the middle:

$\frac{7!}{3!\cdot 2!} = 420$

(ii) Vowel block: I, E, A, E can be arranged in $\frac{4!}{2!} = 12$ ways inside. The block with P, P, P, N, L gives 6 objects ($\frac{6!}{3!} = 120$ ways):

$12\cdot 120 = 1440$

(iii) E's at places (1, 7), (2, 8) or (3, 9): 3 ways. The other 7 letters P, P, P, I, N, A, L:

$3\cdot \frac{7!}{3!} = 3\cdot 840 = 2520$

Answer: (i) 420; (ii) 1440; (iii) 2520

Q24. Find the number of permutations of all the letters of the word COMPLETENESS so that there are exactly 5 letters between 2 S's. [26-Q]

Ans.

COMPLETENESS has 12 letters: E × 3, S × 2, and C, O, M, P, L, T, N once each.

Exactly 5 letters between the S's: the S's are at places (1, 7), (2, 8), ..., (6, 12): 6 ways. The other 10 letters (E × 3 and 7 different letters) fill the remaining places:

$6\cdot \frac{10!}{3!} = 6\cdot 604800 = 3628800$

Answer: 3,628,800

Q25. Find the number of permutations of all the letters of the word INTERNET in such a way that there are exactly the sum of the number of three letters between the 2N's and four letters between the 2E's. [26-Q]

Ans.

INTERNET has 8 letters: N × 2, E × 2, T × 2, I, R. Three letters between the N's means the N's are at places (a, a + 4); four letters between the E's means the E's are at places (b, b + 5).

N places: (1,5), (2,6), (3,7), (4,8)

E places: (1,6), (2,7), (3,8)

Count the non−overlapping combinations: N(1,5) with E(2,7) or E(3,8); N(2,6) with E(3,8); N(3,7) with E(1,6); N(4,8) with E(1,6) or E(2,7) → 6 placements. The other 4 places are filled by T, T, I, R in $\frac{4!}{2!} = 12$ ways.

$6\cdot 12 = 72$

Answer: 72

Q26. From the letters of the word CELESTIAL, 7 letters are to be chosen and arrange them in a line. In how many ways can it be done (a) if there is no other restriction? (b) if there is at least one E in the arrangements? (c) if there is at least one E and at most one L in the arrangements? [5-Q]

Ans.

CELESTIAL has 9 letters: E × 2, L × 2 and C, S, T, I, A once each. Let the 7 chosen letters contain e E's, l L's and k of the 5 single letters ($e + l + k = 7$, so $e + l \geq 2$). The number of arrangements is ${}^5C_k \cdot \frac{7!}{e!\,l!}$.

e	l	k	Arrangements
0	2	5	7!/2! = 2520
1	1	5	1·7!/(1!1!) = 5040
1	2	4	5·7!/2! = 12600
2	0	5	7!/2! = 2520
2	1	4	5·7!/2! = 12600
2	2	3	10·7!/(2!2!) = 12600

(a) $2520 + 5040 + 12600 + 2520 + 12600 + 12600 = 47880$

(b) at least one E: remove (e, l) = (0, 2): $47880 - 2520 = 45360$

(c) at least one E and at most one L: (1,1), (2,0), (2,1): $5040 + 2520 + 12600 = 20160$

Answer: (a) 47,880; (b) 45,360; (c) 20,160

CH.06 Conic Sections

Notes · Questions with Answers

NOTES

Conic Section
A conic section (or simply a conic) is a curve obtained by intersecting a plane with a double right-circular cone. The main conics are circles, ellipses, parabolas and hyperbolas, each satisfying the general equation $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$.

Circle
A circle is the set of all points in a plane that are equidistant from a fixed point, called the **center**, where the fixed distance is called the **radius**.

Parabola
A parabola is the set of all points (x, y) in the plane that are equidistant from a fixed line (the **directrix**) and a fixed point not on that line (the **focus**).

Vertex and Axis of a Parabola
The line through the focus and perpendicular to the directrix is the **axis** of the parabola. The point where the axis meets the parabola is the **vertex** — it lies exactly halfway between the focus and the directrix. A parabola is symmetric about its axis.

Latus Rectum
The line segment that passes through the focus and is parallel to the directrix is called the **latus rectum**. Its endpoints lie on the parabola, and if p is the distance from vertex to focus, the latus rectum has length 4p, extending 2p above and 2p below the axis.

Rotation of Axes
In a rotation of axes, the origin stays fixed while the x- and y-axes are rotated through an angle θ to new positions, denoted x′ and y′. Rotating axes lets us eliminate the xy-term from a conic's general equation.

General (Conic) Equation
$Ax^2+Bxy+Cy^2+Dx+Ey+F=0$

Standard Form of the Equation of a Circle
$(x-h)^2+(y-k)^2=r^2$
Where center = (h, k) and radius = r.

General Form of the Equation of a Circle
$x^2+y^2+2gx+2fy+c=0$
Center = $(-g, -f)$, Radius = $\sqrt{g^2+f^2-c}$

Degenerate Cases of a Circle
For $x^2+y^2+2gx+2fy+c=0$:

1. If $g^2+f^2-c > 0$, the graph is a circle with center $(-g,-f)$ and radius $\sqrt{g^2+f^2-c}$.
2. If $g^2+f^2-c = 0$, the graph is a single point $(-g,-f)$.
3. If $g^2+f^2-c < 0$, the equation has no real solution and no graph.

Standard Forms of a Parabola — Vertex at (0,0)

Equation	Vertex	Focus	Directrix	Axis	Description
$y^2=4px$	(0,0)	(p,0)	x=−p	y=0	opens right
$y^2=-4px$	(0,0)	(−p,0)	x=p	y=0	opens left
$x^2=4py$	(0,0)	(0,p)	y=−p	x=0	opens up
$x^2=-4py$	(0,0)	(0,−p)	y=p	x=0	opens down

Standard Forms of a Parabola — Vertex at (h,k)

Equation	Vertex	Focus	Directrix	Axis	Description
$(y-k)^2=4p(x-h)$	(h,k)	(h+p,k)	x=h−p	y=k	opens right
$(y-k)^2=-4p(x-h)$	(h,k)	(h−p,k)	x=h+p	y=k	opens left
$(x-h)^2=4p(y-k)$	(h,k)	(h,k+p)	y=k−p	x=h	opens up
$(x-h)^2=-4p(y-k)$	(h,k)	(h,k−p)	y=k+p	x=h	opens down

Rotation of Axes Formulas
$x' = x\cos\theta + y\sin\theta$
$y' = -x\sin\theta + y\cos\theta$

$x = x'\cos\theta - y'\sin\theta$
$y = x'\sin\theta + y'\cos\theta$

Eliminating the x′y′-Term (Rotation Angle & Invariants)
To remove the xy-term from $Ax^2+Bxy+Cy^2+Dx+Ey+F=0$ ($B\neq0$), rotate by θ where:
$\cot 2\theta = \frac{A-C}{B}$ (θ=45° if B(cot2θ)=0, i.e. A=C)

Rotation invariants:
1. $F = F'$
2. $A + C = A' + C'$
3. $B^2-4AC = B'^2-4A'C'$

Calculator tip: On the calculator, press **SHIFT** then **+** for Pol(x,y) and enter A−C,B (in that order) — it returns r and θ=2θ directly, without computing cos2θ by hand first. If B is negative, enter C−A,−B instead (flip both signs), so the angle comes back in the usual 0°–180° range.

1 MARK

6.2 Circles

Q1. The standard equation of the circle satisfying the center (2, −2) and radius 5 is [26-Q]

A. $(x-2)^2+(y+2)^2=25$ **B.** $(x+2)^2+(y-2)^2=25$ **C.** $(x-2)^2+(y-2)^2=25$
D. $(x+2)^2+(y+2)^2=25$

Ans.
Standard form: $(x-h)^2+(y-k)^2=r^2$ with center (h, k).

$(x-2)^2+(y-(-2))^2 = 5^2$
$(x-2)^2+(y+2)^2 = 25$

Answer: A

Q2. The equation of the circle with center at (1, −3) and radius 5 is [24-Q]

A. $x^2+y^2+2x-6y+5=0$ **B.** $x^2+y^2-2x+6y-15=0$ **C.** $x^2+y^2-2x+6y+5=0$
D. $x^2+y^2+2x-6y-15=0$

Ans.
Standard form, then expand:

$(x-1)^2+(y+3)^2 = 25$
$x^2-2x+1+y^2+6y+9 = 25$
$x^2+y^2-2x+6y-15 = 0$

Answer: B

Q3. The equation of the circle with center (2, −1) that passes through the point (3, −2) is [25-Q]

A. $(x-3)^2+(y+2)^2=2$ **B.** $(x+2)^2+(y-1)^2=2$ **C.** $(x-2)^2+(y+1)^2=2$
D. $(x-2)^2+(y+1)^2=\sqrt{2}$

Ans.
The radius is the distance from the center to the given point:

$r^2 = (3-2)^2+(-2+1)^2 = 1+1 = 2$
$(x-2)^2+(y+1)^2 = 2$

Answer: C

Q4. Which of the following points lies inside the circle defined by the equation $(x+3)^2+(y-2)^2=100$? [26-Q]

A. (−2, −8) **B.** (6, −4) **C.** (8, 10) **D.** (1, 10)

Ans.
Center (−3, 2), $r^2=100$. A point is inside if $(x+3)^2+(y-2)^2<100$.

Point	$(x+3)^2+(y-2)^2$	Compare with 100
(−2, −8)	1 + 100 = 101	outside
(6, −4)	81 + 36 = 117	outside
(8, 10)	121 + 64 = 185	outside
(1, 10)	16 + 64 = 80	**inside**

Answer: D

Q5. The radius of the circle $x^2+y^2-4x+2y+k=0$ is 7. Find the value of k. [25-Q]

A. 49 **B.** 44 **C.** −44 **D.** −49

Ans.
Here 2g = −4, 2f = 2, c = k, so g = −2, f = 1.

$r = \sqrt{\{g^2+f^2-c\}}$
$7^2 = 4+1-k$
$49 = 5-k$
$k = -44$

Answer: C

Q6. The distance between point (1, 1) and the center of the circle $(x-2)^2+(y+3)^2=16$ is

[25-Q]

A. $\sqrt{5}$ **B.** $\sqrt{13}$ **C.** 4 **D.** $\sqrt{17}$

Ans.

The center is (2, −3). Distance formula:

$d = \sqrt{\{(2-1)^2 + (-3-1)^2\}}$

$d = \sqrt{\{1+16\}} = \sqrt{17}$

Answer: D

Q7. The graph of the equation $(x-1)^2 + (y+4)^2 + 9 = 0$ represents [25-Q]

A. a single point **B.** a circle **C.** a parabola **D.** nothing

Ans.

Rewrite: $(x-1)^2 + (y+4)^2 = -9$. The left side is never negative, so no real point (x, y) satisfies it ($g^2 + f^2 - c < 0$). The graph is nothing.

Answer: D

Q8. Which of the equation represents a circle? [24-Q]

A. $x^2 + y^2 - 10x - 2y + 29 = 0$ **B.** $x^2 + y^2 - 4x - 6y + 13 = 0$ **C.** $x^2 + y^2 - 6x + 2y + 9 = 0$ **D.** $x^2 + y^2 - 2x + 8y + 18 = 0$

Ans.

For $x^2 + y^2 + 2gx + 2fy + c = 0$ it is a circle only if $g^2 + f^2 - c > 0$.

Option	g, f, c	$g^2 + f^2 - c$	Graph
A	−5, −1, 29	25 + 1 − 29 = −3	no graph
B	−2, −3, 13	4 + 9 − 13 = 0	a point
C	−3, 1, 9	9 + 1 − 9 = 1	**circle**
D	−1, 4, 18	1 + 16 − 18 = −1	no graph

Answer: C

6.3 Parabola – vertex (0, 0)

Q9. The directrix of the parabola $x^2 = 16y$ is [24-Q]

A. y = −16 **B.** y = 16 **C.** y = 4 **D.** y = −4

Ans.

$x^2 = 4py$ with 4p = 16, so p = 4. It opens up, so the directrix is y = −p.

$y = -4$

Answer: D

Q10. The directrix of the parabola $x^2 = 16y$ is [25-Q]

A. y = −3 **B.** y = 4 **C.** y = −4 **D.** y = 3

Ans.

$x^2 = 4py$, 4p = 16, p = 4. Directrix: y = −p = −4.

Answer: C

Q11. The equation of a parabola is $y = kx^2$ and it contains the point (8, 2). The equation of the directrix is [25-Q]

A. y = −16 **B.** y = −8 **C.** y = −4 **D.** y = −2

Ans.

Substitute (8, 2):

$2 = k(8)^2 \rightarrow k = \frac{1}{32}$

$y = \frac{1}{32}x^2 \rightarrow x^2 = 32y$

4p = 32, so p = 8. It opens up, so the directrix is y = −p.

$y = -8$

Answer: B

Q12. The end points of latus rectum of the parabola $y^2 - 8x = 0$ are [24-Q]

A. (2, 4) and (−2, 4) **B.** (−2, 4) and (−2, −4) **C.** (2, 4) and (2, −4) **D.** (2, −4) and (−2, −4)

Ans.

$y^2 = 8x$, so 4p = 8 and p = 2. Focus F(p, 0) = (2, 0). The latus rectum is the chord through the focus with end points (p, ±2p):

(2, 4) and (2, −4)

Answer: C

6.3 Parabola – vertex (h, k)

Q13. The equation of the parabola $x^2 - 4x - 12y + 16 = 0$ has vertex [24-Q]

A. (2, 2) **B.** (2, 1) **C.** (1, 2) **D.** (2, 0)

Ans.

Complete the square in x:

$x^2 - 4x = 12y - 16$

$x^2 - 4x + 4 = 12y - 12$

$(x-2)^2 = 12y - 12$

$(x-2)^2 = 12(y-1)$

Vertex (h, k) = (2, 1).

Answer: B

Q14. The focus of the parabola $(y+5)^2 = 12(x-4)$ is [26-Q]

A. (7, −5) **B.** (3, −5) **C.** (1, −5) **D.** (4, −5)

Ans.

Form $(y-k)^2 = 4p(x-h)$: h = 4, k = −5, 4p = 12, p = 3 (opens right).

Focus (h + p, k) = (4 + 3, −5) = (7, −5)

Answer: A

Q15. Find the focus of the parabola $y^2 - 6y + 2x + 7 = 0$. [24-Q]

A. (0, 3) **B.** $(\frac{1}{2}, 3)$ **C.** (1, 3) **D.** $(\frac{3}{2}, 3)$

Ans.

Complete the square in y:

$y^2 - 6y = -2x - 7$

$y^2 - 6y + 9 = -2x + 2$

$(y-3)^2 = -2x + 2$

$(y-3)^2 = -2(x-1)$

h = 1, k = 3, 4p = 2, p = $\frac{1}{2}$. It opens left, so the focus is (h − p, k).

$F = (1 - \frac{1}{2}, 3) = (\frac{1}{2}, 3)$

Answer: B

Q16. Find the focus of the parabola $y^2 + 12x - 6y - 3 = 0$. [25-Q]

A. (−2, 3) **B.** (3, −1) **C.** (3, 1) **D.** (2, 3)

Ans.

Complete the square in y:

$y^2 - 6y = -12x + 3$

$y^2 - 6y + 9 = -12x + 12$

$(y-3)^2 = -12x + 12$

$(y-3)^2 = -12(x-1)$

h = 1, k = 3, 4p = 12, p = 3, opens left.

F = (h − p, k) = (1 − 3, 3) = (−2, 3)

Answer: A

Q17. The focus for the equation $(x+1)^2 = -12(y-2)$ is [25*-Q]

A. (1, 1) **B.** (1, −1) **C.** (−1, −1) **D.** (−1, 1)

Ans.

Form $(x-h)^2 = -4p(y-k)$: h = −1, k = 2, 4p = 12, p = 3 (opens down).

F = (h, k − p) = (−1, 2 − 3) = (−1, −1)

Answer: C

Q18. The focus of the parabola $x^2 - 2x + 8y - 23 = 0$ is [26-Q]

A. (−1, 3) **B.** (1, 1) **C.** (1, 5) **D.** (3, 3)

Ans.

Complete the square in x:

$x^2 - 2x = -8y + 23$

$x^2 - 2x + 1 = -8y + 24$

$(x-1)^2 = -8y + 24$

$(x-1)^2 = -8(y-3)$

h = 1, k = 3, 4p = 8, p = 2, opens down.

F = (h, k − p) = (1, 3 − 2) = (1, 1)

Answer: B

Q19. The directrix for the equation $(y-k)^2 = -4p(x-h)$ is [24-Q]

A. x = h − p **B.** x = h + p **C.** y = k − p **D.** y = k + p

Ans.

The parabola $(y-k)^2 = -4p(x-h)$ opens to the left, with vertex (h, k) and focus (h − p, k). The directrix lies on the other side of the vertex:

x = h + p

Answer: B

Q20. The directrix of the equation $y^2 - 12x - 6y + 3 = 0$. [24-Q]

A. $x = -\frac{7}{2}$ **B.** $x = \frac{7}{2}$ **C.** $x = \frac{3}{2}$ **D.** $x = -\frac{3}{2}$

Ans.

Complete the square in y:

$y^2 - 6y = 12x - 3$

$y^2 - 6y + 9 = 12x + 6$

$(y-3)^2 = 12x + 6$

$(y-3)^2 = 12(x + \frac{1}{2})$

h = $-\frac{1}{2}$, k = 3, 4p = 12, p = 3, opens right. Directrix x = h − p:

$x = -\frac{1}{2} - 3 = -\frac{7}{2}$

Answer: A

Q21. The equation of the directrix of the parabola $y^2 + 8x + 6y + 25 = 0$. [25-Q]

A. x = −5 **B.** x = 0 **C.** x = −4 **D.** x = 4

Ans.

Complete the square in y:

$y^2 + 6y = -8x - 25$

$y^2 + 6y + 9 = -8x - 16$

$(y + 3)^2 = -8x - 16$

$(y + 3)^2 = -8(x + 2)$

h = −2, k = −3, 4p = 8, p = 2, opens left. Directrix x = h + p:

$x = -2 + 2 = 0$

Answer: B

Q22. The equation of the directrix of the parabola $y^2 - 4x - 4y = 0$. [25*-Q]

A. x = −2 **B.** x = 2 **C.** x = 1 **D.** x = −1

Ans.

Complete the square in y:

$y^2 - 4y = 4x$

$y^2 - 4y + 4 = 4x + 4$

$(y - 2)^2 = 4x + 4$

$(y - 2)^2 = 4(x + 1)$

h = −1, k = 2, 4p = 4, p = 1, opens right. Directrix x = h − p:

$x = -1 - 1 = -2$

Answer: A

Q23. The directrix for the equation $(x - h)^2 = 4p(y - k)$ is [S-Q]

A. x = h − p **B.** x = h + p **C.** y = k − p **D.** y = k + p

Ans.

$(x - h)^2 = 4p(y - k)$ opens up, with focus (h, k + p). The directrix is below the vertex:

$y = k - p$

Answer: C

Q24. The directrix for the equation $(x + 2)^2 = -4p(y + 2)$ is [25*-Q]

A. y = −3 **B.** y = 2 **C.** y = −1 **D.** y = 0

Ans.

For $(x - h)^2 = -4p(y - k)$ the directrix is y = k + p. Here h = k = −2, so y = −2 + p.
The printed equation does not give a value of p. The book key is A, which corresponds to the parabola $(x + 2)^2 = 4(y + 2)$: it opens up with p = 1, so the directrix is

$y = k - p = -2 - 1 = -3$

Answer: A (book key)

Q25. The equation of the directrix of the parabola $x^2 + 4x + 4y + 8 = 0$ is [S-Q]

A. y = −1 **B.** y = 1 **C.** y = 0 **D.** $y = \frac{3}{2}$

Ans.

Complete the square in x:

$x^2 + 4x = -4y - 8$

$x^2 + 4x + 4 = -4y - 4$

$(x + 2)^2 = -4y - 4$

$(x + 2)^2 = -4(y + 1)$

h = −2, k = −1, 4p = 4, p = 1, opens down. Directrix y = k + p:

$y = -1 + 1 = 0$

Answer: C

Q26. The equation of the directrix of the parabola $x^2 + 4x + 4y + 32 = 0$ is [25-Q]

A. y = −6 **B.** y = 0 **C.** y = 6 **D.** $y = \frac{3}{2}$

Ans.

Complete the square in x:

$x^2 + 4x = -4y - 32$

$x^2 + 4x + 4 = -4y - 28$

$(x + 2)^2 = -4y - 28$

$(x + 2)^2 = -4(y + 7)$

h = −2, k = −7, 4p = 4, p = 1, opens down. Directrix y = k + p:

$y = -7 + 1 = -6$

Answer: A

2 MARKS

6.2 Circles

Q1. Write down the equation of the unit circle. [24-Q]

Ans.

The unit circle has center (0, 0) and radius 1.

$x^2 + y^2 = 1$

Answer: $x^2 + y^2 = 1$

Q2. Find the equation in standard form of the circle with center at (1, 4) and tangent to the line y = −3. [24-Q]

Ans.

Center (h, k) = (1, 4) and tangent to y = b: radius r = |k − b|.

$r = |4 - (-3)| = 7$

$(x - 1)^2 + (y - 4)^2 = 49$

Answer: $(x - 1)^2 + (y - 4)^2 = 49$

Q3. Find the general form of the equation of the circle with diameter has endpoints (7, −13) and (1, 3). [25-Q]

Ans.

The center is the midpoint of the diameter; the radius is half of its length.

$\text{Center} = \left(\frac{7+1}{2}, \frac{-13+3}{2}\right) = (4, -5)$

$r^2 = (7 - 4)^2 + (-13 + 5)^2 = 9 + 64 = 73$

$(x - 4)^2 + (y + 5)^2 = 73$

$x^2 - 8x + 16 + y^2 + 10y + 25 - 73 = 0$

$x^2 + y^2 - 8x + 10y - 32 = 0$

Answer: $x^2 + y^2 - 8x + 10y - 32 = 0$

Q4. Find the center and radius of the circle $(x - 2)^2 + (y - 4)^2 = 16$. [24-Q]

Ans.

Compare with $(x - h)^2 + (y - k)^2 = r^2$:

$h = 2, k = 4, r^2 = 16$

Answer: center (2, 4), radius r = 4

Q5. Find the center and radius of the equation $x^2 + y^2 + 2gx + 2fy + c = 0$. [24-Q]

Ans.

Complete the square:

$(x^2 + 2gx + g^2) + (y^2 + 2fy + f^2) = g^2 + f^2 - c$

$(x + g)^2 + (y + f)^2 = g^2 + f^2 - c$

Answer: center (−g, −f), radius r = $\sqrt{g^2 + f^2 - c}$

Q6. Find the center and radius of the circle $x^2 + 2x + y^2 + 4y + 4 = 0$. [24-Q]

Ans.

Complete the square:

$(x^2 + 2x + 1) + (y^2 + 4y + 4) = -4 + 1 + 4$

$(x + 1)^2 + (y + 2)^2 = 1$

Answer: center (−1, −2), radius r = 1

Q7. The equation of the circle is $x^2 + y^2 - 4x + 2y - 11 = 0$. The center and radius of circle are (x_1, y_1) and r, respectively, then find $r(x_1 + y_1)$. [25-Q]

Ans.

2g = −4, 2f = 2, c = −11, so g = −2, f = 1.

$\text{Center} = (-g, -f) = (2, -1) \rightarrow x_1 = 2, y_1 = -1$

$r = \sqrt{\{4 + 1 + 11\}} = \sqrt{16} = 4$

$r(x_1 + y_1) = 4(2 - 1) = 4$

Answer: 4

6.3 Parabola – vertex (0, 0)

Q8. Write down the focus and directrix of the parabola $x^2 + 8y = 0$. [24-Q, 25-Q, 26-Q]

Ans.

$x^2 = -8y$, which has the form $x^2 = -4py$ with 4p = 8, p = 2. It opens down.

Focus F(0, −p) = (0, −2)

Directrix: y = p = 2

Answer: Focus (0, −2), directrix y = 2

Q9. The line y + 4 = 0 is the directrix of the parabola $x^2 = ky$. Find the value of k. [25-Q]

Ans.

Directrix y = −4. For $x^2 = 4py$ the directrix is y = −p, so p = 4 and the parabola opens up.

$k = 4p = 16$

Answer: k = 16

Q10. The line y − 1 = 0 is the directrix of the parabola $x^2 = ky$. Find the value of k. [25-Q]

Ans.

Directrix y = 1, so −p = 1 and p = −1 (the parabola opens down).

$k = 4p = -4$

Answer: k = −4

Q11. The line x + 1 = 0 is the directrix of the parabola $y^2 = -kx$. Find the value of k. [25-Q]

Ans.

Directrix x = −1. For $y^2 = 4px$ the directrix is x = −p, so p = 1 and the parabola opens to the right: $y^2 = 4x$.

$-k = 4 \rightarrow k = -4$

Answer: k = −4

Q12. If the line $y - \frac{1}{4} = 0$ is the directrix of the parabola $x^2 + ky = 0$, find the value of k. [25-Q]

Ans.

$x^2 = -ky$. Directrix $y = \frac{1}{4}$: for $x^2 = -4py$ the directrix is $y = p$, so $p = \frac{1}{4}$ (opens down).

$-k = -4p = -1$

$k = 1$

Answer: k = 1

Q13. Find the equation of the parabola with vertex (0, 0), focus on the positive y axis and passing through (−3, 3). [25-Q]

Ans.

Focus on the positive y-axis, so the parabola opens up: $x^2 = 4py$. Substitute (−3, 3):

$(-3)^2 = 4p(3) \rightarrow 9 = 12p$

$p = \frac{3}{4} \rightarrow 4p = 3$

$x^2 = 3y$

Answer: $x^2 = 3y$

Q14. Find an equation for the parabola satisfying the vertex (0, 0), symmetric about the x-axis, passes through (3, 3). [26-Q]

Ans.

Symmetric about the x-axis with vertex (0, 0): $y^2 = 4px$ (opening right, since the point has $x > 0$).

$3^2 = 4p(3) \rightarrow 4p = 3$

$y^2 = 3x$

Answer: $y^2 = 3x$

Q15. Find the equation of the directrix of parabola which is the vertex (0, 0), open down and passes through the point $(-4\sqrt{3}, -4)$. [26-Q]

Ans.

Opens down: $x^2 = -4py$. Substitute $(-4\sqrt{3}, -4)$:

$(-4\sqrt{3})^2 = -4p(-4)$

$48 = 16p \rightarrow p = 3$

For $x^2 = -4py$ the directrix is $y = p$.

$y = 3$

Answer: y = 3

6.3 Parabola – vertex (h, k)

Q16. The line $x - 1 = 0$ is the directrix of the parabola $(y - 1)^2 = k(x + 2)$. Find the value of k. [24-Q, 25-Q]

Ans.

Vertex (h, k′) = (−2, 1). Directrix x = 1. For $(y - 1)^2 = 4p(x - h)$ the directrix is $x = h - p$:

$1 = -2 - p \rightarrow p = -3$

$p < 0$ means the parabola opens to the left.

$k = 4p = -12$

Answer: k = −12

Q17. The line $x = -1$ is the directrix of the parabola $(y - 2)^2 = k(x + 3)$. Find the value of k. [25*-Q]

Ans.

Vertex x-coordinate h = −3. Directrix $x = h - p$:

$-1 = -3 - p \rightarrow p = -2$

$k = 4p = -8$

Answer: k = −8

Q18. The line $x = 2$ is the directrix of a parabola $(y + 1)^2 = k(x - 1)$. Find the focus of the parabola. [26-Q]

Ans.

Vertex (1, −1). Directrix $x = h - p$:

$2 = 1 - p \rightarrow p = -1$

$k = 4p = -4$

Focus $(h + p, k) = (1 - 1, -1) = (0, -1)$

Answer: Focus (0, −1)

Q19. Find the standard form of the equation of the parabola with vertex (3, 2) and focus at (3, 5). [25*-Q]

Ans.

The focus is directly above the vertex, so the parabola opens up with $(x - h)^2 = 4p(y - k)$. $p = 5 - 2 = 3$.

$(x - 3)^2 = 12(y - 2)$

Answer: $(x - 3)^2 = 12(y - 2)$

Q20. Find the general form of equation of parabola with directrix $y - 1 = 0$ and vertex (1, 3). [26-Q]

Ans.

Directrix $y = 1$ is below the vertex (1, 3), so the parabola opens up. $p = 3 - 1 = 2$, $4p = 8$.

$(x - 1)^2 = 8(y - 3)$

$x^2 - 2x + 1 = 8y - 24$

$x^2 - 2x - 8y + 25 = 0$

Answer: $x^2 - 2x - 8y + 25 = 0$

6.4 Rotation of Axes

Q21. Find the new coordinate of the point (2, 3) if the coordinate axes are rotated through an angle of $\theta = 60°$. [25-Q]

Ans.

Rotation formulas (axes rotated through θ):

$x' = x\cos\theta + y\sin\theta$

$y' = -x\sin\theta + y\cos\theta$

$\theta = 60°$: $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$.

$x' = (2)(\frac{1}{2}) + (3)(\frac{\sqrt{3}}{2}) = \frac{2 + 3\sqrt{3}}{2}$

$y' = -(2)(\frac{\sqrt{3}}{2}) + (3)(\frac{1}{2}) = \frac{3 - 2\sqrt{3}}{2}$

Answer: $(x', y') = (\frac{2 + 3\sqrt{3}}{2}, \frac{3 - 2\sqrt{3}}{2})$

Q22. If the coordinates are rotated through the angle of $\theta = 45°$, then the new coordinate of the point (x, y) is (2, 0). Find the point (x, y). [25-Q]

Ans.

Here $(x', y') = (2, 0)$ and $\theta = 45°$ ($\cos\theta = \sin\theta = \frac{\sqrt{2}}{2}$). Use the formulas that give the old coordinates:

$x = x'\cos\theta - y'\sin\theta = 2\cdot\frac{\sqrt{2}}{2} - 0 = \sqrt{2}$

$y = x'\sin\theta + y'\cos\theta = 2\cdot\frac{\sqrt{2}}{2} + 0 = \sqrt{2}$

Answer: $(x, y) = (\sqrt{2}, \sqrt{2})$

3 MARKS

6.2 Circles

Q1. Find an equation of for circle with center (1, −2) that passes through (4, 2). [25*-Q]

Ans.

Radius = distance from the center to (4, 2):

$r^2 = (4 - 1)^2 + (2 + 2)^2 = 9 + 16 = 25$

$(x - 1)^2 + (y + 2)^2 = 25$

Answer: $(x - 1)^2 + (y + 2)^2 = 25$

Q2. Find the equation of a circle that pass through (3, 2) and (0, 3), given that its center lies on the y-axis. [24-Q]

Ans.

The center is (0, b). Both points are at the same distance r from it:

$(3 - 0)^2 + (2 - b)^2 = (0 - 0)^2 + (3 - b)^2$

$9 + 4 - 4b + b^2 = 9 - 6b + b^2$

$13 - 4b = 9 - 6b$

$2b = -4 \rightarrow b = -2$

$r^2 = (3 + 2)^2 = 25$

$x^2 + (y + 2)^2 = 25$

Check: (3, 2): 9 + 16 = 25 ✓; (0, 3): 0 + 25 = 25 ✓.

Answer: $x^2 + (y + 2)^2 = 25$

Q3. Find the equation of the circle with radius 5 whose center lies on y-axis and pass through the point (3, 2). [24-Q]

Ans.

Center (0, b), r = 5:

$(3 - 0)^2 + (2 - b)^2 = 25$

$(2 - b)^2 = 16$

$2 - b = \pm 4 \rightarrow b = -2$ or $b = 6$

Two circles are possible:

$x^2 + (y + 2)^2 = 25$

$x^2 + (y - 6)^2 = 25$

Answer: $x^2 + (y - 6)^2 = 25$ and $x^2 + (y + 2)^2 = 25$

Q4. Find the standard equation of the circle with radius $5\sqrt{2}$ whose center lies on y-axis and passes through the point (5, 3). [25-Q]

Ans.

Center (0, b), $r^2 = (5\sqrt{2})^2 = 50$:

$(5 - 0)^2 + (3 - b)^2 = 50$

$(3 - b)^2 = 25$

$3 - b = \pm 5 \rightarrow b = -2$ or $b = 8$

$x^2 + (y + 2)^2 = 50$

$x^2 + (y - 8)^2 = 50$

Answer: $x^2 + (y - 8)^2 = 50$ and $x^2 + (y + 2)^2 = 50$

Q5. Find the general form of the equation of the circle which passes through the points (0, 0), (4, 0) and (0, 2). [25-Q]

Ans.

Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$.

$(0, 0)$: $c = 0$

$(4, 0)$: $16 + 8g + c = 0 \rightarrow g = -2$

$(0, 2)$: $4 + 4f + c = 0 \rightarrow f = -1$

$x^2 + y^2 - 4x - 2y = 0$

Answer: $x^2 + y^2 - 4x - 2y = 0$

Q6. Determine whether the given equation $4x^2 + 4y^2 - 24x + 16y = 48$ represents a circle, a point or no graph. If the equation represents a circle, find the center and radius. [25-Q]

Ans.

Divide by 4 to get the form $x^2 + y^2 + 2gx + 2fy + c = 0$:

$x^2 + y^2 - 6x + 4y - 12 = 0$

$g = -3, f = 2, c = -12$

$g^2 + f^2 - c = 9 + 4 + 12 = 25 > 0$

Since $g^2 + f^2 - c > 0$, it is a circle.

Center $(-g, -f) = (3, -2)$

$r = \sqrt{25} = 5$

Answer: a circle with center (3, −2) and radius 5

6.3 Parabola – vertex (0, 0)

Q7. Sketch the graph of parabola $y^2 + 4x = 0$. Identify its vertex, focus, directrix and end points of latus rectum. [24-Q]

Ans.

$y^2 = -4x$ has the form $y^2 = -4px$ with $4p = 4$, $p = 1$. It opens to the left.

$4p = 4 \rightarrow p = 1$

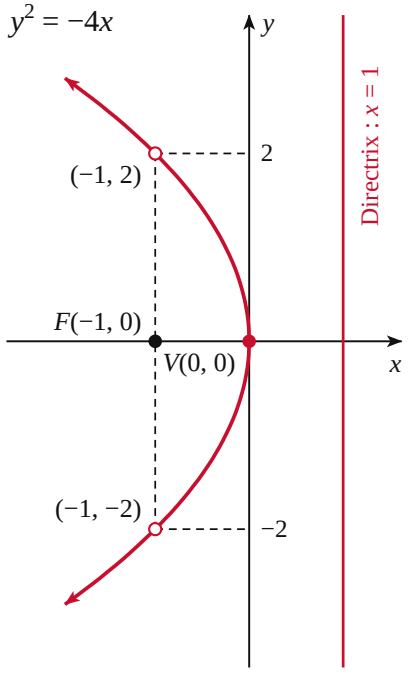
Vertex V = (0, 0)

Focus F = (−1, 0)

Directrix: x = 1

Axis of symmetry: y = 0

End points of latus rectum: (−1, 2) and (−1, −2)



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: V(0, 0), F(−1, 0), directrix x = 1, latus rectum end points (−1, ±2)

Q8. Sketch the graph of the parabola $x^2 - 16y = 0$, showing the vertex, focus, directrix and end points of latus rectum of the equation. [26-Q]

Ans.

$x^2 = 16y$ has the form $x^2 = 4py$ with $4p = 16$, $p = 4$. It opens up.

$4p = 16 \rightarrow p = 4$

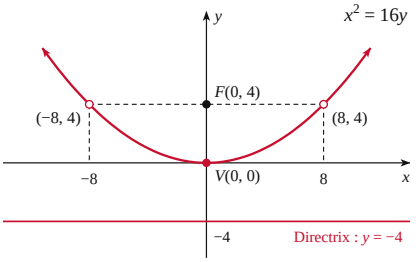
Vertex V = (0, 0)

Focus F = (0, 4)

Directrix: y = −4

Axis of symmetry: x = 0

End points of latus rectum: (−8, 4) and (8, 4)



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: V(0, 0), F(0, 4), directrix y = −4, latus rectum end points (±8, 4)

Q9. Find an equation of the parabola that is symmetric about the y–axis with its vertex at the origin, and passes through (−1, 3). [26-Q]

Ans.

Symmetric about the y–axis with vertex at the origin: $x^2 = ay$. Substitute (−1, 3):

$(-1)^2 = a(3) \rightarrow a = \frac{1}{3}$

$x^2 = \frac{1}{3} y$

Answer: $x^2 = \frac{1}{3}y$ (i.e. $3x^2 = y$)

Q10. Find the general equation of the parabola whose vertex is at the origin and passes through the points (−2, −4) and (−2, 4). [25-Q]

Ans.

The two points have the same x and opposite y, so the parabola is symmetric about the x–axis: $y^2 = ax$. Substitute (−2, 4):

$4^2 = a(-2) \rightarrow a = -8$

$y^2 = -8x$

$y^2 + 8x = 0$

Check (−2, −4): $16 = -8(-2) = 16$ ✓.

Answer: $y^2 + 8x = 0$

6.3 Parabola – vertex (h, k)

Q11. Find the vertex, axis of symmetry, focus, directrix, and the end points of the latus rectum of the parabola $(y - 1)^2 = 4(x + 2)$. [26-Q]

Ans.

Compare with $(y - k)^2 = 4p(x - h)$: $h = -2$, $k = 1$, $4p = 4$, $p = 1$. It opens to the right.

$4p = 4 \rightarrow p = 1$

Vertex V = (−2, 1)

Focus F = (−1, 1)

Directrix: x = −3

Axis of symmetry: y = 1

End points of latus rectum: (−1, 3) and (−1, −1)

Answer: V(−2, 1), axis y = 1, F(−1, 1), directrix x = −3, latus rectum end points (−1, 3), (−1, −1)

Q12. Find the equation of the directrix of the parabola $(y - 1)^2 - 8(x + 3) = 0$. [25-Q]

Ans.

$(y - 1)^2 = 8(x + 3)$: $h = -3$, $k = 1$, $4p = 8$, $p = 2$, opens right. Directrix $x = h - p$:

$x = -3 - 2 = -5$

Answer: x = −5

Q13. Sketch the graph of $x = -\frac{1}{2}(y + 1)^2 - 2$ showing the focus and directrix. [25-Q]

Ans.

Rewrite in standard form:

$x + 2 = -\frac{1}{2}(y + 1)^2$

$(y + 1)^2 = -2(x + 2)$

$h = -2$, $k = -1$, $4p = 2$, $p = \frac{1}{2}$. It opens to the left.

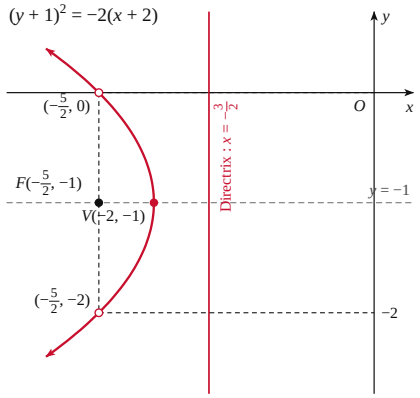
$4p = 2 \rightarrow p = \frac{1}{2}$

Vertex V = (−2, −1)

Focus F = $(-\frac{5}{2}, -1)$

Directrix: $x = -\frac{3}{2}$

Axis of symmetry: y = −1



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: V(−2, −1), F(−$\frac{5}{2}$, −1), directrix x = −$\frac{3}{2}$

Q14. Write the given equation $y^2 - 12x - 6y - 3 = 0$ is standard form. And then find the focus and directrix. [25-Q]

Ans.

Complete the square in y:

$y^2 - 6y = 12x + 3$

$y^2 - 6y + 9 = 12x + 12$

$(y-3)^2 = 12x + 12$

$(y-3)^2 = 12(x+1)$

h = −1, k = 3, 4p = 12, p = 3, opens right.

4p = 12 → p = 3

Vertex V = (−1, 3)

Focus F = (2, 3)

Directrix: x = −4

Answer: $(y-3)^2 = 12(x+1)$; focus (2, 3), directrix x = −4

Q15. Sketch the graph of the parabola $(x+3)^2 = -4(y-1)$ showing the vertex, focus, directrix and end points of latus rectum. [25-Q]

Ans.

Compare with $(x-h)^2 = -4p(y-k)$: h = −3, k = 1, 4p = 4, p = 1. It opens down.

4p = 4 → p = 1

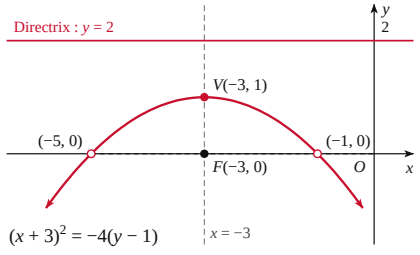
Vertex V = (−3, 1)

Focus F = (−3, 0)

Directrix: y = 2

Axis of symmetry: x = −3

End points of latus rectum: (−5, 0) and (−1, 0)



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: V(−3, 1), F(−3, 0), directrix y = 2, latus rectum end points (−5, 0), (−1, 0)

Q16. Graph $(x+3)^2 = -4(y-2)$. Identify and label the vertex, axis of symmetry, focus, directrix, and end points of latus rectum. [25-Q]

Ans.

Compare with $(x-h)^2 = -4p(y-k)$: h = −3, k = 2, 4p = 4, p = 1. It opens down.

4p = 4 → p = 1

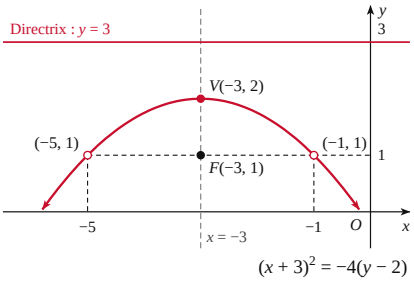
Vertex V = (−3, 2)

Focus F = (−3, 1)

Directrix: y = 3

Axis of symmetry: x = −3

End points of latus rectum: (−5, 1) and (−1, 1)



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: V(−3, 2), axis x = −3, F(−3, 1), directrix y = 3, latus rectum end points (−5, 1), (−1, 1)

Q17. Find the general form of the equation of the parabola with vertex (3, 1) and focus (3, 4). [25-Q]

Ans.

Focus above the vertex: opens up, p = 4 − 1 = 3, 4p = 12.

$(x-3)^2 = 12(y-1)$

$x^2 - 6x + 9 = 12y - 12$

$x^2 - 6x - 12y + 21 = 0$

Answer: $x^2 - 6x - 12y + 21 = 0$

Q18. Find the general form of the equation of the parabola with vertex (2, 1) and focus (2, 4). [24-Q]

Ans.

Focus above the vertex: opens up, p = 3, 4p = 12.

$(x-2)^2 = 12(y-1)$

$x^2 - 4x + 4 = 12y - 12$

$x^2 - 4x - 12y + 16 = 0$

Answer: $x^2 - 4x - 12y + 16 = 0$

Q19. Find the general form of the equation of the parabola with vertex at (−2, 3) and focus at (0, 3). [24-Q]

Ans.

Focus to the right of the vertex: opens right, p = 0 − (−2) = 2, 4p = 8.

$(y-3)^2 = 8(x+2)$

$y^2 - 6y + 9 = 8x + 16$

$y^2 - 6y - 8x - 7 = 0$

Answer: $y^2 - 6y - 8x - 7 = 0$

6.4 Rotation of Axes

Q20. Find the new coordinates of the point (1, 2) if the coordinate axes are rotated through an angle of θ = 60°. [24-Q]

Ans.

Rotation formulas (axes rotated through θ):

$x' = x\cos\theta + y\sin\theta$

$y' = -x\sin\theta + y\cos\theta$

θ = 60°: $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$.

$x' = (1)(\frac{1}{2}) + (2)(\frac{\sqrt{3}}{2}) = \frac{1+2\sqrt{3}}{2}$

$y' = -(1)(\frac{\sqrt{3}}{2}) + (2)(\frac{1}{2}) = \frac{2-\sqrt{3}}{2}$

Answer: $(x', y') = (\frac{1+2\sqrt{3}}{2}, \frac{2-\sqrt{3}}{2})$

Q21. Find the new coordinates of the point (1, 2) if the coordinate axes are rotated through an angle of θ = 120°. [25*-Q]

Ans.

Rotation formulas (axes rotated through θ):

$x' = x\cos\theta + y\sin\theta$

$y' = -x\sin\theta + y\cos\theta$

θ = 120°: $\cos\theta = -\frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$.

$x' = (1)(-\frac{1}{2}) + (2)(\frac{\sqrt{3}}{2}) = \frac{-1+2\sqrt{3}}{2}$

$y' = -(1)(\frac{\sqrt{3}}{2}) + (2)(-\frac{1}{2}) = \frac{-2-\sqrt{3}}{2}$

Answer: $(x', y') = (\frac{-1+2\sqrt{3}}{2}, \frac{-2-\sqrt{3}}{2})$

Q22. Find the new coordinates of the point (1, 3) if the coordinate axes are rotated through an angle of θ = 45°. [25-Q]

Ans.

Rotation formulas (axes rotated through θ):

$x' = x\cos\theta + y\sin\theta$

$y' = -x\sin\theta + y\cos\theta$

$\theta = 45°$: $\cos\theta = \frac{\sqrt{2}}{2}$, $\sin\theta = \frac{\sqrt{2}}{2}$.

$x' = (1)(\frac{\sqrt{2}}{2}) + (3)(\frac{\sqrt{2}}{2}) = 2\sqrt{2}$

$y' = -(1)(\frac{\sqrt{2}}{2}) + (3)(\frac{\sqrt{2}}{2}) = \sqrt{2}$

Answer: $(x', y') = (2\sqrt{2}, \sqrt{2})$

Q23. Find the new coordinate of the point $(2, \sqrt{3})$ if the coordinate axes are rotated through an angle of $\theta = 60°$. [25-Q, 25*-Q]

Ans.

Rotation formulas (axes rotated through θ):

$x' = x\cos\theta + y\sin\theta$

$y' = -x\sin\theta + y\cos\theta$

$\theta = 60°$: $\cos\theta = \frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$.

$x' = (2)(\frac{1}{2}) + (\sqrt{3})(\frac{\sqrt{3}}{2}) = \frac{5}{2}$

$y' = -(2)(\frac{\sqrt{3}}{2}) + (\sqrt{3})(\frac{1}{2}) = -\frac{\sqrt{3}}{2}$

Answer: $(x', y') = (\frac{5}{2}, -\frac{\sqrt{3}}{2})$

Q24. Find the new coordinates of the point (2, 1) if the coordinate axes are rotated through an angle of $\theta = 30°$. [26-Q]

Ans.

Rotation formulas (axes rotated through θ):

$x' = x\cos\theta + y\sin\theta$

$y' = -x\sin\theta + y\cos\theta$

$\theta = 30°$: $\cos\theta = \frac{\sqrt{3}}{2}$, $\sin\theta = \frac{1}{2}$.

$x' = (2)(\frac{\sqrt{3}}{2}) + (1)(\frac{1}{2}) = \frac{1+2\sqrt{3}}{2}$

$y' = -(2)(\frac{1}{2}) + (1)(\frac{\sqrt{3}}{2}) = \frac{-2+\sqrt{3}}{2}$

Answer: $(x', y') = (\frac{1+2\sqrt{3}}{2}, \frac{-2+\sqrt{3}}{2})$

5 MARKS

6.2 Circles

Q1. Find the general form of the equation of the circle which passes through (2, 4), (1, −3) and which has its center on the line $x + 2y = 0$. [26-Q]

Ans.

Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ with center $(-g, -f)$.
Center on $x + 2y = 0$:

$-g + 2(-f) = 0 \rightarrow g = -2f$ (1)

Substitute the two points:

(2, 4): $4 + 16 + 4g + 8f + c = 0 \rightarrow 4g + 8f + c = -20$ (2)
(1, −3): $1 + 9 + 2g - 6f + c = 0 \rightarrow 2g - 6f + c = -10$ (3)

(2) − (3):

$2g + 14f = -10 \rightarrow g + 7f = -5$ (4)

Put (1) in (4): $-2f + 7f = -5$, so $f = -1$ and $g = 2$. From (3): $4 + 6 + c = -10$, so $c = -20$.

$x^2 + y^2 + 4x - 2y - 20 = 0$

Check: (2, 4): $4 + 16 + 8 - 8 - 20 = 0$ ✓; (1, −3): $1 + 9 + 4 + 6 - 20 = 0$ ✓; center $(-2, 1)$: $-2 + 2(1) = 0$ ✓.

Answer: $x^2 + y^2 + 4x - 2y - 20 = 0$

6.3 Parabola – vertex (0, 0)

Q2. The graph of the parabola $y^2 - 12x = 0$, show that the vertex, focus, directrix and end points of latus rectum. [25-Q]

Ans.

$y^2 = 12x$ has the form $y^2 = 4px$ with $4p = 12$, $p = 3$. It opens to the right.

$4p = 12 \rightarrow p = 3$

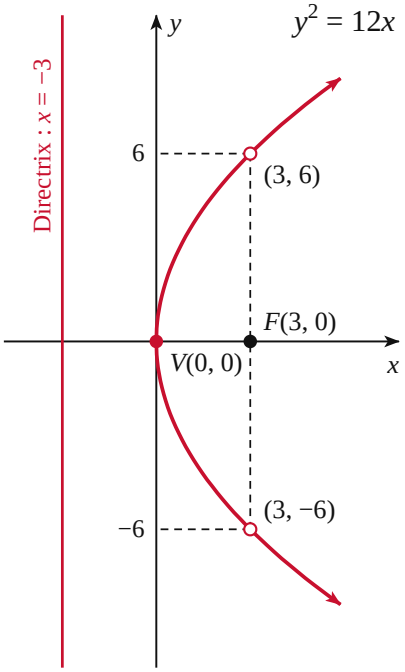
Vertex V = (0, 0)

Focus F = (3, 0)

Directrix: $x = -3$

Axis of symmetry: $y = 0$

End points of latus rectum: (3, 6) and (3, −6)



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: V(0, 0), F(3, 0), directrix x = −3, latus rectum end points (3, ±6)

Q3. The graph of parabola $y - x^2 = 0$, show that the vertex, focus, directrix and end points of the latus rectum. [24-Q]

Ans.

$y = x^2 \rightarrow x^2 = y$, which has the form $x^2 = 4py$ with $4p = 1$, $p = \frac{1}{4}$. It opens up.

$4p = 1 \rightarrow p = \frac{1}{4}$

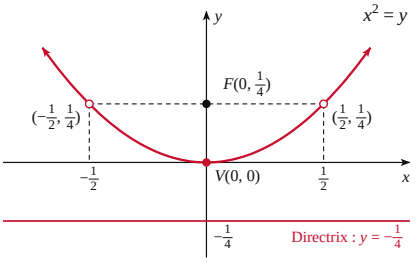
Vertex V = (0, 0)

Focus F = $(0, \frac{1}{4})$

Directrix: $y = -\frac{1}{4}$

Axis of symmetry: $x = 0$

End points of latus rectum: $(-\frac{1}{2}, \frac{1}{4})$ and $(\frac{1}{2}, \frac{1}{4})$



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: V(0, 0), F(0, $\frac{1}{4}$), directrix $y = -\frac{1}{4}$, latus rectum end points $(\pm\frac{1}{2}, \frac{1}{4})$

6.3 Parabola – vertex (h, k)

Q4. Write the standard form and sketch the graph of $y^2 - 8x - 6y - 23 = 0$, showing the vertex, focus, directrix and end points of the latus rectum. [24-Q]

Ans.

Complete the square in y:

$y^2 - 6y = 8x + 23$

$y^2 - 6y + 9 = 8x + 32$

$(y - 3)^2 = 8x + 32$

$(y - 3)^2 = 8(x + 4)$

$h = -4$, $k = 3$, $4p = 8$, $p = 2$, opens right.

$4p = 8 \rightarrow p = 2$

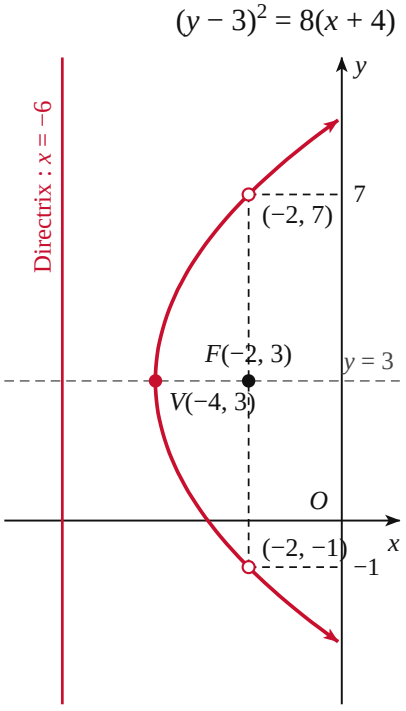
Vertex V = (−4, 3)

Focus F = (−2, 3)

Directrix: $x = -6$

Axis of symmetry: $y = 3$

End points of latus rectum: (−2, 7) and (−2, −1)



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: $(y-3)^2 = 8(x+4)$; V(−4, 3), F(−2, 3), directrix x = −6, latus rectum end points (−2, 7), (−2, −1)

Q5. Write the given equation $4y^2 + 12x - 12y + 39 = 0$ in standard form. Sketch its graph showing the vertex, focus, directrix and end points of latus rectum. [25-Q]

Ans.

Divide by 4 so that the coefficient of y^2 is 1:

$y^2 + 3x - 3y + \frac{39}{4} = 0$

Complete the square in y:

$y^2 - 3y = -3x - \frac{39}{4}$

$y^2 - 3y + \frac{9}{4} = -3x - \frac{15}{2}$

$(y - \frac{3}{2})^2 = -3x - \frac{15}{2}$

$(y - \frac{3}{2})^2 = -3(x + \frac{5}{2})$

h = $-\frac{5}{2}$, k = $\frac{3}{2}$, 4p = 3, p = $\frac{3}{4}$, opens left.

4p = 3 → p = $\frac{3}{4}$

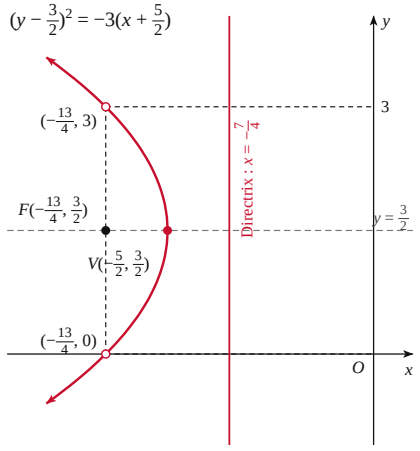
Vertex V = $(-\frac{5}{2}, \frac{3}{2})$

Focus F = $(-\frac{13}{4}, \frac{3}{2})$

Directrix: x = $-\frac{7}{4}$

Axis of symmetry: y = $\frac{3}{2}$

End points of latus rectum: $(-\frac{13}{4}, 3)$ and $(-\frac{13}{4}, 0)$



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: $(y - \frac{3}{2})^2 = -3(x + \frac{5}{2})$; V$(-\frac{5}{2}, \frac{3}{2})$, F$(-\frac{13}{4}, \frac{3}{2})$, directrix x = $-\frac{7}{4}$, latus rectum end points $(-\frac{13}{4}, \frac{3}{2} \pm \frac{3}{2}) = (-\frac{13}{4}, 3), (-\frac{13}{4}, 0)$

Q6. Write the given equation $x^2 - 10x - 8y + 33 = 0$ in standard form. Sketch its graph showing the vertex, focus, directrix and end points of latus rectum. [25*-Q]

Ans.

Complete the square in x:

$x^2 - 10x = 8y - 33$

$x^2 - 10x + 25 = 8y - 8$

$(x - 5)^2 = 8y - 8$

$(x - 5)^2 = 8(y - 1)$

h = 5, k = 1, 4p = 8, p = 2, opens up.

4p = 8 → p = 2

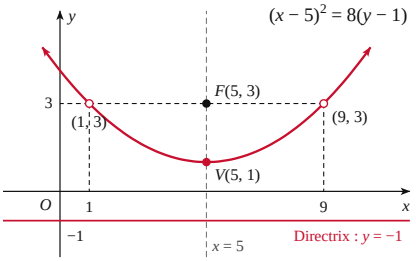
Vertex V = (5, 1)

Focus F = (5, 3)

Directrix: y = −1

Axis of symmetry: x = 5

End points of latus rectum: (1, 3) and (9, 3)



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: $(x - 5)^2 = 8(y - 1)$; V(5, 1), F(5, 3), directrix y = −1, latus rectum end points (1, 3), (9, 3)

Q7. Write the given equation $x^2 - 2x + 8y + 9 = 0$ in standard form. Sketch its graph showing the vertex, focus, directrix and end points of latus rectum. [25-Q, 26-Q]

Ans.

Complete the square in x:

$x^2 - 2x = -8y - 9$

$x^2 - 2x + 1 = -8y - 8$

$(x - 1)^2 = -8y - 8$

$(x - 1)^2 = -8(y + 1)$

h = 1, k = −1, 4p = 8, p = 2, opens down.

4p = 8 → p = 2

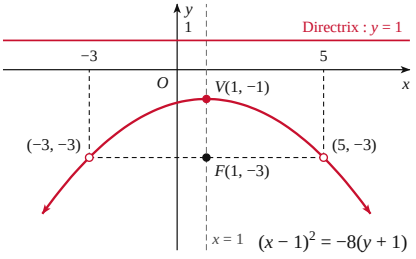
Vertex V = (1, −1)

Focus F = (1, −3)

Directrix: y = 1

Axis of symmetry: x = 1

End points of latus rectum: (−3, −3) and (5, −3)



Sketch: crimson curve = parabola, crimson line = directrix, black dots = **V** (vertex) and **F** (focus), open circles = end points of the latus rectum.

Answer: $(x - 1)^2 = -8(y + 1)$; V(1, −1), F(1, −3), directrix y = 1, latus rectum end points (−3, −3), (5, −3)

Q8. Determine the equation of the parabola, whose axis of symmetry is the line y = 1, passing through the points (1, 3) and (−2, −3). [25-Q]

Ans.

The axis is the horizontal line y = 1, so the parabola has the form $(y - 1)^2 = 4p(x - h)$ with vertex (h, 1).

(1, 3): $(3 - 1)^2 = 4p(1 - h)$ → $4 = 4p(1 - h)$ (1)

$(-2, -3)$: $(-3-1)^2 = 4p(-2-h) \rightarrow 16 = 4p(-2-h)$ (2)

Divide (2) by (1):

$$4 = \frac{-2-h}{1-h}$$

$4 - 4h = -2 - h$

$6 = 3h \rightarrow h = 2$

From (1): $4 = 4p(1-2) = -4p$, so $4p = -4$.

$(y-1)^2 = -4(x-2)$

Check: (1, 3): $4 = -4(-1) = 4$ ✓; $(-2, -3)$: $16 = -4(-4) = 16$ ✓.

Answer: $(y-1)^2 = -4(x-2)$

6.4 Rotation of Axes

Q9. Rotate the coordinate axes to remove the xy-term of the equation $x^2 + 2xy + y^2 + \sqrt{2}x - \sqrt{2}y = 0$ in x′y′-coordinate system. Then sketch the graph. [24-0, 25-0]

Ans.

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$A = 1,\ B = 2,\ C = 1$

$D = \sqrt{2},\ E = -\sqrt{2},\ F = 0$

Angle of rotation:

$$\cot 2\theta = \frac{A-C}{B} = \frac{0}{2} = 0$$

$2\theta = 90° \rightarrow \theta = 45°$

Since B is positive: $A' = A + C = 2$, $B' = 0$, $C' = 0$.

$$D' = D\cos\theta + E\sin\theta = (\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) + (-\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) = 0$$

$$E' = -D\sin\theta + E\cos\theta = -(\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) + (-\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) = -2$$

$F' = F = 0$

Equation in the $x'y'$-system:

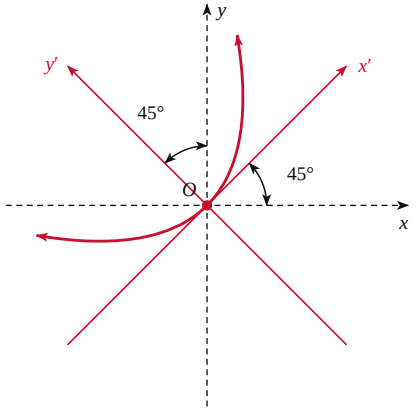
$2x'^2 - 2y' = 0$

Divide by 2 and complete the square:

$x'^2 = y'$

$x'^2 = y'$

This is a parabola with vertex $V' = (0, 0)$, focus $F' = \left(0, \frac{1}{4}\right)$, directrix $y' = -\frac{1}{4}$ (in the $x'y'$-system).



Original: $x^2 + 2xy + y^2 + \sqrt{2}x - \sqrt{2}y = 0$

Rotated: $(x')^2 = y'$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 45°

$x'^2 = y'$

Q10. Rotate the coordinate axes to remove the xy-term of the equation $x^2 + 2xy + y^2 + 4\sqrt{2}x - 4\sqrt{2}y - 8 = 0$ in x′y′-coordinate system. Then sketch the graph. [26-0]

Ans.

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$A = 1,\ B = 2,\ C = 1$

$D = 4\sqrt{2},\ E = -4\sqrt{2},\ F = -8$

Angle of rotation:

$$\cot 2\theta = \frac{A-C}{B} = \frac{0}{2} = 0$$

$2\theta = 90° \rightarrow \theta = 45°$

Since B is positive: $A' = A + C = 2$, $B' = 0$, $C' = 0$.

$$D' = D\cos\theta + E\sin\theta = (4\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) + (-4\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) = 0$$

$$E' = -D\sin\theta + E\cos\theta = -(4\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) + (-4\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) = -8$$

$F' = F = -8$

Equation in the $x'y'$-system:

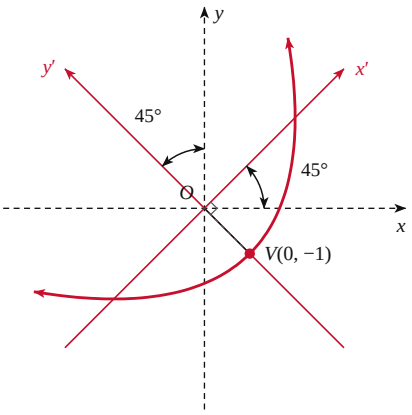
$2x'^2 - 8y' - 8 = 0$

Divide by 2 and complete the square:

$x'^2 = 4y' + 4$

$x'^2 = 4(y' + 1)$

This is a parabola with vertex $V' = (0, -1)$, focus $F' = (0, 0)$, directrix $y' = -2$ (in the $x'y'$-system).



Original: $x^2 + 2xy + y^2 + 4\sqrt{2}x - 4\sqrt{2}y - 8 = 0$

Rotated: $(x')^2 = 4(y' + 1)$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 45°

$x'^2 = 4(y' + 1)$

Q11. Rotate the coordinate axes to remove the xy-term of the equation $3x^2 + 2\sqrt{3}xy + y^2 + 2x - 2\sqrt{3}y - 4 = 0$ in x′y′-coordinate system. Then sketch the graph. [24-0]

Ans.

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$A = 3,\ B = 2\sqrt{3},\ C = 1$

$D = 2,\ E = -2\sqrt{3},\ F = -4$

Angle of rotation:

$$\cot 2\theta = \frac{A-C}{B} = \frac{2}{2\sqrt{3}} = \frac{\sqrt{3}}{3}$$

$$\cos 2\theta = \frac{1}{2}$$

$$\cos\theta = \sqrt{\frac{1+\cos 2\theta}{2}} = \frac{\sqrt{3}}{2}$$

$$\sin\theta = \sqrt{\frac{1-\cos 2\theta}{2}} = \frac{1}{2}$$

$\theta = 30°$

Since B is positive: $A' = A + C = 4$, $B' = 0$, $C' = 0$.

$$D' = D\cos\theta + E\sin\theta = (2)\left(\frac{\sqrt{3}}{2}\right) + (-2\sqrt{3})\left(\frac{1}{2}\right) = 0$$

$$E' = -D\sin\theta + E\cos\theta = -(2)\left(\frac{1}{2}\right) + (-2\sqrt{3})\left(\frac{\sqrt{3}}{2}\right) = -4$$

$F' = F = -4$

Equation in the $x'y'$-system:

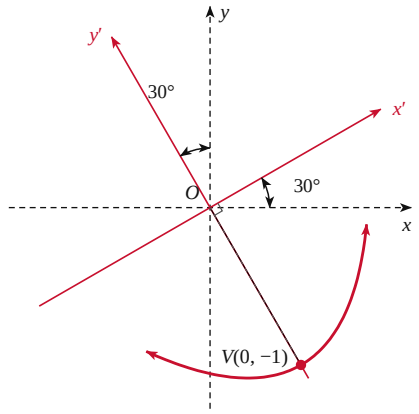
$4x'^2 - 4y' - 4 = 0$

Divide by 4 and complete the square:

$x'^2 = y' + 1$

$x'^2 = (y' + 1)$

This is a parabola with vertex $V' = (0, -1)$, focus $F' = \left(0, -\frac{3}{4}\right)$, directrix $y' = -\frac{5}{4}$ (in the $x'y'$-system).



Original: $3x^2 + 2\sqrt{3}xy + y^2 + 2x - 2\sqrt{3}y - 4 = 0$

Rotated: $(x')^2 = (y' + 1)$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 30°

$x'^2 = (y' + 1)$

Q12. Rotate the coordinate axes to remove the xy−term of the equation $x^2 - 2xy + y^2 - x - y = 0$ in x′y′−coordinate system. Then sketch the graph. [25*-Q]

Ans.

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

A = 1, B = −2, C = 1

D = −1, E = −1, F = 0

Angle of rotation:

$\cot 2\theta = \frac{A - C}{B} = \frac{0}{-2} = 0$

$2\theta = 90° \rightarrow \theta = 45°$

Since B is negative: A′ = 0, B′ = 0, C′ = A + C = 2.

$D' = D\cos\theta + E\sin\theta = (-1)(\frac{\sqrt{2}}{2}) + (-1)(\frac{\sqrt{2}}{2}) = -\sqrt{2}$

$E' = -D\sin\theta + E\cos\theta = -(-1)(\frac{\sqrt{2}}{2}) + (-1)(\frac{\sqrt{2}}{2}) = 0$

F′ = F = 0

Equation in the x′y′−system:

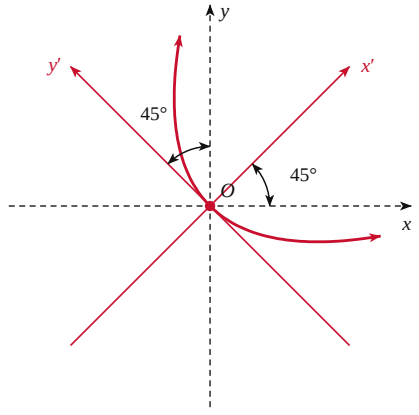
$2y'^2 - \sqrt{2}x' = 0$

Divide by 2 and complete the square:

$y'^2 = \frac{\sqrt{2}}{2}x'$

$y'^2 = \frac{\sqrt{2}}{2}x'$

This is a parabola with vertex V′ = (0, 0), focus $F' = (\frac{\sqrt{2}}{8}, 0)$, directrix $x' = -\frac{\sqrt{2}}{8}$ (in the x′y′−system).



Original: $x^2 - 2xy + y^2 - x - y = 0$

Rotated: $(y')^2 = \frac{\sqrt{2}}{2}x'$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 45°

$y'^2 = \frac{\sqrt{2}}{2}x'$

Q13. Rotate the coordinate axes to remove the xy−term of the equation $x^2 - 2\sqrt{3}xy + 3y^2 + 8\sqrt{3}x + 8y = 0$ in x′y′−coordinate system. Then sketch the graph. [25-Q, 26-Q]

Ans.

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

A = 1, $B = -2\sqrt{3}$, C = 3

$D = 8\sqrt{3}$, E = 8, F = 0

Angle of rotation:

$\cot 2\theta = \frac{A - C}{B} = \frac{-2}{-2\sqrt{3}} = \frac{\sqrt{3}}{3}$

$\cos 2\theta = \frac{1}{2}$

$\cos\theta = \sqrt{\frac{1 + \cos 2\theta}{2}} = \frac{\sqrt{3}}{2}$

$\sin\theta = \sqrt{\frac{1 - \cos 2\theta}{2}} = \frac{1}{2}$

$\theta = 30°$

Since B is negative: A′ = 0, B′ = 0, C′ = A + C = 4.

$D' = D\cos\theta + E\sin\theta = (8\sqrt{3})(\frac{\sqrt{3}}{2}) + (8)(\frac{1}{2}) = 16$

$E' = -D\sin\theta + E\cos\theta = -(8\sqrt{3})(\frac{1}{2}) + (8)(\frac{\sqrt{3}}{2}) = 0$

F′ = F = 0

Equation in the x′y′−system:

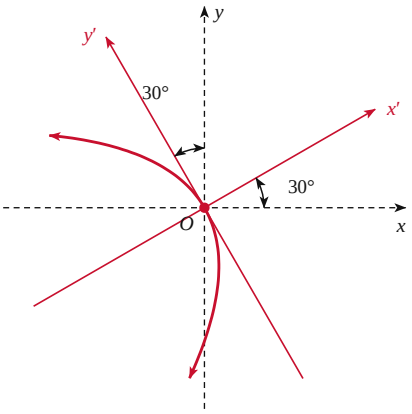
$4y'^2 + 16x' = 0$

Divide by 4 and complete the square:

$y'^2 = -4x'$

$y'^2 = -4x'$

This is a parabola with vertex V′ = (0, 0), focus F′ = (−1, 0), directrix x′ = 1 (in the x′y′−system).



Original: $x^2 - 2\sqrt{3}xy + 3y^2 + 8\sqrt{3}x + 8y = 0$

Rotated: $(y')^2 = -4x'$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 30°

$y'^2 = -4x'$

Q14. Rotate the coordinate axes to remove the xy−term of the equation $x^2 - 2xy + y^2 + \sqrt{2}x + \sqrt{2}y + 2 = 0$ in x′y′−coordinate system. Then sketch the graph. [24-Q, 25-Q]

Ans.

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

A = 1, B = −2, C = 1

$D = \sqrt{2}$, $E = \sqrt{2}$, F = 2

Angle of rotation:

$\cot 2\theta = \frac{A - C}{B} = \frac{0}{-2} = 0$

$2\theta = 90° \rightarrow \theta = 45°$

Since B is negative: A′ = 0, B′ = 0, C′ = A + C = 2.

$D' = D\cos\theta + E\sin\theta = (\sqrt{2})(\frac{\sqrt{2}}{2}) + (\sqrt{2})(\frac{\sqrt{2}}{2}) = 2$

$E' = -D\sin\theta + E\cos\theta = -(\sqrt{2})(\frac{\sqrt{2}}{2}) + (\sqrt{2})(\frac{\sqrt{2}}{2}) = 0$

F′ = F = 2

Equation in the x′y′−system:

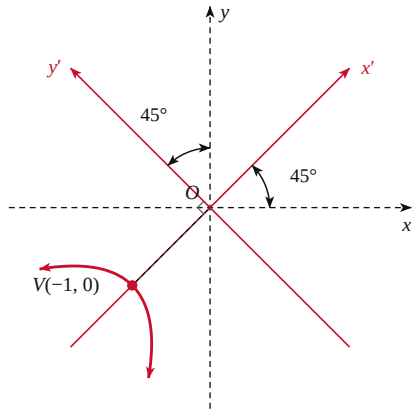
$2y'^2 + 2x' + 2 = 0$

Divide by 2 and complete the square:

$y'^2 = -x' - 1$

$y'^2 = -(x' + 1)$

This is a parabola with vertex V′ = (−1, 0), focus $F' = (-\frac{5}{4}, 0)$, directrix $x' = -\frac{3}{4}$ (in the x′y′−system).



Original: $x^2 - 2xy + y^2 + \sqrt{2}x + \sqrt{2}y + 2 = 0$

Rotated: $(y')^2 = -(x' + 1)$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 45°

$y'^2 = -(x' + 1)$

Q15. Rotate the coordinate axes to remove the xy−term of the equation $(x - y)^2 = -\sqrt{2}(x + y) - 2$ in x′y′−coordinate system. Then sketch the graph. [25-Q]

Ans.

Expand to the general form:

$(x - y)^2 + \sqrt{2}(x + y) + 2 = 0 \rightarrow x^2 - 2xy + y^2 + \sqrt{2}x + \sqrt{2}y + 2 = 0$

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$A = 1,\ B = -2,\ C = 1$

$D = \sqrt{2},\ E = \sqrt{2},\ F = 2$

Angle of rotation:

$$\cot 2\theta = \frac{A - C}{B} = \frac{0}{-2} = 0$$

$2\theta = 90° \rightarrow \theta = 45°$

Since B is negative: $A' = 0,\ B' = 0,\ C' = A + C = 2$.

$$D' = D\cos\theta + E\sin\theta = (\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) + (\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) = 2$$

$$E' = -D\sin\theta + E\cos\theta = -(\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) + (\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) = 0$$

$F' = F = 2$

Equation in the $x'y'$−system:

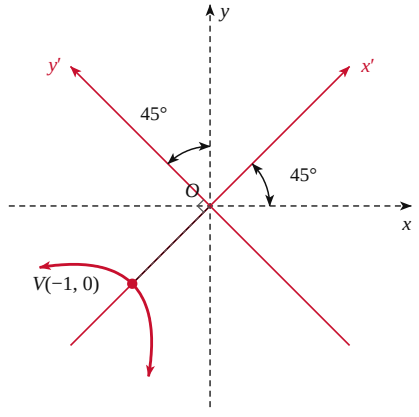
$2y'^2 + 2x' + 2 = 0$

Divide by 2 and complete the square:

$y'^2 = -x' - 1$

$y'^2 = -(x' + 1)$

This is a parabola with vertex $V' = (-1, 0)$, focus $F' = \left(-\frac{5}{4}, 0\right)$, directrix $x' = -\frac{3}{4}$ (in the x′y′−system).



Original: $x^2 - 2xy + y^2 + \sqrt{2}x + \sqrt{2}y + 2 = 0$

Rotated: $(y')^2 = -(x' + 1)$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 45°

$y'^2 = -(x' + 1)$

Q16. Rotate the coordinate axes to remove the xy−term of the equation $(x + \sqrt{3}y)^2 = 2(y - \sqrt{3}x)$ in x′y′−coordinate system. Then sketch the graph. [24-Q]

Ans.

Expand to the general form:

$(x + \sqrt{3}y)^2 - 2(y - \sqrt{3}x) = 0 \rightarrow x^2 + 2\sqrt{3}xy + 3y^2 + 2\sqrt{3}x - 2y = 0$

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$A = 1,\ B = 2\sqrt{3},\ C = 3$

$D = 2\sqrt{3},\ E = -2,\ F = 0$

Angle of rotation:

$$\cot 2\theta = \frac{A - C}{B} = \frac{-2}{2\sqrt{3}} = -\frac{\sqrt{3}}{3}$$

$$\cos 2\theta = -\frac{1}{2}$$

$$\cos\theta = \sqrt{\frac{1 + \cos 2\theta}{2}} = \frac{1}{2}$$

$$\sin\theta = \sqrt{\frac{1 - \cos 2\theta}{2}} = \frac{\sqrt{3}}{2}$$

$\theta = 60°$

Since B is positive: $A' = A + C = 4,\ B' = 0,\ C' = 0$.

$$D' = D\cos\theta + E\sin\theta = (2\sqrt{3})\left(\frac{1}{2}\right) + (-2)\left(\frac{\sqrt{3}}{2}\right) = 0$$

$$E' = -D\sin\theta + E\cos\theta = -(2\sqrt{3})\left(\frac{\sqrt{3}}{2}\right) + (-2)\left(\frac{1}{2}\right) = -4$$

$F' = F = 0$

Equation in the $x'y'$−system:

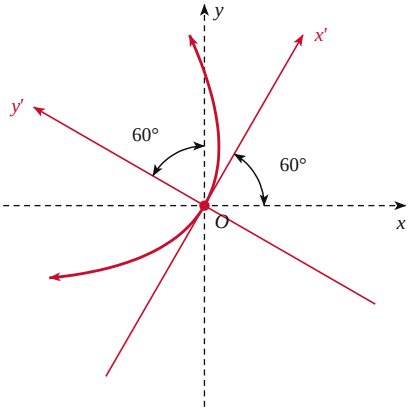
$4x'^2 - 4y' = 0$

Divide by 4 and complete the square:

$x'^2 = y'$

$x'^2 = y'$

This is a parabola with vertex $V' = (0, 0)$, focus $F' = \left(0, \frac{1}{4}\right)$, directrix $y' = -\frac{1}{4}$ (in the $x'y'$−system).



Original: $x^2 + 2\sqrt{3}xy + 3y^2 + 2\sqrt{3}x - 2y = 0$

Rotated: $(x')^2 = y'$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 60°

$x'^2 = y'$

Q17. Rotate the coordinate axes to remove the xy−term of the equation $(\sqrt{3}x - y)^2 = -2(x + \sqrt{3}y)$ in x′y′−coordinate system. Then sketch the graph. [25-Q]

Ans.

Expand to the general form:

$(\sqrt{3}x - y)^2 + 2(x + \sqrt{3}y) = 0 \rightarrow 3x^2 - 2\sqrt{3}xy + y^2 + 2x + 2\sqrt{3}y = 0$

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$A = 3,\ B = -2\sqrt{3},\ C = 1$

$D = 2,\ E = 2\sqrt{3},\ F = 0$

Angle of rotation:

$$\cot 2\theta = \frac{A - C}{B} = \frac{2}{-2\sqrt{3}} = -\frac{\sqrt{3}}{3}$$

$$\cos 2\theta = -\frac{1}{2}$$

$$\cos\theta = \sqrt{\frac{1 + \cos 2\theta}{2}} = \frac{1}{2}$$

$$\sin\theta = \sqrt{\frac{1 - \cos 2\theta}{2}} = \frac{\sqrt{3}}{2}$$

$\theta = 60°$

Since B is negative: $A' = 0,\ B' = 0,\ C' = A + C = 4$.

$$D' = D\cos\theta + E\sin\theta = (2)\left(\frac{1}{2}\right) + (2\sqrt{3})\left(\frac{\sqrt{3}}{2}\right) = 4$$

$E' = -D\sin\theta + E\cos\theta = -(2)\left(\frac{\sqrt{3}}{2}\right) + (2\sqrt{3})\left(\frac{1}{2}\right) = 0$

$F' = F = 0$

Equation in the $x'y'$–system:

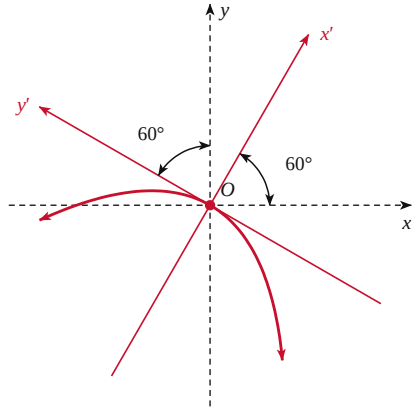
$4y'^2 + 4x' = 0$

Divide by 4 and complete the square:

$y'^2 = -x'$

$y'^2 = -x'$

This is a parabola with vertex $V' = (0, 0)$, focus $F' = \left(-\frac{1}{4}, 0\right)$, directrix $x' = \frac{1}{4}$ (in the $x'y'$–system).



Original: $3x^2 - 2\sqrt{3}xy + y^2 + 2x + 2\sqrt{3}y = 0$

Rotated: $(y')^2 = -x'$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 60°

$y'^2 = -x'$

Q18. Rotate the coordinate axes to remove the xy–term of the equation $(\sqrt{3}x - y)^2 = -2(x - \sqrt{3}y)$ in x'y'–coordinate system. Then sketch the graph. [25-Q]

Ans.

Expand to the general form:

$(\sqrt{3}x - y)^2 + 2(x - \sqrt{3}y) = 0 \rightarrow 3x^2 - 2\sqrt{3}xy + y^2 + 2x - 2\sqrt{3}y = 0$

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$A = 3,\ B = -2\sqrt{3},\ C = 1$

$D = 2,\ E = -2\sqrt{3},\ F = 0$

Angle of rotation:

$\cot 2\theta = \frac{A - C}{B} = \frac{2}{-2\sqrt{3}} = -\frac{\sqrt{3}}{3}$

$\cos 2\theta = -\frac{1}{2}$

$\cos\theta = \sqrt{\frac{1 + \cos 2\theta}{2}} = \frac{1}{2}$

$\sin\theta = \sqrt{\frac{1 - \cos 2\theta}{2}} = \frac{\sqrt{3}}{2}$

$\theta = 60°$

Since B is negative: $A' = 0,\ B' = 0,\ C' = A + C = 4.$

$D' = D\cos\theta + E\sin\theta = (2)\left(\frac{1}{2}\right) + (-2\sqrt{3})\left(\frac{\sqrt{3}}{2}\right) = -2$

$E' = -D\sin\theta + E\cos\theta = -(2)\left(\frac{\sqrt{3}}{2}\right) + (-2\sqrt{3})\left(\frac{1}{2}\right) = -2\sqrt{3}$

$F' = F = 0$

Equation in the $x'y'$–system:

$4y'^2 - 2x' - 2\sqrt{3}y' = 0$

Divide by 4 and complete the square:

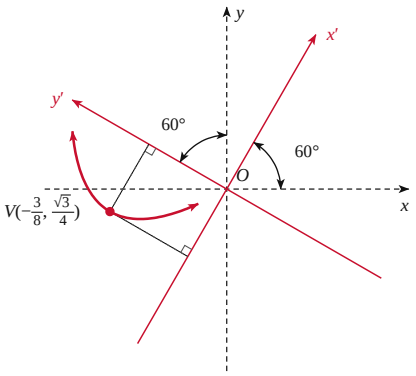
$y'^2 - \frac{\sqrt{3}}{2}y' = \frac{1}{2}x'$

$y'^2 - \frac{\sqrt{3}}{2}y' + \frac{3}{16} = \frac{1}{2}x' + \frac{3}{16}$

$\left(y' - \frac{\sqrt{3}}{4}\right)^2 = \frac{1}{2}x' + \frac{3}{16}$

$\left(y' - \frac{\sqrt{3}}{4}\right)^2 = \frac{1}{2}\left(x' + \frac{3}{8}\right)$

This is a parabola with vertex $V' = \left(-\frac{3}{8}, \frac{\sqrt{3}}{4}\right)$, focus $F' = \left(-\frac{1}{4}, \frac{\sqrt{3}}{4}\right)$, directrix $x' = -\frac{1}{2}$ (in the $x'y'$–system).



Original: $3x^2 - 2\sqrt{3}xy + y^2 + 2x - 2\sqrt{3}y = 0$

Rotated: $\left(y' - \frac{\sqrt{3}}{4}\right)^2 = \frac{1}{2}\left(x' + \frac{3}{8}\right)$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = 60°

$\left(y' - \frac{\sqrt{3}}{4}\right)^2 = \frac{1}{2}\left(x' + \frac{3}{8}\right)$

Q19. Rotate the coordinate axes to remove the xy–term of the equation $9x^2 + 12xy + 4y^2 - 72x + 108y - 36 = 0$ in x'y'–coordinate system. Then sketch the graph. [24-Q, 25-Q, 25*-Q]

Ans.

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$A = 9,\ B = 12,\ C = 4$

$D = -72,\ E = 108,\ F = -36$

Angle of rotation:

$\cot 2\theta = \frac{A - C}{B} = \frac{5}{12} = \frac{5}{12}$

$\cos 2\theta = \frac{5}{13}$

$\cos\theta = \sqrt{\frac{1 + \cos 2\theta}{2}} = \frac{3\sqrt{13}}{13}$

$\sin\theta = \sqrt{\frac{1 - \cos 2\theta}{2}} = \frac{2\sqrt{13}}{13}$

$\theta \approx 33.7°$

Since B is positive: $A' = A + C = 13,\ B' = 0,\ C' = 0.$

$D' = D\cos\theta + E\sin\theta = (-72)\left(\frac{3\sqrt{13}}{13}\right) + (108)\left(\frac{2\sqrt{13}}{13}\right) = 0$

$E' = -D\sin\theta + E\cos\theta = -(-72)\left(\frac{2\sqrt{13}}{13}\right) + (108)\left(\frac{3\sqrt{13}}{13}\right) = 36\sqrt{13}$

$F' = F = -36$

Equation in the $x'y'$–system:

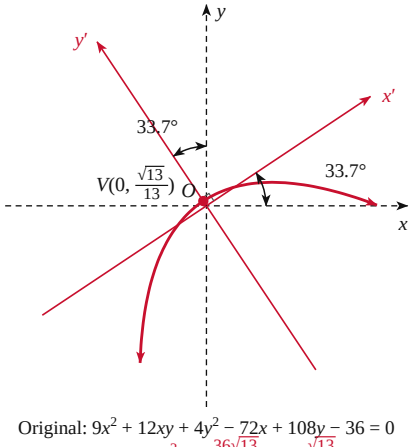
$13x'^2 + 36\sqrt{13}y' - 36 = 0$

Divide by 13 and complete the square:

$x'^2 = -\frac{36\sqrt{13}}{13}y' + \frac{36}{13}$

$x'^2 = -\frac{36\sqrt{13}}{13}\left(y' - \frac{\sqrt{13}}{13}\right)$

This is a parabola with vertex $V' = \left(0, \frac{\sqrt{13}}{13}\right)$, focus $F' = \left(0, -\frac{8\sqrt{13}}{13}\right)$, directrix $y' = \frac{10\sqrt{13}}{13}$ (in the $x'y'$–system).



Original: $9x^2 + 12xy + 4y^2 - 72x + 108y - 36 = 0$

Rotated: $(x')^2 = -\frac{36\sqrt{13}}{13}\left(y' - \frac{\sqrt{13}}{13}\right)$

Sketch: dashed black lines = original x, y axes; crimson lines = new x', y' axes; the curve is drawn in crimson.

Answer: θ = ≈ 33.7°

$x'^2 = -\frac{36\sqrt{13}}{13}\left(y' - \frac{\sqrt{13}}{13}\right)$

Q20. Rotate the coordinate axes to remove the xy−term of the equation $16x^2 - 24xy + 9y^2 + 100x - 200y + 100 = 0$ in x′y′−coordinate system. Then sketch the graph. [S-Q]

Ans.

Compare with $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$A = 16,\ B = -24,\ C = 9$

$D = 100,\ E = -200,\ F = 100$

Angle of rotation:

$$\cot 2\theta = \frac{A - C}{B} = \frac{7}{-24} = -\frac{7}{24}$$

$$\cos 2\theta = -\frac{7}{25}$$

$$\cos\theta = \sqrt{\frac{1 + \cos 2\theta}{2}} = \frac{3}{5}$$

$$\sin\theta = \sqrt{\frac{1 - \cos 2\theta}{2}} = \frac{4}{5}$$

$$\theta = \approx 53.1°$$

Since B is negative: A′ = 0, B′ = 0, C′ = A + C = 25.

$$D' = D\cos\theta + E\sin\theta = (100)\left(\frac{3}{5}\right) + (-200)\left(\frac{4}{5}\right) = -100$$

$$E' = -D\sin\theta + E\cos\theta = -(100)\left(\frac{4}{5}\right) + (-200)\left(\frac{3}{5}\right) = -200$$

$F' = F = 100$

Equation in the x′y′−system:

$25y'^2 - 100x' - 200y' + 100 = 0$

Divide by 25 and complete the square:

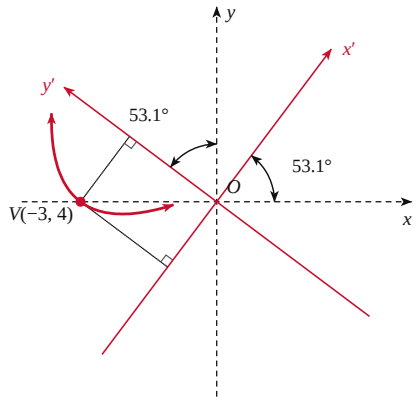
$y'^2 - 8y' = 4x' - 4$

$y'^2 - 8y' + 16 = 4x' + 12$

$(y' - 4)^2 = 4x' + 12$

$(y' - 4)^2 = 4(x' + 3)$

This is a parabola with vertex V′ = (−3, 4), focus F′ = (−2, 4), directrix x′ = −4 (in the x′y′ −system).



Original: $16x^2 - 24xy + 9y^2 + 100x - 200y + 100 = 0$

Rotated: $(y' - 4)^2 = 4(x' + 3)$

Sketch: dashed black lines = original x, y axes; crimson lines = new x′, y′ axes; the curve is drawn in crimson.

Answer: θ = ≈ 53.1°

$(y' - 4)^2 = 4(x' + 3)$

CH.07 Trigonometric Functions

Notes · Questions with Answers

NOTES

Periodic Function and Period

If f(x) = f(x + p), where p is a positive real number, then f is called a **periodic function**. If p is the smallest such number, then p is the **period** of the function f.

The sine function y = sin x is periodic with period 2π, since 2π is the smallest positive real number such that sin x = sin(x + 2π).

Amplitude

The **amplitude** of a periodic function is half the difference between the maximum and minimum values.

So the amplitude of y = sin x is $\frac{1-(-1)}{2} = 1$.

Inverse Trigonometric Functions

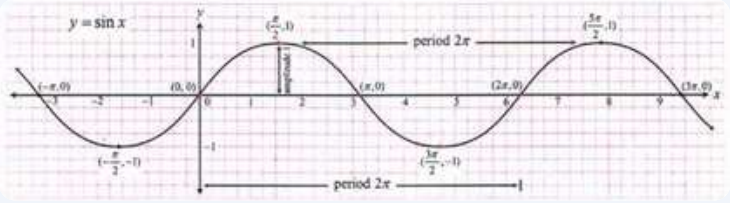
Trigonometric functions are not one-to-one, so the domain of each is restricted to make it one-to-one. In these restricted domains the inverse functions are defined:

Function	Definition	Conditions
$y = \sin^{-1} x$	sin y = x	$-1 \le x \le 1, \ \frac{-\pi}{2} \le y \le \frac{\pi}{2}$
$y = \cos^{-1} x$	cos y = x	$-1 \le x \le 1, \ 0 \le y \le \pi$
$y = \tan^{-1} x$	tan y = x	$x \in R, \ \frac{-\pi}{2} < y < \frac{\pi}{2}$
$y = \cot^{-1} x$	cot y = x	$x \in R, \ 0 < y < \pi$
$y = \sec^{-1} x$	sec y = x	$\lvert x\rvert \ge 1, \ 0 \le y \le \pi, \ y \ne \frac{\pi}{2}$
$y = \csc^{-1} x$	csc y = x	$\lvert x\rvert \ge 1, \ \frac{-\pi}{2} \le y \le \frac{\pi}{2}, \ y \ne 0$

Graph of y = sin x

Domain: R Range: {y | −1 ≤ y ≤ 1} Period: 2π Amplitude: 1

Five key points: x-intercepts (0, 0), (π, 0), (2π, 0); maximum point $(\frac{\pi}{2}, 1)$; minimum point $(\frac{3\pi}{2}, -1)$.



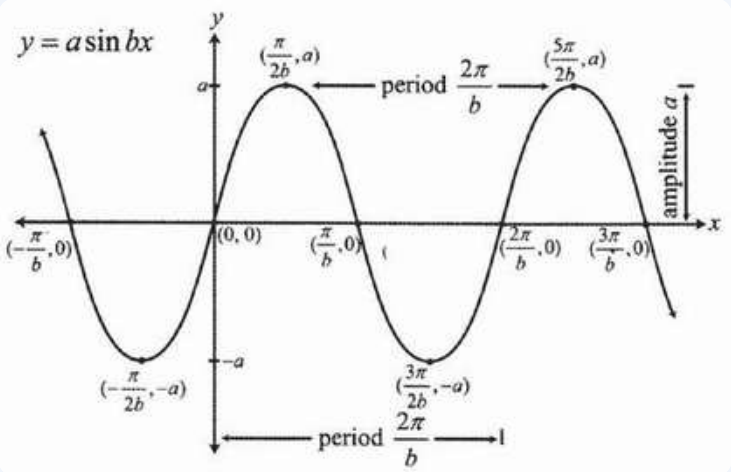
Graph of y = a sin bx (a > 0, b > 0)

Obtained from y = sin x by vertical scaling (scale factor a) and horizontal scaling (scale factor $\frac{1}{b}$), in either order:

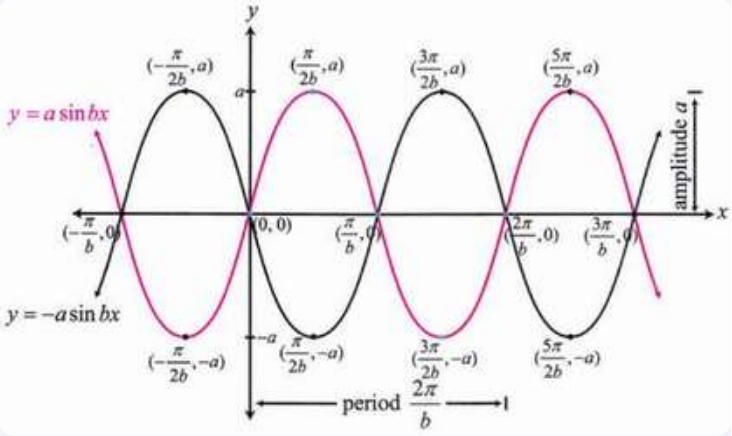
$(x, y) \to (\frac{x}{b}, ay)$

Domain: R Range: {y | −a ≤ y ≤ a} Period: $\frac{2\pi}{b}$ Amplitude: a

Five key points: x-intercepts (0, 0), $(\frac{\pi}{b}, 0)$, $(\frac{2\pi}{b}, 0)$; maximum point $(\frac{\pi}{2b}, a)$; minimum point $(\frac{3\pi}{2b}, -a)$.



y = −a sin bx: reflect y = a sin bx in the x-axis.



Note: a sin(−b)x = −a sin bx, so there is no need to consider b < 0.

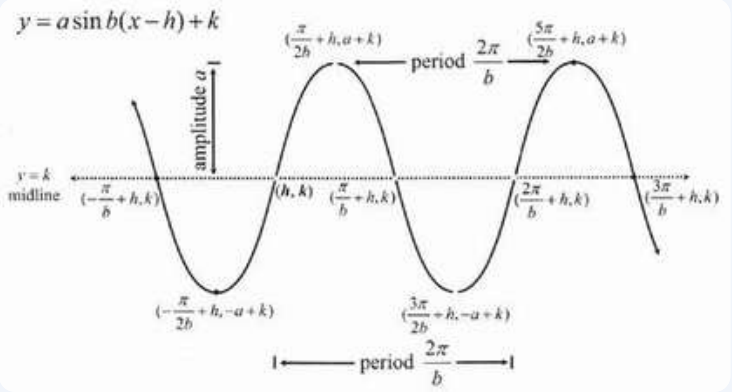
Graph of y = a sin b(x − h) + k

$y = \sin x \rightarrow y = a \sin bx \rightarrow y = a \sin b(x - h) + k$

$(x, y) \rightarrow \left(\frac{x}{b}, ay\right) \rightarrow \left(\frac{x}{b} + h, ay + k\right)$

y = sin x	y = a sin bx	y = a sin b(x − h) + k
(0, 0)	(0, 0)	(h, k)
$\left(\frac{\pi}{2}, 1\right)$	$\left(\frac{\pi}{2b}, a\right)$	$\left(\frac{\pi}{2b} + h, a + k\right)$
(π, 0)	$\left(\frac{\pi}{b}, 0\right)$	$\left(\frac{\pi}{b} + h, k\right)$
$\left(\frac{3\pi}{2}, -1\right)$	$\left(\frac{3\pi}{2b}, -a\right)$	$\left(\frac{3\pi}{2b} + h, -a + k\right)$
(2π, 0)	$\left(\frac{2\pi}{b}, 0\right)$	$\left(\frac{2\pi}{b} + h, k\right)$

The midline is y = k, amplitude a, period $\frac{2\pi}{b}$.

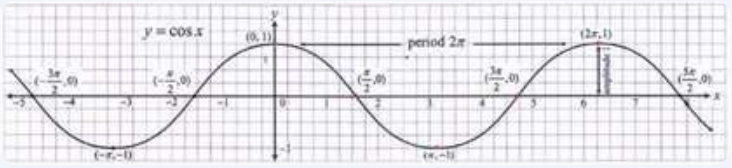


Graph of y = cos x

Domain: R Range: {y | −1 ≤ y ≤ 1} Period: 2π Amplitude: 1

Five key points: x-intercepts $\left(\frac{\pi}{2}, 0\right)$, $\left(\frac{3\pi}{2}, 0\right)$; maximum points (0, 1), (2π, 1); minimum point (π, −1).

Note that $\sin\left(x + \frac{\pi}{2}\right) = \cos x$: translate y = sin x horizontally by $\frac{-\pi}{2}$ units to get y = cos x.

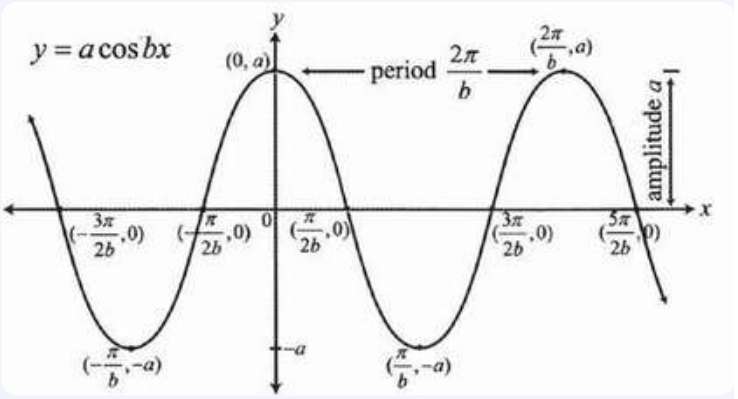


Graph of y = a cos bx and y = a cos b(x − h) + k

From y = cos x: $(x, y) \rightarrow \left(\frac{x}{b}, ay\right)$ gives y = a cos bx (a > 0, b > 0).

Domain: R Range: {y | −a ≤ y ≤ a} Period: $\frac{2\pi}{b}$ Amplitude: a

Five key points: x-intercepts $\left(\frac{\pi}{2b}, 0\right)$, $\left(\frac{3\pi}{2b}, 0\right)$; maximum points (0, a), $\left(\frac{2\pi}{b}, a\right)$; minimum point $\left(\frac{\pi}{b}, -a\right)$.

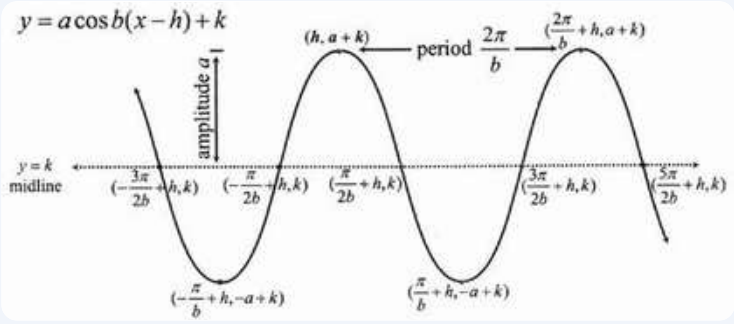


y = −a cos bx: reflect y = a cos bx in the x-axis. Note: a cos(−b)x = a cos bx.

y = a cos b(x − h) + k:

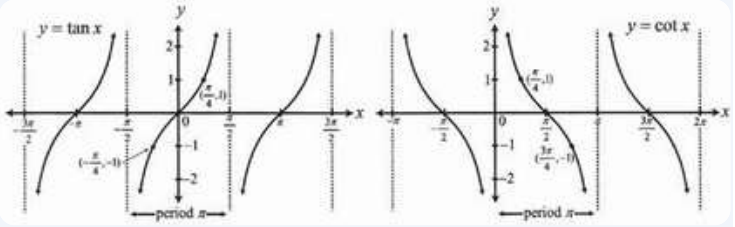
$(x, y) \rightarrow \left(\frac{x}{b}, ay\right) \rightarrow \left(\frac{x}{b} + h, ay + k\right)$

Five key points: points on the midline $\left(\frac{\pi}{2b} + h, k\right)$, $\left(\frac{3\pi}{2b} + h, k\right)$; maximum points (h, a + k), $\left(\frac{2\pi}{b} + h, a + k\right)$; minimum point $\left(\frac{\pi}{b} + h, -a + k\right)$.

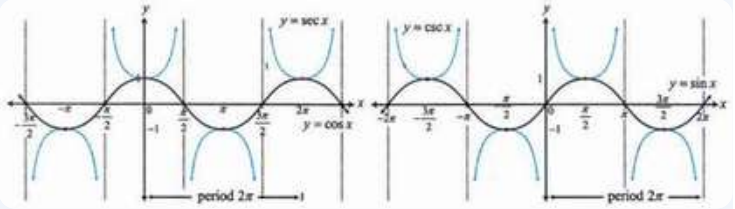


Graphs of tan x, cot x, sec x, csc x

	y = tan x	y = cot x
Domain	$x \neq \frac{\pm\pi}{2}, \frac{\pm 3\pi}{2}, \ldots$	$x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range	R	R
Period	π	π
x-intercepts	$x = 0, \pm\pi, \pm 2\pi, \ldots$	$x = \frac{\pm\pi}{2}, \frac{\pm 3\pi}{2}, \ldots$
Asymptotes	$x = \frac{\pm\pi}{2}, \frac{\pm 3\pi}{2}, \ldots$	$x = 0, \pm\pi, \pm 2\pi, \ldots$

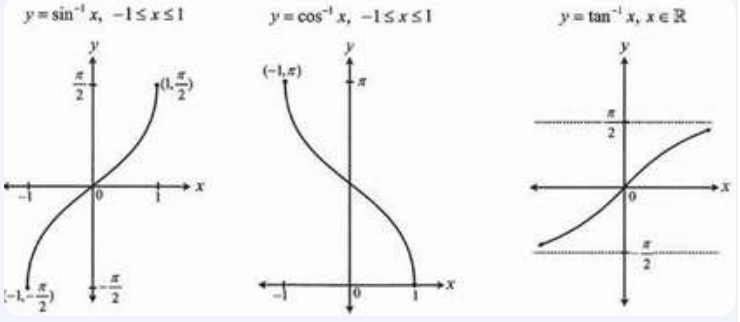


	y = sec x	y = csc x
Domain	$x \neq \frac{\pm\pi}{2}, \frac{\pm 3\pi}{2}, \ldots$	$x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range	$y \leq -1$ or $y \geq 1$	$y \leq -1$ or $y \geq 1$
Period	2π	2π
Asymptotes	$x = \frac{\pm\pi}{2}, \frac{\pm 3\pi}{2}, \ldots$	$x = 0, \pm\pi, \pm 2\pi, \ldots$



Transforming with $(x, y) \to (\frac{x}{b}, ay)$ gives y = a tan bx, y = a cot bx, y = a sec bx and y = a csc bx.

Properties of Inverse Trigonometric Functions



1. $\sin(\sin^{-1} x) = x$ for $-1 \leq x \leq 1$ (but $\sin^{-1}(\sin x) = x$ is not true in general)
2. $\cos(\cos^{-1} x) = x$ for $-1 \leq x \leq 1$
3. $\sin^{-1}(-x) = -\sin^{-1} x$ ($y = \sin^{-1} x$ is an odd function)
4. $\cos^{-1}(-x) = \pi - \cos^{-1} x$, i.e. $\cos^{-1} x + \cos^{-1}(-x) = \pi$

Important Limit

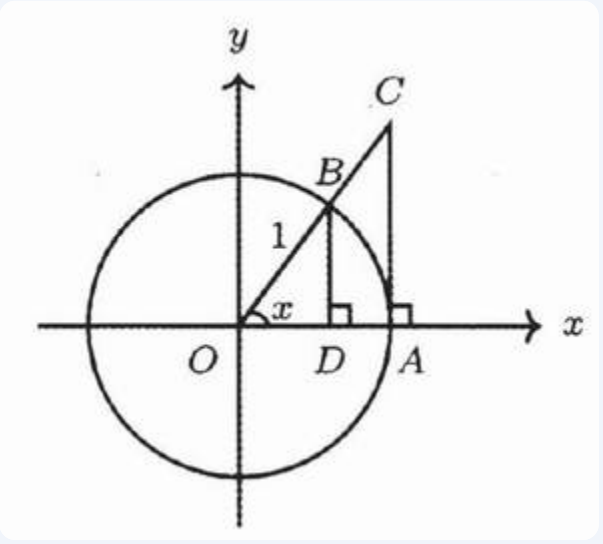
$\lim_{x \to 0} \frac{\sin x}{x} = 1$

Proof: In the unit circle, OA = OB = 1, ∠AOB = x radians, and BD < arc AB < AC. In right △OBD, BD = sin x; in right △OAC, AC = tan x; arc AB = x. So

$\sin x < x < \tan x$

$1 < \frac{x}{\sin x} < \frac{1}{\cos x} \quad \Rightarrow \quad 1 > \frac{\sin x}{x} > \cos x$

When $x \to 0$, $\cos x \to 1$. By the squeeze theorem, $\lim_{x \to 0} \frac{\sin x}{x} = 1$.



Derivative of sin x, cos x, tan x (derivations)

Derivative of sin x. Let y = sin x. Then $\delta y = \sin(x + \delta x) - \sin x = 2\cos(x + \frac{\delta x}{2}) \cdot \sin(\frac{\delta x}{2})$, using $\sin\alpha - \sin\beta = 2\cos(\frac{\alpha+\beta}{2})\sin(\frac{\alpha-\beta}{2})$.

$\frac{dy}{dx} = \lim_{\delta x \to 0} \cos(x + \frac{\delta x}{2}) \cdot \lim_{\delta x \to 0} \frac{\sin(\frac{\delta x}{2})}{\frac{\delta x}{2}} = \cos x \times 1$

$\frac{d}{dx} \sin x = \cos x$

Derivative of cos x. Since $\cos x = \sin(\frac{\pi}{2} + x)$,

$\frac{d}{dx} \cos x = \cos(\frac{\pi}{2} + x) \cdot 1 = -\sin x$

Derivative of tan x. Since $\tan x = \frac{\sin x}{\cos x}$, by the quotient rule

$\frac{d}{dx} \tan x = \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$

Derivatives of Trigonometric Functions

	Basic	Chain rule form (u = u(x))
1	$\frac{d}{dx} \sin x = \cos x$	$\frac{d}{dx} \sin u = \cos u \cdot \frac{du}{dx}$
2	$\frac{d}{dx} \cos x = -\sin x$	$\frac{d}{dx} \cos u = -\sin u \cdot \frac{du}{dx}$
3	$\frac{d}{dx} \tan x = \sec^2 x$	$\frac{d}{dx} \tan u = \sec^2 u \cdot \frac{du}{dx}$
4	$\frac{d}{dx} \cot x = -\csc^2 x$	$\frac{d}{dx} \cot u = -\csc^2 u \cdot \frac{du}{dx}$
5	$\frac{d}{dx} \sec x = \sec x \cdot \tan x$	$\frac{d}{dx} \sec u = \sec u \cdot \tan u \cdot \frac{du}{dx}$
6	$\frac{d}{dx} \csc x = -\csc x \cdot \cot x$	$\frac{d}{dx} \csc u = -\csc u \cdot \cot u \cdot \frac{du}{dx}$

1 MARK

7.1 Graphs of Sine Functions

Q1. The amplitude of the graph y = sin x is [24-Q]

A. 2 **B.** −2 **C.** −1 **D.** 1

Ans.

For y = a sin bx the amplitude is |a|. Here a = 1, so the amplitude is 1.

Answer: D

Q2. The amplitude of the graph y = 6 sin 3(x − 2) + 5 is [25-Q]

A. 3 **B.** 2 **C.** 5 **D.** 6

Ans.
The amplitude is $|a| = |6| = 6$. (The 3 only changes the period and the 5 only moves the midline.)
Answer: D

Q3. The amplitude of $y = -3 \sin 2(x - 1) + 4$ is [26-Q]
A. 2 **B.** 3 **C.** 4 **D.** 1

Ans.
The amplitude is $|a| = |-3| = 3$.
Answer: B

Q4. The period of the function $y = -3 \sin \pi(x + 1) - 2$ is [26-Q]
A. 1 **B.** 2 **C.** 3 **D.** 4

Ans.
Period $= \frac{2\pi}{|b|}$ with $b = \pi$.
Period $= \frac{2\pi}{\pi} = 2$
Answer: B

Q5. The range of the function $y = 2 \sin \frac{\pi}{3}(x - 1) + 3$ is [26-Q]
A. $\{y : 1 \le y \le 3\}$ **B.** $\{y : 2 \le y \le 3\}$ **C.** $\{y : 1 \le y \le 5\}$ **D.** $\{y : 1 < y < 5\}$

Ans.
Amplitude 2 and midline $y = 3$, so y runs from $3 - 2 = 1$ up to $3 + 2 = 5$.
Answer: C (1 ≤ y ≤ 5)

Q6. The range of the function $y = -2 \sin \frac{\pi}{2}(x - 3) + 2$ is [25-Q]
A. $\{2\}$ **B.** $\{y : -1 \le y \le 1\}$ **C.** $\{y : -2 \le y \le 2\}$ **D.** $\{y : 0 \le y \le 4\}$

Ans.
Amplitude $|-2| = 2$ and midline $y = 2$, so $2 - 2 \le y \le 2 + 2$.
Answer: D (0 ≤ y ≤ 4)

Q7. The range of the function $y = -2 \sin \frac{\pi}{2}(x - 3) + 2$ is [25-Q]
A. $\{y \mid 0 \le y \le 4\}$ **B.** $\{y \mid 0 \le y \le 2\}$ **C.** $\{x \mid 0 < x < 4\}$ **D.** $\{y \mid 2 \le y \le 4\}$

Ans.
Amplitude 2, midline $y = 2$, so the range is $0 \le y \le 4$.
Answer: A

Q8. The range of the function $y = 2 \sin \frac{\pi}{3}(x - 1) + 3$ is [25-Q]
A. $\{y \mid 1 < y < 3\}$ **B.** $\{y \mid 2 \le y \le 3\}$ **C.** $\{y \mid 1 \le y \le 5\}$ **D.** $\{y \mid 1 < y < 5\}$

Ans.
Amplitude 2, midline $y = 3$, so the range is $1 \le y \le 5$ (the end values are reached, so the inequalities are not strict).
Answer: C

Q9. If the restricted domain of the function $y = \sin x$ is $0 \le x \le 2\pi$, then the corresponding restricted domain of $y = \sin \frac{\pi}{2} x$ is [24-Q, 25*-Q]
A. $0 \le x \le \pi$ **B.** $0 \le x \le 4$ **C.** $0 \le x \le 2\pi$ **D.** $0 \le x \le 2$

Ans.
Put $\frac{\pi}{2} x$ between the same limits as the original angle:
$0 \le \frac{\pi}{2} x \le 2\pi$
$0 \le x \le 4$
Answer: B

Q10. If the restricted domain of the function $y = \sin x$ is $0 \le x \le 2\pi$, then the corresponding restricted domain of $y = \sin \frac{\pi}{4} x$ is [25-Q]
A. $1 \le x \le \pi$ **B.** $0 \le x \le 8$ **C.** $0 \le x \le 2\pi$ **D.** $0 \le x \le 2$

Ans.
$0 \le \frac{\pi}{4} x \le 2\pi$
$0 \le x \le 8$
Answer: B

Q11. The maximum value of the function $y = -2 \sin(\pi x - 3) + 2$ is [25-Q]
A. 2 **B.** 3 **C.** 4 **D.** 6

Ans.
The amplitude is 2 and the midline is $y = 2$, so the maximum is $2 + 2 = 4$.
Answer: C

Q12. The maximum point of the graph of $y = \frac{1}{2} \sin x$ where $0 \le x \le 2\pi$ is [25*-Q]
A. $(\frac{\pi}{2}, \frac{1}{2})$ **B.** $(\frac{\pi}{2}, 1)$ **C.** $(\frac{3\pi}{2}, -\frac{1}{2})$ **D.** $(\frac{3\pi}{2}, -1)$

Ans.
The maximum of $\sin x$ occurs at $x = \frac{\pi}{2}$. There $y = \frac{1}{2} \cdot 1 = \frac{1}{2}$.
Answer: A $(\frac{\pi}{2}, \frac{1}{2})$

7.2 Graphs of Cosine Functions

Q13. The period of the function $y = 2 \cos \pi(x + 2) + 1$ is [24-Q]
A. 1 **B.** 2 **C.** 3 **D.** π

Ans.
Period $= \frac{2\pi}{|b|}$ with $b = \pi$.
Period $= \frac{2\pi}{\pi} = 2$
Answer: B

Q14. The period of the function $y = 3 \cos \pi(x + 1) + 2$ is [24-Q]
A. 1 **B.** 2 **C.** 3 **D.** π

Ans.
Period $= \frac{2\pi}{\pi} = 2$
Answer: B

Q15. The period of the function $y = 3 \cos \frac{\pi}{2}(x - 3) + 2$ is [25*-Q]
A. 4 **B.** 2 **C.** 1 **D.** 3

Ans.
Period $= \frac{2\pi}{\frac{\pi}{2}} = 4$
Answer: A

Q16. The range of the function $y = 3 \cos 2x$ is [25-Q]
A. $\Re$ **B.** $\{y : -3 \le y \le 3\}$ **C.** $\{y : -2 \le y \le 2\}$ **D.** $\{y : 2 \le y \le 3\}$

Ans.
Amplitude 3 and midline $y = 0$, so $-3 \le y \le 3$.
Answer: B

Q17. The range of the function $y = -2 \cos \frac{\pi}{2}(x - 3) + 3$ is [26-Q]
A. $\{y : 1 \le y \le 5\}$ **B.** $\{y : -1 \le y \le 1\}$ **C.** $\{y : -5 \le y \le -1\}$ **D.** $\{y : -\pi \le y \le \pi\}$

Ans.
Amplitude $|-2| = 2$ and midline $y = 3$, so $3 - 2 \le y \le 3 + 2$.
Answer: A (1 ≤ y ≤ 5)

Q18. How many maximum points of the graph $y = \frac{1}{2} \cos 2x$ in period π? [25-Q]
A. 2 **B.** 3 **C.** 4 **D.** 6

Ans.
The period of $y = \frac{1}{2} \cos 2x$ is $\frac{2\pi}{2} = \pi$. Taking the interval $0 \le x \le \pi$, the maximum value $\frac{1}{2}$ is reached at $x = 0$ and at $x = \pi$, i.e. at the two ends of the period.
Answer: A

7.3 Graphs of Other Trigonometric Functions

Q19. The trigonometric function with a period 3π is [24-Q]
A. $\sin \frac{1}{3} x$ **B.** $\cos 3x$ **C.** $\tan \frac{1}{3} x$ **D.** $\cot 3x$

Ans.
Sine and cosine have period $\frac{2\pi}{b}$; tangent and cotangent have period $\frac{\pi}{b}$.
$\tan \frac{1}{3} x$: period $= \frac{\pi}{\frac{1}{3}} = 3\pi$
The others give 6π, $\frac{2\pi}{3}$ and $\frac{\pi}{3}$.
Answer: C

Q20. The trigonometric function with a period 4π is [25-Q]
A. $\sin \frac{1}{4} x$ **B.** $\cos 4x$ **C.** $\tan \frac{1}{4} x$ **D.** $\cot 4x$

Ans.
$\tan \frac{1}{4} x$: period $= \frac{\pi}{\frac{1}{4}} = 4\pi$
$\sin \frac{1}{4} x$ has period 8π, $\cos 4x$ has $\frac{\pi}{2}$, $\cot 4x$ has $\frac{\pi}{4}$.
Answer: C

Q21. The trigonometric function with a period $\frac{\pi}{2}$ is [25-Q]
A. $\sin \frac{1}{2} x$ **B.** $\cos 2x$ **C.** $\tan \frac{1}{2} x$ **D.** $\cot 2x$

Ans.
$\cot 2x$: period $= \frac{\pi}{2}$
$\cos 2x$ has period π (not $\frac{\pi}{2}$), $\sin \frac{1}{2} x$ has 4π and $\tan \frac{1}{2} x$ has 2π.
Answer: D

Q22. The range of the function $y = 2\sec x + 3$ is [24-Q]

A. $\Re$ **B.** $\{y \mid -1 \le y \le 5\}$ **C.** $\{y \mid y \le 1 \text{ or } y \ge 5\}$ **D.** $\{y \mid -5 \le y \le 0\}$

Ans.

The range of sec x is $y \le -1$ or $y \ge 1$. Multiplying by 2 gives $y \le -2$ or $y \ge 2$, and adding 3 gives $y \le 1$ or $y \ge 5$.

Answer: C

7.4 Inverse Trigonometric Functions

Q23. $\sin^{-1}\frac{\sqrt{3}}{2} + \cos^{-1}\frac{\sqrt{2}}{2} =$ [24-Q]

A. π **B.** $\frac{3\pi}{2}$ **C.** $\frac{7\pi}{12}$ **D.** $\frac{7\pi}{13}$

Ans.

$\sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3}$ and $\cos^{-1}\frac{\sqrt{2}}{2} = \frac{\pi}{4}$.

$\frac{\pi}{3} + \frac{\pi}{4} = \frac{4\pi + 3\pi}{12} = \frac{7\pi}{12}$

Answer: C

Q24. $\cos^{-1} x + \cos^{-1}(-x) =$ [24-Q]

A. π **B.** $-\pi$ **C.** $\frac{\pi}{2}$ **D.** $-\frac{\pi}{2}$

Ans.

From the notes $\cos^{-1}(-x) = \pi - \cos^{-1} x$, so

$\cos^{-1} x + \cos^{-1}(-x) = \cos^{-1} x + \pi - \cos^{-1} x = \pi$

Answer: A

Q25. If $\cot^{-1} x = y$, then [25-Q]

A. $0 < y < \pi$ **B.** $0 \le y \le \pi$ **C.** $-\frac{\pi}{2} < y < \frac{\pi}{2}$ **D.** $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$

Ans.

From the table of inverse trigonometric functions the range of $y = \cot^{-1} x$ is $0 < y < \pi$.

Answer: A

7.5 Differentiation of Trigonometric Functions

Q26. If $y = \sin^2 x$, then $\frac{d^2y}{dx^2} + 4y =$ [25-Q]

A. 1 **B.** −2 **C.** −1 **D.** 2

Ans.

Differentiate twice:

$\frac{dy}{dx} = 2\sin x \cos x = \sin 2x$

$\frac{d^2y}{dx^2} = 2\cos 2x$

Then $\frac{d^2y}{dx^2} + 4y = 2\cos 2x + 4\sin^2 x$. Using $\cos 2x = 1 - 2\sin^2 x$:

$= 2(1 - 2\sin^2 x) + 4\sin^2 x$

$= 2$

Answer: D

Q27. $\frac{d}{dx}\tan^2(x^3) =$ [25-Q]

A. $2\tan x^3$ **B.** $\tan x^3 \sec x^3$ **C.** $6x^2 \tan x^3 \sec^2 x^3$ **D.** $6\tan x \sec x$

Ans.

$\tan^2(x^3) = [\tan(x^3)]^2$. Chain rule (twice):

$\frac{d}{dx}[\tan(x^3)]^2 = 2\tan(x^3) \cdot \sec^2(x^3) \cdot 3x^2$

$= 6x^2 \tan x^3 \sec^2 x^3$

Answer: C

2 MARKS

7.1 Graphs of Sine Functions

Q1. What is the amplitude of a periodic function? [25-Q]

Ans.

The amplitude is half the distance between the maximum value and the minimum value of the periodic function.

$\text{Amplitude} = \frac{\text{maximum} - \text{minimum}}{2}$

Q2. What is the amplitude of a periodic function $y = 2\sin(x - 1) + 3$? [25-Q]

Ans.

For $y = a\sin b(x - h) + k$ the amplitude is $|a|$.

Here $a = 2$, so the amplitude is 2.

Check: the maximum is $3 + 2 = 5$ and the minimum is $3 - 2 = 1$, so $\frac{5 - 1}{2} = 2$.

Answer: 2

Q3. If a and b are period and amplitude of the graph of $y = \sin x$, then find the sum of a and b. [25*-Q]

Ans.

The period of $y = \sin x$ is 2π and its amplitude is 1, so $a = 2\pi$ and $b = 1$.

$a + b = 2\pi + 1$

Answer: $1 + 2\pi$

Q4. Find c + d if c and d are amplitude and period of the graph of $y = 3\sin 2x + 1$ respectively. [25-Q]

Ans.

Amplitude $c = |3| = 3$. Period $d = \frac{2\pi}{2} = \pi$.

$c + d = 3 + \pi$

Answer: $3 + \pi$

7.2 Graphs of Cosine Functions

Q5. Find a and b such that a and b are amplitude and period of the graph of $y = 4\cos 3x + 7$. [24-Q]

Ans.

Amplitude $a = |4| = 4$. Period $b = \frac{2\pi}{3}$.

Answer: $a = 4$, $b = \frac{2\pi}{3}$

Q6. Find the amplitude and period of the graph $y = 4\cos 3x + 7$. [25-Q]

Ans.

Amplitude $= |4| = 4$, and period $= \frac{2\pi}{3}$.

Answer: amplitude 4, period $\frac{2\pi}{3}$

Q7. If a and b are period and amplitude of the graph of $y = \cos x$, then find the sum of a and b. [25-Q]

Ans.

Period $a = 2\pi$ and amplitude $b = 1$.

$a + b = 2\pi + 1$

Answer: $1 + 2\pi$

Q8. Find the sum of amplitude and period of the graph of $y = 3\cos\frac{\pi}{4}x$. [25-Q]

Ans.

Amplitude = 3. Period $= \frac{2\pi}{\frac{\pi}{4}} = 8$.

$3 + 8 = 11$

Answer: 11

Q9. Find the sum of amplitude and period of the graph of $y = 2\cos\frac{\pi}{3}(x - 3) + 4$. [25-Q, 26-Q]

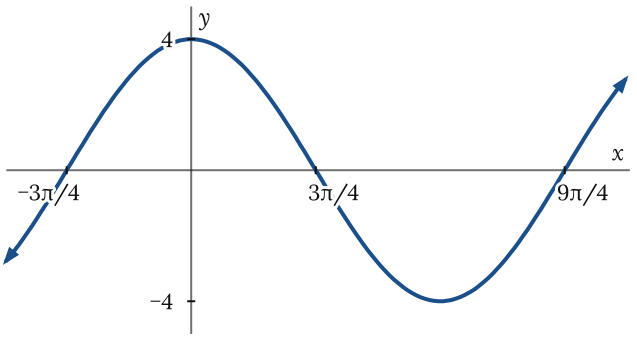
Ans.

Amplitude = 2. Period $= \frac{2\pi}{\frac{\pi}{3}} = 6$.

$2 + 6 = 8$

Answer: 8

Q10. Find an equation for the graph in the following illustration. Give your answer in the form $y = a\cos bx$, where a and b are real numbers.



[24-Q]

Ans.

The graph is an upside-down-free cosine curve starting at its maximum 4 on the y-axis, so $y = a\cos bx$ with $a = 4$ (maximum 4, minimum −4).

The graph first crosses the x-axis at $x = \frac{3\pi}{4}$, which is a quarter of a period ($\cos bx = 0$ when $bx = \frac{\pi}{2}$):

$b \cdot \frac{3\pi}{4} = \frac{\pi}{2}$

$b = \frac{\pi}{2} \times \frac{4}{3\pi} = \frac{2}{3}$

Check with the period: $2\pi \div \frac{2}{3} = 3\pi$. The zeros at $-\frac{3\pi}{4}$, $\frac{3\pi}{4}$, $\frac{9\pi}{4}$ are $\frac{3\pi}{2}$ apart, which is half a period. ✓

Answer: $y = 4\cos\frac{2}{3}x$

Q11. If the point (x, y) is on the graph of y = cos x, then find the respective point on the graph of y = f(3x − 3) + 2. [S-Q]

Ans.

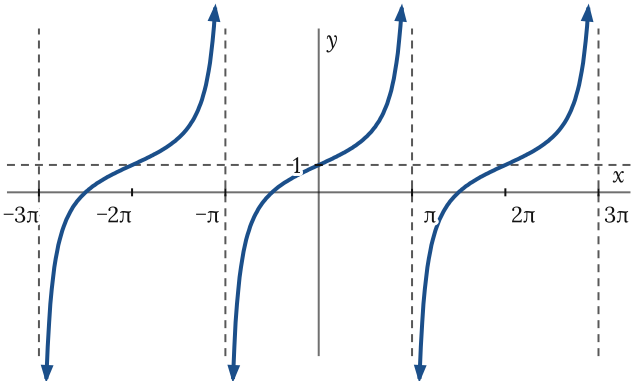
Write y = f(3x − 3) + 2 = f(3(x − 1)) + 2, with b = 3, h = 1, k = 2. The transformation is (x, y) → ($\frac{x}{b}$ + h, y + k).

(x, y) → ($\frac{x}{3}$ + 1, y + 2)

Answer: ($\frac{x}{3}$ + 1, y + 2)

7.3 Graphs of Other Trigonometric Functions

Q12. Find p + q given the following graph of the function y = tan px + q.



[25-Q]

Ans.

The vertical asymptotes are at x = −3π, −π, π, 3π ..., so the distance between neighbouring asymptotes (one period) is 2π.

Period = $\frac{\pi}{p}$ = 2π

p = $\frac{1}{2}$

For y = tan px + q the curve passes through the midpoint (0, q) of each branch. The graph crosses the y-axis at y = 1, so q = 1. (The dashed horizontal line y = 1 passes through the centre of every branch.)

p + q = $\frac{1}{2}$ + 1 = $\frac{3}{2}$

Answer: p + q = 1.5

7.4 Inverse Trigonometric Functions

Q13. Restrict the appropriate domain of y = sin x to be a one−to−one function. [24-Q]

Ans.

On $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ the sine function increases from −1 to 1 and takes each value exactly once, so it is one−to−one there.

Answer: $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

Q14. Restrict the appropriate domain of y = sin x to define the inverse function. [24-Q]

Ans.

An inverse exists only when the function is one−to−one. Restrict sin x to its principal interval:

Answer: $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

Q15. Restrict the appropriate domain of y = cos x to be a one−to−one function. [24-Q, S-Q]

Ans.

On 0 ≤ x ≤ π the cosine function decreases from 1 to −1 and takes each value once.

Answer: 0 ≤ x ≤ π

Q16. Restrict the appropriate domain of y = tan x to be a one−to−one function. [24-Q, 25-Q]

Ans.

Between the asymptotes $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{2}$, tan x increases through all real values exactly once.

Answer: $-\frac{\pi}{2} < x < \frac{\pi}{2}$

Q17. Find $\sin^{-1}\frac{\sqrt{3}}{2}$. [26-Q]

Ans.

We need the angle θ in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ with $\sin\theta = \frac{\sqrt{3}}{2}$.

$\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$

Answer: $\frac{\pi}{3}$

Q18. Evaluate the value of $\sin(\tan^{-1}\frac{1}{\sqrt{3}})$. [25-Q]

Ans.

$\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$ and $\frac{\pi}{6}$ lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$.

$\sin(\tan^{-1}\frac{1}{\sqrt{3}}) = \sin\frac{\pi}{6} = \frac{1}{2}$

Answer: $\frac{1}{2}$

Q19. Evaluate $\cos(\sin^{-1}\frac{\sqrt{3}}{2})$. [26-Q]

Ans.

$\sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3}$

$\cos(\sin^{-1}\frac{\sqrt{3}}{2}) = \cos\frac{\pi}{3} = \frac{1}{2}$

Answer: $\frac{1}{2}$

Q20. Evaluate $\cos^{-1}(\cos(-\frac{1}{2}))$. [24-Q]

Ans.

$\cos^{-1}(\cos x) = x$ only for $0 \le x \le \pi$. Here $\cos(-\frac{1}{2}) = \cos\frac{1}{2}$ and $\frac{1}{2}$ is in $[0, \pi]$, so

$\cos^{-1}(\cos(-\frac{1}{2})) = \cos^{-1}(\cos\frac{1}{2}) = \frac{1}{2}$

Answer: $\frac{1}{2}$

Q21. Evaluate the value of $\sin^{-1}(\cos(\sin^{-1}\frac{\sqrt{3}}{2}))$. [25*-Q, 26-Q]

Ans.

$\sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3}$

$\cos\frac{\pi}{3} = \frac{1}{2}$

$\sin^{-1}(\frac{1}{2}) = \frac{\pi}{6}$

Answer: $\frac{\pi}{6}$

Q22. Find the value of $\tan^{-1}[2\cos(2\sin^{-1}\frac{1}{2})]$. [S-Q, 25*-Q]

Ans.

$\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$

$2\sin^{-1}\frac{1}{2} = \frac{\pi}{3}$

$2\cos\frac{\pi}{3} = 2 \times \frac{1}{2} = 1$

$\tan^{-1}(1) = \frac{\pi}{4}$

Answer: $\frac{\pi}{4}$

Q23. Find the value of $\csc[\cos^{-1}\{\cot(\sin^{-1}(-1))\}]$. [26-Q]

Ans.

$\sin^{-1}(-1) = -\frac{\pi}{2}$

$\cot(-\frac{\pi}{2}) = 0$

$\cos^{-1}(0) = \frac{\pi}{2}$

$\csc\frac{\pi}{2} = 1$

Answer: 1

Q24. Find the value of $\csc[\tan^{-1}(\frac{1}{2}\sec(2\sin^{-1}\frac{1}{2}))]$. [24-Q]

Ans.

$2\sin^{-1}\frac{1}{2} = 2 \times \frac{\pi}{6} = \frac{\pi}{3}$

$\sec\frac{\pi}{3} = 2$

$\frac{1}{2} \times 2 = 1$

$\tan^{-1}(1) = \frac{\pi}{4}$

$\csc \frac{\pi}{4} = \sqrt{2}$

Answer: $\sqrt{2}$

Q25. Find the value of $\sin^{-1}(\cos^{-1}(\cos 0.2)) + \sin^{-1}(-0.2) + \cos^{-1}(\sin^{-1}(\sin 0.2)) + \cos^{-1}(-0.2)$. [25-Q]

Ans.

0.2 lies in $[0, \pi]$ and in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so $\cos^{-1}(\cos 0.2) = 0.2$ and $\sin^{-1}(\sin 0.2) = 0.2$.

Sum $= \sin^{-1}(0.2) + \sin^{-1}(-0.2) + \cos^{-1}(0.2) + \cos^{-1}(-0.2)$

Use $\sin^{-1}(-x) = -\sin^{-1} x$ and $\cos^{-1} x + \cos^{-1}(-x) = \pi$:

$= \sin^{-1}(0.2) - \sin^{-1}(0.2) + \pi$

$= \pi$

Answer: π

7.5 Differentiation of Trigonometric Functions

Q26. If $y = \sin 4x + \cos^4 x$, then find the value of $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$. [25-Q, 26-Q]

Ans.

Differentiate each term (chain rule):

$\frac{dy}{dx} = 4\cos 4x + 4\cos^3 x \cdot (-\sin x)$

$= 4\cos 4x - 4\sin x \cos^3 x$

At $x = \frac{\pi}{4}$: $\cos 4x = \cos \pi = -1$, $\sin x = \cos x = \frac{\sqrt{2}}{2}$.

$\frac{dy}{dx} = 4(-1) - 4 \cdot \frac{\sqrt{2}}{2} \cdot (\frac{\sqrt{2}}{2})^3$

$= -4 - 4 \cdot (\frac{\sqrt{2}}{2})^4$

$= -4 - 4 \cdot \frac{1}{4}$

$= -5$

Answer: −5

3 MARKS

7.5 Differentiation of Trigonometric Functions

Q1. If $y = x^2 + 2x + 3$, show that $(\frac{dy}{dx})^2 + (\frac{d^2y}{dx^2})^3 = 4y$. [24-Q]

Ans.

$\frac{dy}{dx} = 2x + 2$

$\frac{d^2y}{dx^2} = 2$

$(\frac{dy}{dx})^2 + (\frac{d^2y}{dx^2})^3 = (2x + 2)^2 + 2^3$

$= 4x^2 + 8x + 4 + 8$

$= 4(x^2 + 2x + 3)$

$= 4y$

Hence the result is proved. ✓

Q2. Find the limit of $\frac{\sin x}{x}$ as $x \to 0$. [5-Q]

Ans.

Direct substitution gives $\frac{0}{0}$, so a special limit is needed. For small x measured in radians, $\sin x \approx x$. More precisely, for $0 < x < \frac{\pi}{2}$ the areas in the unit circle give $\sin x < x < \tan x$, and dividing by $\sin x$ gives

$\cos x < \frac{\sin x}{x} < 1$

As $x \to 0$, $\cos x \to 1$, so by the squeeze (sandwich) principle $\frac{\sin x}{x}$ is trapped between 1 and 1.

$\lim_{x \to 0} \frac{\sin x}{x} = 1$

Answer: 1

Q3. Differentiate the function $\tan x$ with respect to x. [25-Q]

Ans.

Write $\tan x = \frac{\sin x}{\cos x}$ and use the quotient rule with $u = \sin x$, $v = \cos x$ ($u' = \cos x$, $v' = -\sin x$):

$\frac{d}{dx} \tan x = \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x}$

$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$

$= \frac{1}{\cos^2 x}$

$= \sec^2 x$

Answer: $\sec^2 x$

5 MARKS

7.1 Graphs of Sine Functions

Q1. Draw the graph of $y = -3 \sin \frac{\pi}{4} x$. [26-Q]

Ans.

Compare with $y = a \sin b(x - h) + k$. Here $a = -3$, $b = \frac{\pi}{4}$, $h = 0$, $k = 0$.

Domain: $\mathbb{R}$. Range: $-3 \le y \le 3$. Amplitude $= |a| = 3$. Period $= \frac{2\pi}{|b|} = 8$. Midline: $y = 0$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \sin x$:

$(0, 0) \to (0, 0)$

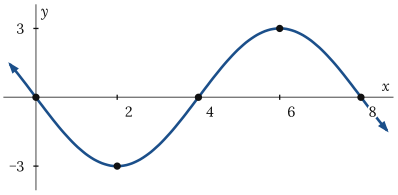
$(\frac{\pi}{2}, 1) \to (2, -3)$

$(\pi, 0) \to (4, 0)$

$(\frac{3\pi}{2}, -1) \to (6, 3)$

$(2\pi, 0) \to (8, 0)$

Key points: $(0, 0), (2, -3), (4, 0), (6, 3), (8, 0)$.



Q2. Draw the graph of $y = 3 \sin \pi(x + 1) - 2$. [25*-Q]

Ans.

Compare with $y = a \sin b(x - h) + k$. Here $a = 3$, $b = \pi$, $h = -1$, $k = -2$.

Domain: $\mathbb{R}$. Range: $-5 \le y \le 1$. Amplitude $= |a| = 3$. Period $= \frac{2\pi}{|b|} = 2$. Midline: $y = -2$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \sin x$:

$(0, 0) \to (-1, -2)$

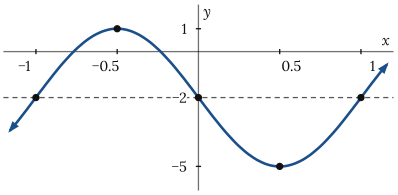
$(\frac{\pi}{2}, 1) \to (-0.5, 1)$

$(\pi, 0) \to (0, -2)$

$(\frac{3\pi}{2}, -1) \to (0.5, -5)$

$(2\pi, 0) \to (1, -2)$

Key points: $(-1, -2), (-0.5, 1), (0, -2), (0.5, -5), (1, -2)$.



Q3. Draw the graph of $y = 3 \sin \pi(x - 1) + 2$. [26-Q]

Ans.

Compare with $y = a \sin b(x - h) + k$. Here $a = 3$, $b = \pi$, $h = 1$, $k = 2$.

Domain: $\mathbb{R}$. Range: $-1 \le y \le 5$. Amplitude $= |a| = 3$. Period $= \frac{2\pi}{|b|} = 2$. Midline: $y = 2$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \sin x$:

$(0, 0) \to (1, 2)$

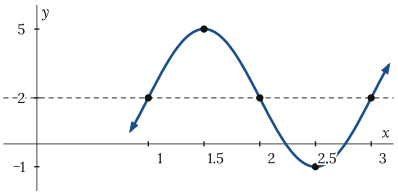
$(\frac{\pi}{2}, 1) \to (1.5, 5)$

$(\pi, 0) \to (2, 2)$

$(\frac{3\pi}{2}, -1) \to (2.5, -1)$

$(2\pi, 0) \to (3, 2)$

Key points: $(1, 2), (1.5, 5), (2, 2), (2.5, -1), (3, 2)$.



Q4. Draw the graph of $y = 2\sin\frac{\pi}{3}(x+1)+3$. [24-Q]

Ans.

Compare with $y = a\sin b(x-h)+k$. Here $a = 2$, $b = \frac{\pi}{3}$, $h = -1$, $k = 3$.

Domain: $\mathbb{R}$. Range: $1 \le y \le 5$. Amplitude $= |a| = 2$. Period $= \frac{2\pi}{|b|} = 6$. Midline: $y = 3$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \sin x$:

$(0, 0) \to (-1, 3)$

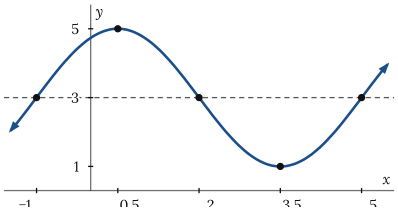
$(\frac{\pi}{2}, 1) \to (0.5, 5)$

$(\pi, 0) \to (2, 3)$

$(\frac{3\pi}{2}, -1) \to (3.5, 1)$

$(2\pi, 0) \to (5, 3)$

Key points: $(-1, 3)$, $(0.5, 5)$, $(2, 3)$, $(3.5, 1)$, $(5, 3)$.



Q5. Draw the graph of $y = 3\sin\frac{\pi}{3}(x-1)+4$. [25-Q]

Ans.

Compare with $y = a\sin b(x-h)+k$. Here $a = 3$, $b = \frac{\pi}{3}$, $h = 1$, $k = 4$.

Domain: $\mathbb{R}$. Range: $1 \le y \le 7$. Amplitude $= |a| = 3$. Period $= \frac{2\pi}{|b|} = 6$. Midline: $y = 4$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \sin x$:

$(0, 0) \to (1, 4)$

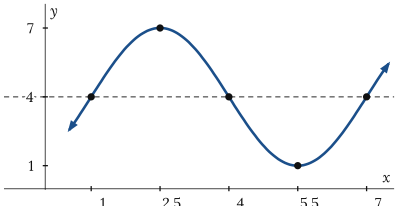
$(\frac{\pi}{2}, 1) \to (2.5, 7)$

$(\pi, 0) \to (4, 4)$

$(\frac{3\pi}{2}, -1) \to (5.5, 1)$

$(2\pi, 0) \to (7, 4)$

Key points: $(1, 4)$, $(2.5, 7)$, $(4, 4)$, $(5.5, 1)$, $(7, 4)$.



Q6. Draw the graph of $y = 2\sin\frac{\pi}{3}(x-2)+1$. [26-Q]

Ans.

Compare with $y = a\sin b(x-h)+k$. Here $a = 2$, $b = \frac{\pi}{3}$, $h = 2$, $k = 1$.

Domain: $\mathbb{R}$. Range: $-1 \le y \le 3$. Amplitude $= |a| = 2$. Period $= \frac{2\pi}{|b|} = 6$. Midline: $y = 1$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \sin x$:

$(0, 0) \to (2, 1)$

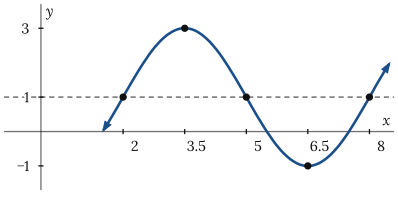
$(\frac{\pi}{2}, 1) \to (3.5, 3)$

$(\pi, 0) \to (5, 1)$

$(\frac{3\pi}{2}, -1) \to (6.5, -1)$

$(2\pi, 0) \to (8, 1)$

Key points: $(2, 1)$, $(3.5, 3)$, $(5, 1)$, $(6.5, -1)$, $(8, 1)$.



Q7. Sketch the graph of $y = 3\sin(\frac{x}{2} + \frac{\pi}{4}) - 5$. [24-Q]

Ans.

Compare with $y = a\sin b(x-h)+k$. Here $a = 3$, $b = 0.5$, $h = -\frac{\pi}{2}$, $k = -5$.

Domain: $\mathbb{R}$. Range: $-8 \le y \le -2$. Amplitude $= |a| = 3$. Period $= \frac{2\pi}{|b|} = 4\pi$. Midline: $y = -5$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \sin x$:

$(0, 0) \to (-\frac{\pi}{2}, -5)$

$(\frac{\pi}{2}, 1) \to (\frac{\pi}{2}, -2)$

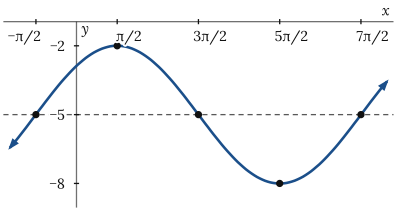
$(\pi, 0) \to (\frac{3\pi}{2}, -5)$

$(\frac{3\pi}{2}, -1) \to (\frac{5\pi}{2}, -8)$

$(2\pi, 0) \to (\frac{7\pi}{2}, -5)$

Key points: $(-\frac{\pi}{2}, -5)$, $(\frac{\pi}{2}, -2)$, $(\frac{3\pi}{2}, -5)$, $(\frac{5\pi}{2}, -8)$, $(\frac{7\pi}{2}, -5)$.

Rewrite the argument: $\frac{x}{2} + \frac{\pi}{4} = \frac{1}{2}(x + \frac{\pi}{2})$, so $b = \frac{1}{2}$ and $h = -\frac{\pi}{2}$.



Q8. Draw the graph of $y = 3\sin 2(x - \frac{\pi}{4}) + 1$. [25*-Q]

Ans.

Compare with $y = a\sin b(x-h)+k$. Here $a = 3$, $b = 2$, $h = \frac{\pi}{4}$, $k = 1$.

Domain: $\mathbb{R}$. Range: $-2 \le y \le 4$. Amplitude $= |a| = 3$. Period $= \frac{2\pi}{|b|} = \pi$. Midline: $y = 1$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \sin x$:

$(0, 0) \to (\frac{\pi}{4}, 1)$

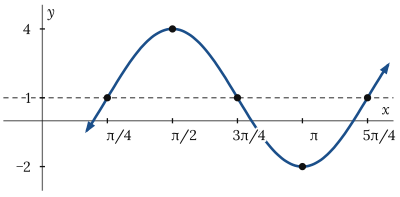
$(\frac{\pi}{2}, 1) \to (\frac{\pi}{2}, 4)$

$(\pi, 0) \to (\frac{3\pi}{4}, 1)$

$(\frac{3\pi}{2}, -1) \to (\pi, -2)$

$(2\pi, 0) \to (\frac{5\pi}{4}, 1)$

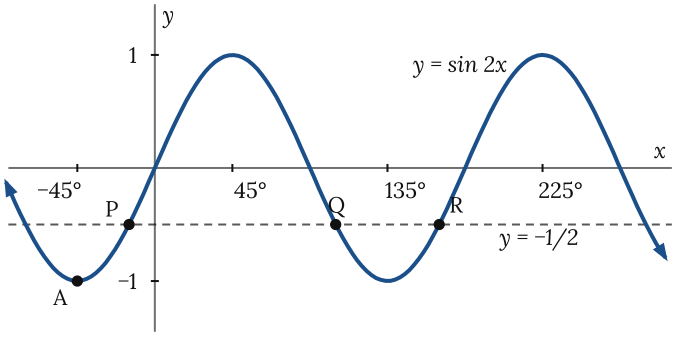
Key points: $(\frac{\pi}{4}, 1)$, $(\frac{\pi}{2}, 4)$, $(\frac{3\pi}{4}, 1)$, $(\pi, -2)$, $(\frac{5\pi}{4}, 1)$.



Q9. The graph below shows the curve $y = \sin 2x$ in the interval $-60° \le x \le 270°$.

(a) Point A has coordinates $(-45°, -1)$ and is the minimum point closest to the origin. Point B is the maximum point closest to the origin. State the coordinates of B.

(b) A straight line with equation $y = -\frac{1}{2}$ meets the graph at three points P, Q and R as shown in the diagram. Given that point P has coordinates $(-15°, -\frac{1}{2})$, use graph symmetries to determine the coordinates of Q and R.



[5-Q]

Ans.

(a) The maximum point closest to the origin: $\sin 2x = 1$ first happens when $2x = 90°$, i.e. $x = 45°$.

B = (45°, 1)

(b) The graph is symmetric about the vertical lines through its turning points. The minimum points are at $2x = -90°$ and $2x = 270°$, i.e. $x = -45°$ (point A) and $x = 135°$.

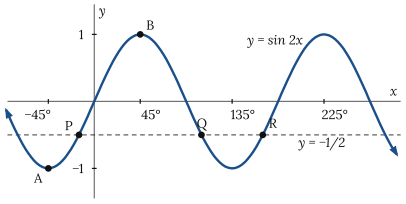
P(−15°, $-\frac{1}{2}$) is 30° to the right of A, because −15° − (−45°) = 30°. By symmetry about x = 135°, Q is 30° to the left of the minimum and R is 30° to the right of it.

x_Q = 135° − 30° = 105°

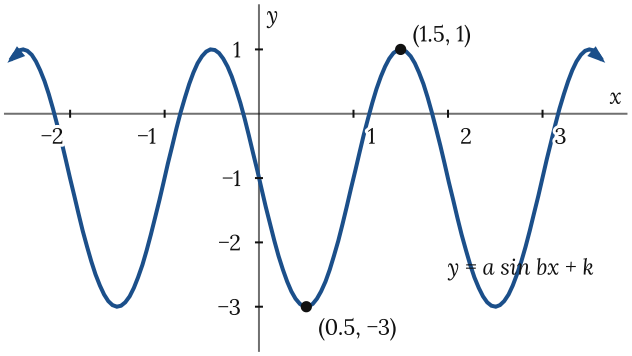
x_R = 135° + 30° = 165°

Check: sin(2 × 105°) = sin 210° = $-\frac{1}{2}$ and sin(2 × 165°) = sin 330° = $-\frac{1}{2}$.

Answer: Q(105°, −0.5), R(165°, −0.5)



Q10. The graph represents y = a sin bx + k, where a and b are constants. Two points – one a minimum and the other a maximum – are indicated on the graph.
(a) Find the values of a, b and k.
(b) For these values of a, b and k, find the step–by–step transformation of the function y = a sin bx + k from the function y = sin x.



[24-Q]

Ans.
(a) The minimum is −3 and the maximum is 1, so

$k = \frac{1+(-3)}{2} = -1$

$|a| = \frac{1-(-3)}{2} = 2$

The minimum is at x = 0.5 and the maximum is at x = 1.5, so half a period is 1 and the period is 2:

$\frac{2\pi}{b} = 2$

$b = \pi$

The graph passes through (0, −1) and then goes **down** to the minimum (0.5, −3). A normal sine curve goes up first, so a is negative: a = −2.
Answer: a = −2, b = π, k = −1
Check: y = −2 sin(π·0.5) − 1 = −3 and y = −2 sin(π·1.5) − 1 = 1. ✓
(b) Step–by–step transformations:
y = sin x → y = 2 sin x → y = 2 sin πx → y = −2 sin πx → y = −2 sin πx − 1
(vertical stretch by the factor 2; horizontal compression by the factor $\frac{1}{\pi}$; reflection in the x–axis; shift down by 1. The reflection must come before the shift, otherwise the midline would move to y = +1.)

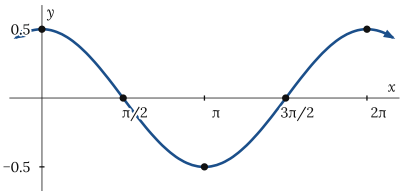
7.2 Graphs of Cosine Functions

Q11. Draw the graph of y = $\frac{1}{2}$ cos x. [26-Q]

Ans.
Compare with y = a cos b(x − h) + k. Here a = 0.5, b = 1, h = 0, k = 0.

Domain: ℝ. Range: −0.5 ≤ y ≤ 0.5. Amplitude = |a| = 0.5. Period = $\frac{2\pi}{|b|} = 2\pi$. Midline: y = 0.

Apply (x, y) → ($\frac{x}{b}$ + h, ay + k) to the five key points of y = cos x:
(0, 1) → (0, 0.5)
($\frac{\pi}{2}$, 0) → ($\frac{\pi}{2}$, 0)
(π, −1) → (π, −0.5)
($\frac{3\pi}{2}$, 0) → ($\frac{3\pi}{2}$, 0)
(2π, 1) → (2π, 0.5)
Key points: (0, 0.5), ($\frac{\pi}{2}$, 0), (π, −0.5), ($\frac{3\pi}{2}$, 0), (2π, 0.5).

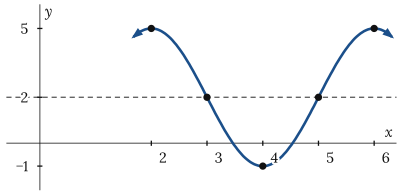


Q12. Draw the graph of y = 3 cos $\frac{\pi}{2}$(x − 2) + 2. Determine the given function is even or odd? [5-Q]

Ans.
Compare with y = a cos b(x − h) + k. Here a = 3, b = $\frac{\pi}{2}$, h = 2, k = 2.

Domain: ℝ. Range: −1 ≤ y ≤ 5. Amplitude = |a| = 3. Period = $\frac{2\pi}{|b|} = 4$. Midline: y = 2.

Apply (x, y) → ($\frac{x}{b}$ + h, ay + k) to the five key points of y = cos x:
(0, 1) → (2, 5)
($\frac{\pi}{2}$, 0) → (3, 2)
(π, −1) → (4, −1)
($\frac{3\pi}{2}$, 0) → (5, 2)
(2π, 1) → (6, 5)
Key points: (2, 5), (3, 2), (4, −1), (5, 2), (6, 5).



Even or odd? Simplify the function: 3 cos $\frac{\pi}{2}$(x − 2) + 2 = 3 cos($\frac{\pi x}{2}$ − π) + 2 = −3 cos $\frac{\pi x}{2}$ + 2.

f(−x) = −3 cos(−$\frac{\pi x}{2}$) + 2

= −3 cos $\frac{\pi x}{2}$ + 2

= f(x)

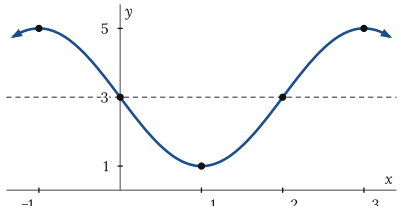
Since f(−x) = f(x), the function is even (its graph is symmetric about the y–axis).

Q13. Draw the graph of y = 2 cos $\frac{\pi}{2}$(x + 1) + 3. [25-Q]

Ans.
Compare with y = a cos b(x − h) + k. Here a = 2, b = $\frac{\pi}{2}$, h = −1, k = 3.

Domain: ℝ. Range: 1 ≤ y ≤ 5. Amplitude = |a| = 2. Period = $\frac{2\pi}{|b|} = 4$. Midline: y = 3.

Apply (x, y) → ($\frac{x}{b}$ + h, ay + k) to the five key points of y = cos x:
(0, 1) → (−1, 5)
($\frac{\pi}{2}$, 0) → (0, 3)
(π, −1) → (1, 1)
($\frac{3\pi}{2}$, 0) → (2, 3)
(2π, 1) → (3, 5)
Key points: (−1, 5), (0, 3), (1, 1), (2, 3), (3, 5).



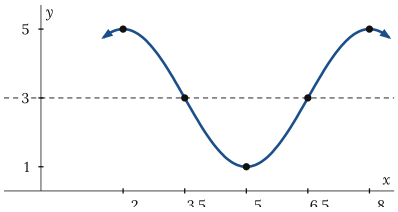
Q14. Draw the graph of y = 2 cos $\frac{\pi}{3}$(x − 2) + 3. [25-Q]

Ans.
Compare with y = a cos b(x − h) + k. Here a = 2, b = $\frac{\pi}{3}$, h = 2, k = 3.

Domain: ℝ. Range: 1 ≤ y ≤ 5. Amplitude = |a| = 2. Period = $\frac{2\pi}{|b|} = 6$. Midline: y = 3.

Apply (x, y) → ($\frac{x}{b}$ + h, ay + k) to the five key points of y = cos x:
(0, 1) → (2, 5)
($\frac{\pi}{2}$, 0) → (3.5, 3)
(π, −1) → (5, 1)

$(\frac{3\pi}{2}, 0) \to (6.5, 3)$
$(2\pi, 1) \to (8, 5)$
Key points: (2, 5), (3.5, 3), (5, 1), (6.5, 3), (8, 5).



Q15. Draw the graph of $y = 2\cos\frac{\pi}{3}(x + 1) + 3$. [25-Q]

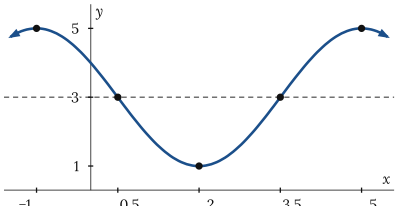
Ans.

Compare with $y = a\cos b(x - h) + k$. Here $a = 2$, $b = \frac{\pi}{3}$, $h = -1$, $k = 3$.

Domain: $\mathbb{R}$. Range: $1 \le y \le 5$. Amplitude $= |a| = 2$. Period $= \frac{2\pi}{|b|} = 6$. Midline: $y = 3$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \cos x$:

$(0, 1) \to (-1, 5)$
$(\frac{\pi}{2}, 0) \to (0.5, 3)$
$(\pi, -1) \to (2, 1)$
$(\frac{3\pi}{2}, 0) \to (3.5, 3)$
$(2\pi, 1) \to (5, 5)$
Key points: (−1, 5), (0.5, 3), (2, 1), (3.5, 3), (5, 5).



Q16. Draw the graph of $y = 2\cos\frac{\pi}{6}(x + 1) + 3$. [25*-Q]

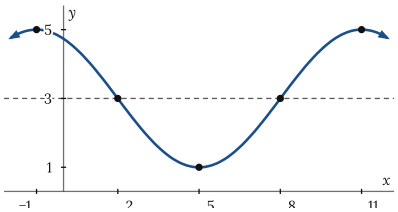
Ans.

Compare with $y = a\cos b(x - h) + k$. Here $a = 2$, $b = \frac{\pi}{6}$, $h = -1$, $k = 3$.

Domain: $\mathbb{R}$. Range: $1 \le y \le 5$. Amplitude $= |a| = 2$. Period $= \frac{2\pi}{|b|} = 12$. Midline: $y = 3$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \cos x$:

$(0, 1) \to (-1, 5)$
$(\frac{\pi}{2}, 0) \to (2, 3)$
$(\pi, -1) \to (5, 1)$
$(\frac{3\pi}{2}, 0) \to (8, 3)$
$(2\pi, 1) \to (11, 5)$
Key points: (−1, 5), (2, 3), (5, 1), (8, 3), (11, 5).



Q17. Draw the graph of $y = 4\cos 3(x + \frac{\pi}{3}) - 2$. [24-Q]

Ans.

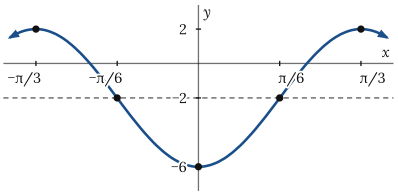
Compare with $y = a\cos b(x - h) + k$. Here $a = 4$, $b = 3$, $h = -\frac{\pi}{3}$, $k = -2$.

Domain: $\mathbb{R}$. Range: $-6 \le y \le 2$. Amplitude $= |a| = 4$. Period $= \frac{2\pi}{|b|} = \frac{2\pi}{3}$. Midline: $y = -2$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \cos x$:

$(0, 1) \to (-\frac{\pi}{3}, 2)$
$(\frac{\pi}{2}, 0) \to (-\frac{\pi}{6}, -2)$
$(\pi, -1) \to (0, -6)$
$(\frac{3\pi}{2}, 0) \to (\frac{\pi}{6}, -2)$
$(2\pi, 1) \to (\frac{\pi}{3}, 2)$
Key points: $(-\frac{\pi}{3}, 2), (-\frac{\pi}{6}, -2), (0, -6), (\frac{\pi}{6}, -2), (\frac{\pi}{3}, 2)$.

(The book key lists the points as $(-\frac{\pi}{3}, 2), (-\frac{2\pi}{9}, -2), (-\frac{\pi}{9}, -6), (0, -2), (\frac{\pi}{9}, 2)$; those x–values do not fit its own period $\frac{2\pi}{3}$, so the values computed here are used.)



Q18. Draw the graph of $y = \cos(x - \frac{3\pi}{4}) + 3$. [25-Q]

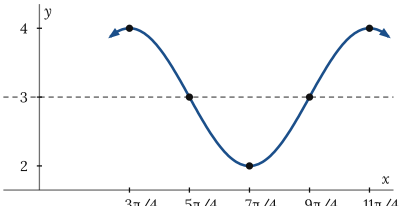
Ans.

Compare with $y = a\cos b(x - h) + k$. Here $a = 1$, $b = 1$, $h = \frac{3\pi}{4}$, $k = 3$.

Domain: $\mathbb{R}$. Range: $2 \le y \le 4$. Amplitude $= |a| = 1$. Period $= \frac{2\pi}{|b|} = 2\pi$. Midline: $y = 3$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \cos x$:

$(0, 1) \to (\frac{3\pi}{4}, 4)$
$(\frac{\pi}{2}, 0) \to (\frac{5\pi}{4}, 3)$
$(\pi, -1) \to (\frac{7\pi}{4}, 2)$
$(\frac{3\pi}{2}, 0) \to (\frac{9\pi}{4}, 3)$
$(2\pi, 1) \to (\frac{11\pi}{4}, 4)$
Key points: $(\frac{3\pi}{4}, 4), (\frac{5\pi}{4}, 3), (\frac{7\pi}{4}, 2), (\frac{9\pi}{4}, 3), (\frac{11\pi}{4}, 4)$.



Q19. Sketch the graph of $y = 5\cos(2x + \frac{\pi}{3}) - 1$. [25-Q]

Ans.

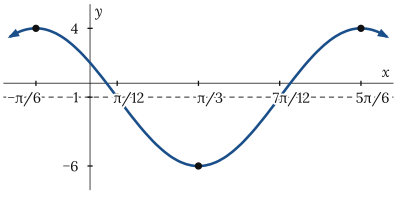
Compare with $y = a\cos b(x - h) + k$. Here $a = 5$, $b = 2$, $h = -\frac{\pi}{6}$, $k = -1$.

Domain: $\mathbb{R}$. Range: $-6 \le y \le 4$. Amplitude $= |a| = 5$. Period $= \frac{2\pi}{|b|} = \pi$. Midline: $y = -1$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \cos x$:

$(0, 1) \to (-\frac{\pi}{6}, 4)$
$(\frac{\pi}{2}, 0) \to (\frac{\pi}{12}, -1)$
$(\pi, -1) \to (\frac{\pi}{3}, -6)$
$(\frac{3\pi}{2}, 0) \to (\frac{7\pi}{12}, -1)$
$(2\pi, 1) \to (\frac{5\pi}{6}, 4)$
Key points: $(-\frac{\pi}{6}, 4), (\frac{\pi}{12}, -1), (\frac{\pi}{3}, -6), (\frac{7\pi}{12}, -1), (\frac{5\pi}{6}, 4)$.

Rewrite the argument: $2x + \frac{\pi}{3} = 2(x + \frac{\pi}{6})$, so $b = 2$ and $h = -\frac{\pi}{6}$.



Q20. Draw the graph of $y = 2\cos(\frac{x}{2} + \frac{\pi}{4}) - 3$. [25-Q]

Ans.

Compare with $y = a\cos b(x - h) + k$. Here $a = 2$, $b = 0.5$, $h = -\frac{\pi}{2}$, $k = -3$.

Domain: $\mathbb{R}$. Range: $-5 \le y \le -1$. Amplitude $= |a| = 2$. Period $= \frac{2\pi}{|b|} = 4\pi$. Midline: $y = -3$.

Apply $(x, y) \to (\frac{x}{b} + h, ay + k)$ to the five key points of $y = \cos x$:

$(0, 1) \to (-\frac{\pi}{2}, -1)$

$(\frac{\pi}{2}, 0) \to (\frac{\pi}{2}, -3)$

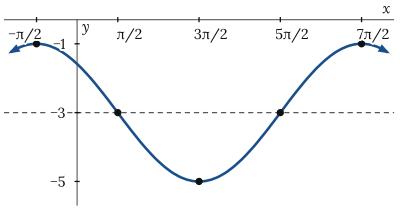
$(\pi, -1) \to (\frac{3\pi}{2}, -5)$

$(\frac{3\pi}{2}, 0) \to (\frac{5\pi}{2}, -3)$

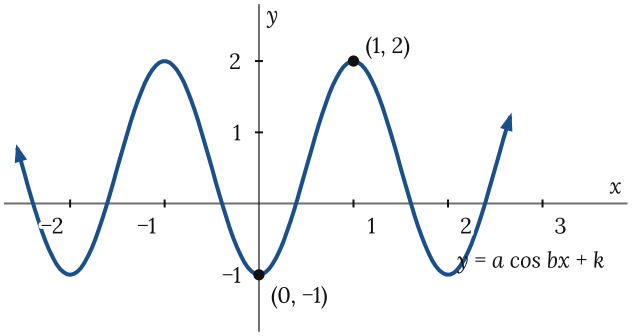
$(2\pi, 1) \to (\frac{7\pi}{2}, -1)$

Key points: $(-\frac{\pi}{2}, -1), (\frac{\pi}{2}, -3), (\frac{3\pi}{2}, -5), (\frac{5\pi}{2}, -3), (\frac{7\pi}{2}, -1)$.

Rewrite the argument: $\frac{x}{2} + \frac{\pi}{4} = \frac{1}{2}(x + \frac{\pi}{2})$, so $b = \frac{1}{2}$ and $h = -\frac{\pi}{2}$. (The book key prints the range as $-5 \le y \le 1$; with midline -3 and amplitude 2 the range is $-5 \le y \le -1$, as shown here.)



Q21. The graph represents $y = a \cos bx + k$, where a, b, and k are constants. Two points – one a minimum and the other a maximum – are indicated on the graph.
(a) Find the values of a, b, and k.
(b) For these values of a, b, and k, find the step–by–step transformation of the function $y = a \cos bx + k$ from the function $y = \cos x$.



[25-Q]

Ans.
(a) The minimum is -1 and the maximum is 2, so

$k = \frac{2 + (-1)}{2} = 0.5$

$|a| = \frac{2 - (-1)}{2} = 1.5$

The minimum is at $x = 0$ and the next maximum is at $x = 1$, so half a period is 1 and the period is 2:

$\frac{2\pi}{b} = 2$

$b = \pi$

The graph starts at a **minimum** when $x = 0$, but a normal cosine curve starts at a maximum, so a is negative: $a = -1.5$.
Answer: a = −1.5, b = π, k = 0.5
Check: $y = -1.5 \cos 0 + 0.5 = -1$ and $y = -1.5 \cos \pi + 0.5 = 2$. ✓
(b) Step–by–step transformations:
$y = \cos x \to y = 1.5 \cos x \to y = 1.5 \cos \pi x \to y = -1.5 \cos \pi x \to y = -1.5 \cos \pi x + 0.5$
(vertical stretch by 1.5; horizontal compression by the factor $\frac{1}{\pi}$; reflection in the x–axis; shift up by 0.5.)

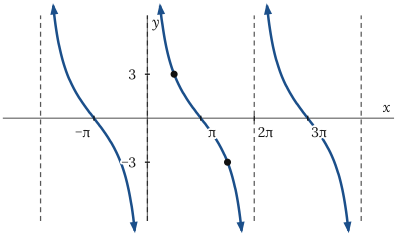
7.3 Graphs of Other Trigonometric Functions

Q22. Draw the graph of $y = 3 \cot \frac{x}{2}$ and determine its domain, range, period and asymptotes. [25-Q]

Ans.
$y = 3 \cot \frac{x}{2}$ is obtained from $y = \cot x$ by the rule $(x, y) \to (\frac{x}{b}, ay)$ with $a = 3$ and $b = \frac{1}{2}$.

Period $= \frac{\pi}{b} = \frac{\pi}{\frac{1}{2}} = 2\pi$

Since $\cot x = \frac{\cos x}{\sin x}$, the function is undefined when $\sin \frac{x}{2} = 0$, i.e. when $\frac{x}{2} = 0, \pm\pi, \pm 2\pi, \pm 4\pi, \ldots$
Asymptotes: $x = 0, \pm 2\pi, \pm 4\pi, \ldots$
x–intercepts: $\cos \frac{x}{2} = 0$ gives $x = \pm\pi, \pm 3\pi, \ldots$ Other points: $(\frac{\pi}{2}, 3)$ and $(\frac{3\pi}{2}, -3)$, because $\cot \frac{\pi}{4} = 1$ and $\cot \frac{3\pi}{4} = -1$.



Domain: $x \ne 0, \pm 2\pi, \pm 4\pi, \ldots$ i.e. $\{x \in \Re : x \ne 0, \pm 2\pi, \pm 4\pi, \ldots\}$
Range: all real numbers $\Re$.
Period: 2π.

7.4 Inverse Trigonometric Functions

Q23. Evaluate (a) $\cos^{-1} \frac{\sqrt{2}}{2}$ (b) $\sin(\cos^{-1} \frac{\sqrt{3}}{2})$ (c) $\cos^{-1}(\cos(-\frac{1}{2}))$. [24-Q]

Ans.
(a) $\cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$ and $\frac{\pi}{4} \in [0, \pi]$, so $\cos^{-1} \frac{\sqrt{2}}{2} = \frac{\pi}{4}$.

(b) $\cos^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{6}$, so $\sin(\cos^{-1} \frac{\sqrt{3}}{2}) = \sin \frac{\pi}{6} = \frac{1}{2}$.

(c) $\cos(-\frac{1}{2}) = \cos \frac{1}{2}$ and $\frac{1}{2} \in [0, \pi]$, so $\cos^{-1}(\cos(-\frac{1}{2})) = \frac{1}{2}$.

Answer: (a) $\frac{\pi}{4}$, (b) $\frac{1}{2}$, (c) $\frac{1}{2}$

Q24. Find the value of $\cos(\sin^{-1} x)$. Then use the result and find the value of $\cos(\sin^{-1} x + \frac{\pi}{3})$ for $-1 \le x \le 1$. [24-Q]

Ans.
Let $\theta = \sin^{-1} x$, so $\sin \theta = x$ and $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$. On this interval $\cos \theta \ge 0$.

$\cos^2\theta = 1 - \sin^2\theta = 1 - x^2$

Since $\cos \theta \ge 0$, $\cos \theta = \sqrt{1 - x^2}$.
$\cos(\sin^{-1} x) = \sqrt{1 - x^2}$

Now use $\cos(A + B) = \cos A \cos B - \sin A \sin B$ with $A = \theta$ and $B = \frac{\pi}{3}$:

$$\cos(\theta + \tfrac{\pi}{3}) = \cos\theta \cos\tfrac{\pi}{3} - \sin\theta \sin\tfrac{\pi}{3}$$
$$= \sqrt{1 - x^2} \cdot \frac{1}{2} - x \cdot \frac{\sqrt{3}}{2}$$
$$= \frac{\sqrt{1 - x^2} - \sqrt{3}\,x}{2}$$

Answer: $\frac{\sqrt{1 - x^2} - x\sqrt{3}}{2}$

Q25. Prove that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$. [25-Q]

Ans.
Let $\theta = \sin^{-1} x$. Then $\sin \theta = x$ and $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$.

Using $\sin \theta = \cos(\frac{\pi}{2} - \theta)$, we get $\cos(\frac{\pi}{2} - \theta) = x$.

From $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ we get $0 \le \frac{\pi}{2} - \theta \le \pi$, which is exactly the range of $\cos^{-1}$. Hence

$\cos^{-1} x = \frac{\pi}{2} - \theta = \frac{\pi}{2} - \sin^{-1} x$

Therefore $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$. ✓

7.5 Differentiation of Trigonometric Functions

Q26. Differentiate the following functions with respect to x. (a) $y = 2\pi x + 2 \cos \pi x$ (b) $y = \sin(1 - x^2)$ (c) $y = x \sin x$ [24-Q]

Ans.
(a) Differentiate term by term; for $\cos \pi x$ use the chain rule, $\frac{d}{dx}(\pi x) = \pi$.

$$\frac{dy}{dx} = 2\pi + 2(-\sin \pi x) \cdot \pi$$
$$= 2\pi - 2\pi \sin \pi x$$
$$= 2\pi(1 - \sin \pi x)$$

(b) Chain rule with inner function $1 - x^2$, whose derivative is $-2x$:

$$\frac{dy}{dx} = \cos(1 - x^2) \cdot (-2x)$$
$$= -2x \cos(1 - x^2)$$

(c) Product rule with $u = x$, $v = \sin x$ ($u' = 1$, $v' = \cos x$):

$$\frac{dy}{dx} = x \cos x + \sin x \cdot 1$$

= sin x + x cos x

Answer: (a) $2\pi(1 - \sin \pi x)$ (b) $-2x \cos(1 - x^2)$ (c) sin x + x cos x

CH.08 Logarithmic and Exponential Functions

Notes · Questions with Answers

NOTES

Logarithmic Function

For a number $b > 0$, $b \neq 1$, the **logarithmic function** $y = \log_b x$ is defined as: y is the number such that $x = b^y$, for $x > 0$.

Exponential Function

A function of the form $y = b^x$, where $b > 0$, $b \neq 1$, is called an **exponential function** of x. $y = \log_b x$ and $y = b^x$ are inverse of each other.

Exponential Growth and Decay

For $b > 1$, $y = b^x$ is an **exponential growth** function, such as $y = 2^x$ and $y = 3^x$.

For $0 < b < 1$, $y = b^x$ is an **exponential decay** function, such as $y = 2^{-x} = (\frac{1}{2})^x$ and $y = 3^{-x} = (\frac{1}{3})^x$.

Exponential Number e

The limit $\lim_{t \to 0} (1 + t)^{1/t} = 2.7183\ldots$ is denoted by **e**, called the exponential number. e is an irrational number.

Natural (Napierian) Logarithm

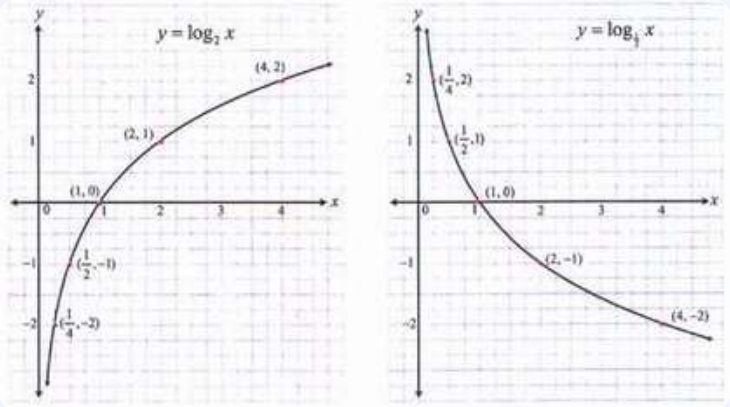
The logarithm of x to the base e is called the **natural logarithm** (Napierian logarithm), denoted $\log_e x = \ln x$.

Logarithmic Function

Domain of $y = \log_b x$: $x > 0$ Range: R Asymptote: y-axis ($x = 0$)

For $b > 1$, $y = \log_b x$ is an **increasing** function (if $x_1 > x_2$ then $\log_b x_1 > \log_b x_2$), e.g. $y = \log_2 x$.

For $0 < b < 1$, $y = \log_b x$ is a **decreasing** function (if $x_1 > x_2$ then $\log_b x_1 < \log_b x_2$), e.g. $y = \log_{1/2} x$.



b > 1	0 < b < 1
$\log_b x < 0$ if $0 < x < 1$ $\log_b x = 0$ if $x = 1$ $\log_b x > 0$ if $x > 1$	$\log_b x > 0$ if $0 < x < 1$ $\log_b x = 0$ if $x = 1$ $\log_b x < 0$ if $x > 1$

Transformations of Logarithmic Graphs

Since $\log_{1/b} x = -\log_b x$:

$y = \log_b x \rightarrow$ (reflection on x-axis) $\rightarrow y = \log_{1/b} x$

For $b > 1$: $\log_b x = \frac{\log_2 x}{\log_2 b}$, so the graph of $y = \log_b x$ is a vertical scaling of $y = \log_2 x$ with scale factor $\frac{1}{\log_2 b}$.

For $0 < b < 1$: $\log_b x = \frac{\log_{1/2} x}{\log_{1/2} b}$, a vertical scaling of $y = \log_{1/2} x$ with scale factor $\frac{1}{\log_{1/2} b}$.

So every graph of $y = \log_b x$ is a vertical scaling of the graph of $y = \log_2 x$ or $y = \log_{1/2} x$.

Derivative of Logarithmic Functions

$$\frac{d}{dx} \log_b x = \frac{1}{x} \log_b e$$

$$\frac{d}{dx} \log_b u(x) = \frac{1}{u(x)} \cdot \log_b e \cdot \frac{du}{dx}$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} \ln u(x) = \frac{1}{u(x)} \cdot \frac{du}{dx}$$

Derivation. Let $y = \log_b x$ and let δx be a small increment. Then $\delta y = \log_b(x + \delta x) - \log_b x = \log_b \frac{x + \delta x}{x}$.

$$\frac{\delta y}{\delta x} = \frac{1}{\delta x} \log_b\left(1 + \frac{\delta x}{x}\right) = \frac{1}{x} \log_b (1 + t)^{1/t}, \quad t = \frac{\delta x}{x}$$

When $\delta x \to 0$, $t \to 0$, so $\frac{dy}{dx} = \frac{1}{x} \log_b \lim_{t \to 0} (1 + t)^{1/t} = \frac{1}{x} \log_b e$.

t	1	0.5	0.25	0.10	0.01	0.001	0.0001	0.00001
$\frac{1}{t}$	1	2	4	10	100	1000	10000	100000
$(1 + t)^{1/t}$	2	2.25	2.4414	2.5937	2.7048	2.7169	2.7181	2.7183

The table shows $(1 + t)^{1/t} \to 2.7183$ as $t \to 0$; this limit is denoted by e. Since $\log_e e = 1$, $\frac{d}{dx} \ln x = \frac{1}{x}$.

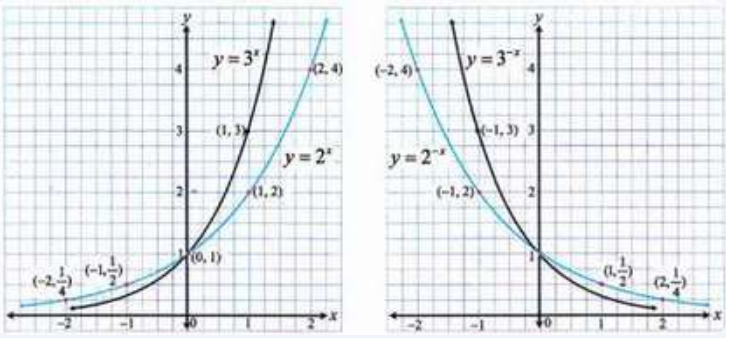
Logarithmic Differentiation

The derivatives of positive functions can be found by taking the natural logarithm of both sides before differentiating:

1. Take ln of both sides: $\ln y = \ln f(x)$, and simplify using the laws of logarithms.
2. Differentiate both sides with respect to x: $\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx} [\ln f(x)]$.
3. Multiply by y: $\frac{dy}{dx} = y \cdot \frac{d}{dx} [\ln f(x)]$, and replace y by f(x).

Exponential Function

Domain of $y = b^x$: R Range: $y > 0$ Asymptote: x-axis ($y = 0$)



$y = b^x \rightarrow$ (reflection on y-axis) $\rightarrow y = b^{-x}$

From $y = b^x$: $y = ab^x$ (vertical scaling, factor $a > 0$); $y = ab^{x-h} + k$ (horizontal translation h units, vertical translation k units).

Key Points of y = ab^(x − h) + k

Points (0, 1) and (1, b) on $y = b^x$ correspond to

$(0, 1) \rightarrow (0, a) \rightarrow (h, a + k)$

$(1, b) \rightarrow (1, ab) \rightarrow (1 + h, ab + k)$

Asymptote of $y = ab^{x-h} + k$: $y = k$.

Range: if $a > 0$, $\{y : y > k\}$; if $a < 0$, $\{y : y < k\}$.

Derivative of Exponential Functions

$$\frac{d}{dx}\, b^x = b^x \ln b$$

$$\frac{d}{dx}\, b^{u(x)} = b^{u(x)} \cdot \ln b \cdot \frac{du}{dx}$$

$$\frac{d}{dx}\, e^x = e^x$$

$$\frac{d}{dx}\, e^{u(x)} = e^{u(x)} \cdot \frac{du}{dx}$$

Derivation. Let $y = b^x$, so $x = \log_b y$. Differentiate both sides with respect to x: $1 = \frac{1}{y} \log_b e \cdot \frac{dy}{dx}$, so

$$\frac{dy}{dx} = y \cdot \frac{1}{\log_b e} = \frac{b^x}{\log_e b} = b^x \ln b$$

1 MARK

8.1 Logarithmic Functions

Q1. The function $y = \log_b x$ is an increasing function if [25-Q]

A. $0 < b \geq 1$ **B.** $b > 0$ **C.** $b > 1$ **D.** $0 < b < 1$

Ans.
$y = \log_b x$ is increasing when its base is greater than 1, and decreasing when $0 < b < 1$.
Answer: C (b > 1)

Q2. The function $y = \log_b x$ is a decreasing function if [26-Q]

A. $0 < b < 1$ **B.** $0 < b \leq 1$ **C.** $b > 0$ **D.** $b > 1$

Ans.
The base of a logarithm must satisfy $b > 0$, $b \neq 1$. The graph falls as x increases exactly when $0 < b < 1$.
Answer: A

Q3. The reflection in the x–axis of the graph of $y = \log_b x$ is [24-Q]

A. $\log_b x$ **B.** $\log_{1/b} x$ **C.** $\log_{10} x$ **D.** $\log_b 2$

Ans.
Reflection in the x–axis gives $y = -\log_b x$. Using $\log_b x = \frac{\log x}{\log b}$:

$$-\log_b x = -\frac{\log x}{\log b} = \frac{\log x}{-\log b} = \frac{\log x}{\log \frac{1}{b}} = \log_{1/b} x$$

Answer: B

Q4. The scale factor in the horizontal/vertical stretch from the graph $y = \log_{1/2} x$ to $y = \log_b x$ is [24-Q]

A. $\log b$ **B.** $\log \frac{1}{2}$ **C.** $\log_b \frac{1}{2}$ **D.** $\log_{1/2} b$

Ans.
Change of base: $\log_b x = \frac{\log_{1/2} x}{\log_{1/2} b}$. Since $\frac{1}{\log_{1/2} b} = \log_b \frac{1}{2}$:

$$\log_b x = \log_b \frac{1}{2} \cdot \log_{1/2} x$$

so $y = \log_{1/2} x$ is multiplied by the factor $\log_b \frac{1}{2}$.
Answer: C

Q5. The scale factor from $y = \log_2 x$ to $y = \log_{10} x$ (vertical stretch) is [24-Q]

A. $\frac{1}{\log_{1/2} 10}$ **B.** $\log_2 10$ **C.** $\frac{1}{\log_2 10}$ **D.** $\log_{1/2} 10$

Ans.
Change of base: $\log_{10} x = \frac{\log_2 x}{\log_2 10}$.

$$\log_{10} x = \frac{1}{\log_2 10} \cdot \log_2 x$$

The scale factor is $\frac{1}{\log_2 10}$.
Answer: C

Q6. The scale factor from $y = \log_3 x$ to $y = \log_{10} x$ (vertical stretch) is [25-Q]

A. $\frac{1}{\log_{1/3} 10}$ **B.** $\log_3 10$ **C.** $\frac{1}{\log_3 10}$ **D.** $\log_{1/3} 10$

Ans.
Change of base: $\log_{10} x = \frac{\log_3 x}{\log_3 10}$.

$$\log_{10} x = \frac{1}{\log_3 10} \cdot \log_3 x$$

The scale factor is $\frac{1}{\log_3 10}$.
Answer: C

Q7. The graph of $y = \log_2 x$ is transformed to $y = \log_2(x + 2) + 3$. The vertical translation is [24-Q]

A. 1 **B.** 2 **C.** 3 **D.** 4

Ans.
In $y = \log_2(x - h) + k$ the number k is the vertical translation. Here $k = 3$ (and the horizontal translation is 2 units to the left).
Answer: C

Q8. The graph of $y = \log_2 x$ is transformed to $y = \log_2(x + 2) - 2$. The vertical translation is [25-Q, 25*-Q]

A. 1 **B.** −2 **C.** 3 **D.** 4

Ans.
Here $k = -2$, so the graph moves 2 units down.
Answer: B

Q9. The graph of $f(x) = -\log_2(x - 2) + 1$ is translated 2 units to the left and 3 units up to give g(x). Then g(2) = [25-Q]

A. 4 **B.** −3 **C.** 3 **D.** −2

Ans.
Translate 2 left: replace x by x + 2; translate 3 up: add 3.

$$g(x) = -\log_2((x + 2) - 2) + 1 + 3 = -\log_2 x + 4$$

$$g(2) = -\log_2 2 + 4 = -1 + 4 = 3$$

Note: the printed key gives B (−3), which would be the answer if the graph were moved 3 units **down** ($g(x) = -\log_2 x - 2$). With the question as printed, the computed answer is 3.
Answer: C (3)

Q10. The domain of $y = 3 \log_{1/2}(-x)$ is [24-Q]

A. $\Re$ **B.** $x < 0$ **C.** $x > 0$ **D.** none of these

Ans.
The argument of a logarithm must be positive:

$$-x > 0$$
$$x < 0$$

Answer: B

Q11. The domain of $y = \log_{1/2}(x + 2) + 1$ is [25*-Q]

A. $\Re$ **B.** $x > -2$ **C.** $x > -1$ **D.** $x < -1$

Ans.

$$x + 2 > 0$$
$$x > -2$$

Answer: B

Q12. The domain of $y = \log_2(x - 2) + 1$ is [25-Q]

A. $0 < x < \pi$ **B.** $\Re$ **C.** $x > 2$ **D.** $x > 1$

Ans.

$$x - 2 > 0$$
$$x > 2$$

Answer: C

Q13. The domain of $y = 3 \ln(x - \pi) + 2$ is [24-Q]

A. $x > \pi$ **B.** $\Re$ **C.** $x > 2$ **D.** $x > \pi - 2$

Ans.

$$x - \pi > 0$$
$$x > \pi$$

Answer: A

Q14. The domain of $y = \ln(4 - x^2)$ is [26-Q]

A. $x < -2$ **B.** $x > 2$ **C.** $-2 < x < 2$ **D.** $0 < x < 2$

Ans.

$$4 - x^2 > 0$$
$$x^2 < 4$$
$$-2 < x < 2$$

Answer: C

Q15. The domain of $y = \log_2 x^2$ is [24-Q]

A. $\Re \setminus \{0, 1\}$ **B.** $\Re \setminus \{0\}$ **C.** $\Re$ **D.** $\Re \setminus \{1\}$

Ans.
We need $x^2 > 0$. This is true for every x except $x = 0$.
Answer: B ($\Re \setminus \{0\}$)

Q16. The domain of $y = \log_2 x^4$ is [25-Q]

A. $\Re$ **B.** $x > 0$ **C.** $x \neq 0$ **D.** $x < 0$

Ans.
We need $x^4 > 0$, which holds for all $x \neq 0$.
Answer: C

Q17. The domain for which $\log_b x^2 = 2 \log_b x$ is true is [25-Q]

A. $0 < x < 2$ **B.** $\Re$ **C.** $x > 2$ **D.** $x > 0$

Ans.
The power rule $\log_b x^n = n \log_b x$ requires $\log_b x$ on the right–hand side to exist, i.e. $x > 0$.
Answer: D

Q18. The domain of $y = 3\ln|x-3| + \pi$ is [24-Q]
A. $0 \le x \le \pi$ **B.** $\Re$ **C.** $x > 3$ **D.** $x \neq 3$

Ans.
We need $|x-3| > 0$, which holds whenever $x \neq 3$.
Answer: D

Q19. The domain of $y = 3\ln|x+3| + \pi$ is [24-Q]
A. $\Re$ **B.** $0 < x < \pi$ **C.** $x \neq -3$ **D.** $x > 3$

Ans.
We need $|x+3| > 0$, which holds whenever $x \neq -3$.
Answer: C

Q20. The asymptote of the graph $y = \log_2(x-1)$ is [25-Q]
A. $y = 1$ **B.** $y = 2$ **C.** $x = 1$ **D.** $x = 2$

Ans.
The vertical asymptote of $y = \log_b(x-h) + k$ is $x = h$. Here $h = 1$.
Answer: C

Q21. The asymptote of the graph $y = \log_2(x-1) + 2$ is [24-Q]
A. $x = 1$ **B.** $x = -1$ **C.** $x = -2$ **D.** $x = 2$

Ans.
The asymptote is $x = h = 1$.
Answer: A

Q22. The asymptote of the graph $y = \log_2(x-1) + 2$ is [25-Q]
A. $y = 1$ **B.** $y = 2$ **C.** $x = 1$ **D.** $x = 2$

Ans.
The asymptote is the vertical line $x = h = 1$.
Answer: C

Q23. The asymptote of the graph $y = \log_2(x+1) + 2$ is [24-Q]
A. $x = 0$ **B.** $x = 1$ **C.** $x = -1$ **D.** $x = 2$

Ans.
$x + 1 = x - (-1)$, so $h = -1$ and the asymptote is $x = -1$.
Answer: C

Q24. The asymptote of the graph $y = \log_2(x+1) + 2$ is [25*-Q]
A. $x = 2$ **B.** $x = -1$ **C.** $x = 1$ **D.** $x = -2$

Ans.
The asymptote is $x = -1$.
Answer: B

Q25. The asymptote of the graph $y = \log_{1/2}(x-2) + 3$ is [25-Q]
A. $x = 0$ **B.** $x = 1$ **C.** $x = 2$ **D.** $x = 3$

Ans.
$h = 2$, so the asymptote is $x = 2$.
Answer: C

Q26. The range of $y = 2\log_{1/2} x$ is [26-Q]
A. $\Re$ **B.** $-\Re$ **C.** $x > 0$ **D.** $\Re \setminus \{0\}$

Ans.
A logarithmic function takes every real value; stretching by 2 does not change this. Range = $\Re$.
Answer: A

Q27. The range of $f(x) = \log_{1/2}(x+1) - 2$ is [26-Q]
A. $x > 0$ **B.** $x \ge 0$ **C.** $\Re$ **D.** $x < 0$

Ans.
Translations do not change the range of a logarithmic function: Range = $\Re$.
Answer: C

8.3 Exponential Functions

Q28. For the exponential function $y = b^x$ to be a decay function, b must satisfy [24-Q]
A. $b = 0$ **B.** $b = 1$ **C.** $b = \frac{1}{2}$ **D.** $0 < b < 1$

Ans.
$y = b^x$ decays (decreases) when $0 < b < 1$ and grows when $b > 1$.
Answer: D

Q29. For the function $y = 3^{-x}$, as x becomes very large, y gets close to [S-Q]
A. 3^{1000} **B.** 0 **C.** 1 **D.** $\frac{1}{3}$

Ans.
$3^{-x} = \frac{1}{3^x}$, and this tends to 0 as $x \to \infty$.
Answer: B

Q30. For the function $y = 4^{-x}$, as x becomes very large, y gets close to [24-Q]
A. 1 **B.** 0 **C.** 2 **D.** none of these

Ans.
$4^{-x} = \frac{1}{4^x} \to 0$ as $x \to \infty$.
Answer: B

Q31. The range of $y = 3^{x+1} - 2$ is [25*-Q]
A. $\{-2\}$ **B.** $y > -2$ **C.** $y \le -2$ **D.** $y \ge 2$

Ans.
$3^{x+1} > 0$ for all x, so subtracting 2 gives $y > -2$.
Answer: B

Q32. The range of $y = 2e^{-x+1} - 3$ is [25-Q]
A. $y \le -3$ **B.** $y > -3$ **C.** $y \ge -3$ **D.** $y < -3$

Ans.
$e^{-x+1} > 0$, so $2e^{-x+1} > 0$ and $y > -3$.
Answer: B

Q33. If $a < 0$, the range of $y = ab^{x-h} + k$ is [25-Q]
A. $y \ge k$ **B.** $y \le k$ **C.** $y < k$ **D.** $y > k$

Ans.
$b^{x-h} > 0$ and $a < 0$, so $ab^{x-h} < 0$ and $y < k$.
Answer: C

Q34. The range of $y = -2^{x-1} + 3$ is [24-Q]
A. $y \ge 3$ **B.** $y > 3$ **C.** $y \le 3$ **D.** $y < 3$

Ans.
$-2^{x-1} < 0$, so $y < 3$.
Answer: D

Q35. The range of $y = -2e^{-x+1} + 3$ is [S-Q]
A. $\{3\}$ **B.** $y > 3$ **C.** $y < 3$ **D.** $y \ge 3$

Ans.
$-2e^{-x+1} < 0$, so $y < 3$.
Answer: C

Q36. The horizontal asymptote of $y = ab^{x-h} + k$ is [26-Q]
A. $y = 0$ **B.** $x = 0$ **C.** $y = k$ **D.** $y = h$

Ans.
As the exponential part tends to 0, $y \to k$. The asymptote is $y = k$.
Answer: C

Q37. The asymptote of the graph $y = 2^{x+1} + 3$ is [24-Q]
A. $y = 0$ **B.** $y = 3$ **C.** $x = 0$ **D.** $y = -3$

Ans.
For $y = ab^{x-h} + k$ the horizontal asymptote is $y = k$.
Here $k = 3$.
Answer: B (y = 3)

Q38. The asymptote of the graph $y = 3\cdot 2^{x+1} - 2$ is [25-Q]
A. $y = 0$ **B.** $x = 0$ **C.** $y = -2$ **D.** $y = 2$

Ans.
For $y = ab^{x-h} + k$ the horizontal asymptote is $y = k$.
Here $k = -2$.
Answer: C (y = −2)

Q39. The asymptote of the graph $y = 2e^{x-1} + 3$ is [25*-Q]
A. $y = 0$ **B.** $x = 0$ **C.** $y = -3$ **D.** $y = 3$

Ans.
For $y = ab^{x-h} + k$ the horizontal asymptote is $y = k$.
Here $k = 3$.
Answer: D (y = 3)

Q40. The asymptote of the graph $y = 3e^{x-1} + 2$ is [24-Q]
A. $y = 0$ **B.** $x = 0$ **C.** $y = -2$ **D.** $y = 2$

Ans.
For $y = ab^{x-h} + k$ the horizontal asymptote is $y = k$.
Here $k = 2$.
Answer: D (y = 2)

Q41. The asymptote of the graph $y = -2^{-x+4} + 3$ is [S-Q]
A. $y = 0$ **B.** $x = 0$ **C.** $y = 3$ **D.** $y = -3$

Ans.
For $y = ab^{x-h} + k$ the horizontal asymptote is $y = k$.
Here $k = 3$.
Answer: C (y = 3)

Q42. The asymptote of the graph $y = -2^{x+4} - 1$ is [25-Q]
A. $y = 0$ **B.** $y = -1$ **C.** $x = 0$ **D.** $y = 1$

Ans.
For $y = ab^{x-h} + k$ the horizontal asymptote is $y = k$.
Here $k = -1$.
Answer: B (y = −1)

Q43. The asymptote of the graph $y = -3^{x-4} - 2$ is [24-Q]
A. $y = 0$ **B.** $x = 0$ **C.** $y = 2$ **D.** $y = -2$

Ans.
For $y = ab^{x-h} + k$ the horizontal asymptote is $y = k$.
Here $k = -2$.
Answer: D (y = −2)

Q44. The asymptote of the graph $y = -2^{-x+5} + 2$ is [26-Q]
A. $y = 0$ **B.** $x = 0$ **C.** $y = 2$ **D.** $y = -2$

Ans.
For $y = ab^{x-h} + k$ the horizontal asymptote is $y = k$.
Here $k = 2$.

Answer: C (y = 2)

Q45. The asymptote of the graph $y = -2^{x-2} - 3$ is [25-Q]

A. y = 0 **B.** y = −3 **C.** x = 0 **D.** y = 3

Ans.

For $y = ab^{x-h} + k$ the horizontal asymptote is y = k.
Here k = −3.
Answer: B (y = −3)

Q46. The asymptote of the graph $y = -3^{-x+2} - 4$ is [24-Q]

A. y = 0 **B.** x = 0 **C.** y = −4 **D.** y = 4

Ans.

For $y = ab^{x-h} + k$ the horizontal asymptote is y = k.
Here k = −4.
Answer: C (y = −4)

Q47. The asymptote of the graph $y = -5e^{x-1} + 4$ is [25-Q]

A. y = 4 **B.** y = 0 **C.** x = 0 **D.** y = −4

Ans.

For $y = ab^{x-h} + k$ the horizontal asymptote is y = k.
Here k = 4.
Answer: A (y = 4)

Q48. The asymptote of the graph $y = -5e^{-x+3} + 7$ is [26-Q]

A. y = 7 **B.** y = 0 **C.** x = 0 **D.** y = −7

Ans.

For $y = ab^{x-h} + k$ the horizontal asymptote is y = k.
Here k = 7.
Answer: A (y = 7)

Q49. The asymptote of the graph $y = 2^{|x|}$ is [25-Q]

A. y = 0 **B.** x = 0 **C.** y = 1 **D.** none of these

Ans.

$2^{|x|} \geq 1$ for all x and it grows without bound on both sides; the graph never approaches a line. So there is no asymptote.
Answer: D

Q50. The y−intercept of $y = 2 \cdot 3^{x+1} - 2$ is [24-Q]

A. 2 **B.** 4 **C.** 6 **D.** −2

Ans.

Put x = 0:

$y = 2 \cdot 3^1 - 2 = 6 - 2 = 4$

Answer: B (4)

Q51. The y−intercept of $y = 4e^{-x} + 7$ is [25-Q]

A. 4 **B.** 7 **C.** 11 **D.** 3

Ans.

Put x = 0:

$y = 4e^0 + 7 = 4 + 7 = 11$

Answer: C (11)

Q52. The point (1, 2) lies on $y = 2^x$. The corresponding point on $y = 2 \cdot 2^{x-1} + 1$ is [24-Q]

A. (1, 5) **B.** (0, 3) **C.** (2, 5) **D.** (2, 3)

Ans.

For $y = ab^{x-h} + k$ the rule is $(x, y) \to (x + h, ay + k)$. Here a = 2, h = 1, k = 1:

$(1, 2) \to (1 + 1, 2 \cdot 2 + 1) = (2, 5)$

Answer: C (2, 5)

Q53. The graph of $f(x) = -2^{x-4} + 1$ is translated 3 units to the left and 1 unit up to give g(x). Then g(2) = [25-Q]

A. 0 **B.** 1 **C.** −1 **D.** 2

Ans.

Translate 3 left: replace x by x + 3; then add 1:

$g(x) = -2^{(x+3)-4} + 1 + 1 = -2^{x-1} + 2$

$g(2) = -2^1 + 2 = 0$

Answer: A (0)

2 MARKS

8.1 Logarithmic Functions

Q1. Find the domain of $y = \log_2(x - 2) + 1$. [S-Q, 24-Q]

Ans.

$x - 2 > 0$
$x > 2$

Domain: x > 2

Q2. Find the domain of $y = \log_3(x - 2) + 4$. [25*-Q]

Ans.

$x - 2 > 0$
$x > 2$

Domain: x > 2

Q3. Find the domain of $y = \log_3(x - 2) + 4$. [26-Q]

Ans.

$x - 2 > 0$
$x > 2$

Domain: x > 2

Q4. Find the domain of $y = \log_2 x^2$ if x > 0. [26-Q]

Ans.

We need $x^2 > 0$, and the problem already restricts x > 0.
Domain: x > 0

Q5. Find the domain of $y = \ln(x^2 - 9)$. [25-Q]

Ans.

We need $x^2 - 9 > 0$.

$(x - 3)(x + 3) > 0$

The product is positive when both factors have the same sign: x < −3 or x > 3.
Domain: x < −3 or x > 3

Q6. Find the range of $y = \log_2 x$ for $1 \leq x \leq 8$. [26-Q]

Ans.

$y = \log_2 x$ is increasing, so the smallest and largest values occur at the end points:
x = 1: $y = \log_2 1 = 0$; x = 8: $y = \log_2 8 = 3$.

Range: 0 ≤ y ≤ 3

Q7. Find the range of $y = \log_2(x - 1)$. [24-Q]

Ans.

The range of a logarithmic function is not changed by horizontal translation.
Range: ℜ (all real numbers)

Q8. Find the range of $y = \log_3(x - 2) + 4$. [25-Q]

Ans.

Translations do not restrict the values of a logarithmic function.
Range: ℜ (all real numbers)

Q9. Find a if the graph of $y = \log_a x$ passes through the point (e, 2). [Sample]

Ans.

Substitute x = e, y = 2:

$2 = \log_a e$

$e = a^2$

$a = \sqrt{e}$

(a > 0).
$a = \sqrt{e}$

Q10. Find a if the graph of $y = \log_a x$ passes through the point (4, 2). [24-Q]

Ans.

$2 = \log_a 4$

$4 = a^2$

$a = 2$

a = 2

Q11. Find b if the graph of $y = \log_b(x + 1) - 2$ passes through the point $(-\frac{3}{4}, 0)$. [25*-Q]

Ans.

Substitute $x = -\frac{3}{4}$, y = 0:

$0 = \log_b(-\frac{3}{4} + 1) - 2$

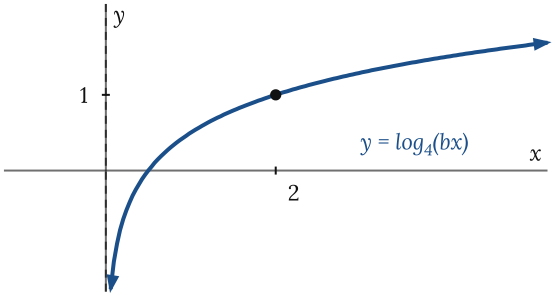
$2 = \log_b \frac{1}{4}$

$\frac{1}{4} = b^2$

$b = \frac{1}{2}$

(b > 0).
$b = \frac{1}{2}$

Q12. The graph of $f(x) = \log_4(bx)$ is shown with the point (2, 1) on it. Find b.



[25-Q]

Ans.

The point (2, 1) lies on the graph, so substitute x = 2, y = 1:

$1 = \log_4(2b)$

$2b = 4^1 = 4$

$b = 2$

b = 2

8.3 Exponential Functions

Q13. Write the domain of $y = b^x$ and its asymptote. [24-Q]

Ans.

The exponent x can be any real number, and $b^x > 0$ gets close to 0 as x decreases (b > 1) or increases (0 < b < 1).

Domain: ℜ; Asymptote: y = 0

Q14. Find the range of $y = -2^{x-1} + 3$. [25-Q]

Ans.

$2^{x-1} > 0$, so $-2^{x-1} < 0$ and

$y = -2^{x-1} + 3 < 3$

Range: y < 3

Q15. Find the range of $y = e^{|x|} + 2$. [24-Q]

Ans.

$|x| \geq 0$, so $e^{|x|} \geq e^0 = 1$ (minimum at x = 0).

$y = e^{|x|} + 2 \geq 1 + 2 = 3$

Range: y ≥ 3

Q16. Find the range and the asymptote of $y = -2^{x-1} + 3$. [26-Q]

Ans.

$-2^{x-1} < 0$, so y < 3. The horizontal asymptote is y = k = 3.

Range: y < 3; Asymptote: y = 3

Q17. Find the asymptote of $y = 2^{|x|}$. [25-Q]

Ans.

The graph of $y = 2^{|x|}$ is V-shaped with y ≥ 1 and it increases without bound as $x \to \pm\infty$.

There is no asymptote.

Q18. Find the y-intercept a of the graph of $y = 2e^{x-1} + 3$. [5-Q]

Ans.

Put x = 0:

$a = 2e^{-1} + 3 = \frac{2}{e} + 3$

$a = 2e^{-1} + 3$

Q19. The point (0, 1) lies on $y = 2^x$. Find the corresponding point on $y = 3 \cdot 2^{x-1} + 2$. [24-Q]

Ans.

(x, y) → (x + h, ay + k) with a = 3, h = 1, k = 2:

(0, 1) → (0 + 1, 3 · 1 + 2) = (1, 5)

(1, 5)

Q20. The point (0, 1) lies on $y = 2^x$. Find the corresponding point on $y = 5 \cdot 2^{x+3} - 1$. [25-Q]

Ans.

Here a = 5, h = −3, k = −1:

(0, 1) → (0 − 3, 5 · 1 − 1) = (−3, 4)

(−3, 4)

Q21. The point (1, 3) lies on $y = 3^x$. Find the corresponding point on $y = 2 \cdot 3^{x+1} + 1$. [24-Q]

Ans.

Here a = 2, h = −1, k = 1:

(1, 3) → (1 − 1, 2 · 3 + 1) = (0, 7)

(0, 7)

Q22. The point (1, 3) lies on $y = 3^x$. Find the corresponding point on $y = 2 \cdot 3^{x+1} - 2$. [25-Q]

Ans.

(1, 3) → (1 − 1, 2 · 3 − 2) = (0, 4)

(0, 4)

Q23. The point (0, 1) lies on $y = 5^x$. Find the corresponding point on $y = 3 \cdot 5^{x-2} + 1$. [26-Q]

Ans.

Here a = 3, h = 2, k = 1:

(0, 1) → (0 + 2, 3 · 1 + 1) = (2, 4)

(2, 4)

Q24. The point (0, 7) lies on $y = 2 \cdot 3^{x+1} + 1$. Find the corresponding point on $y = 3^x$. [25-Q]

Ans.

Go back from the transformed graph to $y = 3^x$: (x, y) → $(x - h, \frac{y-k}{a})$ with a = 2, h = −1, k = 1:

$(0, 7) \to (0 + 1, \frac{7-1}{2}) = (1, 3)$

(1, 3)

Q25. Find b if the graph of $y = 2b^{3x} + 4$ passes through the point (1, 20). [24-Q]

Ans.

Substitute x = 1, y = 20:

$20 = 2b^3 + 4$

$2b^3 = 16$

$b^3 = 8$

$b = 2$

b = 2

8.4 Differentiation of Logarithmic and Exponential Functions

Q26. Find $\frac{dy}{dx}$ if $3^x - \ln(xy) = 1$. [25-Q]

Ans.

Differentiate both sides with respect to x. Use $(3^x)' = 3^x \ln 3$ and $(\ln u)' = \frac{u'}{u}$ with u = xy, $u' = y + x\frac{dy}{dx}$.

$3^x \ln 3 - \frac{1}{xy}(y + x\frac{dy}{dx}) = 0$

$3^x \ln 3 - \frac{1}{x} - \frac{1}{y}\frac{dy}{dx} = 0$

$\frac{1}{y}\frac{dy}{dx} = 3^x \ln 3 - \frac{1}{x}$

$\frac{dy}{dx} = y(3^x \ln 3 - \frac{1}{x})$

$$\frac{dy}{dx} = \frac{y(x\,3^x \ln 3 - 1)}{x}$$

3 MARKS

8.1 Logarithmic Functions

Q1. Sketch the graph of $y = \log_2(-x)$. Label the key points, domain and asymptote. [26-Q]

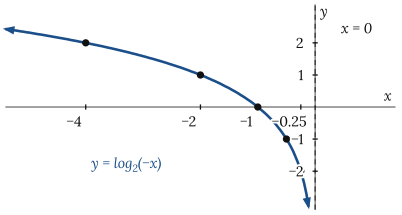
Ans.

$y = \log_2(-x)$ is the reflection of $y = \log_2 x$ in the y-axis.

Apply the rule (x, y) → (−x, y) (reflection in the y-axis) to five key points of the basic graph:

(4, 2) → (−4, 2)
(2, 1) → (−2, 1)
(1, 0) → (−1, 0)
(0.5, −1) → (−0.5, −1)
(0.25, −2) → (−0.25, −2)

Domain: x < 0
Range: ℜ
Asymptote: x = 0



Q2. Sketch the graph of $y = 3 \log_2 x$. Label the key points, domain and asymptote. [24-Q]

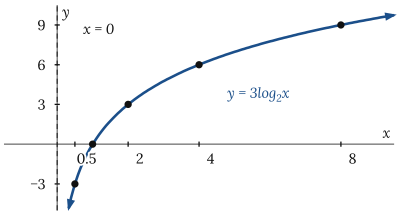
Ans.

$y = 3 \log_2 x$ is $y = \log_2 x$ stretched vertically by the factor 3.

Apply the rule (x, y) → (x, 3y) (vertical stretch, factor 3) to five key points of the basic graph:

(0.5, −1) → (0.5, −3)
(1, 0) → (1, 0)
(2, 1) → (2, 3)
(4, 2) → (4, 6)
(8, 3) → (8, 9)

Domain: x > 0
Range: ℜ
Asymptote: x = 0



Q3. Sketch the graph of $y = \log_2(x - 3)$. State its domain and range. [24-Q]

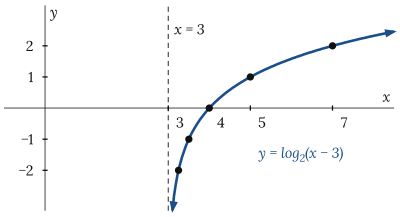
Ans.

$y = \log_2(x - 3)$ is $y = \log_2 x$ translated 3 units to the right.

Apply the rule $(x, y) \to (x + 3, y)$ (translation 3 units to the right) to five key points of the basic graph:

$(0.25, -2) \to (3.25, -2)$
$(0.5, -1) \to (3.5, -1)$
$(1, 0) \to (4, 0)$
$(2, 1) \to (5, 1)$
$(4, 2) \to (7, 2)$

Domain: $x > 3$ $(x - 3 > 0)$
Range: $\Re$
Asymptote: $x = 3$



Domain: x > 3; Range: $\Re$

Q4. Sketch the graph of $y = \log_2(x - 2) + 2$. Label the key points, domain and asymptote. [24-Q]

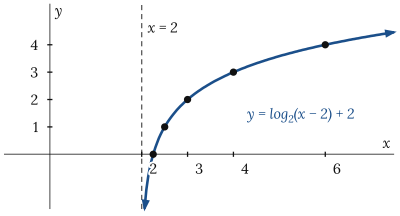
Ans.

$y = \log_2(x - 2) + 2$ is $y = \log_2 x$ translated 2 units right and 2 units up.

Apply the rule $(x, y) \to (x + 2, y + 2)$ to five key points of the basic graph:

$(0.25, -2) \to (2.25, 0)$
$(0.5, -1) \to (2.5, 1)$
$(1, 0) \to (3, 2)$
$(2, 1) \to (4, 3)$
$(4, 2) \to (6, 4)$

Domain: $x > 2$
Range: $\Re$
Asymptote: $x = 2$



Q5. Sketch the graph of $y = \log_2(x - 2) + 1$. State its domain, range and asymptote. [26-Q]

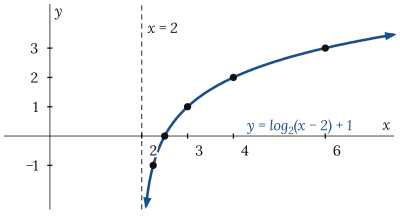
Ans.

$y = \log_2(x - 2) + 1$ is $y = \log_2 x$ translated 2 units right and 1 unit up.

Apply the rule $(x, y) \to (x + 2, y + 1)$ to five key points of the basic graph:

$(0.25, -2) \to (2.25, -1)$
$(0.5, -1) \to (2.5, 0)$
$(1, 0) \to (3, 1)$
$(2, 1) \to (4, 2)$
$(4, 2) \to (6, 3)$

Domain: $x > 2$
Range: $\Re$
Asymptote: $x = 2$



Domain: x > 2; Range: $\Re$; Asymptote: x = 2

Q6. Sketch the graph of $y = \log_2(x + 2) - 1$. Label the key points, domain and asymptote. [25*-Q]

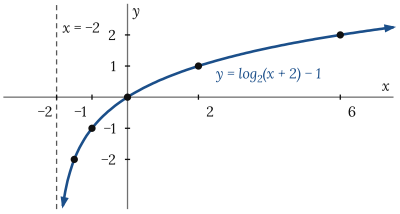
Ans.

$y = \log_2(x + 2) - 1$ is $y = \log_2 x$ translated 2 units left and 1 unit down.

Apply the rule $(x, y) \to (x - 2, y - 1)$ to five key points of the basic graph:

$(0.5, -1) \to (-1.5, -2)$
$(1, 0) \to (-1, -1)$
$(2, 1) \to (0, 0)$
$(4, 2) \to (2, 1)$
$(8, 3) \to (6, 2)$

Domain: $x > -2$
Range: $\Re$
Asymptote: $x = -2$



Q7. Find the domain and range of $f(x) = 2\log_2(2x - 4) + 5$. [24-Q]

Ans.

The argument of the logarithm must be positive:

$2x - 4 > 0$
$2x > 4$
$x > 2$

The stretch by 2, the horizontal compression and the translations do not restrict the values of a logarithmic function.

Domain: x > 2; Range: $\Re$

Q8. Use transformations of $y = \log_{1/2} x$ to sketch $y = \log_{1/2}(x + 1) - 1$. Label the key points, domain and asymptote. [25-Q]

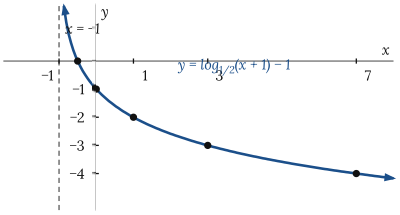
Ans.

Start from $y = \log_{1/2} x$ (a decreasing graph through (1, 0) with asymptote $x = 0$). Translate it 1 unit left (horizontal translation) and 1 unit down (vertical translation) to obtain $y = \log_{1/2}(x + 1) - 1$.

Apply the rule $(x, y) \to (x - 1, y - 1)$ to five key points of the basic graph:

$(0.5, 1) \to (-0.5, 0)$
$(1, 0) \to (0, -1)$
$(2, -1) \to (1, -2)$
$(4, -2) \to (3, -3)$
$(8, -3) \to (7, -4)$

Domain: $x > -1$
Range: $\Re$
Asymptote: $x = -1$



Q9. Sketch the graph of $y = \log_{1/2}(x + 1) - 2$. Label the key points, domain and asymptote. [26-Q]

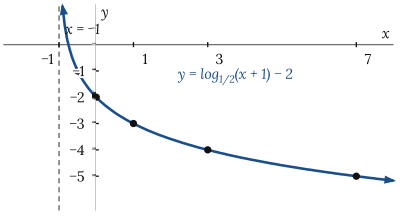
Ans.

$y = \log_{1/2}(x + 1) - 2$ is $y = \log_{1/2} x$ translated 1 unit left and 2 units down.

Apply the rule $(x, y) \to (x - 1, y - 2)$ to five key points of the basic graph:

$(0.5, 1) \to (-0.5, -1)$
$(1, 0) \to (0, -2)$
$(2, -1) \to (1, -3)$
$(4, -2) \to (3, -4)$
$(8, -3) \to (7, -5)$

Domain: $x > -1$
Range: $\Re$
Asymptote: $x = -1$

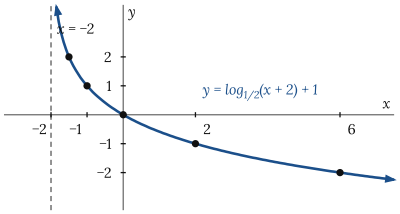


Q10. Sketch the graph of $y = \log_{1/2}(x + 2) + 1$. Label the axis intercepts and give the equation of the asymptote. [24-Q]

Ans.

$y = \log_{1/2}(x + 2) + 1$ is $y = \log_{1/2} x$ translated 2 units left and 1 unit up. The intercepts: x-intercept where $y = 0$ and y-intercept where $x = 0$ both give the origin (0, 0), because $\log_{1/2} 2 = -1$.

Apply the rule (x, y) → (x − 2, y + 1) to five key points of the basic graph:
(0.5, 1) → (−1.5, 2)
(1, 0) → (−1, 1)
(2, −1) → (0, 0)
(4, −2) → (2, −1)
(8, −3) → (6, −2)
Domain: x > −2
Range: ℜ
Asymptote: x = −2



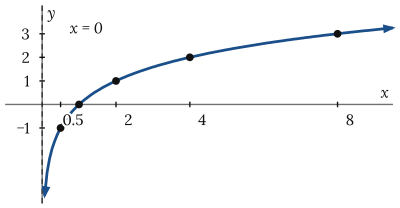
Intercepts: (0, 0); Asymptote: x = −2

Q11. Starting from the graph of $y = \log_2 x$, draw step by step the transformed graphs to obtain $y = \log_2(x - 1) + 2$. [24-Q]

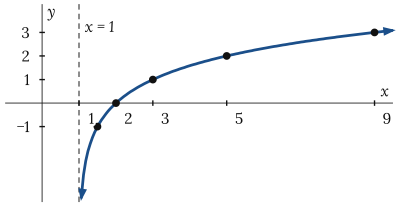
Ans.

Use key points (x, y) of the basic graph $y = \log_2 x$: $(\frac{1}{2}, -1)$, (1, 0), (2, 1), (4, 2), (8, 3).

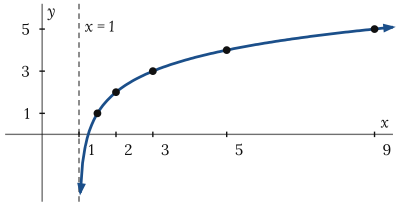
Step 1: graph of the basic function $y = \log_2 x$ (asymptote x = 0).



Step 2: Horizontal translation 1 unit to the right: $y = \log_2(x - 1)$. Rule (x, y) → (x + 1, y). The asymptote moves to x = 1.



Step 3: Vertical translation 2 units up: $y = \log_2(x - 1) + 2$. Rule (x, y) → (x, y + 2). The asymptote stays x = 1.



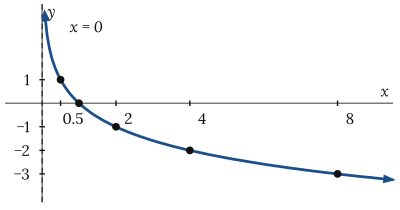
Final key points: (1.5, 1), (2, 2), (3, 3), (5, 4), (9, 5). Domain: x > 1; range ℜ; asymptote x = 1.

Q12. Starting from the graph of $y = \log_{1/2} x$, draw step by step the transformed graphs to obtain $y = \log_{1/2}(x - 1) + 3$. [25-Q]

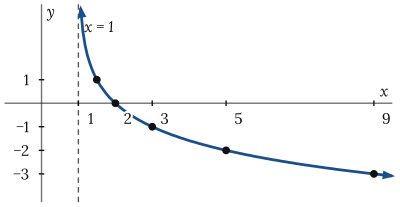
Ans.

Use key points (x, y) of the basic graph $y = \log_{1/2} x$: $(\frac{1}{2}, 1)$, (1, 0), (2, −1), (4, −2), (8, −3).

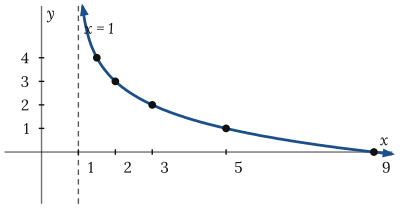
Step 1: graph of the basic function $y = \log_{1/2} x$ (asymptote x = 0).



Step 2: Horizontal translation 1 unit to the right: $y = \log_{1/2}(x - 1)$. Rule (x, y) → (x + 1, y). The asymptote moves to x = 1.



Step 3: Vertical translation 3 units up: $y = \log_{1/2}(x - 1) + 3$. Rule (x, y) → (x, y + 3). The asymptote stays x = 1.



Final key points: (1.5, 4), (2, 3), (3, 2), (5, 1), (9, 0). Domain: x > 1; range ℜ; asymptote x = 1.

8.3 Exponential Functions

Q13. Sketch the graphs of $y_1 = 4^x$ and $y_2 = 5^{-x}$ on the same axes. Label the intercept and the asymptotes. [5-Q]

Ans.

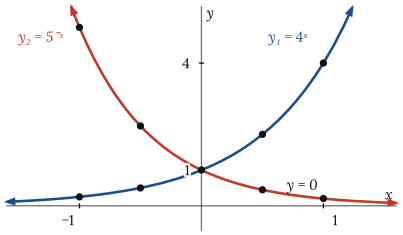
Both graphs have the basic exponential shape and pass through the common point (0, 1) (the y−intercept), because $b^0 = 1$.

$\mathbf{y_1 = 4^x}$ is increasing (b = 4 > 1). $\mathbf{y_2 = 5^{-x} = (\frac{1}{5})^x}$ is decreasing (0 < b < 1). It is the reflection of $y = 5^x$ in the y−axis.

Key points of y_1: $(-1, \frac{1}{4})$, $(-\frac{1}{2}, \frac{1}{2})$, (0, 1), $(\frac{1}{2}, 2)$, (1, 4).

Key points of y_2: $(-1, 5)$, $(-\frac{1}{2}, \sqrt{5})$, (0, 1), $(\frac{1}{2}, \frac{1}{\sqrt{5}})$, $(1, \frac{1}{5})$.

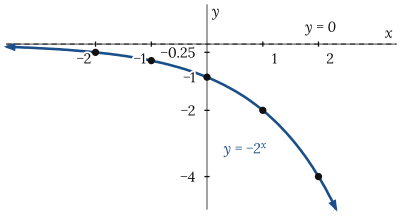
Asymptote (both): y = 0 (the x−axis). **Intercept:** (0, 1).



Q14. Sketch the graph of $y = -2^x$. State its range and asymptote. [24-Q]

Ans.

$y = -2^x$ is the reflection of $y = 2^x$ in the x−axis.
Apply the rule (x, y) → (x, −y) (reflection in the x−axis) to five key points of the basic graph:
(−2, 0.25) → (−2, −0.25)
(−1, 0.5) → (−1, −0.5)
(0, 1) → (0, −1)
(1, 2) → (1, −2)
(2, 4) → (2, −4)
Range: y < 0
Asymptote: y = 0
Domain: ℜ



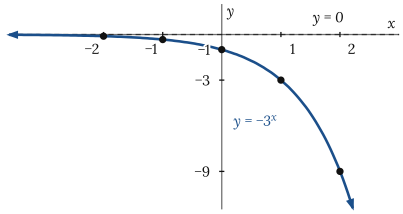
Range: y < 0; Asymptote: y = 0

Q15. Sketch the graph of $y = -3^x$. State its range and asymptote. [25-Q]

Ans.

$y = -3^x$ is the reflection of $y = 3^x$ in the x−axis.
Apply the rule (x, y) → (x, −y) to five key points of the basic graph:
$(-2, \frac{1}{9})$ → $(-2, -\frac{1}{9})$
$(-1, \frac{1}{3})$ → $(-1, -\frac{1}{3})$
(0, 1) → (0, −1)
(1, 3) → (1, −3)
(2, 9) → (2, −9)
Range: y < 0
Asymptote: y = 0

Domain: $\Re$

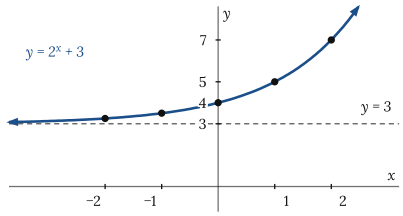


Range: y < 0; Asymptote: y = 0

Q16. Sketch the graph of $y = 2^x + 3$. State its range and asymptote. [26-Q]

Ans.

$y = 2^x + 3$ is $y = 2^x$ translated 3 units up.
Apply the rule (x, y) → (x, y + 3) to five key points of the basic graph:
(−2, 0.25) → (−2, 3.25)
(−1, 0.5) → (−1, 3.5)
(0, 1) → (0, 4)
(1, 2) → (1, 5)
(2, 4) → (2, 7)
Range: y > 3
Asymptote: y = 3
Domain: $\Re$

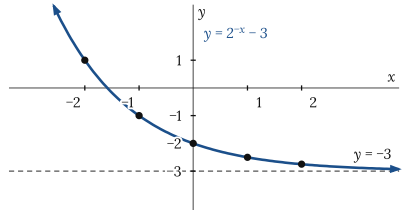


Range: y > 3; Asymptote: y = 3

Q17. Sketch the graph of $y = 2^{-x} - 3$. State its range and asymptote. [24-Q]

Ans.

$y = 2^{-x} - 3$: reflect $y = 2^x$ in the y−axis, then translate 3 units down.
Apply the rule (x, y) → (−x, y − 3) applied to $y = 2^x$ (reflection in the y−axis, then 3 units down) to five key points of the basic graph:
(2, 4) → (−2, 1)
(1, 2) → (−1, −1)
(0, 1) → (0, −2)
(−1, 0.5) → (1, −2.5)
(−2, 0.25) → (2, −2.75)
Range: y > −3
Asymptote: y = −3
Domain: $\Re$

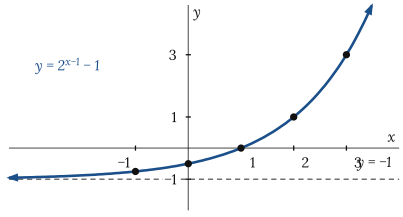


Range: y > −3; Asymptote: y = −3

Q18. Sketch the graph of $y = 2^{x-1} - 1$. State its range and asymptote. [25*-Q]

Ans.

$y = 2^{x-1} - 1$ is $y = 2^x$ translated 1 unit right and 1 unit down.
Apply the rule (x, y) → (x + 1, y − 1) to five key points of the basic graph:
(−2, 0.25) → (−1, −0.75)
(−1, 0.5) → (0, −0.5)
(0, 1) → (1, 0)
(1, 2) → (2, 1)
(2, 4) → (3, 3)
Range: y > −1
Asymptote: y = −1
Domain: $\Re$

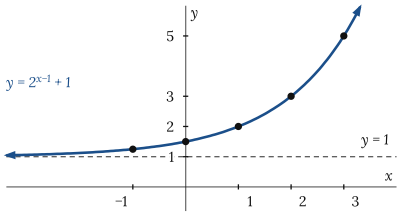


Range: y > −1; Asymptote: y = −1

Q19. Sketch the graph of $y = 2^{x-1} + 1$. State its range and asymptote. [24-Q]

Ans.

$y = 2^{x-1} + 1$ is $y = 2^x$ translated 1 unit right and 1 unit up.
Apply the rule (x, y) → (x + 1, y + 1) to five key points of the basic graph:
(−2, 0.25) → (−1, 1.25)
(−1, 0.5) → (0, 1.5)
(0, 1) → (1, 2)
(1, 2) → (2, 3)
(2, 4) → (3, 5)
Range: y > 1
Asymptote: y = 1
Domain: $\Re$

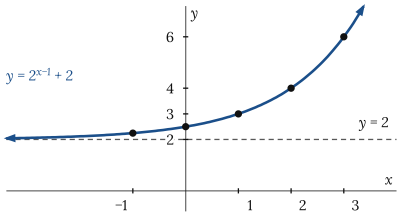


Range: y > 1; Asymptote: y = 1

Q20. Sketch the graph of $y = 2^{x-1} + 2$. State its range and asymptote. [24-Q]

Ans.

$y = 2^{x-1} + 2$ is $y = 2^x$ translated 1 unit right and 2 units up.
Apply the rule (x, y) → (x + 1, y + 2) to five key points of the basic graph:
(−2, 0.25) → (−1, 2.25)
(−1, 0.5) → (0, 2.5)
(0, 1) → (1, 3)
(1, 2) → (2, 4)
(2, 4) → (3, 6)
Range: y > 2
Asymptote: y = 2
Domain: $\Re$

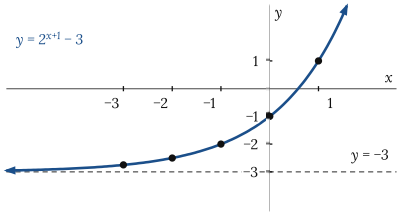


Range: y > 2; Asymptote: y = 2

Q21. Sketch the graph of $y = 2^{x+1} - 3$. State its range and asymptote. [26-Q]

Ans.

$y = 2^{x+1} - 3$ is $y = 2^x$ translated 1 unit left and 3 units down.
Apply the rule (x, y) → (x − 1, y − 3) to five key points of the basic graph:
(−2, 0.25) → (−3, −2.75)
(−1, 0.5) → (−2, −2.5)
(0, 1) → (−1, −2)
(1, 2) → (0, −1)
(2, 4) → (1, 1)
Range: y > −3
Asymptote: y = −3
Domain: $\Re$

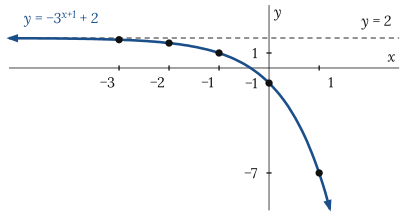


Range: y > −3; Asymptote: y = −3

Q22. Sketch the graph of $y = -3^{x+1} + 2$. State its range and asymptote. [25-Q]

Ans.

$y = -3^{x+1} + 2$: reflect $y = 3^x$ in the x−axis, then translate 1 unit left and 2 units up.
Apply the rule (x, y) → (x − 1, −y + 2) (reflect in the x−axis, shift 1 left and 2 up) to five key points of the basic graph:
$(-2, \frac{1}{9}) \rightarrow (-3, \frac{17}{9})$
$(-1, \frac{1}{3}) \rightarrow (-2, \frac{5}{3})$
(0, 1) → (−1, 1)
(1, 3) → (0, −1)
(2, 9) → (1, −7)
Range: y < 2
Asymptote: y = 2
Domain: $\Re$



Range: y < 2; Asymptote: y = 2

Q23. Sketch the graph of $y = -2\cdot3^{x+1} + 2$. State its range and asymptote. [24-Q]

Ans.

$y = -2\cdot3^{x+1} + 2$: reflect in the x−axis, stretch vertically by 2, translate 1 unit left and 2 units up.

Apply the rule $(x, y) \to (x - 1, -2y + 2)$ to five key points of the basic graph:

$(-2, \frac{1}{9}) \to (-3, \frac{16}{9})$

$(-1, \frac{1}{3}) \to (-2, \frac{4}{3})$

$(0, 1) \to (-1, 0)$

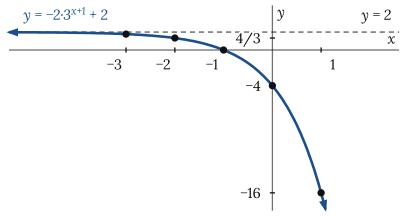
$(1, 3) \to (0, -4)$

$(2, 9) \to (1, -16)$

Range: $y < 2$

Asymptote: $y = 2$

Domain: $\mathbb{R}$



Range: y < 2; Asymptote: y = 2

Q24. Sketch the graph of $y = e^{x-2} + 3$. State the domain, range and asymptote. [25-Q]

Ans.

Compare with $y = ab^{x-h} + k$: $a = 1, b = e, h = 2, k = 3$.

Domain: $\mathbb{R}$. Since $e^{x-2} > 0$, $y > 3$: **Range:** $y > 3$. **Asymptote:** $y = 3$.

Apply $(x, y) \to (x + 2, y + 3)$ to the key points of $y = e^x$:

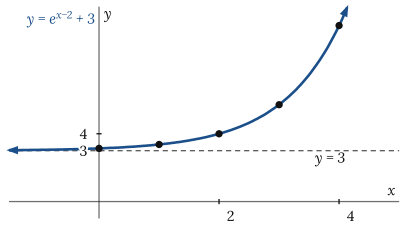
$(-2, e^{-2}) \to (0, e^{-2} + 3)$

$(-1, e^{-1}) \to (1, e^{-1} + 3)$

$(0, 1) \to (2, 4)$

$(1, e) \to (3, e + 3)$

$(2, e^2) \to (4, e^2 + 3)$



Q25. Find the y−intercept, the asymptote and the range of $y = 8e^{x-1} + 5$. [24-Q]

Ans.

y−intercept: put $x = 0$:

$y = 8e^{-1} + 5 = \frac{8}{e} + 5$

Asymptote: $y = k = 5$.

Range: $8e^{x-1} > 0$, so $y > 5$.

y−intercept: $8e^{-1} + 5$; Asymptote: $y = 5$; Range: $y > 5$

Q26. The point (0, 1) lies on $y = 3^x$. Find the corresponding points on (a) $y = -2\cdot3^x$, (b) $y = -2\cdot3^{x-2} + 1$, (c) $y = -2\cdot3^{-2x-2} + 1$. [26-Q]

Ans.

(a) $a = -2$, no shift: $(x, y) \to (x, -2y)$

$(0, 1) \to (0, -2)$

(b) $a = -2, h = 2, k = 1$: $(x, y) \to (x + 2, -2y + 1)$

$(0, 1) \to (2, -1)$

(c) $y = -2\cdot3^{-2(x+1)} + 1$: $b = -2, h = -1$. $(x, y) \to (\frac{x}{b} + h, ay + k) = (\frac{x}{-2} - 1, -2y + 1)$

$(0, 1) \to (0 - 1, -2 + 1) = (-1, -1)$

(a) (0, −2); (b) (2, −1); (c) (−1, −1)

8.4 Differentiation of Logarithmic and Exponential Functions

Q27. Find $\frac{dy}{dx}$ if $y = x \ln 2x^2$. [26-Q]

Ans.

Use the product rule with $u = x$, $v = \ln 2x^2$.

$v' = \frac{4x}{2x^2} = \frac{2}{x}$.

$\frac{dy}{dx} = 1 \cdot \ln 2x^2 + x \cdot \frac{2}{x}$

$\frac{dy}{dx} = \ln 2x^2 + 2$

$$\boxed{\frac{dy}{dx} = \ln 2x^2 + 2}$$

Q28. Find $\frac{dy}{dx}$ if $y^2e^x + \ln(xy) = 1$. [24-Q]

Ans.

Differentiate term by term (product rule for y^2e^x, chain rule for $\ln(xy)$):

$2y\frac{dy}{dx}e^x + y^2e^x + \frac{1}{x} + \frac{1}{y}\frac{dy}{dx} = 0$

$\frac{dy}{dx}(2ye^x + \frac{1}{y}) = -(y^2e^x + \frac{1}{x})$

$\frac{dy}{dx} \cdot \frac{2y^2e^x + 1}{y} = -\frac{xy^2e^x + 1}{x}$

$$\boxed{\frac{dy}{dx} = -\frac{y(xy^2e^x + 1)}{x(2y^2e^x + 1)}}$$

Q29. Find $\frac{dy}{dx}$ if $x \ln y + e^{xy} = 3^xy^3$. [25-Q]

Ans.

Differentiate both sides with respect to x:

$\ln y + \frac{x}{y}\frac{dy}{dx} + e^{xy}(y + x\frac{dy}{dx}) = 3^x \ln 3 \cdot y^3 + 3^x \cdot 3y^2\frac{dy}{dx}$

Collect the $\frac{dy}{dx}$ terms on the left:

$\frac{dy}{dx}(\frac{x}{y} + xe^{xy} - 3^{x+1}y^2) = y^3 3^x \ln 3 - \ln y - ye^{xy}$

Multiply the bracket and the right side by y:

$$\boxed{\frac{dy}{dx} = \frac{y(y^3 3^x \ln 3 - \ln y - ye^{xy})}{x + xye^{xy} - y^3 3^{x+1}}}$$

Q30. Find $\frac{dy}{dx}$ if $x \ln y + e^{xy} = 7$. [26-Q]

Ans.

$\ln y + \frac{x}{y}\frac{dy}{dx} + e^{xy}(y + x\frac{dy}{dx}) = 0$

$\frac{dy}{dx}(\frac{x}{y} + xe^{xy}) = -\ln y - ye^{xy}$

$\frac{dy}{dx} \cdot \frac{x(1 + ye^{xy})}{y} = -(\ln y + ye^{xy})$

$$\boxed{\frac{dy}{dx} = -\frac{y(\ln y + ye^{xy})}{x(1 + ye^{xy})}}$$

Q31. Find $\frac{dy}{dx}$ if $xe^y + \ln(xy) = \sin x$. [24-Q]

Ans.

$e^y + xe^y\frac{dy}{dx} + \frac{1}{x} + \frac{1}{y}\frac{dy}{dx} = \cos x$

$\frac{dy}{dx}(xe^y + \frac{1}{y}) = \cos x - e^y - \frac{1}{x}$

$\frac{dy}{dx} \cdot \frac{xye^y + 1}{y} = \frac{x\cos x - xe^y - 1}{x}$

$$\boxed{\frac{dy}{dx} = \frac{y(x\cos x - xe^y - 1)}{x(xye^y + 1)}}$$

Q32. Find $\frac{dy}{dx}$ if $y = x^{2x}$, $x > 0$. [25*-Q]

Ans.

Take ln of both sides:

$\ln y = 2x \ln x$

Differentiate with respect to x (product rule on the right):

$\frac{1}{y}\frac{dy}{dx} = 2 \ln x + 2x \cdot \frac{1}{x} = 2 \ln x + 2$

$\frac{dy}{dx} = y(2 \ln x + 2)$

$$\boxed{\frac{dy}{dx} = 2x^{2x}(\ln x + 1)}$$

Q33. Find $\frac{dy}{dx}$ if $y = x^{3x}$. [25-Q]

Ans.

Take ln of both sides:

$\ln y = 3x \ln x$

$\frac{1}{y}\frac{dy}{dx} = 3\ln x + 3x \cdot \frac{1}{x} = 3\ln x + 3$

$\frac{dy}{dx} = y(3\ln x + 3)$

$$\boxed{\frac{dy}{dx} = 3x^{3x}(\ln x + 1)}$$

Q34. Find $\frac{dy}{dx}$ if $y = (x+1)^x$. [24-Q]

Ans.

Take ln of both sides:

$\ln y = x\ln(x+1)$

$\frac{1}{y}\frac{dy}{dx} = \ln(x+1) + x \cdot \frac{1}{x+1}$

$\frac{dy}{dx} = (x+1)^x\left[\ln(x+1) + \frac{x}{x+1}\right]$

$$\boxed{\frac{dy}{dx} = (x+1)^x\left[\ln(x+1) + \frac{x}{x+1}\right]}$$

Q35. Find $\frac{dy}{dx}$ if $y^x = x^y$. [25-Q]

Ans.

Take ln of both sides:

$x\ln y = y\ln x$

Differentiate with respect to x:

$\ln y + \frac{x}{y}\frac{dy}{dx} = \frac{dy}{dx}\ln x + \frac{y}{x}$

$\frac{dy}{dx}\left(\frac{x}{y} - \ln x\right) = \frac{y}{x} - \ln y$

$\frac{dy}{dx} \cdot \frac{x - y\ln x}{y} = \frac{y - x\ln y}{x}$

$$\boxed{\frac{dy}{dx} = \frac{y(y - x\ln y)}{x(x - y\ln x)}}$$

Q36. Use logarithmic differentiation to find $\frac{dy}{dx}$ if $y = x^{\cos 2x}$. [25-Q]

Ans.

Take ln of both sides:

$\ln y = \cos 2x \cdot \ln x$

Differentiate (product rule, chain rule on cos 2x):

$\frac{1}{y}\frac{dy}{dx} = -2\sin 2x \cdot \ln x + \frac{\cos 2x}{x}$

$$\boxed{\frac{dy}{dx} = x^{\cos 2x}\left(\frac{\cos 2x}{x} - 2\sin 2x \ln x\right)}$$

5 MARKS

8.4 Differentiation of Logarithmic and Exponential Functions

Q1. Find the derivative of each: (i) $\tan^2(5x+6)$ (ii) $3^x x^3$ (iii) $e^{\sqrt{2x+3}}$ [25-Q]

Ans.

(i) Chain rule twice: $y = [\tan(5x+6)]^2$.

$\frac{dy}{dx} = 2\tan(5x+6) \cdot \sec^2(5x+6) \cdot 5$

$$\boxed{\frac{dy}{dx} = 10\tan(5x+6)\sec^2(5x+6)}$$

(ii) Product rule with $u = 3^x$, $v = x^3$:

$\frac{dy}{dx} = 3^x \ln 3 \cdot x^3 + 3^x \cdot 3x^2$

$\frac{dy}{dx} = x^2 3^x(x\ln 3 + 3)$

$$\boxed{\frac{dy}{dx} = x^2 3^x(x\ln 3 + 3)}$$

(iii) $(e^u)' = e^u u'$ with $u = \sqrt{2x+3}$, $u' = \frac{2}{2\sqrt{2x+3}} = \frac{1}{\sqrt{2x+3}}$:

$$\boxed{\frac{dy}{dx} = \frac{e^{\sqrt{2x+3}}}{\sqrt{2x+3}}}$$

Q2. Find the derivative of each: (a) $\tan 3x + e^{7-2x^2}$ (b) $\frac{\ln 7x}{\sin(x^2+5)}$ (c) $\frac{x^2}{\log_{10} x}$ [5-Q]

Ans.

(a) Differentiate each term:

$\frac{dy}{dx} = 3\sec^2 3x + e^{7-2x^2} \cdot (-4x)$

$$\boxed{\frac{dy}{dx} = 3\sec^2 3x - 4xe^{7-2x^2}}$$

(b) Quotient rule with $u = \ln 7x$ ($u' = \frac{1}{x}$) and $v = \sin(x^2+5)$ ($v' = 2x\cos(x^2+5)$):

$\frac{dy}{dx} = \frac{\frac{1}{x}\sin(x^2+5) - \ln 7x \cdot 2x\cos(x^2+5)}{\sin^2(x^2+5)}$

$$\boxed{\frac{dy}{dx} = \frac{\sin(x^2+5) - 2x^2\ln 7x\cos(x^2+5)}{x\sin^2(x^2+5)}}$$

(c) Quotient rule with $u = x^2$, $v = \log_{10} x$ ($v' = \frac{1}{x\ln 10} = \frac{\log_{10} e}{x}$):

$\frac{dy}{dx} = \frac{2x\log_{10} x - x^2 \cdot \frac{\log_{10} e}{x}}{(\log_{10} x)^2}$

$\frac{dy}{dx} = \frac{x(2\log_{10} x - \log_{10} e)}{(\log_{10} x)^2}$

$$\boxed{\frac{dy}{dx} = \frac{x(2\log_{10} x - \log_{10} e)}{(\log_{10} x)^2}}$$

CH.09 Application of Differentiation

Notes · Questions with Answers

NOTES

Inclination of a line
The **inclination** of a line that intersects the x-axis is the measure of the smallest nonnegative angle θ which the line makes with the positive end of the x-axis. The slope of the line is the tangent of the inclination:
$m = \tan\theta$
A vertical line has inclination 90° but no slope; a horizontal line has inclination 0° and slope 0.

Normal line
The line which is perpendicular to the tangent at (x_1, y_1) is called the **normal** to the curve at (x_1, y_1). Its gradient is $\frac{-1}{y'(x_1)}$, where $y'(x_1) \neq 0$.

Increasing function
A function f is **increasing** on an interval (a, b) if for any two numbers x_1 and x_2 in (a, b),
$x_2 > x_1$ implies $f(x_2) > f(x_1)$
namely, an increase in x produces an increase in y.

Decreasing function
A function f is **decreasing** on an interval (a, b) if for any two numbers x_1 and x_2 in (a, b),
$x_2 > x_1$ implies $f(x_2) < f(x_1)$
namely, an increase in x produces a decrease in y.

Critical point
A **critical point** of a function f(x) is a point in the domain of the function such that either $f'(x) = 0$ or $f'(x)$ is undefined.

Global (absolute) maximum and minimum
A point where f(x) has the minimum value on the entire domain is a **global minimum**; a point where f(x) has the maximum value on the entire domain is a **global maximum**.

Local maximum and minimum
A turning point on the graph where f(x) has a maximum value compared to the neighboring points is a **local maximum**; a turning point where f(x) has a minimum value compared to the neighboring points is a **local minimum**.

Concave up / Concave down
Let f be differentiable on an open interval I. The graph of f is
1. **concave up** on I if f' is increasing on the interval.
2. **concave down** on I if f' is decreasing on the interval.

Point of inflection
A point where the graph of a function has a tangent line and where the concavity changes is a **point of inflection**. At a point of inflection (c, f(c)), either $f''(c) = 0$ or $f''(c)$ is undefined.
Remark. We have an exceptional case that a point c is not an inflection point even if $f''(c) = 0$. For example, $f(x) = (x - c)^{2k}$ where the integer $k \geq 2$.

Linearization
If f is differentiable at x = a, then the approximation function
$L(x) = f(a) + f'(a)(x - a)$
is the **linearization** of f at a. The approximation $f(x) \approx L(x)$ is the **standard linear approximation** of f at a. The point x = a is the **center** of the approximation. The difference of f(x) and L(x) is the **error** of the linearization.

Slope and inclination
$m = \tan\theta$
For two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ on a line:
$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan\theta$$

Equation of tangent at (x1, y1)
For the curve y = f(x), the gradient of the tangent at (x_1, y_1) is $y'(x_1)$, so
$y - y_1 = y'(x_1)(x - x_1)$

Equation of normal at (x1, y1)
The gradient of the normal is $\frac{-1}{y'(x_1)}$ where $y'(x_1) \neq 0$, so
$$y - y_1 = -\frac{1}{y'(x_1)}(x - x_1)$$

Test for increasing and decreasing functions
Let y = f(x) be differentiable on the open interval (a, b).
1. If $f'(x) > 0$ for each $x \in (a, b)$, then f is increasing on (a, b).
2. If $f'(x) < 0$ for each $x \in (a, b)$, then f is decreasing on (a, b).

Steps: increasing / decreasing on an interval
1. Find $f'(x)$.
2. Find the critical points of f.
3. Find the open intervals determined by the critical points.
4. Test the signs of $f'(x)$ for each open interval.
5. Decide whether f is increasing or decreasing on each interval.

Critical point
$f'(x) = 0$ or $f'(x)$ is undefined (at a point in the domain of f).

Test for global maximum and minimum on a closed interval
Let f(x) be a continuous function on a closed interval.
1. Find the critical points of f.
2. Find f(x) at all critical points and endpoints of the interval.
3. Take the smallest value as the global minimum and the largest value as the global maximum.

First derivative test (local maximum / minimum)
Let c be a critical point of a continuous function f.
1. If $f'(x)$ changes from positive to negative at c, then f has a local maximum at x = c.
2. If $f'(x)$ changes from negative to positive at c, then f has a local minimum at x = c.
3. If $f'(x)$ does not change sign at c, then f has neither a local maximum nor a local minimum at x = c.

Test for concavity
Let f be twice differentiable on an open interval I.
1. If $f''(x) > 0$ for all x in I, then f is concave up on I.
2. If $f''(x) < 0$ for all x in I, then f is concave down on I.

Point of inflection
A point (c, f(c)) where the concavity changes: either $f''(c) = 0$ or $f''(c)$ is undefined, and f'' changes sign at c.

Second derivative test
Suppose f'' is continuous on an open interval that contains x = c.
1. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at x = c.
2. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at x = c.
3. If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.

Linearization (standard linear approximation)
$y = f(a) + f'(a)(x - a)$
$L(x) = f(a) + f'(a)(x - a)$
$f(x) \approx L(x)$
Error = true value − approximate value.

1 MARK

9.1 Tangent Line and Normal Line

Q1. Tangents are drawn from the curve $y = \sin x$ from the origin. Then the point of contact lies on the curve $\frac{1}{y^2} - \frac{1}{x^2} = \lambda$. The value of λ is equal to [S-Q]

A. −1 **B.** $\frac{1}{2}$ **C.** 1 **D.** 2

Ans.
Let the point of contact be $(x_1, y_1) = (x_1, \sin x_1)$. Gradient $y' = \cos x_1$.
Tangent: $y - y_1 = y'(x_1)(x - x_1)$. It passes through the origin (0, 0):

$0 - \sin x_1 = \cos x_1(0 - x_1)$

$\sin x_1 = x_1 \cos x_1 \rightarrow y_1 = x_1 \cos x_1$

Then

$$\frac{1}{y_1^2} - \frac{1}{x_1^2} = \frac{1}{x_1^2\cos^2 x_1} - \frac{1}{x_1^2} = \frac{1 - \cos^2 x_1}{x_1^2\cos^2 x_1}$$

$$= \frac{\sin^2 x_1}{\sin^2 x_1} = 1$$

(because $x_1^2\cos^2 x_1 = \sin^2 x_1$).
λ = 1. Answer: C

Q2. Tangents are drawn from the curve $y = \sec x$ from the origin. Then the point of contact lies on the curve $\frac{1}{x^2} - y^2 = \lambda$. The value of λ is equal to [24-Q]

A. −1 **B.** $\frac{1}{2}$ **C.** 1 **D.** 2

Ans.
Let the point of contact be $(x_1, y_1) = (x_1, \sec x_1)$. Gradient $y' = \sec x_1 \tan x_1$.
Tangent through the origin:

$0 - \sec x_1 = \sec x_1 \tan x_1(0 - x_1)$

$1 = x_1 \tan x_1 \rightarrow \tan x_1 = \frac{1}{x_1}$

Then, using $\sec^2 x = 1 + \tan^2 x$:

$$\frac{1}{x_1^2} - y_1^2 = \frac{1}{x_1^2} - \sec^2 x_1$$

$= \frac{1}{x_1^2} - 1 - \tan^2 x_1$

$= \frac{1}{x_1^2} - 1 - \frac{1}{x_1^2} = -1$

λ = −1. Answer: A

Q3. The equation of the tangent to the curve y = sin x at the origin is [24-Q]

A. y = −x **B.** y = x **C.** −y = x **D.** −y = −x

Ans.

$y' = \cos x$, so the gradient at $x_1 = 0$ is cos 0 = 1. The point is (0, 0).

$y - 0 = 1(x - 0)$

$y = x$

Answer: B

Q4. Equation of the tangent to the curve y = sin x + 9 at (0, 9) is [26-Q]

A. y = −x + 9 **B.** −y = x + 9 **C.** y = x + 9 **D.** y = 9x

Ans.

$y' = \cos x$, gradient at $x_1 = 0$ is 1.

$y - 9 = 1(x - 0)$

$y = x + 9$

Answer: C

Q5. The equation of the tangent to the curve y = cos x at (0, 1) is [25*-Q]

A. y = 0 **B.** x = 0 **C.** y = 1 **D.** y = 1 − x

Ans.

$y' = -\sin x$, gradient at $x_1 = 0$ is −sin 0 = 0.

$y - 1 = 0(x - 0)$

$y = 1$

Answer: C

Q6. The equation of the tangent to the curve y = 4 ln x at (1, 0) is [25-Q]

A. $y = \frac{1}{4}(x - 1)$ **B.** y = 4(x − 1) **C.** y = −4(x − 1) **D.** y = 4x

Ans.

$y' = \frac{4}{x}$, gradient at $x_1 = 1$ is 4.

$y - 0 = 4(x - 1)$

$y = 4(x - 1)$

Answer: B

Q7. The gradient of the normal to the curve $y = x^2 + kx + 7$ at the point (−2, −1) is $-\frac{1}{2}$, then k = [25*-Q, 26-Q]

A. 3 **B.** 4 **C.** 5 **D.** 6

Ans.

The normal gradient is $-\frac{1}{y'(x_1)} = -\frac{1}{2}$, so the tangent gradient is y′(−2) = 2.

$y' = 2x + k$

$y'(-2) = -4 + k = 2 \rightarrow k = 6$

Check that (−2, −1) is on the curve: 4 − 12 + 7 = −1 ✓.

Answer: D (k = 6)

9.2 Critical Point

Q8. The critical point of $y = x^2 - 6x + 4$ is [25-Q]

A. x = 3 **B.** x = −3 **C.** x = 4 **D.** x = −4

Ans.

A critical point occurs where f′(x) = 0 or f′(x) is undefined.

$y' = 2x - 6 = 0$

$x = 3$

Answer: A

Q9. The critical points of $f(x) = \sqrt[3]{x^4} - 4\sqrt[3]{x}$ are [25-Q]

A. (1, −3) and (8, 0) **B.** (8, 0) and (0, 0) **C.** (1, 3) and (0, 0) **D.** (1, −3) and (0, 0)

Ans.

$f(x) = x^{4/3} - 4x^{1/3}$

$f'(x) = \frac{4}{3}x^{1/3} - \frac{4}{3}x^{-2/3} = \frac{4(x-1)}{3x^{2/3}}$

f′(x) = 0 when x = 1; f′(x) is undefined at x = 0 (but x = 0 is in the domain of f). Both are critical points.

f(1) = 1 − 4 = −3

f(0) = 0

Critical points (1, −3) and (0, 0). Answer: D

Q10. The curve y = cos x sin x has a critical point at x = [25-Q]

A. $\frac{3\pi}{4}$ **B.** $\frac{\pi}{2}$ **C.** $\frac{\pi}{3}$ **D.** 0

Ans.

$y = \cos x \sin x = \frac{1}{2}\sin 2x$.

$y' = \cos 2x = 0$

$2x = \frac{\pi}{2} + n\pi \rightarrow x = \frac{\pi}{4} + \frac{n\pi}{2}$

For n = 1: $x = \frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4}$. (Check with $y' = \cos^2 x - \sin^2 x$: at $x = \frac{3\pi}{4}$, $\cos^2 = \sin^2 = \frac{1}{2}$, so y′ = 0 ✓.)

Answer: A

Q11. The critical points of the function $f(x) = x^3(x - 4)$ are [26-Q]

A. (0, 0) and (3, −27) **B.** (0, 0) and (4, 0) **C.** (0, 0) and (2, −16) **D.** (3, −27) and (4, 0)

Ans.

$f(x) = x^4 - 4x^3$

$f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3) = 0$

$x = 0$ or $x = 3$

f(0) = 0, f(3) = 81 − 108 = −27

Critical points (0, 0) and (3, −27). Answer: A

9.2 Global Maximum and Minimum

Q12. Find the global minimum value of the function $f(x) = x^3 - 3x - 4$ on [−1, 1]. [24-Q]

A. −6 **B.** 6 **C.** 2 **D.** −2

Ans.

Global extrema on a closed interval: evaluate f at the critical points inside the interval and at the end points; the smallest value is the global minimum and the largest is the global maximum.

$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1) = 0 \rightarrow x = 1, -1$

Both critical points are the end points.

f(−1) = −1 + 3 − 4 = −2

f(1) = 1 − 3 − 4 = −6

Global minimum = −6. Answer: A

Q13. The sum of global minimum and global maximum values of the function $f(x) = x^2 - 4x + 2$ on [−3, 5] is [24-Q]

A. 23 **B.** 25 **C.** 21 **D.** 20

Ans.

Global extrema on a closed interval: evaluate f at the critical points inside the interval and at the end points; the smallest value is the global minimum and the largest is the global maximum.

$f'(x) = 2x - 4 = 0 \rightarrow x = 2$

f(−3) = 9 + 12 + 2 = 23

f(2) = 4 − 8 + 2 = −2

f(5) = 25 − 20 + 2 = 7

Global minimum = −2, global maximum = 23. Sum = 21.

Answer: C

Q14. Find the sum of the global minimum and global maximum value of $f(x) = x^2 - 4x - 3$ on [−1, 3]. [26-Q]

A. 2 **B.** −5 **C.** 5 **D.** −2

Ans.

Global extrema on a closed interval: evaluate f at the critical points inside the interval and at the end points; the smallest value is the global minimum and the largest is the global maximum.

$f'(x) = 2x - 4 = 0 \rightarrow x = 2$

f(−1) = 1 + 4 − 3 = 2

f(2) = 4 − 8 − 3 = −7

f(3) = 9 − 12 − 3 = −6

Global minimum = −7, global maximum = 2. Sum = −5.

Answer: B

9.2 Concavity and Inflection Point

Q15. How many inflection points are there in the graph of $x^4 + 2x^2 + 5$? [5-Q]

A. 0 **B.** 1 **C.** 2 **D.** 3

Ans.

Test for concavity: f″(x) > 0 → concave up; f″(x) < 0 → concave down. An inflection point (c, f(c)) needs f″(c) = 0 (or undefined) **and** f″ changes sign at c.

$f'(x) = 4x^3 + 4x$

$f''(x) = 12x^2 + 4 > 0$ for all x

f″ is never 0 and never changes sign, so there is no inflection point.

Answer: A (0)

Q16. How many inflection points are there in the graph of $x^4 + 4x^3 + 5$? [24-Q]

A. 0 **B.** 1 **C.** 2 **D.** 3

Ans.

Test for concavity: f″(x) > 0 → concave up; f″(x) < 0 → concave down. An inflection point (c, f(c)) needs f″(c) = 0 (or undefined) **and** f″ changes sign at c.

$f''(x) = 12x^2 + 24x = 12x(x + 2) = 0$

$x = 0, x = -2$

f″ > 0 for x < −2, < 0 for −2 < x < 0, > 0 for x > 0: it changes sign at both points.

2 inflection points. Answer: C

Q17. How many inflection points are there in the graph of $x^4 - 24x^2 + 12$? [24-Q]

A. 0 **B.** 1 **C.** 2 **D.** 3

Ans.

Test for concavity: $f''(x) > 0 \rightarrow$ concave up; $f''(x) < 0 \rightarrow$ concave down. An inflection point $(c, f(c))$ needs $f''(c) = 0$ (or undefined) **and** f'' changes sign at c.

$f''(x) = 12x^2 - 48 = 12(x - 2)(x + 2) = 0$

$x = 2, x = -2$

f'' changes sign at both $x = -2$ and $x = 2$.

2 inflection points. Answer: C

Q18. How many inflection points are there in the graph of $f(x) = -x^3 + x^2 - x + 1$? [25, 25*-Q]

A. 0 **B.** 1 **C.** 2 **D.** 3

Ans.

Test for concavity: $f''(x) > 0 \rightarrow$ concave up; $f''(x) < 0 \rightarrow$ concave down. An inflection point $(c, f(c))$ needs $f''(c) = 0$ (or undefined) **and** f'' changes sign at c.

$f'(x) = -3x^2 + 2x - 1$

$f''(x) = -6x + 2 = 0 \rightarrow x = \frac{1}{3}$

f'' is positive for $x < \frac{1}{3}$ and negative for $x > \frac{1}{3}$: the sign changes once.

1 inflection point. Answer: B

Q19. The inflection point for the function $y = x^3 - 3x^2 + 2$ is [26-Q]

A. (1, 1) **B.** (1, 0) **C.** (0, 1) **D.** (−1, 0)

Ans.

Test for concavity: $f''(x) > 0 \rightarrow$ concave up; $f''(x) < 0 \rightarrow$ concave down. An inflection point $(c, f(c))$ needs $f''(c) = 0$ (or undefined) **and** f'' changes sign at c.

$y' = 3x^2 - 6x$

$y'' = 6x - 6 = 0 \rightarrow x = 1$

$y(1) = 1 - 3 + 2 = 0$

y'' changes sign at $x = 1$.

Inflection point (1, 0). Answer: B

9.3 Linearization

Q20. The linearization of $f(x) = (1 + x)^3$ at $x = 0$ is [24-Q]

A. $L(x) = 1 + 3x$ **B.** $L(x) = 1 + x$ **C.** $L(x) = x$ **D.** $L(x) = 1$

Ans.

Linearization of f at $x = a$: $L(x) = f(a) + f'(a)(x - a)$. Here $a = 0$.

$f(0) = 1$

$f'(x) = 3(1 + x)^2 \rightarrow f'(0) = 3$

$L(x) = 1 + 3(x - 0) = 1 + 3x$

Answer: A

Q21. The linearization of $f(x) = (1 + x)^5$ at $x = 1$ is [5-Q]

A. $L(x) = 1 + 5x$ **B.** $L(x) = 32 + 80x$ **C.** $L(x) = 32 + 80(x - 1)$ **D.** $L(x) = 1 + 32x$

Ans.

Linearization of f at $x = a$: $L(x) = f(a) + f'(a)(x - a)$. Here $a = 1$.

$f(1) = 2^5 = 32$

$f'(x) = 5(1 + x)^4 \rightarrow f'(1) = 5(16) = 80$

$L(x) = 32 + 80(x - 1)$

Answer: C

Q22. The linearization of $f(x) = (1 + x)^7$ at $x = 0$ is [24-Q]

A. $L(x) = 1 - 7x$ **B.** $L(x) = 7x - 1$ **C.** $L(x) = 1 + 7x$ **D.** $L(x) = 7 + x$

Ans.

Linearization of f at $x = a$: $L(x) = f(a) + f'(a)(x - a)$. Here $a = 0$.

$f(0) = 1$

$f'(x) = 7(1 + x)^6 \rightarrow f'(0) = 7$

$L(x) = 1 + 7x$

Answer: C

Q23. The linearization of $f(x) = \sqrt{x + 3}$ at $x = 1$ is [25-Q]

A. $L(x) = \frac{1}{4}x - \frac{7}{4}$ **B.** $L(x) = \frac{1}{4}x + \frac{7}{4}$ **C.** $L(x) = \frac{1}{2}x - \frac{7}{2}$ **D.** $L(x) = \frac{1}{2}x + \frac{7}{2}$

Ans.

Linearization of f at $x = a$: $L(x) = f(a) + f'(a)(x - a)$. Here $a = 1$.

$f(1) = \sqrt{4} = 2$

$f'(x) = \{1|2\sqrt{x + 3}\} \rightarrow f'(1) = \frac{1}{4}$

$L(x) = 2 + \frac{1}{4}(x - 1) = \frac{1}{4}x + \frac{7}{4}$

Answer: B

Q24. The linearization of $f(x) = \sin x$ at $x = 0$ is $L(x) =$ [25*-Q]

A. x **B.** $-x$ **C.** $x - \pi$ **D.** $x + \pi$

Ans.

Linearization of f at $x = a$: $L(x) = f(a) + f'(a)(x - a)$. Here $a = 0$.

$f(0) = \sin 0 = 0$

$f'(x) = \cos x \rightarrow f'(0) = 1$

$L(x) = 0 + 1(x - 0) = x$

Answer: A

Q25. If $f(x) = \frac{2x + 3}{x}$, then the linearization of f at $x = 1$ is [26-Q]

A. $L(x) = 2x + 3$ **B.** $L(x) = -3x + 8$ **C.** $L(x) = -2x + 7$ **D.** $L(x) = 3x + 2$

Ans.

Linearization of f at $x = a$: $L(x) = f(a) + f'(a)(x - a)$. Here $a = 1$. Write $f(x) = 2 + 3x^{-1}$.

$f(1) = 5$

$f'(x) = -3x^{-2} \rightarrow f'(1) = -3$

$L(x) = 5 - 3(x - 1) = -3x + 8$

Answer: B

2 MARKS

9.1 Tangent Line and Normal Line

Q1. The gradient of the tangent to the curve $y = ax^{3/2} + x$ at the point (4, −12) is −5. Find the value of a. [25*-Q]

Ans.

Gradient of the tangent = $y'(4) = -5$.

$y' = \frac{3}{2}ax^{1/2} + 1$

$y'(4) = \frac{3}{2}a(2) + 1 = 3a + 1$

$3a + 1 = -5 \rightarrow a = -2$

Check: the point (4, −12) is on the curve: $-2(8) + 4 = -12$ ✓.

Answer: a = −2

Q2. Find the equation of the tangent to the curve $y = \frac{x}{x - 2}$ at the point (3, 3). Give your answer in the form $y = mx + c$, where m and c are real numbers. [24-Q]

Ans.

Quotient rule:

$y' = \frac{(x - 2)(1) - x(1)}{(x - 2)^2} = \frac{-2}{(x - 2)^2}$

$y'(3) = \frac{-2}{1} = -2$

Tangent: $y - y_1 = y'(x_1)(x - x_1)$

$y - 3 = -2(x - 3)$

$y = -2x + 9$

Answer: y = −2x + 9

Q3. The curve $y = 2e^{2x} + e^{-x}$ cuts the y−axis at the point P. Find the equation of the tangent to the curve at the point P. [25-Q]

Ans.

On the y−axis $x = 0$: $y = 2e^0 + e^0 = 3$, so P = (0, 3).

$y' = 4e^{2x} - e^{-x}$

$y'(0) = 4 - 1 = 3$

Tangent:

$y - 3 = 3(x - 0)$

$y = 3x + 3$

Answer: y = 3x + 3

Q4. If the equation of the tangent to the curve $y = (px + 3)^2$ at (0, 9) is $y = 6x + 9$, find the value of p. [25*-Q]

Ans.

The gradient of the tangent is 6, so $y'(0) = 6$.

$y' = 2p(px + 3)$

$y'(0) = 2p(3) = 6p$

$6p = 6 \rightarrow p = 1$

Answer: p = 1

Q5. Find the equations of the tangent and normal lines to the curve $y = x^2 + 2x + 3$ at (0, 3). [25-Q]

Ans.

$y' = 2x + 2$

$y'(0) = 2$

Tangent: $y - y_1 = y'(x_1)(x - x_1)$

$y - 3 = 2(x - 0)$

$y = 2x + 3$

Normal gradient $= -\frac{1}{y'(x_1)} = -\frac{1}{2}$:

$y - 3 = -\frac{1}{2}(x - 0)$

$2y - 6 = -x$

$x + 2y - 6 = 0$

Answer: tangent y = 2x + 3; normal x + 2y − 6 = 0

Q6. Find the equation of the normal line to the curve $y = \ln x$ at $(1, 0)$. [24-Q]

Ans.

$y' = \frac{1}{x}$

$y'(1) = 1$

Normal gradient $= -\frac{1}{1} = -1$.

$y - 0 = -1(x - 1)$

$x + y - 1 = 0$

Answer: $x + y - 1 = 0$

9.2 Critical Point

Q7. Find the critical points of the function $f(x) = x^3(x - 4)$. [24-Q]

Ans.

$f(x) = x^4 - 4x^3$

$f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$

$f'(x) = 0 \rightarrow x = 0$ or $x = 3$ (f' is defined everywhere).

$f(0) = 0$

$f(3) = 3^3(3 - 4) = -27$

Answer: (0, 0) and (3, −27)

Q8. Find the critical points of $f(x) = 3 - (2x - 1)^4$. [25-Q]

Ans.

$f'(x) = -4(2x - 1)^3 \cdot 2 = -8(2x - 1)^3 = 0$

$2x - 1 = 0 \rightarrow x = \frac{1}{2}$

$f(\frac{1}{2}) = 3 - 0 = 3$

Answer: $(\frac{1}{2}, 3)$

9.2 Increasing and Decreasing Functions

Q9. Find the open interval on which the function $f(x) = 2x^3 - 6x + 15$ is decreasing. [25-Q]

Ans.

f is decreasing where $f'(x) < 0$.

$f'(x) = 6x^2 - 6 = 6(x - 1)(x + 1)$

Interval	$x < -1$	$-1 < x < 1$	$x > 1$
$x - 1$	−	−	+
$x + 1$	−	+	+
sign of $f'(x)$	+	−	+
Behavior of f	increasing	decreasing	increasing

Answer: $-1 < x < 1$

Q10. Determine the open interval on which the graph of the function $f(x) = 3x - x^3 - 7$ is increasing. [25-Q]

Ans.

f is increasing where $f'(x) > 0$.

$f'(x) = 3 - 3x^2 = 3(1 - x)(1 + x)$

Interval	$x < -1$	$-1 < x < 1$	$x > 1$
$1 - x$	+	+	−
$1 + x$	−	+	+
sign of $f'(x)$	−	+	−
Behavior of f	decreasing	increasing	decreasing

Answer: $-1 < x < 1$

9.2 Global Maximum and Minimum

Q11. Find the global maximum value of the function $f(x) = x^3 + 3x^2 - 9x + 1$ on $[-4, 2]$. [24-Q]

Ans.

Global extrema on a closed interval: evaluate f at the critical points inside the interval and at the end points; the smallest value is the global minimum and the largest is the global maximum.

$f'(x) = 3x^2 + 6x - 9 = 3(x + 3)(x - 1) = 0$

$x = -3, x = 1$ (both in $[-4, 2]$)

$f(-4) = -64 + 48 + 36 + 1 = 21$

$f(-3) = -27 + 27 + 27 + 1 = 28$

$f(1) = 1 + 3 - 9 + 1 = -4$

$f(2) = 8 + 12 - 18 + 1 = 3$

Answer: global maximum = 28 (at $x = -3$)

Q12. Find the sum of global minimum and global maximum values of the function $f(x) = x^2 + 2x - 12$ on $[-3, 3]$. [25-Q]

Ans.

Global extrema on a closed interval: evaluate f at the critical points inside the interval and at the end points; the smallest value is the global minimum and the largest is the global maximum.

$f'(x) = 2x + 2 = 0 \rightarrow x = -1$

$f(-3) = 9 - 6 - 12 = -9$

$f(-1) = 1 - 2 - 12 = -13$

$f(3) = 9 + 6 - 12 = 3$

Global minimum $= -13$, global maximum $= 3$.

$p + q = -13 + 3 = -10$

Answer: −10

Q13. If p and q are the global minimum and global maximum values of given equation $f(x) = x^3$ on $[-2, 1]$, then find the sum of p and q. [25-Q]

Ans.

$f'(x) = 3x^2 = 0 \rightarrow x = 0$

$f(-2) = -8$

$f(0) = 0$

$f(1) = 1$

$p = -8, q = 1$.

$p + q = -7$

Answer: −7

Q14. If p and q are the global minimum and global maximum values of given equation $f(x) = \frac{x}{x^2 + 1}$ on $[-2, 2]$, then find the sum of p and q. [25*-Q]

Ans.

$f'(x) = \frac{(x^2 + 1)(1) - x(2x)}{(x^2 + 1)^2} = \frac{1 - x^2}{(x^2 + 1)^2}$

$f'(x) = 0 \rightarrow x = \pm 1$ (the denominator is never 0).

$f(-2) = -\frac{2}{5}$

$f(-1) = -\frac{1}{2}$

$f(1) = \frac{1}{2}$

$f(2) = \frac{2}{5}$

$p = -\frac{1}{2}, q = \frac{1}{2}$.

$p + q = 0$

Answer: 0

9.2 Concavity and Inflection Point

Q15. State the point of inflection of the graph of the function $f(x) = x^3 - 6x^2 + 9x + 1$. [24-Q]

Ans.

$f'(x) = 3x^2 - 12x + 9$

$f''(x) = 6x - 12 = 0 \rightarrow x = 2$

$f(2) = 8 - 24 + 18 + 1 = 3$

$f'' < 0$ for $x < 2$ and $f'' > 0$ for $x > 2$, so the concavity changes at $x = 2$.

Answer: (2, 3)

Q16. Find the point of inflection of the function $f(x) = x^3 + 3x^2 - 2x + 1$. [26-Q]

Ans.

$f'(x) = 3x^2 + 6x - 2$

$f''(x) = 6x + 6 = 0 \rightarrow x = -1$

$f(-1) = -1 + 3 + 2 + 1 = 5$

$f'' < 0$ for $x < -1$ and $f'' > 0$ for $x > -1$, so the concavity changes at $x = -1$.

Answer: (−1, 5)

9.2 Second Derivative Test and Optimization

Q17. Find the number in the interval $[0, 3]$ such that the number minus its square is as large as possible. [24-Q]

Ans.

Let the number be x and $f(x) = x - x^2$.

$f'(x) = 1 - 2x = 0 \rightarrow x = \frac{1}{2}$

$f''(x) = -2 < 0$

$f'' < 0$, so $x = \frac{1}{2}$ is a local maximum. Compare with the end points:

$f(0) = 0$

$f(\frac{1}{2}) = \frac{1}{4}$

$f(3) = -6$

Answer: $x = \frac{1}{2}$ (maximum value $\frac{1}{4}$)

9.3 Linearization

Q18. Find the linearization of $f(x) = x^2 - 3x + 2$ at $x = 3$. [24-Q]

Ans.

Linearization of f at x = a: $L(x) = f(a) + f'(a)(x - a)$. Here a = 3.

$f(3) = 9 - 9 + 2 = 2$

$f'(x) = 2x - 3 \rightarrow f'(3) = 3$

$L(x) = 2 + 3(x - 3) = 3x - 7$

Answer: L(x) = 3x − 7

3 MARKS

9.1 Tangent Line and Normal Line

Q1. Find the equation of the tangent line to the curve $y = 2x^2 - 6x + 8$ at the point (2, 4). [24-Q]

Ans.

$y' = 4x - 6$

$y'(2) = 8 - 6 = 2$

Tangent: $y - y_1 = y'(x_1)(x - x_1)$

$y - 4 = 2(x - 2)$

$y = 2x$

Answer: y = 2x

Q2. Find the equation of the normal line to the curve $y = \sqrt{x - 1} / x$ at the point x = 2. [25-Q]

Ans.

When x = 2: $y = \frac{\sqrt{1}}{2} = \frac{1}{2}$, so the point is $(2, \frac{1}{2})$.

Quotient rule with $u = (x - 1)^{1/2}$, $v = x$:

$$y' = \frac{\frac{1}{2}(x - 1)^{-1/2} \cdot x - (x - 1)^{1/2} \cdot 1}{x^2}$$

$$y'(2) = \frac{\frac{1}{2}(1)(2) - 1}{4} = \frac{0}{4} = 0$$

The tangent is horizontal (gradient 0), so the normal is a vertical line through the point $(2, \frac{1}{2})$.

$x = 2$

Answer: x = 2

Q3. Find the equation of normal to the curve $y = x^2 - 5x + 6$ at the point (1, 2). [26-Q]

Ans.

$y' = 2x - 5$

$y'(1) = -3$

Normal gradient $= -\frac{1}{y'(x_1)} = -\frac{1}{-3} = \frac{1}{3}$.

$y - 2 = \frac{1}{3}(x - 1)$

$3y - 6 = x - 1$

$x - 3y + 5 = 0$

Answer: x − 3y + 5 = 0

9.2 Critical Point

Q4. Find the value of the x-coordinates of the critical point on the curve $y = x^x$, $x > 0$. [25-Q]

Ans.

Use logarithmic differentiation:

$\ln y = x \ln x$

$\frac{1}{y} \cdot \frac{dy}{dx} = \ln x + x \cdot \frac{1}{x} = \ln x + 1$

$\frac{dy}{dx} = x^x(\ln x + 1)$

Since $x^x > 0$ for $x > 0$, $dy/dx = 0$ only when

$\ln x + 1 = 0 \rightarrow \ln x = -1$

$x = e^{-1} = \frac{1}{e}$

Answer: $x = \frac{1}{e}$

9.2 Global Maximum and Minimum

Q5. Find the global minimum and global maximum values of the function $f(x) = x^3 - 6x^2 - 15x + 8$ on $[-2, 6]$. [24-Q]

Ans.

Global extrema on a closed interval: evaluate f at the critical points inside the interval and at the end points; the smallest value is the global minimum and the largest is the global maximum.

$f'(x) = 3x^2 - 12x - 15 = 3(x - 5)(x + 1) = 0$

$x = 5, x = -1$

$f(-2) = -8 - 24 + 30 + 8 = 6$

$f(-1) = -1 - 6 + 15 + 8 = 16$

$f(5) = 125 - 150 - 75 + 8 = -92$

$f(6) = 216 - 216 - 90 + 8 = -82$

Answer: global minimum = −92 (at x = 5); global maximum = 16 (at x = −1)

Q6. Use the first derivative test to find the global maximum and global minimum values of the function $f(x) = x^3 - 3x^2 - 9x + 5$ on $[-2, 6]$. [25*-Q]

Ans.

$f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1) = 0$

$x = -1, x = 3$

Sign of f′(x) on [−2, 6]:

Interval	$-2 \le x < -1$	$-1 < x < 3$	$3 < x \le 6$
x − 3	−	−	+
x + 1	−	+	+
sign of f′(x)	+	−	+
Behavior of f	increasing	decreasing	increasing

f changes from increasing to decreasing at x = −1 (local maximum) and from decreasing to increasing at x = 3 (local minimum). Compare with the end points:

$f(-2) = -8 - 12 + 18 + 5 = 3$ *(end point)*

$f(-1) = -1 - 3 + 9 + 5 = 10$ *(local max)*

$f(3) = 27 - 27 - 27 + 5 = -22$ *(local min)*

$f(6) = 216 - 108 - 54 + 5 = 59$ *(end point)*

Answer: global minimum = −22 (at x = 3); global maximum = 59 (at x = 6)

9.2 First Derivative Test

Q7. Use the first derivative test to find the local maximum and local minimum of $f(x) = (x^2 - 4)^{2/3}$. [25*-Q]

Ans.

$$f'(x) = \frac{2}{3}(x^2 - 4)^{-1/3}(2x) = \frac{4x}{3(x^2 - 4)^{1/3}}$$

Critical points: f′(x) = 0 at x = 0, and f′(x) is undefined at x = ±2 (these are in the domain of f).

Interval	$x < -2$	$-2 < x < 0$	$0 < x < 2$	$x > 2$
x	−	−	+	+
$x^2 - 4$	+	−	−	+
sign of f′(x)	−	+	−	+
Behavior of f	decreasing	increasing	decreasing	increasing

At x = −2 and x = 2: decreasing → increasing, so local minimum. At x = 0: increasing → decreasing, so local maximum.

$f(\pm 2) = 0$

$f(0) = (-4)^{2/3} = \sqrt[3]{16} = 2\sqrt[3]{2}$

Answer: local maximum $(0, 2\sqrt[3]{2})$; local minimum (−2, 0) and (2, 0)

9.2 Second Derivative Test and Optimization

Q8. Find the range of $g(x) = x - \ln x$. [25-Q]

Ans.

The domain is x > 0.

$g'(x) = 1 - \frac{1}{x} = 0 \rightarrow x = 1$

$g''(x) = \frac{1}{x^2} > 0$

$g''(1) > 0$, so x = 1 is a local (and global) minimum: $g(1) = 1 - \ln 1 = 1$. As $x \to 0^+$, $-\ln x \to +\infty$ and as $x \to \infty$, $x - \ln x \to \infty$, so g takes every value from 1 upwards.

Answer: range $y \ge 1$, i.e. $[1, \infty)$

Q9. Find the minimum value of the sum of a positive number and its reciprocal. [26-Q]

Ans.

Let the number be x > 0 and $S(x) = x + \frac{1}{x}$.

$S'(x) = 1 - \frac{1}{x^2} = 0 \rightarrow x = 1 \ (x > 0)$

$S''(x) = \frac{2}{x^3}$

$S''(1) = 2 > 0 \rightarrow$ minimum

$S(1) = 1 + 1 = 2$

Answer: minimum value = 2

9.3 Linearization

Q10. Use linearization to approximate the value $\sqrt{120}$. [25-Q]

Ans.

Take $f(x) = \sqrt{x}$ and choose $a = 121$ (the nearest perfect square to 120), so $f(a) = 11$ is easy.

$f'(x) = \frac{1}{2\sqrt{x}} \rightarrow f'(121) = \frac{1}{22}$

$L(x) = 11 + \frac{1}{22}(x - 121)$

Then $f(120) \approx L(120)$:

$\sqrt{120} \approx 11 + \frac{1}{22}(120 - 121) = 11 - \frac{1}{22} = \frac{241}{22} \approx 10.9545$

Answer: $\sqrt{120} \approx 10.9545$

5 MARKS

9.2 Concavity and Inflection Point

Q1. For the function $f(x) = x^3 - 27x$, find: (i) the open intervals on which f is increasing or decreasing; (ii) the function's local maximum and local minimum; (iii) the open intervals on which the graph of f is concave up or concave down; (iv) the point of inflection. [24-Q]

Ans.

(i) Increasing / decreasing

$f'(x) = 3x^2 - 27 = 3(x - 3)(x + 3) = 0$

$x = -3, x = 3$

Interval	$x < -3$	$-3 < x < 3$	$x > 3$
$x - 3$	−	−	+
$x + 3$	−	+	+
sign of $f'(x)$	+	−	+
Behavior of f	increasing	decreasing	increasing

f is increasing on $x < -3$ and $x > 3$; decreasing on $-3 < x < 3$.
(ii) Local extrema (first derivative test): at $x = -3$, increasing → decreasing: local maximum; at $x = 3$, decreasing → increasing: local minimum.

$f(-3) = -27 + 81 = 54$

$f(3) = 27 - 81 = -54$

Local maximum $(-3, 54)$; local minimum $(3, -54)$.
(iii) Concavity

$f''(x) = 6x = 0 \rightarrow x = 0$

$f'' > 0$ for $x > 0$: concave up on $x > 0$; $f'' < 0$ for $x < 0$: concave down on $x < 0$.
(iv) Inflection point: concavity changes at $x = 0$, $f(0) = 0$.
Answer: IP (0, 0)

Q2. For the function $f(x) = 2x^3 - 3x^2 - 12x + 4$, find: (i) the open intervals on which f is increasing or decreasing; (ii) the function's local maximum and local minimum; (iii) the open intervals on which the graph of f is concave up or concave down; (iv) the point of inflection. [25-Q]

Ans.

(i) Increasing / decreasing

$f'(x) = 6x^2 - 6x - 12 = 6(x - 2)(x + 1) = 0$

$x = -1, x = 2$

Interval	$x < -1$	$-1 < x < 2$	$x > 2$
$x - 2$	−	−	+
$x + 1$	−	+	+
sign of $f'(x)$	+	−	+
Behavior of f	increasing	decreasing	increasing

Increasing on $x < -1$ and $x > 2$; decreasing on $-1 < x < 2$.
(ii) Local extrema: local maximum at $x = -1$, local minimum at $x = 2$.

$f(-1) = -2 - 3 + 12 + 4 = 11$

$f(2) = 16 - 12 - 24 + 4 = -16$

Local maximum $(-1, 11)$; local minimum $(2, -16)$.
(iii) Concavity

$f''(x) = 12x - 6 = 0 \rightarrow x = \frac{1}{2}$

Concave up on $x > \frac{1}{2}$; concave down on $x < \frac{1}{2}$.
(iv) Inflection point

$f(\frac{1}{2}) = 2 \cdot \frac{1}{8} - 3 \cdot \frac{1}{4} - 6 + 4 = \frac{1}{4} - \frac{3}{4} - 2 = -\frac{5}{2}$

Answer: IP $(\frac{1}{2}, -\frac{5}{2})$

Q3. For the function $f(x) = -x^3 + 3x^2 + 24x - 1$, find: (i) the open intervals on which f is increasing or decreasing; (ii) the function's local maximum point and local minimum point; (iii) the open intervals on which the graph of f is concave up or concave down. [26-Q]

Ans.

(i) Increasing / decreasing

$f'(x) = -3x^2 + 6x + 24 = -3(x - 4)(x + 2) = 0$

$x = -2, x = 4$

Interval	$x < -2$	$-2 < x < 4$	$x > 4$
$x - 4$	−	−	+
$x + 2$	−	+	+
sign of $f'(x)$	−	+	−
Behavior of f	decreasing	increasing	decreasing

Increasing on $-2 < x < 4$; decreasing on $x < -2$ and $x > 4$.
(ii) Local extrema: decreasing → increasing at $x = -2$ (local minimum); increasing → decreasing at $x = 4$ (local maximum).

$f(-2) = 8 + 12 - 48 - 1 = -29$

$f(4) = -64 + 48 + 96 - 1 = 79$

Local maximum point $(4, 79)$; local minimum point $(-2, -29)$.
(iii) Concavity

$f''(x) = -6x + 6 = 0 \rightarrow x = 1$

Answer: concave up on $x < 1$ ($f'' > 0$); concave down on $x > 1$ ($f'' < 0$)

Q4. For the function $f(x) = x^4 - 4x^3$, find: (a) the open intervals on which the graph of f is concave up or concave down; (b) the points of inflection. [25-Q]

Ans.

$f'(x) = 4x^3 - 12x^2$

$f''(x) = 12x^2 - 24x = 12x(x - 2) = 0$

$x = 0, x = 2$

Interval	$x < 0$	$0 < x < 2$	$x > 2$
x	−	+	+
$x - 2$	−	−	+
sign of $f''(x)$	+	−	+
Behavior of f	concave up	concave down	concave up

(a) Concave up on $x < 0$ and $x > 2$; concave down on $0 < x < 2$.
(b) The concavity changes at $x = 0$ and $x = 2$:

$f(0) = 0$

$f(2) = 16 - 32 = -16$

Answer: IPs (0, 0) and (2, −16)

Q5. Find the intervals of concavity and the inflection points of $f(x) = x(x - 4)^3$. [25*-Q]

Ans.

$f'(x) = (x - 4)^3 + 3x(x - 4)^2 = (x - 4)^2(4x - 4)$

$f''(x) = 2(x - 4)(4x - 4) + 4(x - 4)^2 = (x - 4)(12x - 24) = 12(x - 4)(x - 2)$

$f''(x) = 0$ at $x = 2$ and $x = 4$.

Interval	$x < 2$	$2 < x < 4$	$x > 4$
$x - 4$	−	−	+
$x - 2$	−	+	+
sign of $f''(x)$	+	−	+
Behavior of f	concave up	concave down	concave up

$f(2) = 2(-2)^3 = -16$

$f(4) = 0$

Answer: concave up on $x < 2$ and $x > 4$; concave down on $2 < x < 4$; inflection points (2, −16) and (4, 0)

Q6. Determine the open intervals on which the graph of the function $f(x) = x^4 - 2x^3$ is concave up or concave down. State their inflection points. [26-Q]

Ans.

$f'(x) = 4x^3 - 6x^2$

$f''(x) = 12x^2 - 12x = 12x(x-1) = 0$

$x = 0, x = 1$

Interval	$x < 0$	$0 < x < 1$	$x > 1$
x	−	+	+
$x - 1$	−	−	+
sign of $f''(x)$	+	−	+
Behavior of f	concave up	concave down	concave up

$f(0) = 0$

$f(1) = 1 - 2 = -1$

Answer: concave up on $x < 0$ and $x > 1$; concave down on $0 < x < 1$; inflection points (0, 0) and (1, −1)

Q7. Determine the inflection points for the function $f(x) = x^4 - 24x^2 + 11$. [24-Q]

Ans.

$f'(x) = 4x^3 - 48x$

$f''(x) = 12x^2 - 48 = 12(x-2)(x+2) = 0$

$x = -2, x = 2$

$f'' > 0$ for $x < -2$, $f'' < 0$ for $-2 < x < 2$, $f'' > 0$ for $x > 2$, so the concavity changes at both points.

$f(\pm 2) = 16 - 96 + 11 = -69$

Answer: (−2, −69) and (2, −69)

Q8. Find all points of inflection of the function $f(x) = \frac{1}{20}x^5 - \frac{1}{6}x^4 - \frac{1}{2}x^3$. [25-Q]

Ans.

$f'(x) = \frac{1}{4}x^4 - \frac{2}{3}x^3 - \frac{3}{2}x^2$

$f''(x) = x^3 - 2x^2 - 3x = x(x^2 - 2x - 3) = x(x-3)(x+1) = 0$

$x = -1, x = 0, x = 3$

Interval	$x < -1$	$-1 < x < 0$	$0 < x < 3$	$x > 3$
x	−	−	+	+
$x - 3$	−	−	−	+
$x + 1$	−	+	+	+
sign of $f''(x)$	−	+	−	+
Behavior of f	concave down	concave up	concave down	concave up

f'' changes sign at all three values. The y-coordinates:

$f(-1) = -\frac{1}{20} - \frac{1}{6} + \frac{1}{2} = \frac{-3 - 10 + 30}{60} = \frac{17}{60}$

$f(0) = 0$

$f(3) = \frac{243}{20} - \frac{81}{6} - \frac{27}{2} = \frac{243}{20} - \frac{27}{2} - \frac{27}{2} = -\frac{297}{20}$

Answer: (0, 0), $(-1, \frac{17}{60})$, $(3, -\frac{297}{20})$

Q9. Given $f(x) = \sin x + 1.2$, find the points of inflection along the curve $y = f(x)$ for $-\frac{3\pi}{2} \le x \le \frac{5\pi}{2}$. [25*-Q]

Ans.

$f'(x) = \cos x$

$f''(x) = -\sin x = 0 \rightarrow x = n\pi$

In the interval $-\frac{3\pi}{2} \le x \le \frac{5\pi}{2}$ (about $-4.71 \le x \le 7.85$) the values are $x = -\pi, 0, \pi, 2\pi$. Since $f'' = -\sin x$ changes sign at each multiple of π, each is an inflection point.

$f(n\pi) = \sin(n\pi) + 1.2 = 1.2$

Answer: $(-\pi, 1.2)$, $(0, 1.2)$, $(\pi, 1.2)$, $(2\pi, 1.2)$

Q10. Show that there are two points of inflection on the curve with the equation $f(x) = x^2e^{0.3x}$. [5-Q]

Ans.

$f'(x) = 2xe^{0.3x} + 0.3x^2e^{0.3x} = e^{0.3x}(0.3x^2 + 2x)$

$f''(x) = 0.3e^{0.3x}(0.3x^2 + 2x) + e^{0.3x}(0.6x + 2)$

$= e^{0.3x}(0.09x^2 + 0.6x + 0.6x + 2)$

$= e^{0.3x}(0.09x^2 + 1.2x + 2)$

Since $e^{0.3x} > 0$, $f''(x) = 0$ only when $0.09x^2 + 1.2x + 2 = 0$. The discriminant is

$b^2 - 4ac = 1.2^2 - 4(0.09)(2) = 1.44 - 0.72 = 0.72 > 0$

so there are two distinct real roots:

$x = \frac{-1.2 \pm \sqrt{0.72}}{0.18} \rightarrow x \approx -1.95$ or $x \approx -11.38$

The quadratic factor changes sign at each distinct root, so f'' changes sign at both. Hence the curve has exactly two points of inflection.

Shown: 2 inflection points

9.2 Second Derivative Test and Optimization

Q11. Find and classify the critical points of $f(x) = x^3 - 3x + 2$. [24-Q]

Ans.

$f'(x) = 3x^2 - 3 = 3(x-1)(x+1) = 0$

$x = -1, x = 1$

$f''(x) = 6x$

Second derivative test: if $f'(c) = 0$ and $f''(c) > 0$, f has a local minimum at c; if $f'(c) = 0$ and $f''(c) < 0$, a local maximum. If $f''(c) = 0$ the test fails.

$f''(-1) = -6 < 0 \rightarrow$ **local maximum**

$f(-1) = -1 + 3 + 2 = 4$

$f''(1) = 6 > 0 \rightarrow$ **local minimum**

$f(1) = 1 - 3 + 2 = 0$

Answer: local maximum at (−1, 4); local minimum at (1, 0)

Q12. Find and classify the critical points of $f(x) = x^4 - 4x^3 + 7$. [25-Q]

Ans.

$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3) = 0$

$x = 0, x = 3$

$f''(x) = 12x^2 - 24x = 12x(x-2)$

At $x = 3$: $f''(3) = 36 > 0$, so a local minimum: $f(3) = 81 - 108 + 7 = -20$.

At $x = 0$: $f''(0) = 0$, so the second derivative test fails. Use the first derivative test: $f'(x) = 4x^2(x-3) < 0$ on both sides of 0 (for $x < 3$), so f is decreasing through $x = 0$ — no local extremum. Since $f''(x) = 12x(x-2)$ changes sign at 0, (0, 7) is a point of inflection.

Answer: local minimum at (3, −20); (0, 7) is a point of inflection (not an extremum)

Q13. Find and classify the critical points of $f(x) = xe^x$. [24-Q]

Ans.

$f'(x) = e^x + xe^x = e^x(x+1) = 0 \rightarrow x = -1$

$f''(x) = e^x(x+1) + e^x = e^x(x+2)$

$f''(-1) = e^{-1} > 0$

$f''(-1) > 0$, so $x = -1$ is a local minimum.

$f(-1) = -e^{-1} = -\frac{1}{e}$

Answer: local minimum at $(-1, -\frac{1}{e})$

Q14. Find the x-coordinates of the critical and the inflection points of the function $y = x^4e^{0.5x}$. What is the minimum point of the function? [24-Q]

Ans.

$y' = 4x^3e^{0.5x} + 0.5x^4e^{0.5x} = 0.5x^3e^{0.5x}(x+8)$

$y' = 0 \rightarrow x = 0$ or $x = -8$. **Critical points: x = 0, x = −8.**

$y'' = (2x^3 + 12x^2)e^{0.5x} + 0.5(0.5x^4 + 4x^3)e^{0.5x}$

$= e^{0.5x}(0.25x^4 + 4x^3 + 12x^2)$

$= 0.25x^2e^{0.5x}(x^2 + 16x + 48)$

$= 0.25x^2e^{0.5x}(x+4)(x+12)$

$y'' = 0$ at $x = 0, -4, -12$. At $x = 0$ the factor x^2 does not change sign, so it is not an inflection point. **Inflection points: x = −4 and x = −12.**

Classification of the critical points:

$y''(-8) = 0.25(64)e^{-4}(-4)(4) < 0 \rightarrow$ local maximum at $x = -8$

At $x = 0$, $y''(0) = 0$ (test fails). First derivative test: $y' = 0.5x^3e^{0.5x}(x+8)$ is negative for $-8 < x < 0$ and positive for $x > 0$, so y has a local minimum at $x = 0$: $y(0) = 0$.

Answer: critical points x = 0, −8; inflection points x = −4, −12; minimum point (0, 0)

Q15. Find the minimum value of the sum of a positive number and its reciprocal. [25-Q]

Ans.

Let the number be $x > 0$ and $S(x) = x + \frac{1}{x}$.

$S'(x) = 1 - \frac{1}{x^2} = 0 \rightarrow x = 1$

$S''(x) = \frac{2}{x^3}$, $S''(1) = 2 > 0 \rightarrow$ minimum

$S(1) = 2$

Answer: minimum value = 2

Q16. Two positive numbers x and y are such that $xy = 18$. Another number z is defined by $z = 2x + y$. Find the values of x and y for which z has a critical value and show that this value of z is minimum. [25-Q]

Ans.

From $xy = 18$: $y = \frac{18}{x}$. Then

$z = 2x + \frac{18}{x}$

$z' = 2 - \frac{18}{x^2} = 0$

$x^2 = 9 \rightarrow x = 3\ (x > 0),\ y = \frac{18}{3} = 6$

Second derivative test:

$z'' = \frac{36}{x^3},\ z''(3) = \frac{36}{27} = \frac{4}{3} > 0$

$z''(3) > 0$, so z is minimum at x = 3. Minimum value $z = 2(3) + 6 = 12$.

Answer: x = 3, y = 6 (z = 12 is a minimum)

Q17. A farmer has 8 km of fencing wire, and wishes to fence a rectangular piece of land. One boundary of the land is the bank of a straight river. What are the dimensions of the rectangle so that the area is maximized? [25*-Q]

Ans.

Let the two sides perpendicular to the river be x km each and the side parallel to the river be y km (no fence along the river).

$2x + y = 8 \rightarrow y = 8 - 2x$

$A = xy = x(8 - 2x) = 8x - 2x^2$

$A' = 8 - 4x = 0 \rightarrow x = 2$

$A'' = -4 < 0 \rightarrow$ maximum

Then $y = 8 - 4 = 4$. Maximum area $= 2 \times 4 = 8\ km^2$.

Answer: 4 km (parallel to the river) × 2 km

Q18. A cylinder has a height of h meters and a radius of r meters. If the sum of the height and radius of the cylinder is 3 meters, find an expression for the volume of the cylinder in terms of r. Find also the maximum volume of the cylinder. [25-Q]

Ans.

$h + r = 3 \rightarrow h = 3 - r$

$V = \pi r^2 h = \pi r^2(3 - r) = 3\pi r^2 - \pi r^3$

Maximum volume:

$V' = 6\pi r - 3\pi r^2 = 3\pi r(2 - r) = 0$

$r = 2$ (r = 0 gives no volume)

$V'' = 6\pi - 6\pi r,\ V''(2) = -6\pi < 0 \rightarrow$ maximum

r = 2, h = 1:

$V = \pi(2)^2(1) = 4\pi$

Answer: $V = 3\pi r^2 - \pi r^3$; maximum volume = $4\pi\ m^3$

Q19. A rectangle is to be inscribed in a semicircle of radius 2. What is the largest area the rectangle can have? What are its dimensions? [25-Q]

Ans.

Place the semicircle $y = \sqrt{4 - x^2}$. A rectangle with a vertex (x, y) on the arc has base 2x and height y ($0 < x < 2$).

$A = 2xy = 2x\sqrt{4 - x^2}$

$A' = 2\sqrt{4 - x^2} - \{2x^2/\sqrt{4 - x^2}\} = \{2(4 - x^2) - 2x^2/\sqrt{4 - x^2}\} = \{8 - 4x^2/\sqrt{4 - x^2}\} = 0$

$x^2 = 2 \rightarrow x = \sqrt{2}$

$A' > 0$ for $0 < x < \sqrt{2}$ and $A' < 0$ for $\sqrt{2} < x < 2$, so $x = \sqrt{2}$ gives the maximum (first derivative test).

Base $= 2x = 2\sqrt{2}$

Height $= \sqrt{4 - 2} = \sqrt{2}$

Area $= 2\sqrt{2} \cdot \sqrt{2} = 4$

Answer: largest area = 4; base $2\sqrt{2}$, height $\sqrt{2}$

Q20. Find the point on the curve $y^2 = 4x$ which is nearest to the point (2, 1). [24-Q]

Ans.

For a point (x, y) on the curve, $x = \frac{y^2}{4}$. The squared distance to (2, 1) is

$D = (x - 2)^2 + (y - 1)^2 = (\frac{y^2}{4} - 2)^2 + (y - 1)^2$

$D' = 2(\frac{y^2}{4} - 2)(\frac{y}{2}) + 2(y - 1) = \frac{y^3}{4} - 2y + 2y - 2 = \frac{y^3}{4} - 2$

$D' = 0 \rightarrow y^3 = 8 \rightarrow y = 2,\ x = \frac{4}{4} = 1$

$D'' = \frac{3y^2}{4},\ D''(2) = 3 > 0 \rightarrow$ minimum

Minimum distance $= \sqrt{(1 - 2)^2 + (2 - 1)^2} = \sqrt{2}$.

Answer: (1, 2)

Q21. A rectangular field is surrounded by a fence on three of its sides and a straight hedge on the fourth side. If the length of the fence is 160 meters, find the maximum area of the field enclosed. [26-Q]

Ans.

Let the two sides perpendicular to the hedge be x m each and the side parallel to the hedge be y m.

$2x + y = 160 \rightarrow y = 160 - 2x$

$A = xy = 160x - 2x^2$

$A' = 160 - 4x = 0 \rightarrow x = 40$

$A'' = -4 < 0 \rightarrow$ maximum

$y = 160 - 80 = 80$.

$A = 40 \cdot 80 = 3200$

Answer: maximum area = 3200 m^2

Q22. The profit y of a firm by manufacturing x bicycles is $y = \frac{-x^3}{12} + 5x^2 - 36x - 300$. How many bicycles should the firm manufacture to maximize the profit? [26-Q]

Ans.

$y' = -\frac{x^2}{4} + 10x - 36 = 0$

$x^2 - 40x + 144 = 0$

$(x - 4)(x - 36) = 0 \rightarrow x = 4$ or $x = 36$

$y'' = -\frac{x}{2} + 10$

$y''(4) = -2 + 10 = 8 > 0 \rightarrow$ **local minimum**

$y''(36) = -18 + 10 = -8 < 0 \rightarrow$ **local maximum**

Answer: 36 bicycles

9.3 Linearization

Q23. Use linearization to approximate the value $\sqrt[3]{513}$. [25-Q]

Ans.

Take $f(x) = \sqrt[3]{x} = x^{1/3}$ and a = 512 (since $8^3 = 512$), so f(a) = 8.

$f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}} \rightarrow f'(512) = \frac{1}{3(64)} = \frac{1}{192}$

$L(x) = 8 + \frac{1}{192}(x - 512)$

Then $f(513) \approx L(513)$:

$\sqrt[3]{513} \approx 8 + \frac{1}{192}(1) = 8 + 0.0052 = 8.0052$

Answer: $\sqrt[3]{513} \approx 8.0052$

CH.10 Integration

Notes · Questions with Answers

NOTES

Antiderivative

If F(x) is a function such that $F'(x) = f(x)$ for all x in some interval, then F(x) is called an **antiderivative** of f(x). Antiderivatives are not unique: if F(x) is an antiderivative of f(x), so is F(x) + C for any constant C.

Indefinite Integral

The set of all antiderivatives of f(x) is called the **indefinite integral** of f(x) with respect to x, written $\int f(x)\,dx = F(x) + C$, where $F'(x)=f(x)$.

Constant of Integration

The arbitrary constant C that appears in every indefinite integral is called the **constant of integration**. It arises because the derivative of any constant is zero, so infinitely many functions share the same derivative.

Integrand

In $\int f(x)\,dx$, the function f(x) is called the **integrand**, and dx indicates that the integration is taking place with respect to the variable x.

Rules of Integration

Suppose f(x) and g(x) are continuous functions and $k \in \mathbb{R}$.

1. $\int k\,dx = kx + C$
2. $\int k\,f(x)\,dx = k\int f(x)\,dx$
3. $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

Power Rule

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad (n \neq -1)$$

Linear Substitution Rule

If F(x) is an antiderivative of f(x) (so $F'(x)=f(x)$), and a, b are constants with $a \neq 0$, then:

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$$

Special case — for a derivative $f'(ax+b)$ already in this form:

$$\int f'(ax+b)\,dx = \frac{1}{a}f(ax+b) + C$$

This shortcut avoids a full substitution when the integrand is a simple function of a linear expression ax+b. Common cases:

Integral	Result
$\int (ax+b)^n\,dx\ (n\neq -1)$	$\frac{(ax+b)^{n+1}}{a(n+1)} + C$
$\int \frac{1}{ax+b}\,dx$	$\frac{1}{a}\ln\lvert ax+b\rvert + C$
$\int e^{ax+b}\,dx$	$\frac{1}{a}e^{ax+b} + C$
$\int \sin(ax+b)\,dx$	$-\frac{1}{a}\cos(ax+b) + C$
$\int \cos(ax+b)\,dx$	$\frac{1}{a}\sin(ax+b) + C$
$\int \sec^2(ax+b)\,dx$	$\frac{1}{a}\tan(ax+b) + C$

Standard Integral Forms

Integral	Result
$\int \frac{1}{x}\,dx$	$\ln\lvert x\rvert + C$
$\int e^x\,dx$	$e^x + C$
$\int a^x\,dx\ (a>0, a\neq 1)$	$\frac{a^x}{\ln a} + C$
$\int \sin x\,dx$	$-\cos x + C$
$\int \cos x\,dx$	$\sin x + C$
$\int \sec^2 x\,dx$	$\tan x + C$
$\int \csc^2 x\,dx$	$-\cot x + C$
$\int \sec x\tan x\,dx$	$\sec x + C$
$\int \csc x\cot x\,dx$	$-\csc x + C$
$\int (1+\tan^2 x)\,dx = \int \sec^2 x\,dx$	$\tan x + C$

Additional Results

$\int \sec x\,dx = \ln\lvert\sec x + \tan x\rvert + C$
$\int \csc x\,dx = -\ln\lvert\csc x + \cot x\rvert + C$
$\int \tan x\,dx = \ln\lvert\sec x\rvert + C$
$\int \cot x\,dx = \ln\lvert\sin x\rvert + C$

Trigonometric Formulas (useful for Integration)

Pythagorean Identities

$\sin^2 x + \cos^2 x = 1$
$1 + \tan^2 x = \sec^2 x$
$1 + \cot^2 x = \csc^2 x$

Double-Angle Formulas

$\sin 2x = 2\sin x\cos x$
$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$
$\tan 2x = \frac{2\tan x}{1-\tan^2 x}$

Power-Reducing (Half-Angle) Formulas — used to integrate $\sin^2 x$ and $\cos^2 x$:

$\sin^2 x = \frac{1-\cos 2x}{2}$
$\cos^2 x = \frac{1+\cos 2x}{2}$

Product-to-Sum Formulas — used to integrate products such as $\sin 3x\cos 2x$:

$\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$
$\cos A\sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$
$\cos A\cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$
$\sin A\sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$

Substitution Method — Steps

We use the substitution method when the given integral can be transformed to a simpler integral by a change of variable.

1. Select a substitution $u = g(x)$.
2. Differentiate the substitution and arrange to write dx in terms of du.
3. Substitute and transform the entire integral from x-variable to u-variable form.
4. Integrate with respect to u.
5. Rewrite the answer in terms of x.

Integration by Parts Formula

$\int u\,dv = uv - \int v\,du$

ILATE Rule (Choosing u)

When the integrand is a product of two different types of functions, ILATE gives the priority order for choosing u (the earlier a function's type appears in ILATE, the more likely it should be chosen as u):

Letter	Function type	Example
I	Inverse trigonometric	$\sin^{-1} x$, $\tan^{-1} x$
L	Logarithmic	$\ln x$
A	Algebraic	x, x^2, $\sqrt{x}$
T	Trigonometric	$\sin x$, $\cos x$
E	Exponential	e^x, a^x

The remaining factor (along with dx) is taken as dv. For example, in $\int x \ln x\, dx$, L comes before A, so $u = \ln x$ and $dv = x\, dx$.

Tabular Method (Repeated Integration by Parts)

For a product where one factor (usually a polynomial) eventually differentiates to 0 — e.g. $\int x^2 e^x\, dx$ — repeated integration by parts can be organized in a table instead of applying the formula over and over:

1. Make two columns: D (successive derivatives of the polynomial factor, down to 0) and I (successive integrals of the other factor).
2. Pair each D-row with the I-row directly below it, and attach alternating signs +, −, +, −, ... starting with +.
3. The answer is the sum of each signed D×I product (stop once the D column reaches 0), plus C.

D (differentiate)	I (integrate)
+ x^2	e^x
− $2x$	e^x
+ 2	e^x
0	e^x

$\int x^2 e^x\, dx = x^2e^x - 2xe^x + 2e^x + C$ — read off by multiplying diagonally (each D entry by the I entry one row below it) and adding with the shown signs.

Partial Fraction Method — Approach

We use the partial fraction method to integrate rational functions by decomposing them into simpler fractional components.

1. Factor the denominator completely.
2. Write the rational function as a sum of simpler fractions with unknown constants (A, B, ...) over each factor — a repeated linear factor $(x-a)^2$ gets two terms, $\frac{A}{x-a}$ and $\frac{B}{(x-a)^2}$.
3. Clear denominators and solve for the unknown constants, either by equating coefficients or by substituting convenient values of x.
4. Integrate each simpler fraction separately using standard forms such as $\int \frac{1}{x-a}\, dx = \ln|x-a| + C$.

Cover-Up Method (Partial Fractions)

A quick shortcut for finding the constants when the denominator factors into distinct linear factors, e.g. $\frac{P(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$:

1. To find A: cover up the factor $(x-a)$ in the original fraction, and substitute $x=a$ into what remains.
2. To find B: cover up the factor $(x-b)$, and substitute $x=b$ into what remains.

Example: $\frac{5x-1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$:

$A = \frac{5(1)-1}{1+2} = \frac{4}{3}$ (cover $x-1$, put $x=1$)

$B = \frac{5(-2)-1}{(-2)-1} = \frac{-11}{-3} = \frac{11}{3}$ (cover $x+2$, put $x=-2$)

This only works directly for non-repeated linear factors; repeated or irreducible quadratic factors still need the equate-coefficients approach above.

1 MARK

10.1 Antiderivatives

Q1. The antiderivative of $2^x \ln 2$ is [S-Q]

A. $\ln x$ **B.** $\ln 2$ **C.** $2^x + C$ **D.** 2

Ans.

Basic formula: $\int a^x\, dx = \frac{a^x}{\ln a} + C$.

$\int 2^x \ln 2\, dx = \ln 2 \cdot \int 2^x\, dx = \ln 2 \cdot \frac{2^x}{\ln 2} + C$

$= 2^x + C$

Answer: C

Q2. $\int \frac{1}{3x}\, dx$ is equal to [24-Q]

A. $-\frac{1}{3}\ln|x| + C$ **B.** $\frac{1}{3}\ln|x| + C$ **C.** $\ln|3x| + C$ **D.** $-\ln|3x| + C$

Ans.

Take the constant $\frac{1}{3}$ out and use $\int \frac{1}{x}\, dx = \ln|x| + C$.

$\int \frac{1}{3x}\, dx = \frac{1}{3}\int \frac{1}{x}\, dx$

$= \frac{1}{3}\ln|x| + C$

Answer: B

Q3. $\int \frac{1}{1-2x}\, dx$ is [25-Q]

A. $-\frac{1}{2}\ln|1-2x| + C$ **B.** $-\frac{1}{2}\ln|1-2x|$ **C.** $\frac{1}{2}\ln|1-2x| + C$ **D.** $\frac{1}{2}\ln|1-2x|$

Ans.

Use $\int f'(ax+b)\, dx = \frac{1}{a} f(ax+b) + C$, i.e. $\int \frac{1}{ax+b}\, dx = \frac{1}{a}\ln|ax+b| + C$. Here $a = -2$.

$\int \frac{1}{1-2x}\, dx = \frac{1}{-2}\ln|1-2x| + C$

$= -\frac{1}{2}\ln|1-2x| + C$

Answer: A

Q4. If $\int \frac{15x^4 - 2x}{x^2}\, dx = ax^3 + b\ln|x| + C$, where a and b are integers, then $a + b$ is [25-Q]

A. 1 **B.** 2 **C.** 3 **D.** 4

Ans.

Divide each term by x^2:

$\int \frac{15x^4 - 2x}{x^2}\, dx = \int (15x^2 - \frac{2}{x})\, dx$

$= 15 \cdot \frac{x^3}{3} - 2\ln|x| + C$

$= 5x^3 - 2\ln|x| + C$

So $a = 5$ and $b = -2$.

$a + b = 5 - 2 = 3$

Answer: C

Q5. If $\int \sqrt{1-5x}\, dx = p(1-5x)^{3/2} + C$, then p is [25-Q]

A. $\frac{1}{2}$ **B.** $\frac{15}{2}$ **C.** $\frac{2}{3}$ **D.** $-\frac{2}{15}$

Ans.

Write $\sqrt{1-5x} = (1-5x)^{1/2}$ and use $\int (ax+b)^n\, dx = \frac{1}{a(n+1)}(ax+b)^{n+1} + C$ with $a = -5$, $n = \frac{1}{2}$.

$\int (1-5x)^{1/2}\, dx = \frac{1}{-5 \cdot \frac{3}{2}}(1-5x)^{3/2} + C$

$= -\frac{2}{15}(1-5x)^{3/2} + C$

So $p = -\frac{2}{15}$.

Answer: D

Q6. $\int 2^{3x+1}\, dx =$ [S-Q]

A. $\frac{1}{\ln 2} 2^{3x+1} + C$ **B.** $\frac{1}{3} 2^{3x+1} + C$ **C.** $\frac{1}{3\ln 2} 2^{3x+1} + C$ **D.** $\frac{1}{6\ln 2} 2^{3x+1} + C$

Ans.

Use $\int a^{bx+c}\, dx = \frac{1}{b \ln a} a^{bx+c} + C$ with $a = 2$, $b = 3$ (the factor $f(ax+b)$ rule gives $\frac{1}{3}$).

$\int 2^{3x+1}\, dx = \frac{1}{3} \cdot \frac{2^{3x+1}}{\ln 2} + C$

$= \frac{1}{3\ln 2} 2^{3x+1} + C$

Answer: C

Q7. $\int 3^{2x-1}\, dx =$ [25*-Q, 26-Q]

A. $\frac{1}{2} 3^{2x-1} + C$ **B.** $\frac{1}{6} 3^{2x-1} + C$ **C.** $\frac{1}{2\ln 3} 3^{2x-1} + C$ **D.** $\frac{1}{6\ln 3} 3^{2x-1} + C$

Ans.

$\int a^x\, dx = \frac{a^x}{\ln a} + C$ and the inner function $2x - 1$ has coefficient 2:

$\int 3^{2x-1}\, dx = \frac{1}{2} \cdot \frac{3^{2x-1}}{\ln 3} + C$

$= \frac{1}{2\ln 3}\, 3^{2x-1} + C$

Answer: C

Q8. If $\int e^{-3x}\,dx = p + C$, then p is [24-Q]

A. $-\frac{e^{-3x}}{3}$ **B.** $\frac{e^{-3x}}{3}$ **C.** e^{3x} **D.** e^{-3x}

Ans.

$\int e^{ax}\,dx = \frac{1}{a}\, e^{ax} + C$ with $a = -3$.

$\int e^{-3x}\,dx = \frac{1}{-3}\, e^{-3x} + C = -\frac{e^{-3x}}{3} + C$

Answer: A

Q9. If $\int e^{-3x}\,dx = p + C$, then 3p is [25-Q]

A. $-e^{-3x}$ **B.** e^{3x} **C.** $\frac{1}{3}\, e^{-3x}$ **D.** $3e^{-3x}$

Ans.

From above, $p = -\frac{1}{3}\, e^{-3x}$.

$3p = 3\cdot(-\frac{1}{3}\, e^{-3x}) = -e^{-3x}$

Answer: A

Q10. If $\int e^{-4x}\,dx = p + C$, then 4p is [25*-Q]

A. $-e^{-4x}$ **B.** e^{4x} **C.** $\frac{1}{2}\, e^{-4x}$ **D.** $2e^{-4x}$

Ans.

$\int e^{-4x}\,dx = -\frac{1}{4}\, e^{-4x} + C$, so $p = -\frac{1}{4}\, e^{-4x}$.

$4p = -e^{-4x}$

Answer: A

Q11. The integration $\int (1 + \tan^2(2x + 1))\,dx =$ [5-Q]

A. $\ln\cos(2x + 1)$ **B.** $\frac{1}{2}\ln\cos(2x + 1)$ **C.** $\frac{1}{2}\tan(2x + 1)$ **D.** $2\tan(2x + 1)$

Ans.

Use $1 + \tan^2\theta = \sec^2\theta$, then $\int \sec^2(ax + b)\,dx = \frac{1}{a}\tan(ax + b) + C$.

$\int (1 + \tan^2(2x + 1))\,dx = \int \sec^2(2x + 1)\,dx$

$= \frac{1}{2}\tan(2x + 1) + C$

Answer: C

Q12. The integration $\int (1 - \tan^2(2x + 1))\,dx =$ [24-Q]

A. $\ln\cos(2x + 1)$ **B.** $2x - \frac{1}{2}\tan(2x + 1)$ **C.** $2\tan(2x + 1)$ **D.** $\frac{1}{2}\ln\cos(2x + 1)$

Ans.

Use $\tan^2\theta = \sec^2\theta - 1$, so $1 - \tan^2\theta = 2 - \sec^2\theta$.

$\int (1 - \tan^2(2x + 1))\,dx = \int (2 - \sec^2(2x + 1))\,dx$

$= 2x - \frac{1}{2}\tan(2x + 1) + C$

Answer: B

Q13. $\int \sin 7x \cos 2x\,dx =$ [24-Q]

A. $\frac{1}{10}\sin 5x - \frac{1}{18}\sin 9x$ **B.** $\frac{1}{18}\sin 9x + \frac{1}{10}\sin 5x$

C. $-\frac{1}{18}\cos 9x + \frac{1}{10}\cos 5x$ **D.** $-\frac{1}{18}\cos 9x - \frac{1}{10}\cos 5x$

Ans.

Product formula: $\sin mx \cos nx = \frac{1}{2}[\sin(m - n)x + \sin(m + n)x]$ with $m = 7$, $n = 2$.

$\sin 7x \cos 2x = \frac{1}{2}(\sin 5x + \sin 9x)$

$\int \sin 7x \cos 2x\,dx = \frac{1}{2}(-\frac{\cos 5x}{5} - \frac{\cos 9x}{9}) + C$

$= -\frac{1}{10}\cos 5x - \frac{1}{18}\cos 9x + C$

Answer: D

10.2 Substitution Method

Q14. If $\int x^3 e^{x^4}\,dx =$ [25-Q]

A. e^{x^4} **B.** $\frac{1}{4}e^{x^4}$ **C.** $4e^{x^4}$ **D.** $\frac{1}{4}e^{x^4} + C$

Ans.

Let $u = x^4$, then $du = 4x^3\,dx$, so $x^3\,dx = \frac{1}{4}\,du$.

$\int x^3 e^u\,dx = \frac{1}{4}\int e^u\,du$

$= \frac{1}{4}e^u + C = \frac{1}{4}e^{x^4} + C$

(The answer must include + C.)

Answer: D

Q15. Evaluate $\int \frac{1}{x\ln x}\,dx$. [24-Q]

A. $\ln x$ **B.** $(\ln x)^2$ **C.** $\ln(\ln x)$ **D.** $\ln x \ln x$

Ans.

Let $u = \ln x$, $du = \frac{1}{x}\,dx$.

$\int \frac{1}{x\ln x}\,dx = \int \frac{1}{u}\,du = \ln|u| + C$

$= \ln|\ln x| + C$

Answer: C

Q16. $\int \frac{(1 - \ln x)^2}{x}\,dx =$ [25-Q]

A. $-\frac{1}{3}(1 - \ln x)^3 + C$ **B.** $\frac{1}{3}(1 - \ln x)^3 + C$ **C.** $-2(1 - \ln x) + C$ **D.** $-\frac{(1 - \ln x)^3}{3}$

Ans.

Let $u = 1 - \ln x$, $du = -\frac{1}{x}\,dx$, so $\frac{1}{x}\,dx = -du$.

$\int \frac{(1 - \ln x)^2}{x}\,dx = -\int u^2\,du$

$= -\frac{u^3}{3} + C = -\frac{(1 - \ln x)^3}{3} + C$

Options A and D are the same function; the book key is D (A is the same function with + C written).

Answer: D

Q17. $\int \frac{\sec^2 2x}{\tan 2x}\,dx =$ [25*-Q]

A. $\frac{1}{2}\ln|\tan 2x| + C$ **B.** $\ln|\tan 2x| + C$ **C.** $2\ln|\tan 2x| + C$ **D.** $4\ln|\tan 2x| + C$

Ans.

Let $u = \tan 2x$, $du = 2\sec^2 2x\,dx$, so $\sec^2 2x\,dx = \frac{1}{2}\,du$.

$\int \frac{\sec^2 2x}{\tan 2x}\,dx = \frac{1}{2}\int \frac{1}{u}\,du$

$= \frac{1}{2}\ln|u| + C = \frac{1}{2}\ln|\tan 2x| + C$

Answer: A

Q18. If $\int \frac{2}{\sqrt{x}}\sin\sqrt{x}\,dx = p\cos\sqrt{x} + C$, then p is [25-Q]

A. -6 **B.** -4 **C.** -2 **D.** 4

Ans.

Let $u = \sqrt{x}$, $du = \frac{1}{2\sqrt{x}}\,dx$, so $\frac{dx}{\sqrt{x}} = 2\,du$.

$\int \frac{2}{\sqrt{x}}\sin\sqrt{x}\,dx = \int 2\sin u \cdot 2\,du = 4\int \sin u\,du$

$= -4\cos u + C = -4\cos\sqrt{x} + C$

So $p = -4$.

Answer: B

Q19. If $\int 2xe^{x^2}\,dx = ke^{x^2} + C$, then the value of k is [25^{II}-Q, 26^{II}-Q]

A. $-\frac{1}{2}$ **B.** $\frac{1}{2}$ **C.** -1 **D.** 1

Ans.

Let $u = x^2$, $du = 2x\,dx$.

$\int 2xe^{x^2}\,dx = \int e^u\,du = e^u + C$

$= e^{x^2} + C$

So $k = 1$.

Answer: D

Q20. If $\int \frac{\ln x}{x}\,dx = a(\ln x)^n + C$, where a and n are rational numbers, then $2a + n$ is [26-Q]

A. 1 **B.** 2 **C.** 3 **D.** 4

Ans.

Let $u = \ln x$, $du = \frac{1}{x}\,dx$.

$\int \frac{\ln x}{x}\,dx = \int u\,du = \frac{u^2}{2} + C = \frac{1}{2}(\ln x)^2 + C$

So $a = \frac{1}{2}$, $n = 2$.

$2a + n = 1 + 2 = 3$

Answer: C

Q21. If $\int x\sqrt{1 - x^2}\,dx = k(1 - x^2)^{3/2} + C$, then $k =$ [26-Q]

A. $-\frac{1}{3}$ **B.** $\frac{1}{3}$ **C.** $-\frac{2}{3}$ **D.** $\frac{2}{3}$

Ans.

Let $u = 1 - x^2$, $du = -2x\,dx$, so $x\,dx = -\frac{1}{2}\,du$.

$$\int x\sqrt{1-x^2}\,dx = -\frac{1}{2}\int u^{1/2}\,du$$

$$= -\frac{1}{2}\cdot\frac{u^{3/2}}{3/2} + C = -\frac{1}{3}(1-x^2)^{3/2} + C$$

So $k = -\frac{1}{3}$.

Answer: A

10.3 Integration by Parts

Q22. If $\int x \ln x\,dx = \frac{x^2 \ln x}{p} + \frac{x^2}{q} + C$, then $p + q$ is [5-Q]

A. -2 **B.** -1 **C.** 0 **D.** 1

Ans.

Integration by parts: $\int u\,dv = uv - \int v\,du$.

Let $u = \ln x$, $dv = x\,dx \rightarrow du = \frac{1}{x}\,dx$, $v = \frac{x^2}{2}$.

$$\int x \ln x\,dx = \frac{x^2}{2}\ln x - \int \frac{x^2}{2}\cdot\frac{1}{x}\,dx$$

$$= \frac{x^2 \ln x}{2} - \frac{1}{2}\int x\,dx$$

$$= \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C$$

So $p = 2$, $q = -4$.

$p + q = -2$

Answer: A

Q23. If $\int x \ln x\,dx = \frac{x^2 \ln x}{p} - \frac{x^2}{q} + C$, then $p - q$ is [24-Q]

A. -6 **B.** -2 **C.** 6 **D.** 2

Ans.

From the result $\int x \ln x\,dx = \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C$: $p = 2$, $q = 4$.

$p - q = 2 - 4 = -2$

Answer: B

Q24. If $\int x^2 \ln x\,dx = \frac{x^3 \ln x}{a} + \frac{x^3}{b} + C$, then $a - b$ is [25-Q]

A. -6 **B.** -2 **C.** 6 **D.** 12

Ans.

Integration by parts: $\int u\,dv = uv - \int v\,du$.

Let $u = \ln x$, $dv = x^2\,dx \rightarrow du = \frac{1}{x}\,dx$, $v = \frac{x^3}{3}$.

$$\int x^2 \ln x\,dx = \frac{x^3}{3}\ln x - \frac{1}{3}\int x^2\,dx$$

$$= \frac{x^3 \ln x}{3} - \frac{x^3}{9} + C$$

So $a = 3$, $b = -9$.

$a - b = 3 + 9 = 12$

Answer: D

10.4 Partial Fraction Method

Q25. If $\frac{d}{dx} f(x) = 4x^3 - \frac{3}{x^4}$ such that $f(2) = 0$, then $f(x)$ is [25-Q]

A. $x^4 + \frac{1}{x^3} - \frac{129}{8}$ **B.** $x^3 + \frac{1}{x^4} + \frac{129}{8}$ **C.** $x^4 + \frac{1}{x^3} + \frac{129}{8}$ **D.** $x^3 + \frac{1}{x^4} - \frac{129}{8}$

Ans.

Integrate $f'(x) = 4x^3 - 3x^{-4}$:

$$f(x) = x^4 - 3\cdot\frac{x^{-3}}{-3} + C = x^4 + \frac{1}{x^3} + C$$

Use $f(2) = 0$:

$$16 + \frac{1}{8} + C = 0$$

$$C = -\frac{129}{8}$$

$$f(x) = x^4 + \frac{1}{x^3} - \frac{129}{8}$$

Answer: A

2 MARKS

10.1 Antiderivatives

Q1. Find the antiderivative of $\sqrt[5]{x}$. [25-Q]

Ans.

$\sqrt[5]{x} = x^{1/5}$. Use $\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C$.

$$\int x^{1/5}\,dx = \frac{x^{6/5}}{6/5} + C$$

$$= \frac{5}{6}x^{6/5} + C$$

Answer: $\frac{5}{6}x^{6/5} + C$

Q2. Evaluate the integral $\int (5x^{3/2})\,dx$. [25-Q]

Ans.

$$\int 5x^{3/2}\,dx = 5\cdot\frac{x^{5/2}}{5/2} + C$$

$$= 2x^{5/2} + C$$

Answer: $2x^{5/2} + C$

Q3. Find the integral $\int (1 + e^x)\,dx$. [24-Q]

Ans.

Use the sum rule term by term.

$$\int (1 + e^x)\,dx = \int 1\,dx + \int e^x\,dx$$

$$= x + e^x + C$$

Answer: $x + e^x + C$

Q4. Find $\int (-\frac{2}{x} + 3e^x)\,dx$. [24-Q]

Ans.

$$\int (-\frac{2}{x} + 3e^x)\,dx = -2\int \frac{1}{x}\,dx + 3\int e^x\,dx$$

$$= -2\ln|x| + 3e^x + C$$

Answer: $-2\ln|x| + 3e^x + C$

Q5. Find $\int (\frac{x^4}{x^5} - \frac{1}{x^5})\,dx$. [25-Q]

Ans.

Simplify: $\frac{x^4}{x^5} = \frac{1}{x}$, $\frac{1}{x^5} = x^{-5}$.

$$\int (\frac{1}{x} - x^{-5})\,dx = \ln|x| - \frac{x^{-4}}{-4} + C$$

$$= \ln|x| + \frac{1}{4x^4} + C$$

Answer: $\ln|x| + \frac{1}{4x^4} + C$

Q6. Find $\int \frac{2x^5 - 1}{x^6}\,dx$. [25-Q]

Ans.

Split the fraction:

$$\int (\frac{2}{x} - x^{-6})\,dx = 2\ln|x| - \frac{x^{-5}}{-5} + C$$

$$= 2\ln|x| + \frac{1}{5x^5} + C$$

Answer: $2\ln|x| + \frac{1}{5x^5} + C$

Q7. Write down your answer of $\int \frac{1}{3+4x}\,dx$. [24, Sample]

Ans.

$\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + C$ with $a = 4$.

$$\int \frac{1}{3+4x}\,dx = \frac{1}{4}\ln|3+4x| + C$$

Answer: $\frac{1}{4}\ln|3+4x| + C$

Q8. Find $\int 5\sec^2(5x-3)\,dx$. [25-Q]

Ans.

$\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + C$ with $a = 5$.

$$\int 5\sec^2(5x-3)\,dx = 5\cdot\frac{1}{5}\tan(5x-3) + C$$

$$= \tan(5x-3) + C$$

Answer: $\tan(5x-3) + C$

Q9. Evaluate $\int \sin^2 x\,dx$. [25-Q]

Ans.

Use $\sin^2 x = \frac{1-\cos 2x}{2}$.

$$\int \sin^2 x\,dx = \frac{1}{2}\int (1 - \cos 2x)\,dx$$

$$= \frac{1}{2}(x - \frac{\sin 2x}{2}) + C$$

$= \frac{1}{2}x - \frac{1}{4}\sin 2x + C$

Answer: $\frac{1}{2}x - \frac{1}{4}\sin 2x + C$

Q10. Evaluate $\int (1 + \cot^2 2\pi x)\,dx$. [25-Q]

Ans.

Use $1 + \cot^2\theta = \csc^2\theta$ and $\int \csc^2(ax)\,dx = -\frac{1}{a}\cot(ax) + C$ with $a = 2\pi$.

$\int (1 + \cot^2 2\pi x)\,dx = \int \csc^2 2\pi x\,dx$

$= -\frac{1}{2\pi}\cot 2\pi x + C$

Answer: $-\frac{1}{2\pi}\cot 2\pi x + C$

10.2 Substitution Method

Q11. Find $\int x^2(\frac{x^3}{3} + 1)^3\,dx$. [26-Q]

Ans.

Let $u = \frac{x^3}{3} + 1$, $du = x^2\,dx$.

$\int x^2(\frac{x^3}{3} + 1)^3\,dx = \int u^3\,du$

$= \frac{u^4}{4} + C = \frac{1}{4}(\frac{x^3}{3} + 1)^4 + C$

Answer: $\frac{1}{4}(\frac{x^3}{3} + 1)^4 + C$

Q12. Find $\int \sin^4 x \cos x\,dx$. [26-Q]

Ans.

Let $u = \sin x$, $du = \cos x\,dx$.

$\int \sin^4 x \cos x\,dx = \int u^4\,du = \frac{u^5}{5} + C$

$= \frac{1}{5}\sin^5 x + C$

Answer: $\frac{1}{5}\sin^5 x + C$

Q13. Find $\int e^{x+e^x}\,dx$. [25-Q, 26-Q]

Ans.

Write $e^{x+e^x} = e^x \cdot e^{e^x}$. Let $u = e^x$, $du = e^x\,dx$.

$\int e^x e^{e^x}\,dx = \int e^u\,du = e^u + C$

$= e^{e^x} + C$

Answer: $e^{e^x} + C$

Q14. Find $\int \{2x^2\sqrt[3]{x^3 + 1}\}\,dx$. [24-Q]

Ans.

Let $u = x^3 + 1$, $du = 3x^2\,dx$, so $x^2\,dx = \frac{1}{3}\,du$.

$\int \frac{2x^2}{(x^3+1)^{1/3}}\,dx = \frac{2}{3}\int u^{-1/3}\,du$

$= \frac{2}{3}\cdot\frac{u^{2/3}}{2/3} + C = u^{2/3} + C$

$= (x^3 + 1)^{2/3} + C$

Answer: $(x^3 + 1)^{2/3} + C$

Q15. Find $\int \{x\sqrt{3x^2 - 5}\}\,dx$. [25*-Q]

Ans.

Let $u = 3x^2 - 5$, $du = 6x\,dx$, so $x\,dx = \frac{1}{6}\,du$.

$\int \{x\sqrt{3x^2 - 5}\}\,dx = \frac{1}{6}\int u^{-1/2}\,du$

$= \frac{1}{6}\cdot\frac{u^{1/2}}{1/2} + C = \frac{1}{3}\sqrt{u} + C$

$= \frac{1}{3}\sqrt{3x^2 - 5} + C$

Answer: $\frac{1}{3}\sqrt{3x^2 - 5} + C$

Q16. Find $\int \{e^x\sqrt{2 + 3e^x}\}\,dx$. [25*-Q]

Ans.

Let $u = 2 + 3e^x$, $du = 3e^x\,dx$, so $e^x\,dx = \frac{1}{3}\,du$.

$\int \{e^x\sqrt{2 + 3e^x}\}\,dx = \frac{1}{3}\int u^{-1/2}\,du$

$= \frac{1}{3}\cdot 2\sqrt{u} + C$

$= \frac{2}{3}\sqrt{2 + 3e^x} + C$

Answer: $\frac{2}{3}\sqrt{2 + 3e^x} + C$

Q17. Evaluate the integral $\int 3x\sqrt{x + 1}\,dx$. [25-Q]

Ans.

Let $u = x + 1$, so $x = u - 1$ and $du = dx$.

$\int 3x\sqrt{x+1}\,dx = 3\int (u - 1)u^{1/2}\,du = 3\int (u^{3/2} - u^{1/2})\,du$

$= 3(\frac{u^{5/2}}{5/2} - \frac{u^{3/2}}{3/2}) + C$

$= \frac{6}{5}u^{5/2} - 2u^{3/2} + C$

$= \frac{6}{5}(x + 1)^{5/2} - 2(x + 1)^{3/2} + C$

Answer: $\frac{6}{5}(x + 1)^{5/2} - 2(x + 1)^{3/2} + C$

Q18. Find $\int e^{\sin 2x}\cos 2x\,dx$. [24-Q]

Ans.

Let $u = \sin 2x$, $du = 2\cos 2x\,dx$, so $\cos 2x\,dx = \frac{1}{2}\,du$.

$\int e^{\sin 2x}\cos 2x\,dx = \frac{1}{2}\int e^u\,du$

$= \frac{1}{2}e^u + C = \frac{1}{2}e^{\sin 2x} + C$

Answer: $\frac{1}{2}e^{\sin 2x} + C$

Q19. Find $\int \frac{1}{x(\ln x)^3}\,dx$. [25*-Q]

Ans.

Let $u = \ln x$, $du = \frac{1}{x}\,dx$.

$\int \frac{1}{x(\ln x)^3}\,dx = \int u^{-3}\,du = \frac{u^{-2}}{-2} + C$

$= -\frac{1}{2(\ln x)^2} + C$

Answer: $-\frac{1}{2(\ln x)^2} + C$

Q20. Evaluate $\int \sqrt{x}\sin^2(x^{3/2} - 1)\,dx$. [25-Q]

Ans.

Let $u = x^{3/2} - 1$, $du = \frac{3}{2}x^{1/2}\,dx$, so $\sqrt{x}\,dx = \frac{2}{3}\,du$. Then use $\sin^2 u = \frac{1 - \cos 2u}{2}$.

$\int \sqrt{x}\sin^2(x^{3/2} - 1)\,dx = \frac{2}{3}\int \sin^2 u\,du = \frac{2}{3}\cdot\frac{1}{2}\int (1 - \cos 2u)\,du$

$= \frac{1}{3}(u - \frac{\sin 2u}{2}) + C$

$= \frac{1}{3}(x^{3/2} - 1) - \frac{1}{6}\sin(2x^{3/2} - 2) + C$

Answer: $\frac{1}{3}(x^{3/2} - 1) - \frac{1}{6}\sin(2x^{3/2} - 2) + C$

Q21. Find $\int \frac{2x + 1}{x^2 + x + 6}\,dx$. [24-Q]

Ans.

The numerator is the derivative of the denominator: let $u = x^2 + x + 6$, $du = (2x + 1)\,dx$.

$\int \frac{2x+1}{x^2+x+6}\,dx = \int \frac{1}{u}\,du = \ln|u| + C$

$= \ln|x^2 + x + 6| + C$

Answer: $\ln|x^2 + x + 6| + C$

Q22. Evaluate $\int \frac{2x - 1}{x^2 - x - 6}\,dx$ by using the given substitution $u = x^2 - x - 6$. [24-Q]

Ans.

$u = x^2 - x - 6 \rightarrow du = (2x - 1)\,dx$.

$\int \frac{2x-1}{x^2-x-6}\,dx = \int \frac{1}{u}\,du = \ln|u| + C$

$= \ln|x^2 - x - 6| + C$

Answer: $\ln|x^2 - x - 6| + C$

Q23. Evaluate $\int \frac{xe^x}{(x + 1)^2}\,dx$. [24-Q]

Ans.

Write $x = (x + 1) - 1$:

$\frac{xe^x}{(x+1)^2} = \frac{e^x}{x+1} - \frac{e^x}{(x+1)^2}$

Integrate the second term by parts ($u = e^x$, $dv = (x+1)^{-2}\,dx$, $v = -\frac{1}{x+1}$):

$$\int \frac{e^x}{(x+1)^2}\,dx = -\frac{e^x}{x+1} + \int \frac{e^x}{x+1}\,dx$$

Therefore

$$\int \frac{xe^x}{(x+1)^2}\,dx = \int \frac{e^x}{x+1}\,dx + \frac{e^x}{x+1} - \int \frac{e^x}{x+1}\,dx$$

$$= \frac{e^x}{x+1} + C$$

Answer: $\boldsymbol{\frac{e^x}{x+1} + C}$

10.3 Integration by Parts

Q24. Find $\int 2x \cos x\,dx$. [24-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = 2x$, $dv = \cos x\,dx \to du = 2\,dx$, $v = \sin x$.

$$\int 2x\cos x\,dx = 2x\sin x - \int 2\sin x\,dx$$
$$= 2x\sin x + 2\cos x + C$$

Answer: $2x\sin x + 2\cos x + C$

Q25. Evaluate $\int (2x+1)\cos x\,dx$. [5-Q, 25*-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = 2x+1$, $dv = \cos x\,dx \to du = 2\,dx$, $v = \sin x$.

$$\int (2x+1)\cos x\,dx = (2x+1)\sin x - \int 2\sin x\,dx$$
$$= (2x+1)\sin x + 2\cos x + C$$

Answer: $(2x+1)\sin x + 2\cos x + C$

Q26. Evaluate $\int (2x-1)\sin x\,dx$. [25-Q, 26-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = 2x-1$, $dv = \sin x\,dx \to du = 2\,dx$, $v = -\cos x$.

$$\int (2x-1)\sin x\,dx = -(2x-1)\cos x + \int 2\cos x\,dx$$
$$= (1-2x)\cos x + 2\sin x + C$$

Answer: $(1-2x)\cos x + 2\sin x + C$

Q27. Find $\int x \sin x \cos x\,dx$. [24-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
$\sin x\cos x = \frac{1}{2}\sin 2x$, so the integral is $\frac{1}{2}\int x\sin 2x\,dx$. Let $u = x$, $dv = \sin 2x\,dx \to du = dx$, $v = -\frac{\cos 2x}{2}$.

$$\int x\sin 2x\,dx = -\frac{x\cos 2x}{2} + \frac{1}{2}\int \cos 2x\,dx = -\frac{x\cos 2x}{2} + \frac{\sin 2x}{4}$$

$$\int x\sin x\cos x\,dx = \frac{1}{2}\left(-\frac{x\cos 2x}{2} + \frac{\sin 2x}{4}\right) + C$$
$$= -\frac{x\cos 2x}{4} + \frac{\sin 2x}{8} + C$$

Answer: $\boldsymbol{-\frac{x\cos 2x}{4} + \frac{\sin 2x}{8} + C}$

Q28. Find $\int 2x \sin 2x \cos 2x\,dx$. [25-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
$2\sin 2x\cos 2x = \sin 4x$, so the integral is $\int x\sin 4x\,dx$. Let $u = x$, $dv = \sin 4x\,dx \to du = dx$, $v = -\frac{\cos 4x}{4}$.

$$\int x\sin 4x\,dx = -\frac{x\cos 4x}{4} + \frac{1}{4}\int \cos 4x\,dx$$
$$= -\frac{x\cos 4x}{4} + \frac{\sin 4x}{16} + C$$

Answer: $\boldsymbol{-\frac{x\cos 4x}{4} + \frac{\sin 4x}{16} + C}$

Q29. Evaluate $\int xe^{-x}\,dx$. [26-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = x$, $dv = e^{-x}\,dx \to du = dx$, $v = -e^{-x}$.

$$\int xe^{-x}\,dx = -xe^{-x} + \int e^{-x}\,dx$$
$$= -xe^{-x} - e^{-x} + C$$

Answer: $\boldsymbol{-xe^{-x} - e^{-x} + C}$

Q30. Find $\int x2^x\,dx$. [24-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = x$, $dv = 2^x\,dx \to du = dx$, $v = \frac{2^x}{\ln 2}$.

$$\int x2^x\,dx = \frac{x2^x}{\ln 2} - \frac{1}{\ln 2}\int 2^x\,dx$$
$$= \frac{x2^x}{\ln 2} - \frac{2^x}{(\ln 2)^2} + C$$

Answer: $\boldsymbol{\frac{x2^x}{\ln 2} - \frac{2^x}{(\ln 2)^2} + C}$

Q31. Find $\int x3^x\,dx$. [5-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = x$, $dv = 3^x\,dx \to du = dx$, $v = \frac{3^x}{\ln 3}$.

$$\int x3^x\,dx = \frac{x3^x}{\ln 3} - \frac{1}{\ln 3}\int 3^x\,dx$$
$$= \frac{x3^x}{\ln 3} - \frac{3^x}{(\ln 3)^2} + C$$

Answer: $\boldsymbol{\frac{x3^x}{\ln 3} - \frac{3^x}{(\ln 3)^2} + C}$

Q32. Find $\int x3^x \ln 3\,dx$. [24-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = x$, $dv = 3^x \ln 3\,dx \to du = dx$, $v = 3^x$ (because $\int 3^x \ln 3\,dx = 3^x$).

$$\int x3^x \ln 3\,dx = x3^x - \int 3^x\,dx$$
$$= x3^x - \frac{3^x}{\ln 3} + C$$

Answer: $\boldsymbol{x3^x - \frac{3^x}{\ln 3} + C}$

Q33. Evaluate $\int \ln 2x\,dx$. [25*-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = \ln 2x$, $dv = dx \to du = \frac{1}{x}\,dx$ (since $d/dx \ln 2x = \frac{2}{2x}$), $v = x$.

$$\int \ln 2x\,dx = x\ln 2x - \int x\cdot\frac{1}{x}\,dx$$
$$= x\ln 2x - x + C$$

Answer: $x\ln 2x - x + C$

Q34. Evaluate $\int \ln 3x\,dx$. [25-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = \ln 3x$, $dv = dx \to du = \frac{1}{x}\,dx$, $v = x$.

$$\int \ln 3x\,dx = x\ln 3x - \int 1\,dx$$
$$= x\ln 3x - x + C$$

Answer: $x\ln 3x - x + C$

Q35. Evaluate the integral $\int \ln x^2\,dx$. [25-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
$\ln x^2 = 2\ln x$. Let $u = \ln x$, $dv = dx \to du = \frac{1}{x}\,dx$, $v = x$.

$$\int 2\ln x\,dx = 2(x\ln x - \int 1\,dx)$$
$$= 2x\ln x - 2x + C$$

Answer: $2x\ln x - 2x + C$

Q36. Find $\int x \ln x\,dx$. [2025*]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = \ln x$, $dv = x\,dx \to du = \frac{1}{x}\,dx$, $v = \frac{x^2}{2}$.

$$\int x\ln x\,dx = \frac{x^2\ln x}{2} - \frac{1}{2}\int x\,dx$$
$$= \frac{x^2\ln x}{2} - \frac{x^2}{4} + C$$

Answer: $\boldsymbol{\frac{x^2\ln x}{2} - \frac{x^2}{4} + C}$

Q37. Evaluate $\int \frac{\ln x}{x^2}\,dx$. [25-Q]

Ans.
Integration by parts: $\int u\,dv = uv - \int v\,du$.
Let $u = \ln x$, $dv = x^{-2}\,dx \to du = \frac{1}{x}\,dx$, $v = -\frac{1}{x}$.

$$\int \frac{\ln x}{x^2}\,dx = -\frac{\ln x}{x} + \int \frac{1}{x^2}\,dx$$
$$= -\frac{\ln x}{x} - \frac{1}{x} + C$$

Answer: $-\frac{\ln x}{x} - \frac{1}{x} + C$

10.4 Partial Fraction Method

Q38. Find $\int \frac{1}{x(x-1)}\,dx$. [24-Q]

Ans.

Partial fractions: $\frac{1}{x(x-1)} = \frac{A}{x} + \frac{B}{x-1}$.

$1 = A(x-1) + Bx$

$x = 0$: $1 = -A \rightarrow A = -1$

$x = 1$: $1 = B \rightarrow B = 1$

$\int \frac{1}{x(x-1)}\,dx = \int \left(-\frac{1}{x} + \frac{1}{x-1}\right) dx = -\ln|x| + \ln|x-1| + C$

$= \ln\left|\frac{x-1}{x}\right| + C$

Answer: $\ln\left|\frac{x-1}{x}\right| + C$

Q39. Write down your answer of $\int \frac{1}{x^2-4}\,dx$. [5-Q]

Ans.

$x^2 - 4 = (x-2)(x+2)$. $\frac{1}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2}$.

$1 = A(x+2) + B(x-2)$

$x = 2$: $1 = 4A \rightarrow A = \frac{1}{4}$

$x = -2$: $1 = -4B \rightarrow B = -\frac{1}{4}$

$\int \frac{1}{x^2-4}\,dx = \frac{1}{4}\ln|x-2| - \frac{1}{4}\ln|x+2| + C$

$= \frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$

Answer: $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$

Q40. Evaluate $\int \frac{1}{x^2-x-6}\,dx$. [24-Q]

Ans.

$x^2 - x - 6 = (x-3)(x+2)$.

$\frac{1}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}$

$1 = A(x+2) + B(x-3)$

$x = 3$: $1 = 5A \rightarrow A = \frac{1}{5}$

$x = -2$: $1 = -5B \rightarrow B = -\frac{1}{5}$

$\int \frac{1}{x^2-x-6}\,dx = \frac{1}{5}\ln|x-3| - \frac{1}{5}\ln|x+2| + C$

$= \frac{1}{5}\ln\left|\frac{x-3}{x+2}\right| + C$

Answer: $\frac{1}{5}\ln\left|\frac{x-3}{x+2}\right| + C$

Q41. Evaluate the integral $\int \frac{1}{x^2-5x-6}\,dx$. [26-Q]

Ans.

$x^2 - 5x - 6 = (x-6)(x+1)$.

$\frac{1}{(x-6)(x+1)} = \frac{A}{x-6} + \frac{B}{x+1}$

$1 = A(x+1) + B(x-6)$

$x = 6$: $1 = 7A \rightarrow A = \frac{1}{7}$

$x = -1$: $1 = -7B \rightarrow B = -\frac{1}{7}$

$\int \frac{1}{x^2-5x-6}\,dx = \frac{1}{7}\ln|x-6| - \frac{1}{7}\ln|x+1| + C$

$= \frac{1}{7}\ln\left|\frac{x-6}{x+1}\right| + C$

Answer: $\frac{1}{7}\ln\left|\frac{x-6}{x+1}\right| + C$

Q42. Evaluate $\int \frac{5x-7}{x^2-3x+2}\,dx$. [25-Q, 26-Q]

Ans.

$x^2 - 3x + 2 = (x-1)(x-2)$.

$\frac{5x-7}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$

$5x - 7 = A(x-2) + B(x-1)$

$x = 1$: $-2 = -A \rightarrow A = 2$

$x = 2$: $3 = B \rightarrow B = 3$

$\int \frac{5x-7}{x^2-3x+2}\,dx = 2\ln|x-1| + 3\ln|x-2| + C$

Answer: $3\ln|x-2| + 2\ln|x-1| + C$

Q43. Find $\int \frac{x}{x^2-2x-3}\,dx$. [25-Q]

Ans.

$x^2 - 2x - 3 = (x-3)(x+1)$.

$\frac{x}{(x-3)(x+1)} = \frac{A}{x-3} + \frac{B}{x+1}$

$x = A(x+1) + B(x-3)$

$x = 3$: $3 = 4A \rightarrow A = \frac{3}{4}$

$x = -1$: $-1 = -4B \rightarrow B = \frac{1}{4}$

$\int \frac{x}{x^2-2x-3}\,dx = \frac{3}{4}\ln|x-3| + \frac{1}{4}\ln|x+1| + C$

Answer: $\frac{3}{4}\ln|x-3| + \frac{1}{4}\ln|x+1| + C$

Q44. Evaluate $\int \frac{x^2+4x+1}{(x^2-1)(x+3)}\,dx$. [26-Q]

Ans.

$(x^2-1)(x+3) = (x-1)(x+1)(x+3)$.

$\frac{x^2+4x+1}{(x-1)(x+1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+3}$

$x^2 + 4x + 1 = A(x+1)(x+3) + B(x-1)(x+3) + C(x-1)(x+1)$

$x = 1$: $6 = 8A \rightarrow A = \frac{3}{4}$

$x = -1$: $-2 = -4B \rightarrow B = \frac{1}{2}$

$x = -3$: $-2 = 8C \rightarrow C = -\frac{1}{4}$

$\int (x^2+4x+1)/((x^2-1)(x+3))\,dx = \frac{3}{4}\ln|x-1| + \frac{1}{2}\ln|x+1| - \frac{1}{4}\ln|x+3| + C$

Answer: $\frac{3}{4}\ln|x-1| + \frac{1}{2}\ln|x+1| - \frac{1}{4}\ln|x+3| + C$

Q45. Find $\int \frac{6x+7}{(x+2)^2}\,dx$. [25-Q]

Ans.

Repeated factor: $\frac{6x+7}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2}$.

$6x + 7 = A(x+2) + B$

$x = -2$: $-5 = B$

coefficient of x: $A = 6$

$\int \frac{6x+7}{(x+2)^2}\,dx = \int \left(\frac{6}{x+2} - \frac{5}{(x+2)^2}\right) dx$

$= 6\ln|x+2| + \frac{5}{x+2} + C$

Answer: $6\ln|x+2| + \frac{5}{x+2} + C$

Q46. Find the function $f(x)$ satisfying the equation $f'(x) = x\sqrt{x^2+2}$. [25-Q]

Ans.

$f(x) = \int x\sqrt{x^2+2}\,dx$. Let $u = x^2 + 2$, $du = 2x\,dx$, so $x\,dx = \frac{1}{2}\,du$.

$f(x) = \frac{1}{2}\int u^{1/2}\,du = \frac{1}{2}\cdot\frac{u^{3/2}}{3/2} + C$

$= \frac{1}{3}(x^2+2)^{3/2} + C$

Answer: $f(x) = \frac{1}{3}(x^2+2)^{3/2} + C$

Q47. Find the function y satisfying the equation $\frac{dy}{dx} = 1 + \cos x$ with $y(0) = 4$. [25-Q]

Ans.

Integrate:

$y = \int (1 + \cos x)\,dx = x + \sin x + C$

Use $y(0) = 4$:

$0 + 0 + C = 4 \rightarrow C = 4$

$y = x + \sin x + 4$

Answer: $y = x + \sin x + 4$

Q48. What is the function $f'(x)$ when $f''(x) = \frac{1}{2}(-e^{-x} + e^{x})$ with $f'(0) = 1$? [24-Q]

Ans.

Integrate $f''(x)$:

$f'(x) = \frac{1}{2}\int(-e^{-x}+e^{x})\,dx = \frac{1}{2}(e^{-x}+e^{x})+C$

Use $f'(0)=1$:

$\frac{1}{2}(1+1)+C = 1 \rightarrow C = 0$

$f'(x) = \frac{1}{2}(e^{x}+e^{-x})$

Answer: $f'(x) = \frac{1}{2}(e^{x}+e^{-x})$

3 MARKS

10.1 Antiderivatives

Q1. Find $\int \frac{1}{1+\sin x}\,dx$. [25-Q]

Ans.

Multiply the numerator and denominator by $(1-\sin x)$:

$\int \frac{1}{1+\sin x}\,dx = \int \frac{1-\sin x}{(1+\sin x)(1-\sin x)}\,dx = \int \frac{1-\sin x}{1-\sin^2 x}\,dx$

$= \int \frac{1-\sin x}{\cos^2 x}\,dx = \int(\sec^2 x - \sec x\tan x)\,dx$

$= \tan x - \sec x + C$

Answer: $\tan x - \sec x + C$

10.2 Substitution Method

Q2. Evaluate $\int 3x\sqrt{x^2-5}\,dx$. [24-Q]

Ans.

Let $u = x^2-5$, $du = 2x\,dx$, so $x\,dx = \frac{1}{2}\,du$.

$\int 3x\sqrt{x^2-5}\,dx = \frac{3}{2}\int u^{1/2}\,du$

$= \frac{3}{2}\cdot\frac{u^{3/2}}{3/2}+C = u^{3/2}+C$

$= (x^2-5)^{3/2}+C$

Answer: $(x^2-5)^{3/2}+C$

Q3. Use the substitution method to integrate $\int \cot x \csc^2 x\,dx$. [26-Q]

Ans.

Let $u = \cot x$, $du = -\csc^2 x\,dx$, so $\csc^2 x\,dx = -du$.

$\int \cot x\csc^2 x\,dx = -\int u\,du = -\frac{u^2}{2}+C$

$= -\frac{1}{2}\cot^2 x + C$

(Alternatively, with $u = \csc x$ the result is $-\frac{1}{2}\csc^2 x + C$, which differs only by a constant.)

Answer: $-\frac{1}{2}\cot^2 x + C$

10.3 Integration by Parts

Q4. Evaluate $\int(xe^{x}+2\sin x)\,dx$. [5-Q]

Ans.

Integration by parts: $\int u\,dv = uv - \int v\,du$.

Split into two integrals. For $\int xe^{x}\,dx$ let $u = x$, $dv = e^{x}\,dx \rightarrow du = dx$, $v = e^{x}$.

$\int xe^{x}\,dx = xe^{x} - \int e^{x}\,dx = xe^{x} - e^{x}$

$\int 2\sin x\,dx = -2\cos x$

$\int(xe^{x}+2\sin x)\,dx = xe^{x} - e^{x} - 2\cos x + C$

Answer: $xe^{x} - e^{x} - 2\cos x + C$

Q5. Find $\int x\sin x\cos x\,dx$. [5-Q]

Ans.

Integration by parts: $\int u\,dv = uv - \int v\,du$.

$\sin x\cos x = \frac{1}{2}\sin 2x$. Let $u = x$, $dv = \sin 2x\,dx \rightarrow du = dx$, $v = -\frac{\cos 2x}{2}$.

$\int x\sin x\cos x\,dx = \frac{1}{2}\left(-\frac{x\cos 2x}{2}+\frac{1}{2}\int\cos 2x\,dx\right)$

$= \frac{1}{2}\left(-\frac{x\cos 2x}{2}+\frac{\sin 2x}{4}\right)+C$

$= -\frac{x\cos 2x}{4}+\frac{\sin 2x}{8}+C$

Answer: $-\frac{x\cos 2x}{4}+\frac{\sin 2x}{8}+C$

Q6. Use the integration by parts to evaluate the integral $\int x\cos 5x\,dx$. [25-Q]

Ans.

Integration by parts: $\int u\,dv = uv - \int v\,du$.

Let $u = x$, $dv = \cos 5x\,dx \rightarrow du = dx$, $v = \frac{\sin 5x}{5}$.

$\int x\cos 5x\,dx = \frac{x\sin 5x}{5} - \frac{1}{5}\int\sin 5x\,dx$

$= \frac{x\sin 5x}{5}+\frac{\cos 5x}{25}+C$

Answer: $\frac{x\sin 5x}{5}+\frac{\cos 5x}{25}+C$

Q7. Evaluate the integral $\int x^2 2^{x}\,dx$. [24-Q]

Ans.

Integration by parts: $\int u\,dv = uv - \int v\,du$.

Let $u = x^2$, $dv = 2^{x}\,dx \rightarrow du = 2x\,dx$, $v = \frac{2^{x}}{\ln 2}$.

$\int x^2 2^{x}\,dx = \frac{x^2 2^{x}}{\ln 2} - \frac{2}{\ln 2}\int x2^{x}\,dx$

From the earlier result $\int x2^{x}\,dx = \frac{x2^{x}}{\ln 2} - \frac{2^{x}}{(\ln 2)^2}$:

$= \frac{x^2 2^{x}}{\ln 2} - \frac{2}{\ln 2}\left(\frac{x2^{x}}{\ln 2} - \frac{2^{x}}{(\ln 2)^2}\right)+C$

$= \frac{x^2 2^{x}}{\ln 2} - \frac{2x2^{x}}{(\ln 2)^2} + \frac{2\cdot 2^{x}}{(\ln 2)^3}+C$

Answer: $\frac{x^2 2^{x}}{\ln 2} - \frac{2x2^{x}}{(\ln 2)^2} + \frac{2\cdot 2^{x}}{(\ln 2)^3}+C$

Q8. Evaluate $\int \frac{(\ln x)^2}{x^3}\,dx$. [25-Q]

Ans.

Integration by parts: $\int u\,dv = uv - \int v\,du$.

Let $u = (\ln x)^2$, $dv = x^{-3}\,dx \rightarrow du = \frac{2\ln x}{x}\,dx$, $v = -\frac{1}{2x^2}$.

$\int \frac{(\ln x)^2}{x^3}\,dx = -\frac{(\ln x)^2}{2x^2} + \int \frac{\ln x}{x^3}\,dx$

Again by parts with $u = \ln x$, $dv = x^{-3}\,dx$ ($du = \frac{1}{x}\,dx$, $v = -\frac{1}{2x^2}$):

$\int \frac{\ln x}{x^3}\,dx = -\frac{\ln x}{2x^2} + \frac{1}{2}\int x^{-3}\,dx = -\frac{\ln x}{2x^2} - \frac{1}{4x^2}$

$\int \frac{(\ln x)^2}{x^3}\,dx = -\frac{(\ln x)^2}{2x^2} - \frac{\ln x}{2x^2} - \frac{1}{4x^2}+C$

Answer: $-\frac{(\ln x)^2}{2x^2} - \frac{\ln x}{2x^2} - \frac{1}{4x^2}+C$

Q9. Find $\int e^{2x}\sin 3x\,dx$. [25*-Q]

Ans.

Integration by parts: $\int u\,dv = uv - \int v\,du$.

Let $J = \int e^{2x}\sin 3x\,dx$. First: $u = \sin 3x$, $dv = e^{2x}\,dx \rightarrow du = 3\cos 3x\,dx$, $v = \frac{e^{2x}}{2}$.

$J = \frac{e^{2x}\sin 3x}{2} - \frac{3}{2}\int e^{2x}\cos 3x\,dx$

Second: $u = \cos 3x$, $dv = e^{2x}\,dx \rightarrow du = -3\sin 3x\,dx$, $v = \frac{e^{2x}}{2}$:

$\int e^{2x}\cos 3x\,dx = \frac{e^{2x}\cos 3x}{2} + \frac{3}{2}J$

$J = \frac{e^{2x}\sin 3x}{2} - \frac{3}{2}\left(\frac{e^{2x}\cos 3x}{2} + \frac{3}{2}J\right)$

$J = \frac{e^{2x}\sin 3x}{2} - \frac{3e^{2x}\cos 3x}{4} - \frac{9}{4}J$

$\frac{13}{4}J = \frac{e^{2x}(2\sin 3x - 3\cos 3x)}{4}$

$J = \frac{e^{2x}}{13}(2\sin 3x - 3\cos 3x)+C$

Answer: $\frac{e^{2x}}{13}(2\sin 3x - 3\cos 3x)+C$

Q10. Evaluate $\int e^{x}\sin 3x\cos 3x\,dx$. [5-Q]

Ans.

Integration by parts: $\int u\,dv = uv - \int v\,du$.

$\sin 3x\cos 3x = \frac{1}{2}\sin 6x$, so the integral is $\frac{1}{2}J$ with $J = \int e^{x}\sin 6x\,dx$.

$u = \sin 6x$, $dv = e^{x}\,dx \rightarrow du = 6\cos 6x\,dx$, $v = e^{x}$:

$J = e^{x}\sin 6x - 6\int e^{x}\cos 6x\,dx$

$u = \cos 6x$, $dv = e^{x}\,dx \rightarrow du = -6\sin 6x\,dx$:

$\int e^{x}\cos 6x\,dx = e^{x}\cos 6x + 6J$

$J = e^{x}\sin 6x - 6e^{x}\cos 6x - 36J$

$37J = e^{x}(\sin 6x - 6\cos 6x)$

$J = \frac{e^{x}}{37}(\sin 6x - 6\cos 6x)$

$\int e^x \sin 3x \cos 3x\, dx = \frac{1}{2} J = \frac{e^x}{74}(\sin 6x - 6\cos 6x) + C$

Answer: $\frac{e^x}{74}(\sin 6x - 6\cos 6x) + C$

Q11. Find $\int \sin \sqrt{x}\, dx$. [25-Q]

Ans.

Integration by parts: $\int u\, dv = uv - \int v\, du$.

Substitute $t = \sqrt{x}$, so $x = t^2$ and $dx = 2t\, dt$.

$\int \sin \sqrt[n]{x}\, dx = \int 2t \sin t\, dt$

By parts: $u = 2t$, $dv = \sin t\, dt \rightarrow du = 2\, dt$, $v = -\cos t$.

$\int 2t \sin t\, dt = -2t \cos t + \int 2 \cos t\, dt$

$= -2t \cos t + 2 \sin t + C$

$= 2 \sin \sqrt{x} - 2\sqrt{x} \cos \sqrt{x} + C$

Answer: $2 \sin \sqrt{x} - 2\sqrt{x} \cos \sqrt{x} + C$

10.4 Partial Fraction Method

Q12. Evaluate $\int \frac{1}{x^2 - 1}\, dx$ using the concept of partial fraction. [24-Q]

Ans.

$x^2 - 1 = (x - 1)(x + 1)$.

$\frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$

$1 = A(x + 1) + B(x - 1)$

$x = 1$: $1 = 2A \rightarrow A = \frac{1}{2}$

$x = -1$: $1 = -2B \rightarrow B = -\frac{1}{2}$

$\int \frac{1}{x^2-1}\, dx = \frac{1}{2}\ln|x - 1| - \frac{1}{2}\ln|x + 1| + C$

$= \frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$

Answer: $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$

Q13. Evaluate $\int \frac{1}{(x+4)(x-3)}\, dx$ using the partial fraction method. [25-Q]

Ans.

$\frac{1}{(x+4)(x-3)} = \frac{A}{x+4} + \frac{B}{x-3}$

$1 = A(x - 3) + B(x + 4)$

$x = -4$: $1 = -7A \rightarrow A = -\frac{1}{7}$

$x = 3$: $1 = 7B \rightarrow B = \frac{1}{7}$

$\int \frac{1}{(x+4)(x-3)}\, dx = -\frac{1}{7}\ln|x + 4| + \frac{1}{7}\ln|x - 3| + C$

$= \frac{1}{7}\ln\left|\frac{x-3}{x+4}\right| + C$

Answer: $\frac{1}{7}\ln\left|\frac{x-3}{x+4}\right| + C$

Q14. Use the partial fraction method to evaluate the $\int \frac{1}{x^2 + 2x - 3}\, dx$. [24-Q]

Ans.

$x^2 + 2x - 3 = (x - 1)(x + 3)$.

$1 = A(x + 3) + B(x - 1)$

$x = 1$: $1 = 4A \rightarrow A = \frac{1}{4}$

$x = -3$: $1 = -4B \rightarrow B = -\frac{1}{4}$

$\int \frac{1}{x^2+2x-3}\, dx = \frac{1}{4}\ln|x - 1| - \frac{1}{4}\ln|x + 3| + C$

$= \frac{1}{4}\ln\left|\frac{x-1}{x+3}\right| + C$

Answer: $\frac{1}{4}\ln\left|\frac{x-1}{x+3}\right| + C$

Q15. Use the partial fraction method to evaluate the $\int \frac{1}{x^2 - 2x - 3}\, dx$. [25*-Q]

Ans.

$x^2 - 2x - 3 = (x - 3)(x + 1)$.

$1 = A(x + 1) + B(x - 3)$

$x = 3$: $1 = 4A \rightarrow A = \frac{1}{4}$

$x = -1$: $1 = -4B \rightarrow B = -\frac{1}{4}$

$\int \frac{1}{x^2-2x-3}\, dx = \frac{1}{4}\ln|x - 3| - \frac{1}{4}\ln|x + 1| + C$

$= \frac{1}{4}\ln\left|\frac{x-3}{x+1}\right| + C$

Answer: $\frac{1}{4}\ln\left|\frac{x-3}{x+1}\right| + C$

Q16. Evaluate $\int \frac{1}{2x^2 + 3x + 1}\, dx$, by using the partial fraction method. [26-Q]

Ans.

$2x^2 + 3x + 1 = (2x + 1)(x + 1)$.

$\frac{1}{(2x+1)(x+1)} = \frac{A}{2x+1} + \frac{B}{x+1}$

$1 = A(x + 1) + B(2x + 1)$

$x = -1$: $1 = -B \rightarrow B = -1$

$x = -\frac{1}{2}$: $1 = \frac{1}{2}A \rightarrow A = 2$

$\int \frac{1}{2x^2+3x+1}\, dx = \int \left(\frac{2}{2x+1} - \frac{1}{x+1}\right) dx$

$= \ln|2x + 1| - \ln|x + 1| + C$

$= \ln\left|\frac{2x+1}{x+1}\right| + C$

Answer: $\ln\left|\frac{2x+1}{x+1}\right| + C$

Q17. Find $\int \frac{x^2 + 6}{x^2 - 5x - 6}\, dx$. [24-Q]

Ans.

The degree of the numerator equals the degree of the denominator (improper fraction), so divide first:

$\frac{x^2+6}{x^2-5x-6} = 1 + \frac{5x+12}{x^2-5x-6}$

$\frac{5x+12}{(x-6)(x+1)} = \frac{A}{x-6} + \frac{B}{x+1}$

$5x + 12 = A(x + 1) + B(x - 6)$

$x = 6$: $42 = 7A \rightarrow A = 6$

$x = -1$: $7 = -7B \rightarrow B = -1$

$\int \frac{x^2+6}{x^2-5x-6}\, dx = \int \left(1 + \frac{6}{x-6} - \frac{1}{x+1}\right) dx$

$= x + 6\ln|x - 6| - \ln|x + 1| + C$

Answer: $x + 6\ln|x - 6| - \ln|x + 1| + C$

Q18. Evaluate $\int \frac{2x^3 - 4x^2 - x - 3}{x^2 - 2x - 3}\, dx$ using the concept of partial fraction. [25-Q]

Ans.

Improper fraction: divide the numerator by $x^2 - 2x - 3$.

$2x^3 - 4x^2 - x - 3 = 2x(x^2 - 2x - 3) + (5x - 3)$

$\frac{2x^3-4x^2-x-3}{x^2-2x-3} = 2x + \frac{5x-3}{(x-3)(x+1)}$

$\frac{5x-3}{(x-3)(x+1)} = \frac{A}{x-3} + \frac{B}{x+1}$

$5x - 3 = A(x + 1) + B(x - 3)$

$x = 3$: $12 = 4A \rightarrow A = 3$

$x = -1$: $-8 = -4B \rightarrow B = 2$

$\int \ldots dx = \int \left(2x + \frac{3}{x-3} + \frac{2}{x+1}\right) dx = x^2 + 3\ln|x - 3| + 2\ln|x + 1| + C$

Answer: $x^2 + 3\ln|x - 3| + 2\ln|x + 1| + C$

Q19. Integrate $\int \frac{2x+3}{(x-1)^2}\, dx$ using the concept of partial fraction. [25-Q]

Ans.

$\frac{2x+3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$.

$2x + 3 = A(x - 1) + B$

$x = 1$: $5 = B$

coefficient of x: $A = 2$

$\int \frac{2x+3}{(x-1)^2}\, dx = \int \left(\frac{2}{x-1} + \frac{5}{(x-1)^2}\right) dx$

$= 2\ln|x - 1| - \frac{5}{x-1} + C$

Answer: $2\ln|x - 1| - \frac{5}{x-1} + C$

Q20. Integrate $\int \frac{2x-1}{(x+1)^2}\, dx$ using the concept of partial fraction. [25-Q, 25*-Q]

Ans.

$\frac{2x-1}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2}$.

$2x - 1 = A(x+1) + B$

$x = -1$: $-3 = B$

coefficient of x: $A = 2$

$\int \frac{2x-1}{(x+1)^2}\,dx = \int \left(\frac{2}{x+1} - \frac{3}{(x+1)^2}\right) dx$

$= 2\ln|x+1| + \frac{3}{x+1} + C$

Answer: $2\ln|x+1| + \frac{3}{x+1} + C$

Q21. Find the function f(x) that satisfying the equation $f'(x) = x + \frac{1}{x^2}$, $x > 0$ with $f(2) = 1$.

[24-Q]

Ans.

$f(x) = \int (x + x^{-2})\,dx = \frac{x^2}{2} - \frac{1}{x} + C$

Use $f(2) = 1$:

$2 - \frac{1}{2} + C = 1 \rightarrow C = -\frac{1}{2}$

$f(x) = \frac{x^2}{2} - \frac{1}{x} - \frac{1}{2}$

Answer: $f(x) = \frac{x^2}{2} - \frac{1}{x} - \frac{1}{2}$

Q22. Find the function g(x) that satisfying the equation $g'(x) = x\sqrt{1+x^2}$ with $g(0) = -2$.

[24-Q]

Ans.

Let $u = 1 + x^2$, $du = 2x\,dx$.

$g(x) = \frac{1}{2}\int u^{1/2}\,du = \frac{1}{3}(1+x^2)^{3/2} + C$

Use $g(0) = -2$:

$\frac{1}{3} + C = -2 \rightarrow C = -\frac{7}{3}$

$g(x) = \frac{1}{3}(1+x^2)^{3/2} - \frac{7}{3}$

Answer: $g(x) = \frac{1}{3}(1+x^2)^{3/2} - \frac{7}{3}$

Q23. Find the function g(x) that satisfying the equation $g'(x) = x\sqrt{1+x^2}$ with $g(0) = 2$.

[26-Q]

Ans.

As in the previous question, $g(x) = \frac{1}{3}(1+x^2)^{3/2} + C$. Use $g(0) = 2$:

$\frac{1}{3} + C = 2 \rightarrow C = \frac{5}{3}$

$g(x) = \frac{1}{3}(1+x^2)^{3/2} + \frac{5}{3}$

Answer: $g(x) = \frac{1}{3}(1+x^2)^{3/2} + \frac{5}{3}$

Q24. Find the function f(x) that satisfying the equation $f'(x) = \frac{x}{x^2-1}$ with $f(2) = \frac{1}{2}$.

[26-Q]

Ans.

Let $u = x^2 - 1$, $du = 2x\,dx$.

$f(x) = \frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln|x^2-1| + C$

Use $f(2) = \frac{1}{2}$:

$\frac{1}{2}\ln 3 + C = \frac{1}{2} \rightarrow C = \frac{1}{2}(1 - \ln 3)$

$f(x) = \frac{1}{2}\ln|x^2-1| + \frac{1}{2}(1 - \ln 3)$

Answer: $f(x) = \frac{1}{2}\ln|x^2-1| + \frac{1}{2}(1 - \ln 3)$

Q25. Find the function h(x) that satisfying the equation $h'(x) = 12x(3x^2-1)^3$ with $h(1) = 3$.

[25*-Q]

Ans.

Let $u = 3x^2 - 1$, $du = 6x\,dx$, so $12x\,dx = 2\,du$.

$h(x) = 2\int u^3\,du = \frac{u^4}{2} + C = \frac{1}{2}(3x^2-1)^4 + C$

Use $h(1) = 3$:

$\frac{1}{2}(2)^4 + C = 3 \rightarrow 8 + C = 3 \rightarrow C = -5$

$h(x) = \frac{1}{2}(3x^2-1)^4 - 5$

Answer: $h(x) = \frac{1}{2}(3x^2-1)^4 - 5$

Q26. Find the function g(x) satisfying the equation $g'(x) = e^x \sin(e^x - 2)$ with $g(\ln 2) = 0$.

[25-Q]

Ans.

Let $u = e^x - 2$, $du = e^x\,dx$.

$g(x) = \int \sin u\,du = -\cos u + C = -\cos(e^x - 2) + C$

Use $g(\ln 2) = 0$ ($e^{\ln 2} = 2$, so $\cos 0 = 1$):

$-1 + C = 0 \rightarrow C = 1$

$g(x) = -\cos(e^x - 2) + 1$

Answer: $g(x) = -\cos(e^x - 2) + 1$

Q27. Find the function g(x) satisfying the equation $g'(x) = e^{-x}\sec^2(\pi e^{-x})$ with $g(\ln 4) = \frac{2}{\pi}$.

[25-Q]

Ans.

Let $u = \pi e^{-x}$, $du = -\pi e^{-x}\,dx$, so $e^{-x}\,dx = -\frac{1}{\pi}\,du$.

$g(x) = -\frac{1}{\pi}\int \sec^2 u\,du = -\frac{1}{\pi}\tan u + C$

$= -\frac{1}{\pi}\tan(\pi e^{-x}) + C$

Use $g(\ln 4) = \frac{2}{\pi}$: $e^{-\ln 4} = \frac{1}{4}$, so $\tan(\frac{\pi}{4}) = 1$.

$-\frac{1}{\pi} + C = \frac{2}{\pi} \rightarrow C = \frac{3}{\pi}$

$g(x) = -\frac{1}{\pi}\tan(\pi e^{-x}) + \frac{3}{\pi}$

Answer: $g(x) = -\frac{1}{\pi}\tan(\pi e^{-x}) + \frac{3}{\pi}$

CH.11 Application of Integration

Notes · Questions with Answers

NOTES

Definite Integral

If $F'(x) = f(x)$, where $f(x)$ is continuous on the interval $a \le x \le b$, then the **definite integral** of $f(x)$ on $[a, b]$ is defined by

$\int_a^b f(x)\,dx = F(b) - F(a)$

We write $F(b) - F(a) = [F(x)]_a^b$, or $F(x)\Big|_a^b$. The constant of integration C cancels out in the subtraction, so it is omitted.

Limits of Integration

In $\int_a^b f(x)\,dx$, the numbers a and b are called the **limits of integration**. Here a is the **lower limit** and b is the **upper limit**.

The symbol $\int_a^b f(x)\,dx$ is read "the integral from x = a to x = b of f(x) with respect to x".

Cross-section of a Solid

A **cross-section** of a solid is the plane region generated by intersecting the solid with a plane. If A(x) is the (integrable) cross-section area at x, for x from a to b, the volume of the solid is found by integrating A(x).

Solid of Revolution

If the region under the curve $y = f(x)$ for $a \le x \le b$ is rotated through 360° about the x-axis, a three-dimensional solid is formed. This solid is called a **solid of revolution**.

Definite Integral

$$\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a)$$

where $F'(x) = f(x)$.

Fundamental Theorem of Calculus

Let f be continuous on $[a, b]$. Then for each $x \in [a, b]$,

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$

Let F' be continuous on $[a, b]$. Then for each $x \in [a, b]$,

$$\int_a^x F'(t)\,dt = F(x) - F(a)$$

Rules for Definite Integration

Suppose f and g are continuous on $[a, b]$ and $k \in R$.

1. $\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$
2. $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
3. $\int_a^a f(x)\,dx = 0$
4. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
5. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, where $c \in (a, b)$

Area between the Curve and x-axis

If $f(x)$ is positive and continuous on $[a, b]$, the area bounded by $y = f(x)$, the x-axis and the vertical lines $x = a$ and $x = b$ is

Area $= \int_a^b f(x)\,dx$

To find the area between the graph of y = f(x) and the x-axis over [a, b]:

1. Find the intersection point(s) of the graph $y = f(x)$ and the x-axis.
2. Divide the interval $[a, b]$ at the zeros of f.
3. Integrate f over each subinterval.
4. Add the absolute values of the integrals.

Area between Two Curves

If $f(x) \ge g(x)$ on $[a, b]$, the area between $y = f(x)$ and $y = g(x)$ is

Area $= \int_a^b [f(x) - g(x)]\,dx$

To find the area between y = f(x) and y = g(x) over [a, b]:

1. Find the intersection points of the graphs $y = f(x)$ and $y = g(x)$.
2. Divide the interval $[a, b]$ at the intersection points.
3. Investigate which function is on the top in each subinterval.
4. Integrate $(f - g)$ if f is on the top, or $(g - f)$ if g is on the top, over each subinterval.

Volume by Method of Slicing

If A(x) is the integrable cross-section area from $x = a$ to $x = b$, then

Volume $= \int_a^b A(x)\,dx$

Volume of Revolution (Disc Method)

For the curve $y = f(x)$, $a \le x \le b$, rotated through 360° about the x-axis:

Volume $= \pi\int_a^b [f(x)]^2\,dx$

Volume of Revolution (Washer Method)

If the region bounded by $y = f(x)$ (on top), $y = g(x)$ and the lines $x = a$, $x = b$ is revolved about the x-axis, then

Volume $= \pi\int_a^b ([f(x)]^2 - [g(x)]^2)\,dx$

Steps:

1. Find the intersection points of the graphs $y = f(x)$ and $y = g(x)$.
2. Investigate which function is on the top.
3. Integrate $f^2 - g^2$ if f is on the top, or $g^2 - f^2$ if g is on the top, on the interval.

1 MARK

11.2 Definite Integral

Q1. $\int_1^2 x\,dx =$ [24-Q]

A. $\frac{3}{2}$ **B.** $\frac{1}{2}$ **C.** 2 **D.** 3

Ans.

Fundamental Theorem: $\int_a^b f(x)\,dx = F(b) - F(a)$. Here $F(x) = \frac{x^2}{2}$.

$$\int_1^2 x\,dx = \left[\frac{x^2}{2}\right]_1^2 = \frac{4}{2} - \frac{1}{2} = \frac{3}{2}$$

Answer: A

Q2. If $a\int_0^3 (\sqrt{x})^2\,dx = \frac{9\pi}{2}$, then $a =$ [24-Q]

A. 9 **B.** 9π **C.** $\frac{\pi}{2}$ **D.** π

Ans.

$(\sqrt{x})^2 = x$ for $x \ge 0$.

$$\int_0^3 (\sqrt{x})^2\,dx = \int_0^3 x\,dx = \left[\frac{x^2}{2}\right]_0^3 = \frac{9}{2}$$

So $a \cdot \frac{9}{2} = \frac{9\pi}{2}$, which gives $a = \pi$.

Answer: D

Q3. $\int_{-1}^1 |x|\,dx =$ [24-Q]

A. 2 **B.** 1 **C.** 3 **D.** 5

Ans.

$|x| = -x$ for $x < 0$ and $|x| = x$ for $x \ge 0$, so split the interval at 0 (rule 5).

$$\int_{-1}^1 |x|\,dx = \int_{-1}^0 (-x)\,dx + \int_0^1 x\,dx = \left[-\frac{x^2}{2}\right]_{-1}^0 + \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2} + \frac{1}{2} = 1$$

Answer: B

Q4. $\int_1^2 \frac{x^3+1}{x^2}\,dx =$ [25-Q]

A. $\frac{3}{4}$ **B.** $\frac{1}{4}$ **C.** 2 **D.** -2

Ans.

Divide each term by x^2: $\frac{x^3+1}{x^2} = x + x^{-2}$.

$$\int_1^2 (x + x^{-2})\,dx = \left[\frac{x^2}{2} - \frac{1}{x}\right]_1^2 = \left(\frac{4}{2} - \frac{1}{2}\right) - \left(\frac{1}{2} - 1\right) = \frac{3}{2} + \frac{1}{2} = 2$$

Answer: C

Q5. $\int_1^2 \frac{x^3+1}{x^2}\,dx =$ [26-Q]

A. 2 **B.** $\frac{1}{4}$ **C.** $\frac{3}{2}$ **D.** -2

Ans.

$\frac{x^3+1}{x^2} = x + x^{-2}$.

$$\int_1^2 (x + x^{-2})\,dx = \left[\frac{x^2}{2} - \frac{1}{x}\right]_1^2 = \left(\frac{4}{2} - \frac{1}{2}\right) - \left(\frac{1}{2} - 1\right) = \frac{3}{2} + \frac{1}{2} = 2$$

Answer: A

Q6. $\int_{-1}^0 \frac{1}{1-x}\,dx =$ [25-Q]

A. $\frac{1}{2}$ **B.** $\frac{2}{3}$ **C.** $\frac{7}{10}$ **D.** $\ln 2$

Ans.

Use $\int \frac{1}{ax+b}\,dx = \frac{1}{a}\ln|ax+b| + C$ with $a = -1$.

$\int_{-1}^{0} \frac{1}{1-x}\,dx = [-\ln|1-x|]_{-1}^{0}$

$= -\ln 1 + \ln 2$

$= \ln 2$

Answer: D

Q7. If $\int_0^{\pi/2} \sin\theta\,d\theta$ is equal to [25-Q]

A. $\frac{\pi}{2}$ **B.** $-\frac{\pi}{2}$ **C.** -1 **D.** 1

Ans.

$\int_0^{\pi/2} \sin\theta\,d\theta = [-\cos\theta]_0^{\pi/2}$

$= -\cos\frac{\pi}{2} + \cos 0$

$= 0 + 1 = 1$

Answer: D

Q8. If $\int_0^{\pi/4} \sin 2x\,dx$ is equal to p, then p^2 is [25-Q]

A. $\frac{\pi}{8}$ **B.** $\frac{1}{4}$ **C.** 1 **D.** $\frac{4}{3}$

Ans.

$\int \sin(ax)\,dx = -\frac{1}{a}\cos(ax) + C.$

$\int_0^{\pi/4} \sin 2x\,dx = [-\frac{1}{2}\cos 2x]_0^{\pi/4}$

$= -\frac{1}{2}\cos\frac{\pi}{2} + \frac{1}{2}\cos 0$

$= 0 + \frac{1}{2} = \frac{1}{2}$

So $p = \frac{1}{2}$ and $p^2 = \frac{1}{4}$.

Answer: B

Q9. $\int_0^{\pi/2} \cos x\,dx$ is equal to p, then p^2 is [24-Q, 25-Q]

A. 2 **B.** $\frac{\pi}{4}$ **C.** $\frac{4}{3}$ **D.** 1

Ans.

$\int_0^{\pi/2} \cos x\,dx = [\sin x]_0^{\pi/2}$

$= \sin\frac{\pi}{2} - \sin 0$

$= 1 - 0 = 1$

So $p = 1$ and $p^2 = 1$.

Answer: D

Q10. $\int_0^{\pi/4} \cos 2x\,dx$ is equal to p, then p^2 is [25*-Q]

A. 1 **B.** $\frac{\pi}{4}$ **C.** $\frac{1}{4}$ **D.** $\frac{4}{3}$

Ans.

$\int_0^{\pi/4} \cos 2x\,dx = [\frac{1}{2}\sin 2x]_0^{\pi/4}$

$= \frac{1}{2}\sin\frac{\pi}{2} - \frac{1}{2}\sin 0$

$= \frac{1}{2}$

So $p = \frac{1}{2}$ and $p^2 = \frac{1}{4}$.

Answer: C

Q11. If $\int_0^{\pi/2} \cos^2 x\,dx$ is equal to p, then $p + \pi$ is [25-Q]

A. $\frac{\pi}{4}$ **B.** $\frac{5\pi}{2}$ **C.** $\frac{5\pi}{3}$ **D.** $\frac{5\pi}{4}$

Ans.

Use $\cos^2 x = \frac{1+\cos 2x}{2}$.

$\int_0^{\pi/2} \cos^2 x\,dx = \int_0^{\pi/2} \frac{1+\cos 2x}{2}\,dx$

$= [\frac{x}{2} + \frac{\sin 2x}{4}]_0^{\pi/2}$

$= \frac{\pi}{4} + 0 - 0 = \frac{\pi}{4}$

So $p = \frac{\pi}{4}$, and $p + \pi = \frac{5\pi}{4}$.

Answer: D

Q12. If $\int_0^{\pi/2} \cos x\,dx$ is equal to P, then $P + \pi$ is [26-Q]

A. $2 + \pi$ **B.** $\frac{\pi}{4}$ **C.** $\frac{4}{3} + \pi$ **D.** $1 + \pi$

Ans.

$\int_0^{\pi/2} \cos x\,dx = [\sin x]_0^{\pi/2}$

$= 1 - 0 = 1$

So $P = 1$ and $P + \pi = 1 + \pi$.

Answer: D

Q13. If $\int_{\pi/4}^{\pi/3} \tan x\,dx$ is equal to $\ln p$, then p^2 is [24-Q]

A. $\frac{\pi}{4}$ **B.** 2 **C.** $\frac{1}{4}$ **D.** $\frac{4}{3}$

Ans.

$\int \tan x\,dx = -\ln|\cos x| + C = \ln|\sec x| + C.$

$\int_{\pi/4}^{\pi/3} \tan x\,dx = [-\ln|\cos x|]_{\pi/4}^{\pi/3}$

$= -\ln\cos\frac{\pi}{3} + \ln\cos\frac{\pi}{4}$

$= -\ln\frac{1}{2} + \ln\frac{\sqrt{2}}{2}$

$= \ln\frac{\sqrt{2}/2}{1/2} = \ln\sqrt{2}$

So $p = \sqrt{2}$ and $p^2 = 2$.

Answer: B

Q14. If $\int_0^{\pi/4} \sin x \sec x\,dx = -\ln a$, then a^2 is [25-Q]

A. $\frac{1}{2}$ **B.** $\frac{1}{3}$ **C.** $\frac{1}{4}$ **D.** $\frac{3}{4}$

Ans.

$\sin x \sec x = \frac{\sin x}{\cos x} = \tan x.$

$\int_0^{\pi/4} \tan x\,dx = [-\ln|\cos x|]_0^{\pi/4}$

$= -\ln\cos\frac{\pi}{4} + \ln\cos 0$

$= -\ln\frac{\sqrt{2}}{2} + 0$

Comparing with $-\ln a$: $a = \frac{\sqrt{2}}{2}$, so $a^2 = \frac{2}{4} = \frac{1}{2}$.

Answer: A

Q15. If $\int_{\pi/6}^{\pi/3} \cot x\,dx$ is equal to $\ln p$, then p^2 is [24-Q]

A. $\frac{\pi}{8}$ **B.** $\frac{\pi}{4}$ **C.** 3 **D.** $\frac{4}{3}$

Ans.

$\int \cot x\,dx = \ln|\sin x| + C.$

$\int_{\pi/6}^{\pi/3} \cot x\,dx = [\ln|\sin x|]_{\pi/6}^{\pi/3}$

$= \ln\sin\frac{\pi}{3} - \ln\sin\frac{\pi}{6}$

$= \ln\frac{\sqrt{3}}{2} - \ln\frac{1}{2}$

$= \ln\sqrt{3}$

So $p = \sqrt{3}$ and $p^2 = 3$.

Answer: C

Q16. If $\int_{\pi/3}^{5\pi/6} \cot x\,dx$ is equal to $\ln p$, then p^2 is [24-Q]

A. $\frac{\sqrt{3}}{3}$ **B.** $\frac{1}{3}$ **C.** 3 **D.** $\sqrt{3}$

Ans.

$\int_{\pi/3}^{5\pi/6} \cot x\,dx = [\ln|\sin x|]_{\pi/3}^{5\pi/6}$

$= \ln\sin\frac{5\pi}{6} - \ln\sin\frac{\pi}{3}$

$= \ln\frac{1}{2} - \ln\frac{\sqrt{3}}{2}$

$= \ln\frac{1/2}{\sqrt{3}/2} = \ln\frac{1}{\sqrt{3}}$

So $p = \frac{1}{\sqrt{3}}$ and $p^2 = \frac{1}{3}$.

Answer: B

Q17. $\int_0^1 2^{3x-1}\,dx =$ [25*-Q]

A. $\frac{1}{3\ln 2}$ **B.** $\frac{7}{6\ln 2}$ **C.** $\frac{7}{3\ln 2}$ **D.** $\frac{1}{6\ln 2}$

Ans.

$\int a^{bx+c}\,dx = \frac{a^{bx+c}}{b\ln a} + C.$

$\int_0^1 2^{3x-1}\,dx = [\frac{2^{3x-1}}{3\ln 2}]_0^1$

$= \frac{2^2}{3\ln 2} - \frac{2^{-1}}{3\ln 2}$

$= \frac{4 - 1/2}{3\ln 2} = \frac{7/2}{3\ln 2}$

$= \frac{7}{6\ln 2}$

Answer: B

Q18. $\int_0^1 (e^x + \sin \pi x)\, dx =$ [25*-Q]

A. $e - \frac{2}{\pi}$ **B.** $e - \pi$ **C.** $e - 1$ **D.** $e + \frac{2}{\pi} - 1$

Ans.

$\int_0^1 (e^x + \sin \pi x)\, dx = [e^x - \frac{1}{\pi} \cos \pi x]_0^1$

$= (e + \frac{1}{\pi}) - (1 - \frac{1}{\pi})$

$= e + \frac{2}{\pi} - 1$

Answer: D

Q19. $\int_0^3 x\sqrt{9 - x^2}\, dx =$ [26-Q]

A. 9 **B.** $\frac{1}{3}$ **C.** $\frac{2}{3}$ **D.** $\frac{8}{3}$

Ans.

Substitution $u = 9 - x^2$, $du = -2x\, dx$. When $x = 0$, $u = 9$; when $x = 3$, $u = 0$.

$\int_0^3 x\sqrt{9 - x^2}\, dx = -\frac{1}{2}\int_9^0 u^{1/2}\, du$

$= \frac{1}{2}\int_0^9 u^{1/2}\, du$

$= \frac{1}{2}[\frac{2}{3} u^{3/2}]_0^9$

$= \frac{1}{3}(27 - 0) = 9$

Answer: A

11.3 Area between the curve and x-axis

Q20. Area of the region bounded by the curve $y = \cos x$ between $x = 0$ and $x = \pi$ is [S-Q]

A. 2 sq. units **B.** 4 sq. units **C.** 3 sq. units **D.** 1 sq. units

Ans.

$\cos x \geq 0$ on $[0, \frac{\pi}{2}]$ and $\cos x \leq 0$ on $[\frac{\pi}{2}, \pi]$, so the area is split into two parts.

$A = \int_0^{\pi/2} \cos x\, dx + \int_{\pi/2}^{\pi} (0 - \cos x)\, dx$

$= [\sin x]_0^{\pi/2} - [\sin x]_{\pi/2}^{\pi}$

$= (1 - 0) - (0 - 1)$

$= 2$ sq. units

Answer: A

Q21. Area of the region bounded by the curve $f(x) = 2 + \cos x$ between $x = 0$ and $x = \pi$ is [25-Q]

A. 2π **B.** π **C.** $-\pi$ **D.** $\frac{\pi}{4}$

Ans.

$2 + \cos x \geq 1 > 0$, so the curve is above the x-axis: $A = \int f(x)\, dx$.

$A = \int_0^{\pi} (2 + \cos x)\, dx$

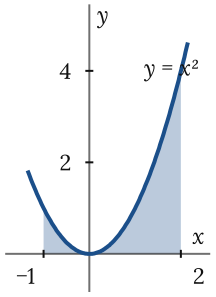
$= [2x + \sin x]_0^{\pi}$

$= (2\pi + 0) - 0$

$= 2\pi$

Answer: A

Q22. Area of shaded region is



[24-Q]

A. $-\frac{10}{3}$ **B.** 3 **C.** 9 **D.** none of these

Ans.

The curve $y = x^2$ is above the x-axis from $x = -1$ to $x = 2$.

$A = \int_{-1}^{2} x^2\, dx$

$= [\frac{x^3}{3}]_{-1}^{2}$

$= \frac{8}{3} - (-\frac{1}{3})$

$= 3$

Answer: B

11.5 Volumes Using Cross-sections

Q23. When the curve $y = \sqrt{x}$ for $0 < x \leq 3$ is rotated through 360° about the x-axis, the volume of revolution is [S-Q]

A. $\frac{9}{2}\pi$ unit^3 **B.** $\frac{9}{2}\pi^2$ unit^3 **C.** $\frac{9}{2}\pi^3$ unit^3 **D.** $\frac{9}{2}$ unit^3

Ans.

Disc method: $V = \pi \int_a^b [f(x)]^2\, dx$.

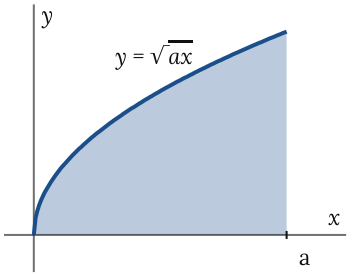
$V = \pi\int_0^3 (\sqrt{x})^2\, dx = \pi\int_0^3 x\, dx$

$= \pi [\frac{x^2}{2}]_0^3$

$= \frac{9\pi}{2}$

Answer: A

Q24. If the volume of the solid generated by revolving the shaded region in the figure through 360° about the x-axis is $\frac{27\pi}{2}$ unit^3, then $a =$



[25-Q]

A. 1 **B.** 2 **C.** 3 **D.** 4

Ans.

$V = \pi\int_0^a (\sqrt{ax})^2\, dx = \pi\int_0^a ax\, dx$

$= \pi [\frac{ax^2}{2}]_0^a$

$= \frac{\pi a^3}{2}$

Given $V = \frac{27\pi}{2}$:

$\frac{\pi a^3}{2} = \frac{27\pi}{2}$

$a^3 = 27$

$a = 3$

Answer: C

Q25. The volume of revolution formed when $f(x) = x + 3$ for $0 \leq x \leq 1$ is rotated through 360° about the x-axis is [26-Q]

A. 4 **B.** $\frac{37}{3}$ **C.** 4π **D.** $\frac{37}{3}\pi$

Ans.

$V = \pi\int_0^1 (x + 3)^2\, dx$

$= \pi [\frac{(x + 3)^3}{3}]_0^1$

$= \pi(\frac{64}{3} - \frac{27}{3})$

$= \frac{37\pi}{3}$

Answer: D

Q26. When the curve $y = x^2$ for $0 \leq x \leq 3$ is rotated through 360° about the x-axis, the volume of revolution is [26-Q]

A. $\frac{9}{5}\pi$ units^3 **B.** $\frac{21}{5}$ units^3 **C.** $\frac{61}{5}$ units^3 **D.** $\frac{243}{5}\pi$ units^3

Ans.

$V = \pi\int_0^3 (x^2)^2\, dx = \pi\int_0^3 x^4\, dx$

$= \pi [\frac{x^5}{5}]_0^3$

$= \frac{243\pi}{5}$

Answer: D

2 MARKS

11.2 Definite Integral

Q1. Evaluate the integral $\int_0^2 (-3x + 2)\, dx$. [25-Q]

Ans.

$\int_0^2 (-3x+2)\,dx = [-\frac{3x^2}{2} + 2x]_0^2$

$= (-6+4) - 0$

$= -2$

Answer: −2

Q2. Find the value of $\int_0^{\pi} |\sin x|\,dx$. [24-Q]

Ans.

$\sin x \geq 0$ on $[0, \pi]$, so $|\sin x| = \sin x$ there.

$\int_0^{\pi} |\sin x|\,dx = \int_0^{\pi} \sin x\,dx = [-\cos x]_0^{\pi}$

$= -\cos\pi + \cos 0$

$= 1 + 1 = 2$

Answer: 2

Q3. Find the value of $\int_0^{\pi} |\sin 2x|\,dx$. [24-Q]

Ans.

$\sin 2x \geq 0$ on $[0, \frac{\pi}{2}]$ and $\sin 2x \leq 0$ on $[\frac{\pi}{2}, \pi]$. Split the interval:

$\int_0^{\pi} |\sin 2x|\,dx = \int_0^{\pi/2} \sin 2x\,dx - \int_{\pi/2}^{\pi} \sin 2x\,dx$

$= [-\frac{1}{2}\cos 2x]_0^{\pi/2} - [-\frac{1}{2}\cos 2x]_{\pi/2}^{\pi}$

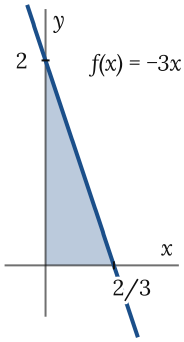
$= (\frac{1}{2} + \frac{1}{2}) - (-\frac{1}{2} - \frac{1}{2})$

$= 1 + 1 = 2$

Answer: 2

11.3 Area between the curve and x-axis

Q4. Find the shaded area.



[24-Q]

Ans.

The line meets the x-axis where $-3x + 2 = 0$, i.e. $x = \frac{2}{3}$. The shaded region lies between $x = 0$ and $x = \frac{2}{3}$.

$A = \int_0^{2/3} (-3x+2)\,dx$

$= [-\frac{3x^2}{2} + 2x]_0^{2/3}$

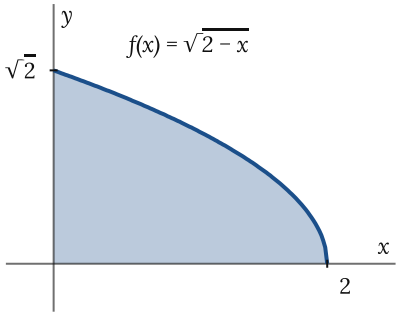
$= -\frac{3}{2}\cdot\frac{4}{9} + \frac{4}{3}$

$= -\frac{2}{3} + \frac{4}{3}$

$= \frac{2}{3}$

Answer: $\frac{2}{3}$

Q5. Find the shaded area.



[25*-Q]

Ans.

The shaded region lies between $x = 0$ and $x = 2$ under $f(x) = (2-x)^{1/2}$.

$A = \int_0^2 (2-x)^{1/2}\,dx$

$= [-\frac{2}{3}(2-x)^{3/2}]_0^2$

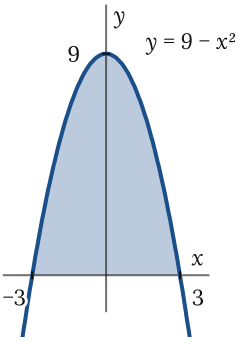
$= 0 + \frac{2}{3}\cdot 2^{3/2}$

$= \frac{2}{3}\cdot 2\sqrt{2}$

$= \frac{4\sqrt{2}}{3}$

Answer: $\frac{4\sqrt{2}}{3}$

Q6. Find the shaded area of the following figure.



[25-Q]

Ans.

The curve cuts the x-axis where $9 - x^2 = 0$, so $x = \pm 3$.

$A = \int_{-3}^{3} (9 - x^2)\,dx$

$= [9x - \frac{x^3}{3}]_{-3}^{3}$

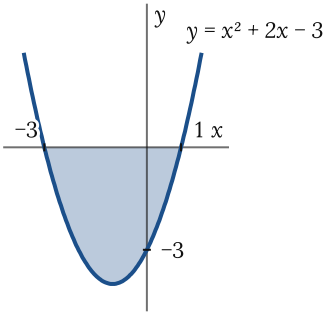
$= (27 - 9) - (-27 + 9)$

$= 18 + 18$

$= 36$

Answer: 36

Q7. Find the shaded area.



[24-Q]

Ans.

$x^2 + 2x - 3 = (x+3)(x-1)$, so the curve cuts the x-axis at $x = -3$ and $x = 1$. Between them the curve is below the x-axis, so $A = \int (0 - f(x))\,dx$.

$A = \int_{-3}^{1} -(x^2 + 2x - 3)\,dx$

$= [-\frac{x^3}{3} - x^2 + 3x]_{-3}^{1}$

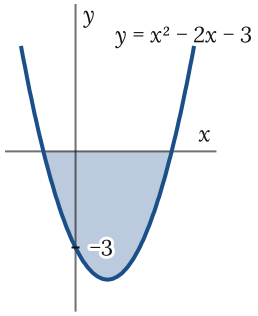
$= (-\frac{1}{3} - 1 + 3) - (9 - 9 - 9)$

$= \frac{5}{3} + 9$

$= \frac{32}{3}$

Answer: $\frac{32}{3}$

Q8. Find the shaded area.



[26-Q]

Ans.

$x^2 - 2x - 3 = (x - 3)(x + 1)$, so the curve cuts the x–axis at $x = -1$ and $x = 3$, and lies below the x–axis between them.

$A = \int_{-1}^{3} -(x^2 - 2x - 3)\,dx$

$= [-\frac{x^3}{3} + x^2 + 3x]_{-1}^{3}$

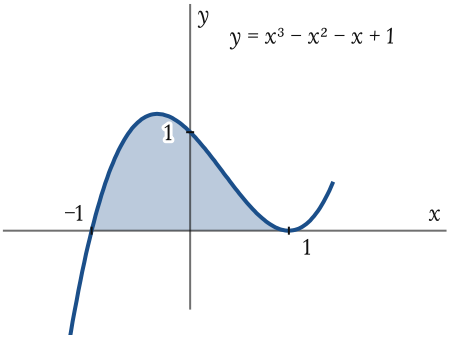
$= (-9 + 9 + 9) - (\frac{1}{3} + 1 - 3)$

$= 9 + \frac{5}{3}$

$= \frac{32}{3}$

Answer: $\frac{32}{3}$

Q9. Find the area of the region enclosed by the curve $y = x^3 - x^2 - x + 1$ and the x–axis.



[26-Q]

Ans.

Factorise: $x^3 - x^2 - x + 1 = x^2(x - 1) - (x - 1) = (x - 1)^2(x + 1)$. The curve meets the x–axis at $x = -1$ and touches it at $x = 1$, and is above the axis on $(-1, 1)$.

$A = \int_{-1}^{1} (x^3 - x^2 - x + 1)\,dx$

$= [\frac{x^4}{4} - \frac{x^3}{3} - \frac{x^2}{2} + x]_{-1}^{1}$

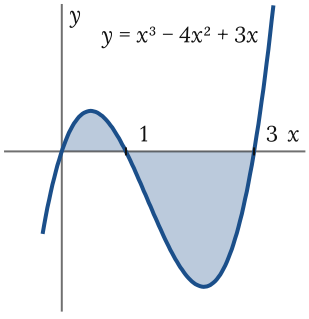
$= (\frac{1}{4} - \frac{1}{3} - \frac{1}{2} + 1) - (\frac{1}{4} + \frac{1}{3} - \frac{1}{2} - 1)$

$= \frac{5}{12} + \frac{11}{12}$

$= \frac{4}{3}$

Answer: $\frac{4}{3}$

Q10. Find the area of the region bounded by the curve $y = x^3 - 4x^2 + 3x$ and the x–axis.



[25-Q]

Ans.

$x^3 - 4x^2 + 3x = x(x - 1)(x - 3)$. The curve is above the x–axis on $[0, 1]$ and below it on $[1, 3]$.

$A = \int_0^1 (x^3 - 4x^2 + 3x)\,dx - \int_1^3 (x^3 - 4x^2 + 3x)\,dx$

$F(x) = \frac{x^4}{4} - \frac{4x^3}{3} + \frac{3x^2}{2}$

$F(0) = 0,\ F(1) = \frac{1}{4} - \frac{4}{3} + \frac{3}{2} = \frac{5}{12}$

$F(3) = \frac{81}{4} - 36 + \frac{27}{2} = -\frac{9}{4}$

$A = (F(1) - F(0)) - (F(3) - F(1))$

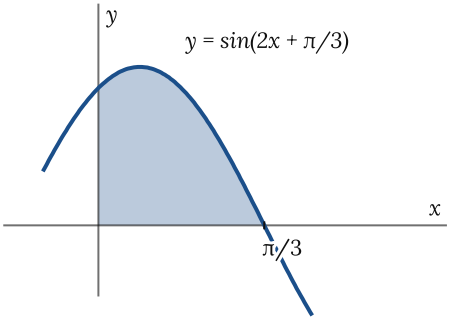
$= \frac{5}{12} - (-\frac{9}{4} - \frac{5}{12})$

$= \frac{5}{12} + \frac{27}{12} + \frac{5}{12}$

$= \frac{37}{12}$

Answer: $\frac{37}{12}$

Q11. The diagram shows part of the curve $y = \sin(2x + \frac{\pi}{3})$. Find the area of the shaded region.



[25-Q]

Ans.

The shaded region lies between $x = 0$ and the first x–intercept, where $2x + \frac{\pi}{3} = \pi$, i.e. $x = \frac{\pi}{3}$.

$A = \int_0^{\pi/3} \sin(2x + \frac{\pi}{3})\,dx$

$= [-\frac{1}{2}\cos(2x + \frac{\pi}{3})]_0^{\pi/3}$

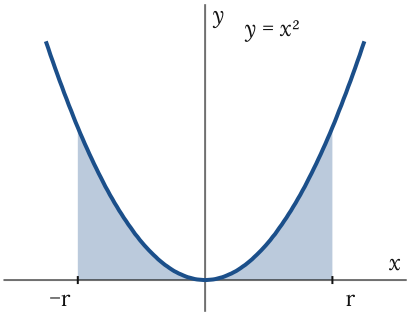
$= -\frac{1}{2}\cos\pi + \frac{1}{2}\cos\frac{\pi}{3}$

$= \frac{1}{2} + \frac{1}{4}$

$= \frac{3}{4}$

Answer: $\frac{3}{4}$

Q12. The shaded area is $6r$ unit^2. Find the value of r.



[25-Q]

Ans.

The region is under $y = x^2$ from $x = -r$ to $x = r$.

$A = \int_{-r}^{r} x^2\,dx = [\frac{x^3}{3}]_{-r}^{r} = \frac{2r^3}{3}$

Given $A = 6r$:

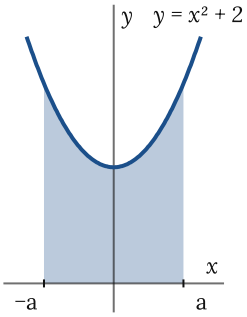
$\frac{2r^3}{3} = 6r$

$r^2 = 9$

Since $r > 0$, $r = 3$.

Answer: r = 3

Q13. The shaded area is $12a$ unit^2. Find the exact value of a.



[25-Q]

Ans.

The region is under $y = x^2 + 2$ from $x = -a$ to $x = a$.

$A = \int_{-a}^{a} (x^2 + 2)\,dx = \left[\frac{x^3}{3} + 2x\right]_{-a}^{a} = \frac{2a^3}{3} + 4a$

Given $A = 12a$:

$\frac{2a^3}{3} + 4a = 12a$

$\frac{2a^3}{3} = 8a$

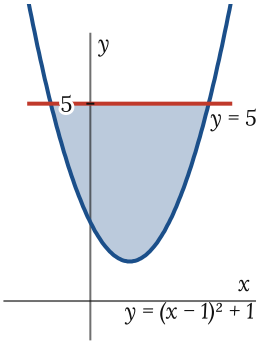
$a^2 = 12$

Since $a > 0$, $a = \sqrt{12} = 2\sqrt{3}$.

Answer: $a = 2\sqrt{3}$

11.4 Area between Two Curves

Q14. Find the shaded area of the following figure.



[24-Q]

Ans.

Find the intersections: $(x-1)^2 + 1 = 5 \rightarrow (x-1)^2 = 4 \rightarrow x = -1$ or $x = 3$. The line $y = 5$ is the upper curve.

$A = \int_{-1}^{3} [5 - ((x-1)^2 + 1)]\,dx$

$= \int_{-1}^{3} [4 - (x-1)^2]\,dx$

$= \left[4x - \frac{(x-1)^3}{3}\right]_{-1}^{3}$

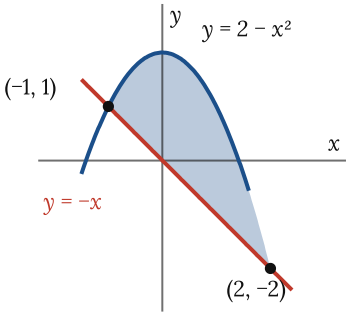
$= \left(12 - \frac{8}{3}\right) - \left(-4 + \frac{8}{3}\right)$

$= 16 - \frac{16}{3}$

$= \frac{32}{3}$

Answer: $\frac{32}{3}$

Q15. In the figure, find the area of the shaded region.



[26-Q]

Ans.

The curves meet at $(-1, 1)$ and $(2, -2)$: $2 - x^2 = -x \rightarrow x^2 - x - 2 = 0 \rightarrow (x-2)(x+1) = 0$. The parabola $y = 2 - x^2$ is on top.

$A = \int_{-1}^{2} [(2 - x^2) - (-x)]\,dx$

$= \int_{-1}^{2} (2 + x - x^2)\,dx$

$= \left[2x + \frac{x^2}{2} - \frac{x^3}{3}\right]_{-1}^{2}$

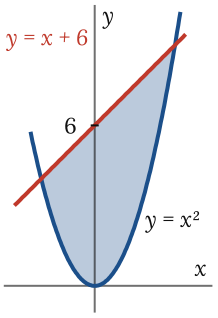
$= \left(4 + 2 - \frac{8}{3}\right) - \left(-2 + \frac{1}{2} + \frac{1}{3}\right)$

$= \frac{10}{3} + \frac{7}{6}$

$= \frac{9}{2}$

Answer: $\frac{9}{2}$

Q16. Find the shaded area.



[25*-Q]

Ans.

Intersections: $x^2 = x + 6 \rightarrow x^2 - x - 6 = 0 \rightarrow (x-3)(x+2) = 0$, so $x = -2$ and $x = 3$.

$A = \int_{-2}^{3} (x + 6 - x^2)\,dx$

$= \left[\frac{x^2}{2} + 6x - \frac{x^3}{3}\right]_{-2}^{3}$

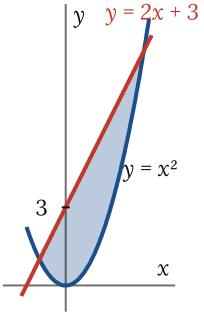
$= \left(\frac{9}{2} + 18 - 9\right) - \left(2 - 12 + \frac{8}{3}\right)$

$= \frac{27}{2} + \frac{22}{3}$

$= \frac{125}{6}$

Answer: $\frac{125}{6}$

Q17. Find the shaded area.



[24-Q]

Ans.

Intersections: $x^2 = 2x + 3 \rightarrow x^2 - 2x - 3 = 0 \rightarrow (x - 3)(x + 1) = 0$, so $x = -1$ and $x = 3$.

$A = \int_{-1}^{3} (2x + 3 - x^2)\,dx$

$= [x^2 + 3x - \frac{x^3}{3}]_{-1}^{3}$

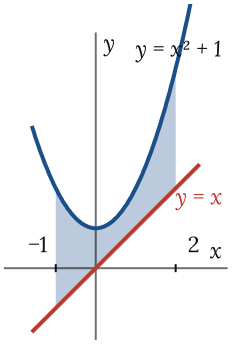
$= (9 + 9 - 9) - (1 - 3 + \frac{1}{3})$

$= 9 + \frac{5}{3}$

$= \frac{32}{3}$

Answer: $\frac{32}{3}$

Q18. Find the area of the shaded region.



[24-Q]

Ans.

The shaded region lies between $x = -1$ and $x = 2$. Since $x^2 + 1 > x$ there, $y = x^2 + 1$ is the upper curve.

$A = \int_{-1}^{2} (x^2 + 1 - x)\,dx$

$= [\frac{x^3}{3} + x - \frac{x^2}{2}]_{-1}^{2}$

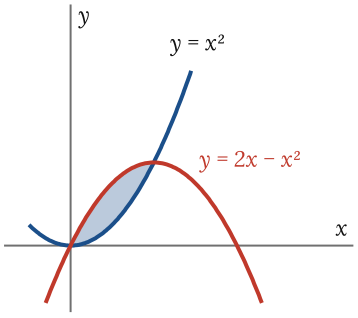
$= (\frac{8}{3} + 2 - 2) - (-\frac{1}{3} - 1 - \frac{1}{2})$

$= \frac{8}{3} + \frac{11}{6}$

$= \frac{9}{2}$

Answer: $\frac{9}{2}$

Q19. Find the shaded area.



[25-Q]

Ans.

Intersections: $x^2 = 2x - x^2 \rightarrow 2x^2 - 2x = 0 \rightarrow x = 0$ or $x = 1$. On $(0, 1)$, $2x - x^2$ is above x^2.

$A = \int_{0}^{1} (2x - 2x^2)\,dx$

$= [x^2 - \frac{2x^3}{3}]_{0}^{1}$

$= 1 - \frac{2}{3}$

$= \frac{1}{3}$

Answer: $\frac{1}{3}$

Q20. Find the area between the line $y = x$ and the curve $y = x^2$. [25*-Q]

Ans.

Intersections: $x = x^2 \rightarrow x(1 - x) = 0$, so $x = 0$ and $x = 1$. On $(0, 1)$ the line $y = x$ is above the curve $y = x^2$.

$A = \int_{0}^{1} (x - x^2)\,dx$

$= [\frac{x^2}{2} - \frac{x^3}{3}]_{0}^{1}$

$= \frac{1}{2} - \frac{1}{3}$

$= \frac{1}{6}$

Answer: $\frac{1}{6}$

Q21. Find the area between the line $y = 2x$ and the curve $y = x^2$. [25-Q]

Ans.

Intersections: $2x = x^2 \rightarrow x(2 - x) = 0$, so $x = 0$ and $x = 2$. On $(0, 2)$ the line is above the parabola.

$A = \int_{0}^{2} (2x - x^2)\,dx$

$= [x^2 - \frac{x^3}{3}]_{0}^{2}$

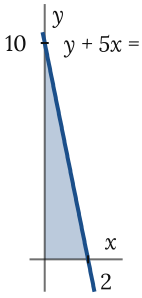
$= 4 - \frac{8}{3}$

$= \frac{4}{3}$

Answer: $\frac{4}{3}$

11.5 Volumes Using Cross-sections

Q22. Find the volume of the solid generated by revolving the shaded region about the x-axis.



[24-Q]

Ans.

The line $y + 5x = 10$ gives $y = 10 - 5x$, which meets the x-axis at $x = 2$.

$V = \pi\int_{0}^{2} (10 - 5x)^2\,dx$

$= \pi\int_{0}^{2} 25(2 - x)^2\,dx$

$= 25\pi\,[-\frac{(2 - x)^3}{3}]_{0}^{2}$

$= 25\pi \cdot \frac{8}{3}$

$= \frac{200\pi}{3}$

Answer: $\frac{200\pi}{3}$

Q23. Find the volume of revolution formed when the curve $f(x) = \sqrt{\sin x}$ for $0 \le x \le \frac{\pi}{2}$ is rotated through 360° about the x-axis. [25-Q, 26-Q]

Ans.

$V = \pi\int_{0}^{\pi/2} (\sqrt{\sin x})^2\,dx = \pi\int_{0}^{\pi/2} \sin x\,dx$

$= \pi\,[-\cos x]_{0}^{\pi/2}$

$= \pi(0 + 1)$

$= \pi$

Answer: π

3 MARKS

11.2 Definite Integral

Q1. Evaluate $\int_{0}^{6} (\frac{1}{3}x + 1)^3\,dx$. [24-Q]

Ans.

Use $\int (ax + b)^n\,dx = \frac{(ax + b)^{n+1}}{a(n + 1)} + C$ with $a = \frac{1}{3}$, $n = 3$.

$\int_{0}^{6} (\frac{1}{3}x + 1)^3\,dx = [\frac{3}{4}(\frac{1}{3}x + 1)^4]_{0}^{6}$

$= \frac{3}{4}(3^4 - 1^4)$

$= \frac{3}{4}(81 - 1)$

$= 60$

Answer: 60

Q2. Evaluate $\int_0^{\sqrt{3}} \{x^3 | \sqrt{x^2+1}\}\, dx$. [25-Q]

Ans.

Substitution $u = x^2 + 1$, $du = 2x\,dx$, $x^2 = u - 1$. When $x = 0$, $u = 1$; when $x = \sqrt{3}$, $u = 4$.

$$\int_0^{\sqrt{3}} \{x^3 | \sqrt{x^2+1}\}\, dx = \frac{1}{2}\int_1^4 \frac{u-1}{\sqrt{u}}\, du$$

$$= \frac{1}{2}\int_1^4 (u^{1/2} - u^{-1/2})\, du$$

$$= \frac{1}{2}\left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^4$$

$$= \frac{1}{2}\left[\left(\frac{16}{3} - 4\right) - \left(\frac{2}{3} - 2\right)\right]$$

$$= \frac{1}{2}\left(\frac{4}{3} + \frac{4}{3}\right)$$

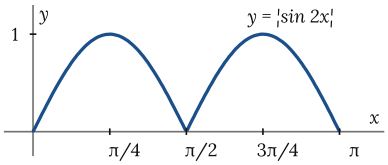
$$= \frac{4}{3}$$

Answer: $\frac{4}{3}$

Q3. Sketch the graph of $|\sin 2x|$ and evaluate $\int_0^{\pi} |\sin 2x|\, dx$. [25-Q]

Ans.

The graph of $|\sin 2x|$ is the graph of $\sin 2x$ with the negative part reflected above the x–axis. Its period is $\frac{\pi}{2}$.



The areas of the two arches are equal, so

$$\int_0^{\pi} |\sin 2x|\, dx = 2\int_0^{\pi/2} \sin 2x\, dx$$

$$= 2\left[-\frac{1}{2}\cos 2x\right]_0^{\pi/2}$$

$$= 2\left(\frac{1}{2} + \frac{1}{2}\right) = 2$$

Answer: 2

Q4. Find the value of $\int_0^{\pi} |\sin 2x|\, dx$. [26-Q]

Ans.

Split at $x = \frac{\pi}{2}$, where $\sin 2x$ changes sign:

$$\int_0^{\pi} |\sin 2x|\, dx = \int_0^{\pi/2} \sin 2x\, dx - \int_{\pi/2}^{\pi} \sin 2x\, dx$$

$$= \left[-\frac{1}{2}\cos 2x\right]_0^{\pi/2} - \left[-\frac{1}{2}\cos 2x\right]_{\pi/2}^{\pi}$$

$$= 1 + 1 = 2$$

Answer: 2

Q5. Find $\int_0^1 x2^x\, dx$. [25-Q]

Ans.

Integration by parts: $u = x$, $dv = 2^x\,dx$, so $du = dx$ and $v = \frac{2^x}{\ln 2}$.

$$\int_0^1 x2^x\, dx = \left[\frac{x2^x}{\ln 2}\right]_0^1 - \int_0^1 \frac{2^x}{\ln 2}\, dx$$

$$= \frac{2}{\ln 2} - \left[\frac{2^x}{(\ln 2)^2}\right]_0^1$$

$$= \frac{2}{\ln 2} - \left(\frac{2}{(\ln 2)^2} - \frac{1}{(\ln 2)^2}\right)$$

$$= \frac{2}{\ln 2} - \frac{1}{(\ln 2)^2}$$

Answer: $\frac{2}{\ln 2} - \frac{1}{(\ln 2)^2}$

Q6. Evaluate the integral $\int_{-e}^{-1} \ln x^2\, dx$. [26-Q]

Ans.

$\ln x^2 = 2\ln|x|$. Put $u = -x$ ($du = -dx$); the limits $-e$ and -1 become e and 1.

$$\int_{-e}^{-1} \ln x^2\, dx = \int_1^e 2\ln u\, du$$

$$= 2[u\ln u - u]_1^e$$

$$= 2[(e - e) - (0 - 1)]$$

$$= 2$$

(Here $\int \ln u\, du = u\ln u - u + C$ by parts.)

Answer: 2

11.3 Area between the curve and x-axis

Q7. Find the area of the region enclosed by $y = x^2 + 2$, the x–axis, $x = 2$ and $x = 3$. [24-Q]

Ans.

$y = x^2 + 2 > 0$, so the curve is above the x–axis.

$$A = \int_2^3 (x^2 + 2)\, dx$$

$$= \left[\frac{x^3}{3} + 2x\right]_2^3$$

$$= (9 + 6) - \left(\frac{8}{3} + 4\right)$$

$$= 15 - \frac{20}{3}$$

$$= \frac{25}{3}$$

Answer: $\frac{25}{3}$

Q8. Find the area of the region enclosed by the curve $y = \sin x$ for $\frac{\pi}{4} \le x \le \pi$. [25-Q]

Ans.

$\sin x \ge 0$ on $[\frac{\pi}{4}, \pi]$, so the curve is above the x–axis.

$$A = \int_{\pi/4}^{\pi} \sin x\, dx$$

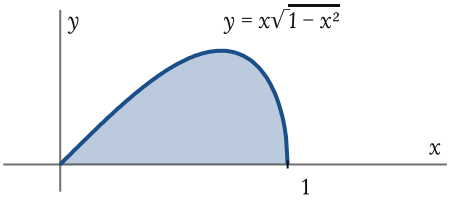
$$= [-\cos x]_{\pi/4}^{\pi}$$

$$= -\cos\pi + \cos\frac{\pi}{4}$$

$$= 1 + \frac{\sqrt{2}}{2}$$

Answer: $1 + \frac{\sqrt{2}}{2}$

Q9. Find the area of the following graph.



[25-Q, 25*-Q]

Ans.

The shaded region lies between $x = 0$ and $x = 1$. Substitution $u = 1 - x^2$, $du = -2x\,dx$; $x = 0 \to u = 1$ and $x = 1 \to u = 0$.

$$A = \int_0^1 x\sqrt{1 - x^2}\, dx = -\frac{1}{2}\int_1^0 u^{1/2}\, du$$

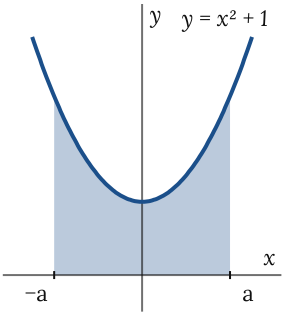
$$= \frac{1}{2}\int_0^1 u^{1/2}\, du$$

$$= \frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_0^1$$

$$= \frac{1}{3}$$

Answer: $\frac{1}{3}$

Q10. The shaded area is $8a$ unit^2. Find the value of a.



[24-Q]

Ans.

The region is under $y = x^2 + 1$ from $x = -a$ to $x = a$.

$A = \int_{-a}^{a} (x^2 + 1)\,dx = \left[\frac{x^3}{3} + x\right]_{-a}^{a} = \frac{2a^3}{3} + 2a$

Given $A = 8a$:

$\frac{2a^3}{3} + 2a = 8a$

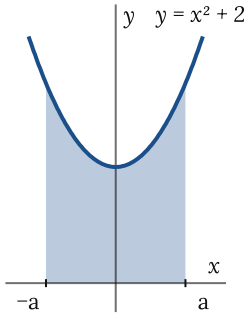
$\frac{2a^3}{3} = 6a$

$a^2 = 9$

Since $a > 0$, $a = 3$.

Answer: a = 3

Q11. The shaded area is $6a$ unit^2. Find the value of $3a^2$.



[25-Q]

Ans.

$A = \int_{-a}^{a} (x^2 + 2)\,dx = \left[\frac{x^3}{3} + 2x\right]_{-a}^{a} = \frac{2a^3}{3} + 4a.$

Given $A = 6a$:

$\frac{2a^3}{3} + 4a = 6a$

$\frac{2a^3}{3} = 2a$

$a^2 = 3$

Therefore $3a^2 = 3 \times 3 = 9$.

Answer: 9

11.4 Area between Two Curves

Q12. Find the area of the region bounded by the curve $y^2 = x$ and the lines $x = 1$, $x = 4$ and the x−axis. [5-Q]

Ans.

For the part above the x−axis, $y = \sqrt{x}$. The region (taken in the first quadrant, above the x−axis) is under $y = \sqrt{x}$ from $x = 1$ to $x = 4$.

$A = \int_1^4 x^{1/2}\,dx$

$= \left[\frac{2}{3} x^{3/2}\right]_1^4$

$= \frac{2}{3}(8 - 1)$

$= \frac{14}{3}$

Answer: $\frac{14}{3}$

Q13. Find the enclosed area between the graphs of $y = x^2$ and $y = x^3$ for $0 \le x \le 1$. [24-Q]

Ans.

For $0 \le x \le 1$, $x^2 \ge x^3$, so $y = x^2$ is the upper curve.

$A = \int_0^1 (x^2 - x^3)\,dx$

$= \left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1$

$= \frac{1}{3} - \frac{1}{4}$

$= \frac{1}{12}$

Answer: $\frac{1}{12}$

Q14. Find the area of the region bounded by the curve $y = x^2$ and $y = 2x - x^2$. [24-Q]

Ans.

Intersections: $x^2 = 2x - x^2 \rightarrow 2x(x - 1) = 0$, so $x = 0$ and $x = 1$. The upper curve is $y = 2x - x^2$.

$A = \int_0^1 [(2x - x^2) - x^2]\,dx$

$= \int_0^1 (2x - 2x^2)\,dx$

$= \left[x^2 - \frac{2x^3}{3}\right]_0^1$

$= 1 - \frac{2}{3}$

$= \frac{1}{3}$

Answer: $\frac{1}{3}$

Q15. Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$. [25-Q]

Ans.

Intersections: $2 - x^2 = -x \rightarrow x^2 - x - 2 = 0 \rightarrow (x - 2)(x + 1) = 0$, so $x = -1$ and $x = 2$. The parabola is above the line between them.

$A = \int_{-1}^{2} (2 - x^2 + x)\,dx$

$= \left[2x - \frac{x^3}{3} + \frac{x^2}{2}\right]_{-1}^{2}$

$= \left(\frac{10}{3}\right) - \left(-2 + \frac{1}{3} + \frac{1}{2}\right)$

$= \frac{10}{3} + \frac{7}{6}$

$= \frac{9}{2}$

Answer: $\frac{9}{2}$

Q16. Find the area between $y = x^2 - 2x + 2$ and $y = 2x - 1$. [26-Q]

Ans.

Intersections: $x^2 - 2x + 2 = 2x - 1 \rightarrow x^2 - 4x + 3 = 0 \rightarrow (x - 1)(x - 3) = 0$, so $x = 1$ and $x = 3$. The line is above the parabola between them.

$A = \int_1^3 [(2x - 1) - (x^2 - 2x + 2)]\,dx$

$= \int_1^3 (-x^2 + 4x - 3)\,dx$

$= \left[-\frac{x^3}{3} + 2x^2 - 3x\right]_1^3$

$= (-9 + 18 - 9) - \left(-\frac{1}{3} + 2 - 3\right)$

$= 0 + \frac{4}{3}$

$= \frac{4}{3}$

Answer: $\frac{4}{3}$

Q17. Find the area of the region bounded by the curve $y = x^4$ and $y = 8x$. [25-Q]

Ans.

Intersections: $x^4 = 8x \rightarrow x(x^3 - 8) = 0$, so $x = 0$ and $x = 2$. On $(0, 2)$ the line is above the curve.

$A = \int_0^2 (8x - x^4)\,dx$

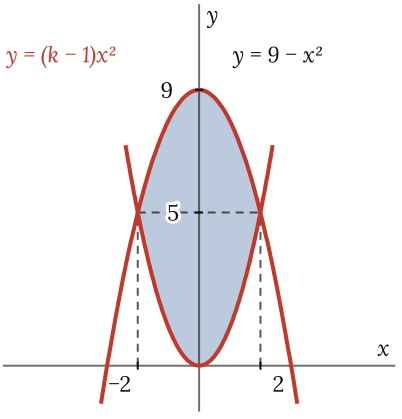
$= \left[4x^2 - \frac{x^5}{5}\right]_0^2$

$= 16 - \frac{32}{5}$

$= \frac{48}{5}$

Answer: $\frac{48}{5}$

Q18. The graphs of $y = (k - 1)x^2$, $k > 1$ and $y = 9 - x^2$ are shown in the diagram. The shaded region has an area of 24. Find the value of k.



[25-Q]

Ans.

From the diagram the two curves meet at $x = \pm 2$. The $9 - x^2$ is the upper curve, and $y = (k-1)x^2$ the lower curve.

$A = \int_{-2}^{2} [(9 - x^2) - (k-1)x^2]\,dx$

$= \int_{-2}^{2} (9 - kx^2)\,dx$

$= [9x - \frac{kx^3}{3}]_{-2}^{2}$

$= (18 - \frac{8k}{3}) - (-18 + \frac{8k}{3})$

$= 36 - \frac{16k}{3}$

Given $A = 24$:

$36 - \frac{16k}{3} = 24$

$\frac{16k}{3} = 12$

$k = \frac{9}{4}$

Check: at $x = 2$ both curves give $9 - 4 = 5$ and $(k-1)\cdot 4 = \frac{5}{4}\cdot 4 = 5$.

Answer: $k = \frac{9}{4}$

11.5 Volumes Using Cross-sections

Q19. Find the volume of revolution formed when the curve $y = \sqrt{4 - x^2}$, for $-2 \le x \le 2$ is rotated through 360° about the x-axis. [25*-Q]

Ans.

$V = \pi\int_{-2}^{2} (4 - x^2)\,dx$

$= \pi [4x - \frac{x^3}{3}]_{-2}^{2}$

$= \pi[(8 - \frac{8}{3}) - (-8 + \frac{8}{3})]$

$= \pi(16 - \frac{16}{3})$

$= \frac{32\pi}{3}$

Answer: $\frac{32\pi}{3}$

Q20. Find the volume of revolution formed when the curve $y = x - x^2$ for $0 \le x \le 1$ is rotated through 360° about the x-axis. [25*-Q]

Ans.

$V = \pi\int_0^1 (x - x^2)^2\,dx = \pi\int_0^1 (x^2 - 2x^3 + x^4)\,dx$

$= \pi [\frac{x^3}{3} - \frac{x^4}{2} + \frac{x^5}{5}]_0^1$

$= \pi(\frac{1}{3} - \frac{1}{2} + \frac{1}{5})$

$= \pi\cdot \frac{10 - 15 + 6}{30}$

$= \frac{\pi}{30}$

Answer: $\frac{\pi}{30}$

Q21. Find the volume of revolution formed when the curve $y = 3x - x^2$ for $1 \le x \le 2$, is rotated through 360° about the x-axis. [26-Q]

Ans.

$V = \pi\int_1^2 (3x - x^2)^2\,dx = \pi\int_1^2 (9x^2 - 6x^3 + x^4)\,dx$

$= \pi [3x^3 - \frac{3x^4}{2} + \frac{x^5}{5}]_1^2$

$= \pi[(24 - 24 + \frac{32}{5}) - (3 - \frac{3}{2} + \frac{1}{5})]$

$= \pi(\frac{32}{5} - \frac{47}{10})$

$= \frac{47\pi}{10}$

(Check: $\frac{32}{5} = \frac{64}{10}$, so $\frac{64}{10} - \frac{17}{10} = \frac{47}{10}$.)

Answer: $\frac{47\pi}{10}$

Q22. Find the volume of revolution formed when the curve $y = 9 - x^2$, for $-3 \le x \le 3$ is rotated through 360° about the x-axis. [25-Q]

Ans.

The integrand is even, so integrate over $[0, 3]$ and double.

$V = \pi\int_{-3}^{3} (9 - x^2)^2\,dx = 2\pi\int_0^3 (81 - 18x^2 + x^4)\,dx$

$= 2\pi [81x - 6x^3 + \frac{x^5}{5}]_0^3$

$= 2\pi(243 - 162 + \frac{243}{5})$

$= 2\pi\cdot \frac{648}{5}$

$= \frac{1296\pi}{5}$

Answer: $\frac{1296\pi}{5}$

Q23. Find the volume of revolution formed when the curve $y = 3x^2 + 2$ for $0 < x < 1$ is rotated through 360° about the x-axis. [25-Q]

Ans.

$V = \pi\int_0^1 (3x^2 + 2)^2\,dx = \pi\int_0^1 (9x^4 + 12x^2 + 4)\,dx$

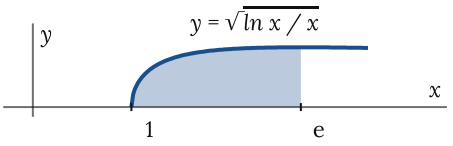
$= \pi [\frac{9x^5}{5} + 4x^3 + 4x]_0^1$

$= \pi(\frac{9}{5} + 4 + 4)$

$= \frac{49\pi}{5}$

Answer: $\frac{49\pi}{5}$

Q24. Find the volume of revolution formed when the curve $y = \sqrt{\frac{\ln x}{x}}$, $1 \le x \le e$, is rotated through 360° about the x-axis.



[24-Q]

Ans.

Substitution $u = \ln x$, $du = \frac{1}{x}\,dx$; $x = 1 \to u = 0$ and $x = e \to u = 1$.

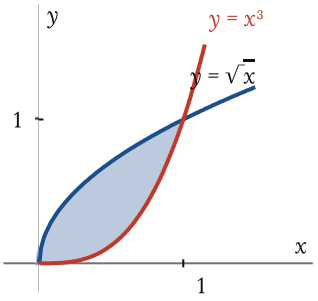
$V = \pi\int_1^e \frac{\ln x}{x}\,dx = \pi\int_0^1 u\,du$

$= \pi [\frac{u^2}{2}]_0^1$

$= \frac{\pi}{2}$

Answer: $\frac{\pi}{2}$

Q25. Sketch the region bounded by the curves $y = \sqrt{x}$, $y = x^3$ and find the volume of the solid generated by revolving this region about the x-axis.



[24-Q]

Ans.

Intersections: $\sqrt{x} = x^3 \to x = x^6 \to x(1 - x^5) = 0$, so $x = 0$ and $x = 1$. On $(0, 1)$, $\sqrt{x}$ is the outer curve and x^3 the inner one. Washer method:

$V = \pi\int_0^1 [(\sqrt{x})^2 - (x^3)^2]\,dx$

$= \pi\int_0^1 (x - x^6)\,dx$

$= \pi [\frac{x^2}{2} - \frac{x^7}{7}]_0^1$

$= \pi(\frac{1}{2} - \frac{1}{7})$

$= \frac{5\pi}{14}$

Answer: $\frac{5\pi}{14}$

5 MARKS

11.2 Definite Integral

Q1. Evaluate $\int_0^{\pi/2} e^{\sin x} \sin 2x\,dx$. [24-Q]

Ans.

$\sin 2x = 2 \sin x \cos x$. Substitution $u = \sin x$, $du = \cos x\,dx$; $x = 0 \to u = 0$ and $x = \frac{\pi}{2} \to u = 1$.

$\int_0^{\pi/2} e^{\sin x} \sin 2x\,dx = 2\int_0^1 ue^u\,du$

By parts: $p = u$, $dq = e^u\,du$, so $dp = du$, $q = e^u$.

$2\int_0^1 ue^u\,du = 2([ue^u]_0^1 - \int_0^1 e^u\,du)$

$= 2(e - [e^u]_0^1)$

$= 2(e - (e - 1))$

$= 2$

Answer: 2

Q2. Find the value of $\int_0^{\pi} x \sin^2 x\,dx$. [25-Q]

Ans.

$\sin^2 x = \frac{1 - \cos 2x}{2}$, so

$\int_0^{\pi} x \sin^2 x\,dx = \frac{1}{2}\int_0^{\pi} x\,dx - \frac{1}{2}\int_0^{\pi} x \cos 2x\,dx$

First part: $\frac{1}{2}\left[\frac{x^2}{2}\right]_0^{\pi} = \frac{\pi^2}{4}$.

Second part by parts ($u = x$, $dv = \cos 2x\,dx$, $du = dx$, $v = \frac{1}{2}\sin 2x$):

$\int_0^{\pi} x \cos 2x\,dx = [\frac{x}{2}\sin 2x]_0^{\pi} - \int_0^{\pi} \frac{1}{2}\sin 2x\,dx$

$= 0 - [-\frac{1}{4}\cos 2x]_0^{\pi}$

$= 0 - (-\frac{1}{4} + \frac{1}{4}) = 0$

Therefore

$\int_0^{\pi} x \sin^2 x\,dx = \frac{\pi^2}{4} - 0$

$= \frac{\pi^2}{4}$

Answer: $\frac{\pi^2}{4}$

11.3 Area between the curve and x-axis

Q3. Find the area of the region bounded by the graph of $y = |x - 2|$ and the x-axis for $1 \le x \le 4$. [25-Q]

Ans.

$|x - 2| = 2 - x$ for $x < 2$ and $x - 2$ for $x \ge 2$, so split the interval at $x = 2$.

$A = \int_1^2 (2 - x)\,dx + \int_2^4 (x - 2)\,dx$

$= [2x - \frac{x^2}{2}]_1^2 + [\frac{x^2}{2} - 2x]_2^4$

$= [(4 - 2) - (2 - \frac{1}{2})] + [(8 - 8) - (2 - 4)]$

$= \frac{1}{2} + 2$

$= \frac{5}{2}$

Answer: $\frac{5}{2}$

11.4 Area between Two Curves

Q4. Find the area between the curve $y = (2 - x)^2 + 1$ and the line $y = 5$. [25-Q]

Ans.

Intersections: $(2 - x)^2 + 1 = 5 \to (2 - x)^2 = 4 \to 2 - x = \pm 2$, so $x = 0$ and $x = 4$. The line $y = 5$ is above the curve between them.

$A = \int_0^4 [5 - (2 - x)^2 - 1]\,dx$

$= \int_0^4 [4 - (2 - x)^2]\,dx$

$= [4x + \frac{(2 - x)^3}{3}]_0^4$

$= (16 - \frac{8}{3}) - (\frac{8}{3})$

$= 16 - \frac{16}{3}$

$= \frac{32}{3}$

Answer: $\frac{32}{3}$

Q5. Find the area of the region enclosed by the curve $y = -x^2 + 2x + 3$ and the line $y = x + 1$. [24-Q]

Ans.

Intersections: $-x^2 + 2x + 3 = x + 1 \to x^2 - x - 2 = 0 \to (x - 2)(x + 1) = 0$, so $x = -1$ and $x = 2$. The parabola (opening downwards) is above the line between them.

$A = \int_{-1}^2 [(-x^2 + 2x + 3) - (x + 1)]\,dx$

$= \int_{-1}^2 (-x^2 + x + 2)\,dx$

$= [-\frac{x^3}{3} + \frac{x^2}{2} + 2x]_{-1}^2$

$= (-\frac{8}{3} + 2 + 4) - (\frac{1}{3} + \frac{1}{2} - 2)$

$= \frac{10}{3} - (-\frac{7}{6})$

$= \frac{9}{2}$

Answer: $\frac{9}{2}$

Q6. Find the area enclosed between the graphs of $y = 4x^2$ and the line $y = x^2 + 3$. [25*-Q]

Ans.

Intersections: $4x^2 = x^2 + 3 \to 3x^2 = 3 \to x = \pm 1$. Between them $x^2 + 3$ is the upper curve (the problem calls it a line, but it is a parabola).

$A = \int_{-1}^1 [(x^2 + 3) - 4x^2]\,dx$

$= \int_{-1}^1 (3 - 3x^2)\,dx$

$= [3x - x^3]_{-1}^1$

$= (3 - 1) - (-3 + 1)$

$= 4$

Answer: 4

Q7. Find the area enclosed between the graphs of $y = \sqrt{x + 3}$ and the line $y = \frac{x + 3}{2}$. [25*-Q]

Ans.

Intersections: $\sqrt{x + 3} = \frac{x + 3}{2}$. Put $u = x + 3$: $\sqrt{u} = \frac{u}{2} \to 4u = u^2 \to u = 0$ or $u = 4$, so $x = -3$ or $x = 1$. For $0 < u < 4$, $\sqrt{u} > \frac{u}{2}$, so the curve is above the line.

$A = \int_{-3}^1 [(x + 3)^{1/2} - \frac{x + 3}{2}]\,dx$

$= [\frac{2}{3}(x + 3)^{3/2} - \frac{(x + 3)^2}{4}]_{-3}^1$

$= (\frac{2}{3} \cdot 8 - \frac{16}{4}) - (0 - 0)$

$= \frac{16}{3} - 4$

$= \frac{4}{3}$

Answer: $\frac{4}{3}$

Q8. (a) Compute the integral $\int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\,dx$.
(b) Find the area bounded between the parabola $y^2 = 4x$ and its latus rectum. [25-Q]

Ans.

(a)

$\int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\,dx = [-\cos x - \sin x]_{\pi/4}^{5\pi/4}$

$= (-\cos\frac{5\pi}{4} - \sin\frac{5\pi}{4}) - (-\cos\frac{\pi}{4} - \sin\frac{\pi}{4})$

$= (\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}) - (-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2})$

$= \sqrt{2} + \sqrt{2}$

$= 2\sqrt{2}$

(b) For $y^2 = 4x$ we have $4a = 4$, so $a = 1$. The focus is $(1, 0)$ and the latus rectum is the line $x = 1$, with end points $(1, \pm 2)$. The region is bounded by the parabola and $x = 1$; by symmetry it is twice the area under $y = 2\sqrt{x}$ for $0 \le x \le 1$.

$A = 2\int_0^1 2x^{1/2}\,dx$

$= 4[\frac{2}{3}x^{3/2}]_0^1$

$= 4 \cdot \frac{2}{3}$

$= \frac{8}{3}$

Answer: (a) $2\sqrt{2}$ (b) $\frac{8}{3}$

11.5 Volumes Using Cross-sections

Q9. The base of the solid is the region bounded by $y = x^2$ and $y = 4$. Find the volume of the solid given that the cross sections perpendicular to the x-axis are equilateral triangles. [24-Q]

Ans.

The curves meet at $x = -2$ and $x = 2$. At position x the base of the triangle is the length $s = 4 - x^2$. The area of an equilateral triangle of side s is $\frac{\sqrt{3}}{4}s^2$, so

$A(x) = \frac{\sqrt{3}}{4}(4 - x^2)^2$

$V = \int_{-2}^{2} A(x)\,dx = \frac{\sqrt{3}}{4}\int_{-2}^{2}(16 - 8x^2 + x^4)\,dx$

$= \frac{\sqrt{3}}{4}\cdot 2\int_{0}^{2}(16 - 8x^2 + x^4)\,dx$

$= \frac{\sqrt{3}}{2}\left[16x - \frac{8x^3}{3} + \frac{x^5}{5}\right]_0^2$

$= \frac{\sqrt{3}}{2}\left(32 - \frac{64}{3} + \frac{32}{5}\right)$

$= \frac{\sqrt{3}}{2}\cdot\frac{256}{15}$

$= \frac{128\sqrt{3}}{15}$

Answer: $\frac{128\sqrt{3}}{15}$

Q10. Find the volume of the solid lies between the planes perpendicular to the x–axis at x = 0 and x = 4. The cross–sections perpendicular to the axis on the interval $0 \le x \le 4$ are squares whose diagonals run from the parabola $y = -\sqrt{x}$ to $y = \sqrt{x}$. [25-Q]

Ans.

At x, the diagonal of the square is $d = \sqrt{x} - (-\sqrt{x}) = 2\sqrt{x}$. Area of a square with diagonal d is $\frac{d^2}{2}$:

$A(x) = \frac{(2\sqrt{x})^2}{2} = \frac{4x}{2} = 2x$

$V = \int_0^4 2x\,dx$

$= [x^2]_0^4$

$= 16$

Answer: 16

Q11. By using the disc method, find the volume of revolution formed when the curve y = f(x) for $a \le x \le b$, is rotated through 360° about the x–axis. (a) f(x) = x + 2; a = 2, b = 3. (b) $f(x) = \frac{1}{x-1}$; a = 2, b = 3. [24-Q]

Ans.

(a)

$V = \pi\int_2^3 (x+2)^2\,dx$

$= \pi\left[\frac{(x+2)^3}{3}\right]_2^3$

$= \pi\left(\frac{125}{3} - \frac{64}{3}\right)$

$= \frac{61\pi}{3}$

(b)

$V = \pi\int_2^3 \frac{1}{(x-1)^2}\,dx$

$= \pi\left[-\frac{1}{x-1}\right]_2^3$

$= \pi\left(-\frac{1}{2} + 1\right)$

$= \frac{\pi}{2}$

Answer: (a) $\frac{61\pi}{3}$ **(b)** $\frac{\pi}{2}$

Q12. Find the volume of revolution formed when the curve y = sin x for $\frac{\pi}{3} \le x \le \pi$, is rotated through 360° about the x axis. [5-Q]

Ans.

Use $\sin^2 x = \frac{1 - \cos 2x}{2}$.

$V = \pi\int_{\pi/3}^{\pi} \sin^2 x\,dx = \frac{\pi}{2}\int_{\pi/3}^{\pi}(1 - \cos 2x)\,dx$

$= \frac{\pi}{2}\left[x - \frac{\sin 2x}{2}\right]_{\pi/3}^{\pi}$

$= \frac{\pi}{2}\left[(\pi - 0) - \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4}\right)\right]$

$= \frac{\pi}{2}\left(\frac{2\pi}{3} + \frac{\sqrt{3}}{4}\right)$

$= \frac{\pi^2}{3} + \frac{\sqrt{3}\pi}{8}$

($\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$, so $\frac{\sin 2x}{2}$ at $x = \frac{\pi}{3}$ equals $\frac{\sqrt{3}}{4}$.)

Answer: $\frac{\pi^2}{3} + \frac{\sqrt{3}\pi}{8}$

Q13. Find the volume of revolution formed when the curve y = sin x for $0 \le x \le \pi$, is rotated through 360° about the x axis. [24-Q]

Ans.

$V = \pi\int_0^{\pi} \sin^2 x\,dx = \frac{\pi}{2}\int_0^{\pi}(1 - \cos 2x)\,dx$

$= \frac{\pi}{2}\left[x - \frac{\sin 2x}{2}\right]_0^{\pi}$

$= \frac{\pi}{2}\cdot\pi$

$= \frac{\pi^2}{2}$

Answer: $\frac{\pi^2}{2}$

Q14. Find the volume of revolution formed when the curve y = sin 2x for $0 \le x \le \frac{\pi}{2}$, is rotated through 360° about the x axis. [24-Q]

Ans.

$\sin^2 2x = \frac{1 - \cos 4x}{2}$.

$V = \pi\int_0^{\pi/2} \sin^2 2x\,dx = \frac{\pi}{2}\int_0^{\pi/2}(1 - \cos 4x)\,dx$

$= \frac{\pi}{2}\left[x - \frac{\sin 4x}{4}\right]_0^{\pi/2}$

$= \frac{\pi}{2}\cdot\frac{\pi}{2}$

$= \frac{\pi^2}{4}$

Answer: $\frac{\pi^2}{4}$

Q15. Find the volume of revolution formed when the curve y = 2x + sin x, the x–axis and the line $x = \pi$ is rotated through 360° about the x axis. [25*-Q]

Ans.

The region lies between x = 0 and $x = \pi$ (the curve meets the x–axis at the origin).

$V = \pi\int_0^{\pi}(2x + \sin x)^2\,dx = \pi\int_0^{\pi}(4x^2 + 4x\sin x + \sin^2 x)\,dx$

Evaluate the three parts:

$\int_0^{\pi} 4x^2\,dx = \left[\frac{4x^3}{3}\right]_0^{\pi} = \frac{4\pi^3}{3}$

$\int_0^{\pi}\sin^2 x\,dx = \frac{1}{2}\int_0^{\pi}(1 - \cos 2x)\,dx = \frac{\pi}{2}$

For the middle part use integration by parts: u = x, dv = sin x dx, du = dx, v = −cos x.

$\int_0^{\pi} 4x\sin x\,dx = 4([-x\cos x]_0^{\pi} + \int_0^{\pi}\cos x\,dx)$

$= 4(\pi + [\sin x]_0^{\pi})$

$= 4(\pi + 0) = 4\pi$

$V = \pi\left(\frac{4\pi^3}{3} + 4\pi + \frac{\pi}{2}\right)$

$= \frac{4\pi^4}{3} + \frac{9\pi^2}{2}$

Answer: $\frac{4\pi^4}{3} + \frac{9\pi^2}{2}$

Q16. Find the volume of revolution generated by revolving the upper half of the circle $x^2 + y^2 = r^2$ about the x–axis. [24-Q]

Ans.

The upper half of the circle is $y = \sqrt{r^2 - x^2}$, for $-r \le x \le r$.

$V = \pi\int_{-r}^{r}(r^2 - x^2)\,dx$

$= \pi\left[r^2x - \frac{x^3}{3}\right]_{-r}^{r}$

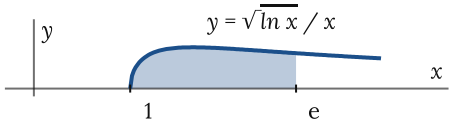
$= \pi\left[\left(r^3 - \frac{r^3}{3}\right) - \left(-r^3 + \frac{r^3}{3}\right)\right]$

$= \pi\cdot\frac{4r^3}{3}$

$= \frac{4\pi r^3}{3}$

Answer: $\frac{4\pi r^3}{3}$ **(volume of a sphere)**

Q17. Find the volume of revolution formed when the curve of $y = \{\sqrt{\ln x}|x\}$, for $1 \le x \le e$ is rotated through 360° about the x–axis.



[26-Q]

Ans.

$V = \pi\int_1^e \frac{\ln x}{x^2}\,dx$. Integration by parts: $u = \ln x$, $dv = x^{-2}\,dx$, so $du = \frac{1}{x}\,dx$ and $v = -\frac{1}{x}$.

$$\int_1^e \frac{\ln x}{x^2}\,dx = [-\frac{\ln x}{x}]_1^e + \int_1^e \frac{1}{x^2}\,dx$$
$$= (-\frac{1}{e} - 0) + [-\frac{1}{x}]_1^e$$
$$= -\frac{1}{e} + (-\frac{1}{e} + 1)$$
$$= 1 - \frac{2}{e}$$

$V = \pi(1 - \frac{2}{e})$

Answer: $\pi(1 - \frac{2}{e})$

Q18. The graph of $y = \ln x$, $x \in [1, e]$ is revolved through 2π about the y-axis. Find the volume of revolution. [5-Q]

Ans.

Rotate about the y-axis, so integrate with respect to y. From $y = \ln x$ we get $x = e^y$; $x = 1$ gives $y = 0$ and $x = e$ gives $y = 1$. The region between the curve $x = e^y$ and the y-axis, for $0 \le y \le 1$, is revolved (disc method with radius $x = e^y$):

$$V = \pi\int_0^1 (e^y)^2\,dy = \pi\int_0^1 e^{2y}\,dy$$
$$= \pi\,[\frac{1}{2}e^{2y}]_0^1$$
$$= \frac{\pi}{2}(e^2 - 1)$$

Answer: $\frac{\pi}{2}(e^2 - 1)$

Q19. Find the volume of revolution generated by revolving the region between $y = x^2$ and $y = 2x$ about the x-axis. [25-Q]

Ans.

Intersections: $x^2 = 2x \to x = 0$ or $x = 2$. On $(0, 2)$, $y = 2x$ is the outer curve and $y = x^2$ the inner curve (washer method).

$$V = \pi\int_0^2 [(2x)^2 - (x^2)^2]\,dx$$
$$= \pi\int_0^2 (4x^2 - x^4)\,dx$$
$$= \pi\,[\frac{4x^3}{3} - \frac{x^5}{5}]_0^2$$
$$= \pi(\frac{32}{3} - \frac{32}{5})$$
$$= \frac{64\pi}{15}$$

Answer: $\frac{64\pi}{15}$

Q20. By using the washer method, find the volume of revolution generated by revolving the regions between $y = x^2$ and $y = 4x - x^2$ about the x-axis. [25-Q, 26-Q]

Ans.

Intersections: $x^2 = 4x - x^2 \to 2x(x - 2) = 0$, so $x = 0$ and $x = 2$. On $(0, 2)$, $y = 4x - x^2$ is the outer curve.

$$V = \pi\int_0^2 [(4x - x^2)^2 - (x^2)^2]\,dx$$
$$= \pi\int_0^2 (16x^2 - 8x^3 + x^4 - x^4)\,dx$$
$$= \pi\int_0^2 (16x^2 - 8x^3)\,dx$$
$$= \pi\,[\frac{16x^3}{3} - 2x^4]_0^2$$
$$= \pi(\frac{128}{3} - 32)$$
$$= \frac{32\pi}{3}$$

Answer: $\frac{32\pi}{3}$

Q21. By using the washer method, find the volume of revolution generated by revolving the region between the semicircle $y = \sqrt{25 - x^2}$ and the line $y = 3$ about the x-axis. [25-Q]

Ans.

Intersections: $25 - x^2 = 9 \to x = \pm 4$. Between them the semicircle is above the line, so the outer radius is $\sqrt{25 - x^2}$ and the inner radius is 3.

$$V = \pi\int_{-4}^4 [(25 - x^2) - 9]\,dx$$
$$= \pi\int_{-4}^4 (16 - x^2)\,dx$$
$$= 2\pi\,[16x - \frac{x^3}{3}]_0^4$$
$$= 2\pi(64 - \frac{64}{3})$$
$$= \frac{256\pi}{3}$$

Answer: $\frac{256\pi}{3}$

Q22. Find the volume of revolution generated by revolving the regions between $y = x^2 + 1$ and $y = -x + 3$ about the x-axis. [25-Q]

Ans.

Intersections: $x^2 + 1 = -x + 3 \to x^2 + x - 2 = 0 \to (x + 2)(x - 1) = 0$, so $x = -2$ and $x = 1$. On $(-2, 1)$ the line $y = 3 - x$ is the outer curve.

$$V = \pi\int_{-2}^1 [(3 - x)^2 - (x^2 + 1)^2]\,dx$$
$$= \pi\int_{-2}^1 (9 - 6x + x^2 - x^4 - 2x^2 - 1)\,dx$$
$$= \pi\int_{-2}^1 (8 - 6x - x^2 - x^4)\,dx$$
$$= \pi\,[8x - 3x^2 - \frac{x^3}{3} - \frac{x^5}{5}]_{-2}^1$$
$$= \pi[(8 - 3 - \frac{1}{3} - \frac{1}{5}) - (-16 - 12 + \frac{8}{3} + \frac{32}{5})]$$
$$= \pi(\frac{67}{15} + \frac{284}{15})$$
$$= \frac{117\pi}{5}$$

(Check: $8 - 3 - \frac{1}{3} - \frac{1}{5} = \frac{67}{15}$, and $-16 - 12 + \frac{8}{3} + \frac{32}{5} = -\frac{284}{15}$.)

Answer: $\frac{117\pi}{5}$

Q23. By using the washer method, find the volume of revolution generated by revolving the regions between $y = x^2 + 2$ and $y = 2x + 5$ about the x-axis. [26-Q]

Ans.

Intersections: $x^2 + 2 = 2x + 5 \to x^2 - 2x - 3 = 0 \to (x - 3)(x + 1) = 0$, so $x = -1$ and $x = 3$. The line is above the parabola between them.

$$V = \pi\int_{-1}^3 [(2x + 5)^2 - (x^2 + 2)^2]\,dx$$
$$= \pi\int_{-1}^3 (4x^2 + 20x + 25 - x^4 - 4x^2 - 4)\,dx$$
$$= \pi\int_{-1}^3 (21 + 20x - x^4)\,dx$$
$$= \pi\,[21x + 10x^2 - \frac{x^5}{5}]_{-1}^3$$
$$= \pi[(63 + 90 - \frac{243}{5}) - (-21 + 10 + \frac{1}{5})]$$
$$= \pi(\frac{522}{5} + \frac{54}{5})$$
$$= \frac{576\pi}{5}$$

(Check: $153 - \frac{243}{5} = \frac{522}{5}$, and $-11 + \frac{1}{5} = -\frac{54}{5}$.)

Answer: $\frac{576\pi}{5}$

Q24. By using the washer method, find the volume of revolution generated by revolving the region between the curve $y^2 = 10x - x^2$ and the straight line $y = 2x$ for $0 \le x \le 2$ about the x-axis. [26-Q]

Ans.

For $0 \le x \le 2$ the upper curve is $y = \sqrt{10x - x^2}$ (outer radius) and the line $y = 2x$ is the inner radius. They meet at $x = 2$, since $10x - x^2 = 4x^2 \to 5x(2 - x) = 0$.

$$V = \pi\int_0^2 [(10x - x^2) - (2x)^2]\,dx$$
$$= \pi\int_0^2 (10x - 5x^2)\,dx$$
$$= \pi\,[5x^2 - \frac{5x^3}{3}]_0^2$$
$$= \pi(20 - \frac{40}{3})$$
$$= \frac{20\pi}{3}$$

Answer: $\frac{20\pi}{3}$